The Unification of Matter, Forces, and Spacetime from the Bost–Connes System

Shao-Wu Hu

Shanghai, China

forest@xindong.com

ORCID: 0009-0009-8300-4691

Abstract

Starting from a finite p-adic reduction of the Bost–Connes system, we investigate the common dynamical origin of matter, mass, and forces. Matter is characterized as stable excitations whose patterns are preserved under propagation and exchange: the relative phases of their persistent evolution yield mass frequencies, exchange between stable states produces resonance responses, and the influence of the background energy distribution on motion produces a metric response.

We construct the complete process from the formation of a stable pattern to its persistent evolution, bringing path spectral responses, node overlaps, and distributions across generations into a single mass generator. The same mass parameter then governs both the rest-frame energy gap and the inertial response at low momentum, while a common background response provides a unified energy scale.

On the basis of these dynamical relations, we use an odd prime p as a structural parameter to establish evaluation relations for mass spectra and coupling constants. With p=7 and the common energy calibration $E_0 = 1$ GeV, we obtain an electroweak scale of 246.19 GeV and an inverse electromagnetic coupling of $\alpha_{\mathrm{EM}}^{-1} = 137.03636$. The absolute relative deviations of the nine charged-fermion masses from the reference central values are all below 1.5%, and those of the three charged leptons are below 0.003%. The neutrino spectrum predicts absolute masses of 5.07, 9.96, and 50.63 meV for the three mass eigenstates, with a mass-squared difference ratio of $\Delta m^2_{31}/\Delta m^2_{21} = 34.52$.

Contents

Chapter 1. Introduction **4**

Chapter 2. Stable States **8**
2.1 Definition and Classification . . . 8
2.2 Structure of Stable States . . . 13
2.3 Formation and Maintenance . . . 20
2.4 Elementary-State Mass . . . 30
2.5 Common Energy Scale . . . 39

Chapter 3. Resonance Forces **47**
3.1 Resonant Exchange . . . 47
3.2 Structure of Resonance Channels . . . 54
3.3 Couplings of Elementary Resonance Forces . . . 58
3.4 Virtual Processes and Effective Couplings . . . 61
3.5 Emergent Resonance Forces . . . 67

Chapter 4. Metric Forces **73**
4.1 Matter Sources . . . 73
4.2 Background Propagation . . . 75
4.3 Maintenance Response . . . 76
4.4 Motion and Force . . . 78
4.5 Strength Calibration . . . 82

Chapter 5. State Space and p-Channels **93**
5.1 Structure of p-Channels . . . 93
5.2 Channel Selection and Physical Correspondence . . . 95

Chapter 6. Energy Scales **99**
6.1 Electroweak Scale . . . 99
6.2 Planck Scale . . . 104
6.3 Nucleon Scale . . . 109

Chapter 7. Force Coupling Constants **113**
7.1 Vertex Spectral Response . . . 113
7.2 Born-Level Couplings . . . 116
7.3 Virtual-Process Corrections . . . 118
7.4 Numerical Values of the Elementary Resonance Forces . . . 122
7.5 Emergent Resonance Forces . . . 124
7.6 The Metric Force Constant . . . 125

Chapter 8. Elementary-State Mass Spectrum **128**
8.1 Phases and Generational Factors . . . 128
8.2 Charged Leptons . . . 132
8.3 Down-Type Quarks . . . 133
8.4 Up-Type Quarks . . . 135
8.5 Neutrinos . . . 137
8.6 Testing the Mass Spectrum . . . 141

Chapter 9. Baryons, Quarks, and Color Confinement **144**
9.1 Collective Baryon Structure . . . 144
9.2 Baryon Quantum Numbers . . . 147
9.3 Total Mass and Maintenance . . . 150
9.4 The Neutron–Proton Mass Difference . . . 153
9.5 Resonance Between Nucleons . . . 156
9.6 Color Confinement and Independent Outputs . . . 157

Chapter 10. Bosons **160**
10.1 Channel Excitations and Condensate Oscillations . . . 160
10.2 The Photon . . . 164
10.3 W and Z Bosons . . . 165
10.4 Gluons . . . 167
10.5 Pions . . . 168
10.6 Condensate Oscillations and the Higgs . . . 170

Chapter 11. Spectral Completeness **175**
11.1 Elementary States and Generation Structure . . . 175
11.2 The Emergent-State Sequence . . . 176
11.3 Forces and Channel Excitations . . . 177
11.4 Condensate Oscillations . . . 178
11.5 Completeness and Testable Conclusions . . . 178

Chapter 12. Conclusions **181**
12.1 Common Dynamics and Evaluations . . . 181
12.2 Numerical Summary . . . 183
12.3 Testable Relations . . . 185
12.4 Closing Remarks . . . 187

Appendix A. Index of Constructions and Dependencies **188**
A.1 Objects and Spectral Inputs . . . 188
A.2 Formation and Common Readings . . . 188
A.3 Participation and Resonance Responses . . . 189
A.4 Elementary-State Mass Spectrum . . . 189
A.5 Collective States and Physical Comparisons . . . 190

References **191**

§1 Introduction

The Bost–Connes system connects the multiplicative structure of number theory with time evolution in quantum statistical mechanics [1]. Within its finite p-adic reduced state space, *Dynamics of the Bost–Connes State Space* (Hu, 2026) [2] establishes a joint cycle centered on exchange between an excitation and the background. The background provides a reference for identifying excitation patterns, whose form is determined by the relative amplitudes and phases at different positions. When a specified propagation action multiplies all components by the same nonzero complex factor, the pattern is preserved. The corresponding mode is called a stable state and takes the form of a stable standing-wave pattern. In actual exchange, the returning component continues the original excitation; its pattern and the share that persists are determined by the return action and the exit process, respectively.

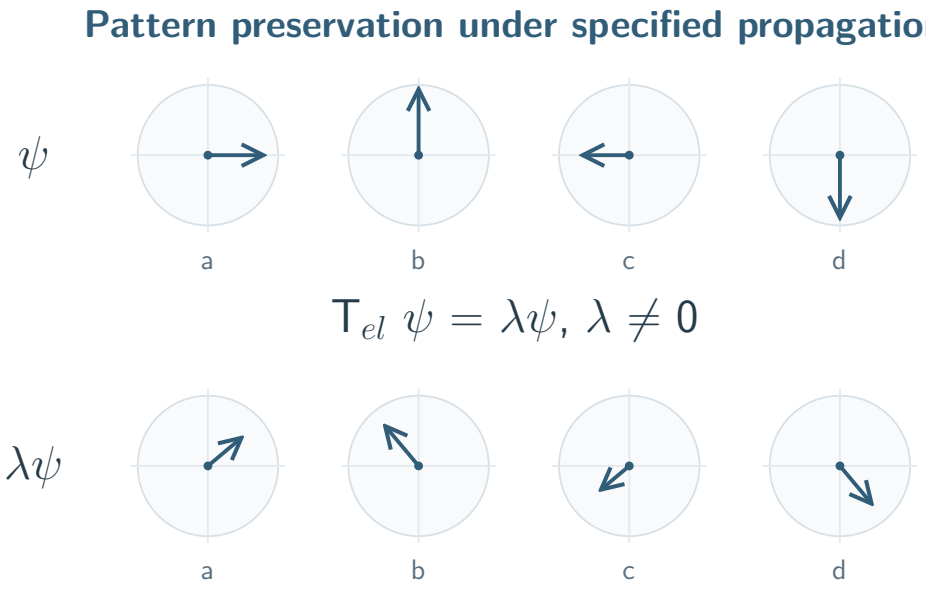


Figure 1-1. Pattern preservation of a stable state. Complex amplitudes at several positions are shown schematically: arrow lengths represent magnitudes, and arrow directions represent phases. The specified propagation action multiplies all components by the same nonzero complex factor, preserving the relative amplitudes and phases between positions.

Free evolution, formation exchange, and pattern reconstruction constitute a complete cycle. Free evolution accumulates phase. Formation exchange allows the original excitation to return or exit into the background, and also allows the background to form new excitations. The returning component, newly formed component, and restored background together constitute the joint final state. Pattern reconstruction gives its position distribution and coherences, and the resulting state continues into the next cycle. Changes in the position distribution provide a reading of motion, while repeated exchange determines formation, survival, and thermalization. The decay products of specific particles are identified in the corresponding physical

processes.

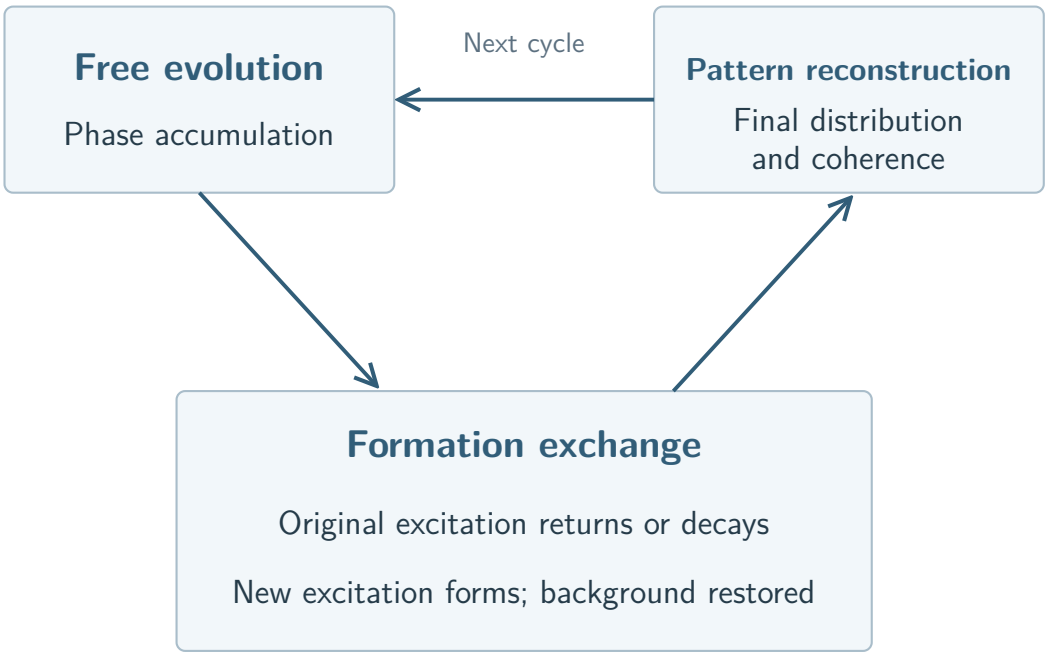


Figure 1-2. The complete cycle of a stable state and the background. Formation exchange includes return and decay of the original excitation, formation of new excitations, and background restoration. Pattern reconstruction expresses the pattern of the same joint final state; it adds neither an interaction nor a duration. The system state becomes the input to the next cycle.

These changes become comparable events through references and records internal to the whole system. A readable record exists when distinctions between processes remain stored in the environment or auxiliaries after a complete cycle and can be distinguished by mutually orthogonal pointer subspaces. Observational spacetime is a readout of the same state-space process: actual elapsed durations calibrate time, and position pointers calibrate space. Its geometric structure is used in Chapter 4. Cycles continue to advance, whereas a selected event may occur only after several cycles; cycle counts and record counts are therefore kept separate.

The pattern established during the formation of a stable state participates in subsequent exchange and determines the maintenance action available to each direction. The rate at which the maintenance phase accumulates relative to the same background gives the mass reading. A stable state exchanges with other excitations through directions that can be matched, producing a resonance response. The change it induces in the available states of the common background propagates to other positions, modifies the maintenance phases of local stable states, and produces a metric response. Mass and the two types of force are thus linked within the same process of formation, maintenance, and exchange. The records retained in the complete output determine how histories compose, while actual elapsed durations

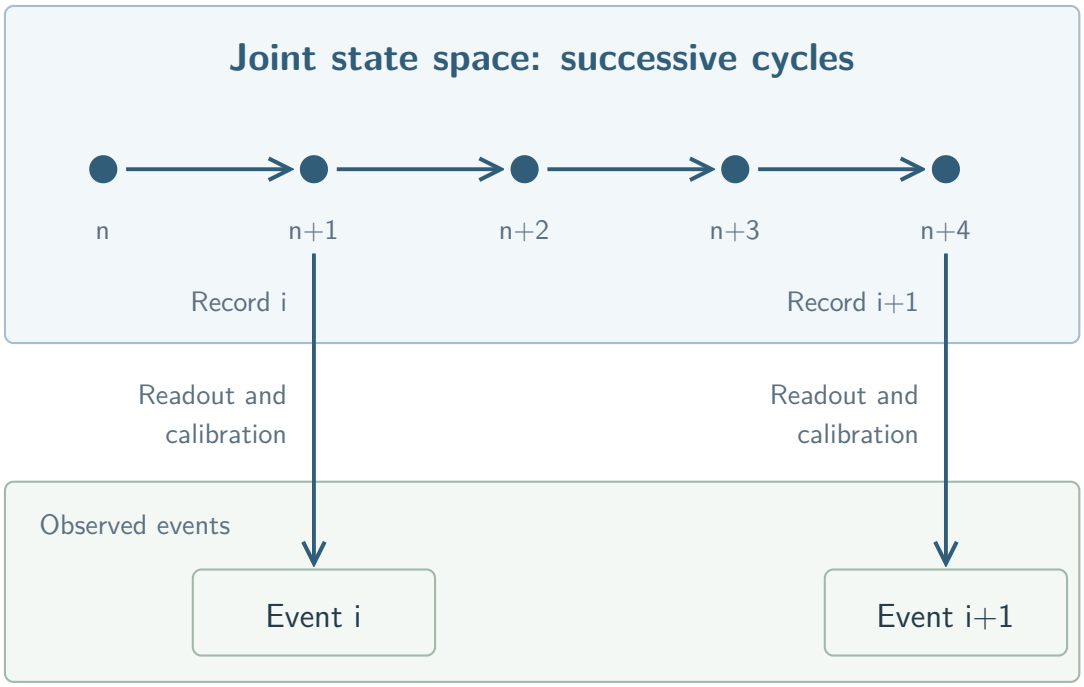


Figure 1-3. Records and events. Records are stored within the joint state space and correspond to events through actual readout and calibration. Only two records of the selected class are shown. Cycles continue between these records, so cycles and selected records must be counted separately.

convert per-cycle actions into comparable frequencies and event rates.

Different stable states have different directions and distributions and participate in exchange in different ways. How do these internal differences become observable differences between specific forms of matter? Evaluating masses and forces therefore involves three questions:

1. **Why do stable states have mass?** Which reading of persistent evolution can characterize mass, and why do different patterns produce different frequencies?

2. **How do stable states interact?** Through which endpoints do the complete carriers exchange, and what determines the form, strength, and range of the interaction?

3. **How can these masses and forces be calculated?** How do paths, participation shares, and a common calibration enter the evaluation of masses and couplings and establish their correspondence with specific forms of matter?

The structure of stable states, parameterized by an odd prime p, together with their formation and exchange conditions, provides the basis for evaluating masses and couplings. Specific numerical values also depend on the structural parameter, the identification with matter, and calibration of the common energy scale. Calculations of mass spectra and coupling constants are thereby connected to the internal structure of stable states and to actual processes.

§2 Stable States

Matter retains a recognizable structure while participating in interactions. In the finitely reduced state space, this structure takes the form of an excitation pattern of amplitudes and phases. Exchange with the background forms the pattern, and subsequent outflow and return sustain its existence; the same process can also leave a phase relative to the background. To obtain mass from this picture, pattern formation, persistent action, and phase readout must be connected within a single joint evolution. The central question is how the spectral structure of an excitation enters actual exchange and becomes the frequency at which its original pattern is maintained.

2.1 Definition and Classification

Fix an odd prime p and a finite reduction depth K≥1. The address ring is the standard finite quotient of the p-adic integers [3]. Each residue class modulo p^K corresponds to a lattice site, and the complex amplitudes at all sites together constitute a pattern in state space:

$$X_{p,K} = \mathbb{Z}/p^K\mathbb{Z}, \qquad \mathcal{H} = L^2(X_{p,K}, \mathbb{C}), \qquad \dim_{\mathbb{C}} \mathcal{H} = p^K. \tag{2.1}$$

The background and excitations belong to the same space. With the normalized counting measure assigning weight $1/p^K$ to each site, the constant function b(x)=1 is a normalized pattern with equal amplitudes and phases; we use it as the background reference. Excitations are distinguished relative to this reference by the amplitude and phase relations between sites. This first specifies the reference for comparing patterns; the background preparation actually used in a cycle is supplied by the particular exchange process.

Nonzero sites are grouped into shells according to their powers of p, with zero forming a separate terminal shell. The equal-amplitude, equal-phase patterns on the shells span the shell-average subspace $\mathcal{C}$; their orthogonal complement is the direction space $\mathcal{M}$. A general directional excitation may contain several components. Different propagation responses change their relative amplitudes and phases. If the entire excitation always acquires only a common nonzero complex factor, these relative relations remain unchanged, producing a stable standing-wave pattern.

Pattern preservation is illustrated in Figure 1-1.

Pattern preservation of a directional excitation is assessed through the shell-preserving response T_{el}. Fix a regular spectral point s_* where it is well defined. A nonzero vector $\psi \in \mathcal{M}$ satisfying

$$T_{\mathrm{el}}(s_*)\psi = \lambda\psi, \qquad \lambda \neq 0, \tag{2.2}$$

is called a **stable state** under this response. The common factor preserves both the shape of the position distribution and the coherence between positions. Representing the complete pattern by its pure-state projector gives

$$P_{[\psi]} = \frac{|\psi\rangle\langle\psi|}{\|\psi\|^2}, \qquad P_{[T_{\mathrm{el}}(s_*)\psi]} = P_{[\psi]}. \tag{2.3}$$

Conversely, a nonzero vector after one response with the same pure-state projector lies in the same one-dimensional subspace as the original vector, and therefore

differs from it by a nonzero complex factor. The condition above is thus equivalent to pattern preservation. Repeating the same response merely accumulates this common factor. A stable state here is a pattern-preserving excitation; a stationary state in the discussion of thermalization is a fixed point of the complete reduced cycle.

This criterion establishes stability under the specified response. In an actual cycle, thermal exchange, retrieval, and control must also be included to determine whether the returned component preserves the original pattern. The fraction of the original excitation surviving from cycle to cycle may decrease, while the background may form new excitations. Pattern preservation, survival of the original excitation, and total occupation are therefore described separately.

Elementary States

An elementary state is a directional excitation with a definite phase arrangement on the unit group. It extends over the entire unit shell: all unit sites have equal amplitude moduli, their relative phases are determined by the chosen direction, and amplitudes outside the unit group vanish. Different elementary states can consequently have the same position probability distribution but different coherent patterns.

To construct this pattern, denote the unit group at depth K by $G_{p,K}$=$(\mathbb{Z}/\mathrm{p}^K\mathbb{Z})\times$, containing N=$\mathrm{p}^{K-1}$(p−1) sites. The natural reduction π:$G_{p,K}\to\mathrm{G}_{p,1}$ maps each unit site to its residue class modulo p. Choose a nontrivial multiplicative character χ of the species-layer unit group $\mathrm{G}_{p,1}$. In the normalized site basis, the corresponding direction vector is

$$|\psi_\chi\rangle = \frac{1}{\sqrt{N}} \sum_{u\in G_{p,K}} \chi(\pi(u))|u\rangle. \tag{2.4}$$

A character satisfies χ(ab)=χ(a)χ(b) and has values of unit modulus, so this vector is normalized. The sum of a nontrivial character over the group is zero. Pullback along π repeats every residue class equally many times, so the sum remains zero. The vector is therefore orthogonal to the unit-shell average state. It vanishes on all other shells and is orthogonal to their average states as well, hence belongs to $\mathcal{M}$.

The phases follow the order of successive powers of a generator of the species-layer unit group. Deeper addresses in the same residue class modulo p share the same phase. Section 2.2.2 develops the coordinates and repetition structure.

This distinction resides in the coherence between positions. The ratio of amplitudes at two unit sites u and v is $\chi(\pi(\mathrm{uv}^{-1}))$, so the full character determines their relative phase. Position probabilities retain only squared amplitude moduli and omit these relations. Identifying an elementary state therefore also requires the coherence information in its pattern.

The multiplicativity of the character makes the pattern invariant up to a common factor under unit-multiplication pullbacks. Applying f(x)$\mapsto$f(ax) multiplies every nonzero component by the same $\chi(\pi(\mathrm{a}))$. The shell-preserving response acts diagonally along these directions, with normalized eigenvalue given by the resonance ratio:

$$T_{\mathrm{el}}(s_*)|\psi_\chi\rangle = R_\chi(s_*)|\psi_\chi\rangle, \qquad R_\chi(s_*) = \frac{L(s_*,\chi)}{\zeta(s_*)}. \tag{2.5}$$

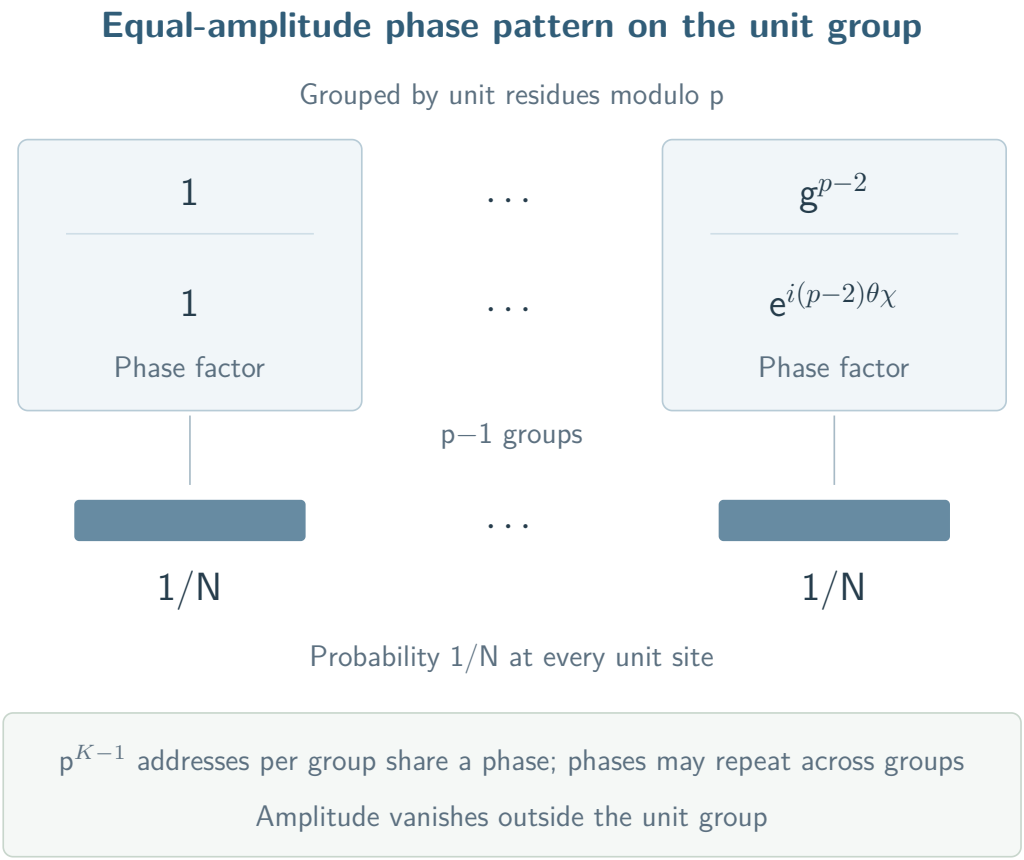


Figure 2-1. Phase arrangement of an elementary state. Each column in the upper row represents a group of unit positions at the species layer. The equal bars below indicate the same position probability at each unit site, not the probability of an entire group. The character determines $\theta\chi$; neither p nor a particular species is fixed in the diagram. Changing the character can change the phase pattern without changing position probabilities.

At the chosen regular spectral point, this direction satisfies the stable-state criterion whenever $R_\chi(s_*) \neq 0$ and is called an **elementary state**. Repeated application of the same shell-preserving response accumulates only the common factor $R_\chi(s_*)^m$, leaving the normalized pattern unchanged.

The species-layer unit group is cyclic of order p−1 and has exactly p−1 characters. Removing the trivial character χ_0, which gives the equal-amplitude, equal-phase pattern on the unit shell, leaves p−2 nontrivial species labels. Their directions are mutually orthogonal. If all these responses are nonzero at the chosen spectral point, there are p−2 elementary-state directions. A zero-response direction still belongs to $\mathcal{M}$ but does not satisfy nonzero pattern preservation at that point.

This count concerns only directions on the unit shell pulled back from the species layer. The complete finite state space also contains directions of higher conductor and on other valuation shells. If several directions have the same nonzero complex response, their coherent superpositions preserve the pattern as well. Spectral degeneracy enlarges the corresponding stable subspace; pattern preservation of an individual elementary direction does not require a simple spectrum.

The character fixes the spectral direction of a species. Section 2.3 prepares shell-average components and conjugate directions as a coherent source, then combines them with the target and auxiliaries into a persistent carrier whose relative phase provides the physical mass reading.

Emergent States

Elementary states distinguish species through nontrivial phase arrangements on the unit shell. The collective picture used for emergent states involves the whole space: units, nonzero nonunits, and zero all participate with equal amplitudes and phases. This coherent pattern over the entire space describes the structure of collective condensation and provides the starting point for emergent states.

Write d=p^K. In the normalized site basis, the collective pattern is

$$|b\rangle = \frac{1}{\sqrt{d}} \sum_{x \in X_{p,K}} |x\rangle, \qquad d = p^K. \tag{2.6}$$

Every position has probability 1/d, and the relative phase between any two positions is zero. This coherent pattern extends over the entire ring. Under the normalized counting measure used here, its function is precisely b(x)=1; the site-basis coefficients $1/\sqrt{\mathrm{d}}$ agree with the normalization convention for function values.

Participation over the whole space can be expanded shell by shell. Let e_j be the normalized equal-amplitude, equal-phase vector on shell j and h_j=$|\mathrm{S}_j|$/d its share of all sites. Then

$$b = \sum_{j=0}^{K} \sqrt{h_j}\, e_j, \qquad h_j = \frac{|S_j|}{d}, \qquad \sum_{j=0}^{K} h_j = 1. \tag{2.7}$$

Thus b belongs to the shell-average subspace $\mathcal{C}$ and is orthogonal to $\mathcal{M}$. The trivial character χ_0 on the unit group gives only e_0; b also includes coherent components on all remaining shells and at zero. The unit-shell share is h_0=1−1/p, and the zero-element share is h_K=p^{-K}. These shares describe how the components form

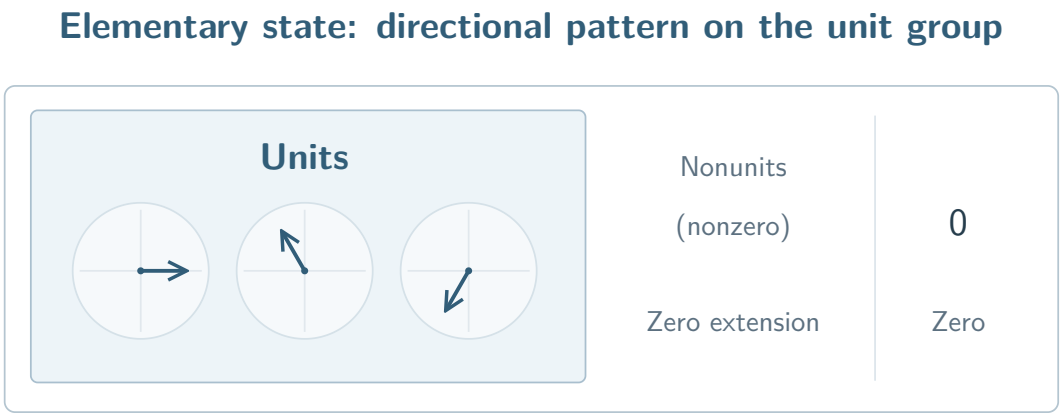


Figure 2-2. An elementary direction has a character phase pattern on the unit group and vanishes outside it. The collective description starts from the equal-amplitude, equal-phase pattern over the entire ring. Each point represents a position of the indicated category; the number of points does not count modes.

one pattern; the trivial direction on the unit shell cannot itself be identified with the whole-space pattern.

Scaling invariance fixes the relative amplitudes of the shells. Let D be the scaling pullback of amplitudes, (Df)(x)=f(px). After K successive applications, every address maps to zero modulo p^K, so D^Kf=f(0)b. A scaling-invariant pattern therefore satisfies f=D^Kf=f(0)b. Conversely, the constant function plainly satisfies Db=b. Hence

$$Db = b, \qquad \ker(D - I) = \mathbb{C}b. \tag{2.8}$$

Unit-group invariance requires equal amplitudes and phases within each shell but allows different amplitudes on different shells. Requiring scaling invariance as well selects the unique constant-function direction over the whole space. This uniqueness concerns the direction of the fixed pattern; overall normalization and common phase still describe the same pure state.

This pattern also has a definite role in the full spectral response. For every positive integer n, the multiplicative pullback T_nf(x)=f(nx) satisfies T_nb=b. In the convergence region Re s>1, the unnormalized spectral transfer and its normalized form therefore obey

$$\widetilde{\mathcal{B}}(s)b = \left(\sum_{n\geq 1} n^{-s}\right) b = \zeta(s)b, \qquad \mathcal{B}(s)b = b. \tag{2.9}$$

The normalized identity continues to hold at regular continuation points where it is well defined. Every multiplicative term gives the same pattern for b, so the spectral sum accumulates only a common factor. Nontrivial elementary directions instead carry the phase weights of their characters. The two types of pattern thus have different spectral responses.

This collective response exhibits the pole structure of ζ near s=1, arising from the infinite spectral sum over all positive integers. The pole disappears if the sum is truncated to finitely many terms.

The actual dynamics also includes phase evolution between shells. With shell energies $H_K e_j$=j log p · e_j, free evolution of the constant-function pattern is

$$b(t) = e^{-itH_K} b = \sum_{j=0}^{K} \sqrt{h_j}\, e^{-itj \log p} e_j. \tag{2.10}$$

Shell occupations remain unchanged, while relative phases generally vary with time. Fixedness under the spectral response and maintenance in the complete cycle therefore have distinct criteria. The latter requires examining the actual exchange, returned components, and their coherence to establish whether the same pattern is recovered at the end of a cycle.

The background reference and a collective excitation may have the same system pattern. Their roles in the joint process and their readings relative to the reference determine their respective physical identities.

2.2 Structure of Stable States

The structure of a stable state determines which components can participate in subsequent exchange. Shells specify the extent of participation, characters arrange

phases within each shell, and position pairing organizes local occupation and coherence. A collective pattern additionally requires coherence between shells. These structures are developed below as the common basis for formation, matching, and mass readings.

2.2.1 Shells and Directions

Shells group sites of equal valuation, allowing participation in a pattern to be described layer by layer. We use the standard structure of discrete valuation and unit decomposition [4]. A nonzero residue class is uniquely written x=p^{j}u with u a unit modulo p^{K-j} and 0≤j<K. Fixing j gives the valuation shell S_j; zero forms the terminal shell $S_K = \{0\}$. Each position belongs to exactly one shell, whose size is

$$|S_j| = (p-1)p^{K-j-1} \quad (0 \le j < K), \qquad |S_K| = 1. \tag{2.11}$$

Different phase patterns are possible within a nonzero shell. The equal-amplitude, equal-phase pattern gives the shell-average state e_j, while nontrivial characters arrange phases according to the multiplicative structure of the unit group. Both are uniformly distributed in amplitude on the shell; their distinction lies in relative phases between sites. In the normalized site basis, the corresponding direction is

$$|f_{j,\chi}\rangle = \frac{1}{\sqrt{|S_j|}} \sum_{u \in G_{p,K-j}} \chi(u)|p^j u\rangle. \tag{2.12}$$

The trivial character gives e_j. Nontrivial characters are orthogonal to it and form an orthonormal basis for the remaining directions on the shell. The component of any pattern on this shell can be expanded in these directions [5]. Different shells have disjoint supports and are mutually orthogonal. All shell-average states span a (K+1)-dimensional subspace, whose orthogonal complement is the direction space:

$$\mathcal{H} = \mathcal{C} \oplus \mathcal{M}, \qquad \dim \mathcal{C} = K+1, \qquad \dim \mathcal{M} = p^K - K - 1. \tag{2.13}$$

A direction $f_{j,\chi}$ thus has both a shell label and a character label. The former determines its support and shell energy, and the latter its phase arrangement within the shell. Superposition coefficients then give the amplitude distribution of a general pattern. Which directions satisfy the stable-state criterion still depends on the nonzero-response condition of Section 2.1; p^K−K−1 counts the entire direction space.

A phase rule at a shallower level can be repeated on deeper unit groups by natural reduction. The smallest modulus p^c on which the rule can be defined is the conductor of the character. If θ denotes its primitive character, $f_{j,\theta}$ can be formed on shells j=0,…,K−c. Species labels, shells, and address resolution therefore have distinct meanings. For elementary states, we first take species-layer directions on the unit shell j=0 with conductor p, retaining the full space for formation and exchange.

Species directions, vacuum-counting addresses, and propagation wave numbers play different roles in the exchanges below. Characters label internal species directions; counting addresses distinguish participating, occupied, and available background modes. Position is a logical degree of freedom of matter and its local background, and propagation wave number is its Fourier description under translations.

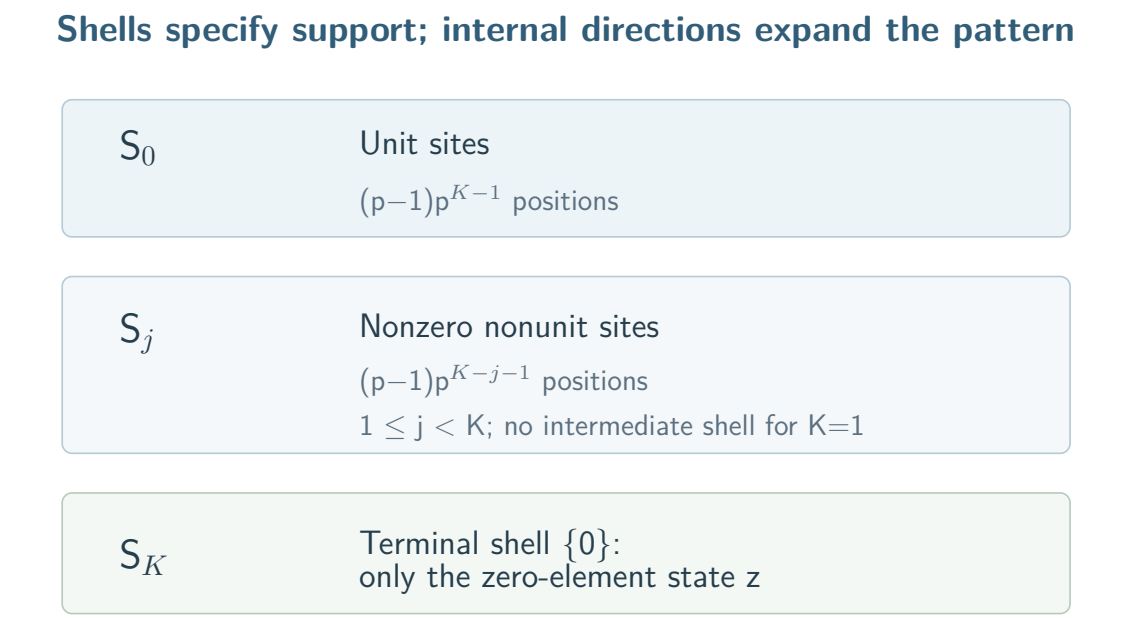


Figure 2-3. The full state space is stratified by valuation shells. Each nonzero shell splits into the direction spanned by its shell-average state and the nontrivial directions. The terminal shell contains a single position. The diagram shows arithmetic supports and orthogonal decomposition; shells are not distance layers in observational space.

Although all three may be constructed from a common group structure, their actual maps and tensor factors must be specified separately. A formation pattern in a propagation zero mode does not mean that matter has no position; species shares at formation do not directly give occupation probabilities at distant positions. Chapter 4 controls contact and transport by actual position, while the configuration encoding in Section 6.2 is read through its own actual matching.

2.2.2 Phase Structure of Elementary States

Elementary directions have the same position occupations on the unit shell, so occupation alone cannot distinguish them. Their difference is how phase changes as one traverses the successive powers of a generator of the unit group. Fix a generator g of the species-layer unit group and label characters by ℓ. At site g^t, the phase factor is

$$\chi_\ell(g^t) = \exp\left(\frac{2\pi i \ell t}{p-1}\right), \qquad \ell \in \mathbb{Z}/(p-1)\mathbb{Z}. \tag{2.14}$$

Here t is the integer index ordering positions along the generator. Once the generator is fixed, ℓ determines the complete phase rule. Changing the generator changes the coordinates, while the group homomorphism represented by the character remains the same. The label ℓ=0 gives the trivial direction on the unit shell; the others give the elementary directions defined in Section 2.1.

Factoring p−1 into pairwise coprime prime powers expands a species label into several coordinates. Write $p-1 = \prod_{i=1}^{r} q_i^{b_i}$, where q_i are distinct primes, i=1,…,r. The Chinese remainder theorem [6] gives

$$\mathbb{Z}/(p-1)\mathbb{Z} \simeq \prod_{i=1}^{r} \mathbb{Z}/q_i^{b_i}\mathbb{Z}, \qquad \ell \longleftrightarrow (\ell_1, \ldots, \ell_r), \quad \ell_i = \ell \bmod q_i^{b_i}. \tag{2.15}$$

These coordinates form the mode vector. Specifying them uniquely determines the species label; changing one coordinate can leave all others fixed. This is independence of the labeling structure. Whether an actual exchange couples different components depends on its operator. The product space of mode vectors is a space of species labels, not a direct specification of positions in observational spacetime.

The order of the character determines the repetition length of the phase arrangement. Write this order as $o\chi$. The first return of the phase to 1 along the generator requires p−1 to divide $\ell o\chi$, so

$$o_\chi = \frac{p-1}{\gcd(\ell, p-1)}, \qquad o_i = \frac{q_i^{b_i}}{\gcd(\ell_i, q_i^{b_i})}, \qquad o_\chi = \operatorname{lcm}(o_1, \ldots, o_r). \tag{2.16}$$

The image of the character is the group of $o\chi$-th roots of unity, so there are $o\chi$ distinct phase values. Positions of equal phase form cosets of ker χ. Each phase occurs (p−1)/$o\chi$ times at the species layer and p^{K-1}(p−1)/$o\chi$ times after pullback to the unit shell at depth K. The kernel describes phase repetition, not nodes of zero amplitude; a single character has constant amplitude modulus throughout its supporting shell.

The order and kernel size do not replace the complete character. Different labels can have the same order and phase multiplicities while assigning phases to different positions. The repetition length here counts generator steps. A temporal oscillation frequency must instead be read from phase accumulation and duration in the actual cycle.

2.2.3 Pairing and Observation Units

A complete pattern usually extends over many sites. Pairing by negation forms two-dimensional observation units describing occupation and coherence within each pair of positions. This local description retains relations within a pair; the complete pattern also contains relations between units.

Position negation pairs a with −a. Complex conjugation of amplitudes leaves position labels fixed and changes the phase rule $\chi\ell$ to $\chi - \ell$. The two operations act on positions and amplitudes, respectively. For a species-layer character,

$$\chi_\ell(-a) = \chi_\ell(-1)\chi_\ell(a), \quad \chi_\ell(-1) = (-1)^\ell; \qquad \overline{\chi_\ell} = \chi_{-\ell}. \tag{2.17}$$

Negation parity is given by $\chi(-1)=\pm 1$. Self-conjugacy requires $2\ell=0$ mod p−1: apart from the trivial label, only the order-two character $\ell=(\mathrm{p}-1)/2$ is real. The remaining nontrivial labels form (p−3)/2 conjugate pairs, an empty set when p=3. Conjugate phase patterns have the same order. At a real regular spectral point, their responses are complex conjugates but are generally unequal. A label counts as an elementary-state direction at that point only if its response is nonzero.

Identifying these two operations with spin or time reversal in observational space requires a separate construction.

Since p is odd, 2 is invertible modulo p^K, and a=−a has only the zero solution. All nonzero positions therefore split into $(\mathrm{p}^K-1)/2$ distinct orbits $D_a = \{a, -a\}$. Each pair is a doublet, and its two-dimensional span is an observation unit. Zero forms a separate one-dimensional space:

$$\mathcal{H}_{D_a} = \mathrm{span}\{|a\rangle, |-a\rangle\}, \qquad Q_{D_a} = |a\rangle\langle a| + |-a\rangle\langle -a|. \tag{2.18}$$

The two positions of a pair lie in the same valuation shell. The unit shell contains $\mathrm{p}^{K-1}(\mathrm{p}-1)/2$ observation units, whereas the species-layer unit group modulo p contains only (p−1)/2 pairs; the latter cannot replace the full-depth observation-unit count. D_a and D_{-a} are the same orbit. Fixing the multiplicative identity 1 as the observation reference point gives the reference unit $\{1, -1\}$. Its role differs from that of the background reference b, which specifies an equal-amplitude, equal-phase pattern.

Let ρ be the density operator obtained by reducing the actual joint state to the system. Its compression to unit D, occupation share, and normalized state are respectively

$$\omega_D = Q_D \rho Q_D, \qquad q_D = \mathrm{Tr}\,\omega_D, \qquad \rho_D = \omega_D/q_D \quad (q_D > 0). \tag{2.19}$$

Here q_D is the occupation share of the unit, and ρ_D is its normalized local state conditional on occupation, including position occupations and coherence. No normalized state is defined when $q_D = 0$. Once compression gives the local state, the actual observation action must still determine which distinctions can be read by a pointer and retained as records.

For an elementary state of Section 2.1, let N=$|S_0|$=$p^{K-1}(p-1)$. Its component in the unit associated with unit position a is

$$Q_{D_a}|\psi_\chi\rangle = \frac{\chi(\pi(a))}{\sqrt{N}}\big(|a\rangle + \epsilon_\chi|-a\rangle\big), \qquad \epsilon_\chi = \chi(-1) = \pm 1. \tag{2.20}$$

Every unit-shell observation unit has occupation 2/N and normalized local pure state $(|a\rangle+\epsilon\chi|-a\rangle)/\sqrt{2}$. The common factor $\chi(\pi(a))$ cancels in its density matrix, so only negation parity remains within the unit. Different elementary states of the same parity give identical occupations and local density matrices in every unit-shell observation unit.

Their global distinctions remain in coherence between different units. Such relations become observable distinctions only when actual joint actions and internal references can read them. Complete species identification therefore requires examining how the whole pattern participates in observation. The connection of position units to physical generations and masses must likewise be established through actual readout and evaluation.

2.2.4 Collective Structure of Emergent States

The collective structure of b consists in participation by all sites with definite relative phases in a single pure state. Knowing that every position has occupation 1/d is insufficient to identify this pattern. The shell expansion displays both shell shares and intershell coherence. Its pure-state density operator is

$$P_b = \sum_{j,\ell=0}^{K} \sqrt{h_j h_\ell}\, |e_j\rangle\langle e_\ell|. \tag{2.21}$$

The diagonal terms give shell shares, and the off-diagonal terms retain intershell coherence. Removing this coherence gives a shell-dephased background while preserving coherence within each shell. Removing all coherence between distinct positions then gives the maximally mixed state I/d. All three have position probability 1/d. Off-diagonal matrix elements directly distinguish the whole-space pure state from the maximally mixed state:

$$\langle x|P_b|y\rangle = \frac{1}{d}, \qquad \langle x|I/d|y\rangle = \frac{\delta_{xy}}{d}. \tag{2.22}$$

In any nonzero observation unit, b has occupation share 2/d and normalized local state $(|a\rangle+|-a\rangle)/\sqrt{2}$; zero contributes another 1/d. Because both positions of a unit always belong to the same shell, the shell-dephased background gives the same local state in each unit. Identifying the full b pattern also requires examining coherence between shells. If an actual cycle uses another background preparation, its unit shares and coherence must be recalculated accordingly.

These structures specify what formation must establish. Shells determine the participation range, characters determine species phases, and observation units give local occupation and coherence; off-diagonal terms across units and shells preserve the complete pattern. Formation must establish these relations, and maintenance must preserve them after exchange and return. We now construct the actual process accordingly.

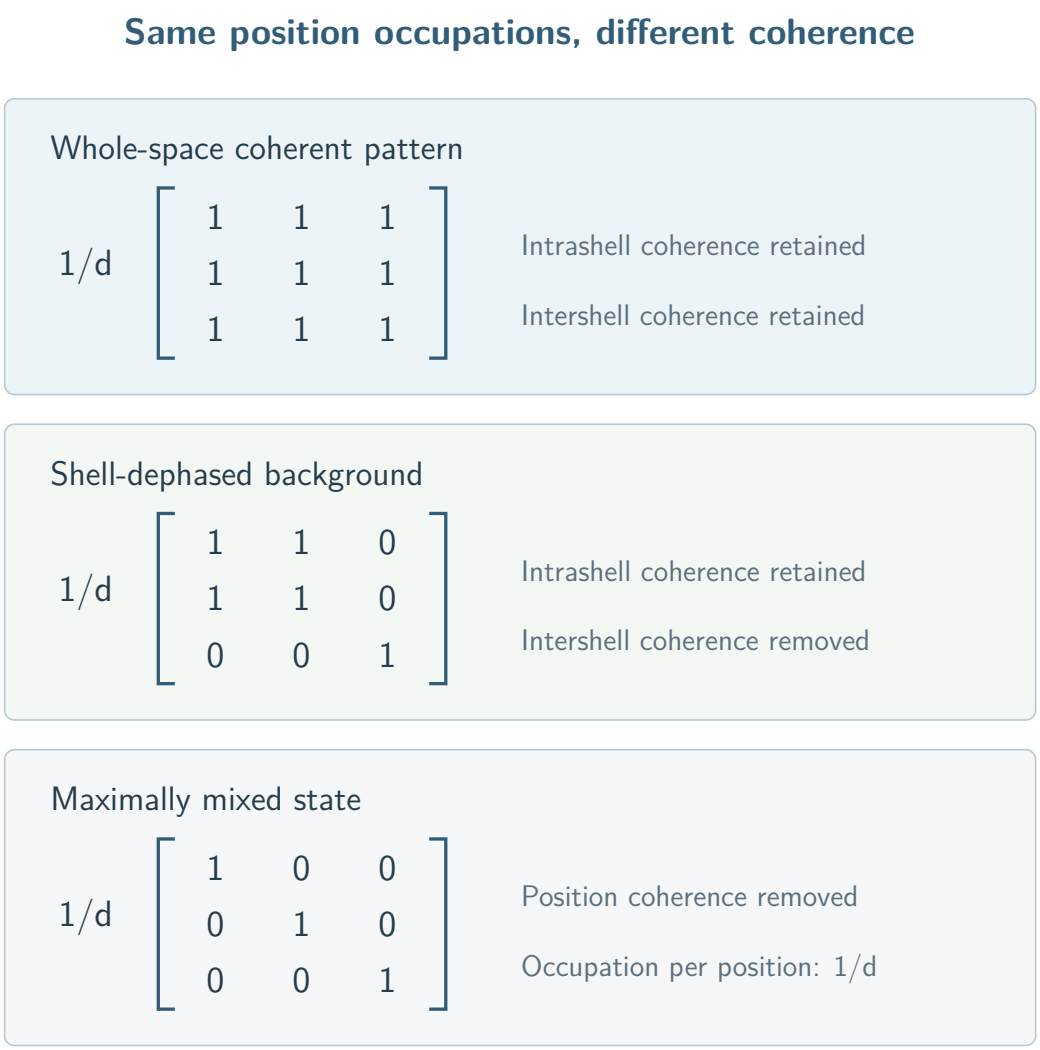


Figure 2-4. Three preparations with identical position occupations retain progressively less coherence. The diagram shows the submatrix for positions x, y, and 0: x and y lie in the same nonzero shell, and 0 in the terminal shell.

2.3 Formation and Maintenance

Exchange between excitation and background produces both formation and exit: the background can supply amplitude to a direction, while an existing direction can flow back to the background. Whether a returned component retains its pattern depends on the complete exchange and its records. We first distinguish these destinations in the original thermal cycle, then use the non-self-conjugate even elementary family to construct coherent pattern preparation and persistent action. The formed source, tracked target, and auxiliary degrees of freedom still involved in return jointly constitute the carrier; mass will be read from this joint object's evolution.

2.3.1 The Thermal Exchange Cycle

A cycle begins with the system state and environment preparation, followed by free evolution, joint exchange, directional retrieval, and background restoration. The final system state enters the next cycle. The background and directions belong to the same finite state space; the environment comprises other degrees of freedom responsible for exchange, retrieval, and records. Each cycle contacts a fresh preparation $\eta_{\mathcal{E}}$ independent of the system and old environment. The old environment retains existing correlations but no longer interacts; this condition is called environment refreshing. The shell-average part is restored to a fixed preparation. The complete order appears in Figure 1-2.

Let $\mathbb{U}$ denote the complete joint unitary, V the shell-preserving input control, and g the specified restoration shares. Then

$$\Omega_n^{\mathrm{out}} = \mathbb{U}(\rho_n \otimes \eta_{\mathcal{E}})\mathbb{U}^\dagger, \qquad \rho_{n+1} = \mathrm{Tr}_{\mathcal{E}}\, \Omega_n^{\mathrm{out}} = \mathcal{F}_{V,g}(\rho_n). \tag{2.23}$$

The joint final state retains all outputs and records. The partial trace gives system position shares and coherence for the next cycle; pattern reconstruction introduces no additional action. The total elapsed duration includes preparation, exchange, retrieval, and restoration. Environment preparation and controls are process data.

The same exchange directly connects formation and exit. Let $|\nu\rangle$ be a directional mode and e_a a shell-average endpoint, with shell energies E_ν, E_a. Pair $F_{a\nu} = |e_a\rangle\langle\nu|$ with an environment matrix element. On two joint states balanced in total energy and total unit-group charge, take

$$\begin{aligned} W_{a\nu}|\nu, u\rangle &= \cos\alpha_{a\nu}|\nu, u\rangle + \sin\alpha_{a\nu}|e_a, v\rangle, \\ W_{a\nu}|e_a, v\rangle &= \cos\alpha_{a\nu}|e_a, v\rangle - \sin\alpha_{a\nu}|\nu, u\rangle. \end{aligned} \tag{2.24}$$

A directional input transferred to a shell-average component represents exit of the original excitation; a background input transferred to a direction represents formation of a new excitation. The environment carries the opposite energy and group-charge changes. The actual records determine how events are distinguished.

Write the exit share as $\epsilon_{a\nu}$, formation share as $f_{\nu a}$, and inverse thermal environment temperature as β. The occupation probabilities of the two preparations multiply the same squared transfer modulus $\sin^2\alpha_{a\nu}$. On a nonzero channel at finite β,

$$\frac{\epsilon_{a\nu}}{f_{\nu a}} = e^{-\beta(E_a - E_\nu)}. \tag{2.25}$$

Thermal preparation constrains the forward-to-reverse ratio. The transfer magnitude also depends on channel occupation and exchange angle, and the angle depends on the actual generator and interaction time. Channel coefficients are given in Supplementary Proof S1.3.

Components remaining in the direction space after exchange are coherently retrieved through zero. Let $z = |0\rangle$, with retrieval auxiliary input g and temporary label a_ν. The round trip executes $|\nu, g\rangle \to w_\nu |z, a_\nu\rangle \to w_\nu |\nu, g\rangle$, where $|w_\nu| = 1$. The auxiliary temporarily stores directional information and returns to the common input at the end, permitting coherence between directions to survive. The net phase must also include free evolution.

The shell-average part is restored to $\sigma_g^{\mathrm{sh}} = \sum_a g_a P_{e_a}$, where $g_a > 0$ and $\sum_a g_a = 1$. The actual isometry maps e_a to $Y_a = \sum_j \sqrt{g_j}\, e_j \otimes r_{ja}$. Restoration labels are mutually orthogonal and orthogonal to the directional-return input. Thus $\mathrm{Tr}_R\, |Y_a\rangle\langle Y_b| = \delta_{ab}\sigma_g^{\mathrm{sh}}$, and cross terms between directional and restored outputs vanish as well. The system loses the corresponding coherence while correlations remain in the joint output. The auxiliary state g and restoration share g_a are used separately here.

Let $\epsilon_\nu = \sum_a \epsilon_{a\nu}$ and $P_\nu = |\nu\rangle\langle\nu|$, and denote the original excitation's return subchannel by $\mathcal{Q}_I$. Without input control, $V = I$,

$$\begin{aligned} \mathcal{F}_{I,g}(P_\nu) &= (1-\epsilon_\nu)P_\nu + \epsilon_\nu \sigma_g^{\mathrm{sh}}, \\ \mathcal{F}_{I,g}(P_{e_a}) &= \sum_\nu f_{\nu a} P_\nu + \left(1 - \sum_\nu f_{\nu a}\right) \sigma_g^{\mathrm{sh}}, \\ \mathcal{Q}_I^n(P_\nu) &= (1-\epsilon_\nu)^n P_\nu. \end{aligned} \tag{2.26}$$

The first two equations retain every destination of directional and background inputs, respectively, each with total trace one. The third tracks the original excitation continuously before its exit. Its normalized pattern after every return remains P_ν, while its survival share can decrease each cycle. New excitations formed later from the background are counted separately. Stationary total occupation and permanent survival of the original excitation are therefore different statements. A lifetime additionally requires the complete cycle period and exit channels.

The original cycle provides directional formation, pattern-preserving return, background restoration, and the thermal ratio of formation to exit shares. The next subsection constructs a common source comprising background and conjugate directions through label alignment and feedback, then generates phase through actual return. Coherence recurrences and conserved-energy conditions for the fixed thermal cycle are in Supplementary Proof S1; thermalization follows Section 4.4 of Ref. [2].

2.3.2 Formation of a Coherent Pattern

Subsequent exchange requires a coherent pattern containing background and conjugate-direction components, allowing constructive or destructive interference within position groups. Fix a non-self-conjugate even direction $\theta, \bar{\theta}$ and write $\sigma_0 = (\psi_\theta + \psi_{\bar{\theta}})/\sqrt{2}$. At the background input $e_0 \otimes b_s$ accepted by the common thermal preparation, use conjugate equal-angle exchange, label alignment, and positive-branch feedback.

The bare exchange on the bright plane $\mathcal{B} = \mathrm{span}\{e_0, \sigma_0\}$, the two-dimensional

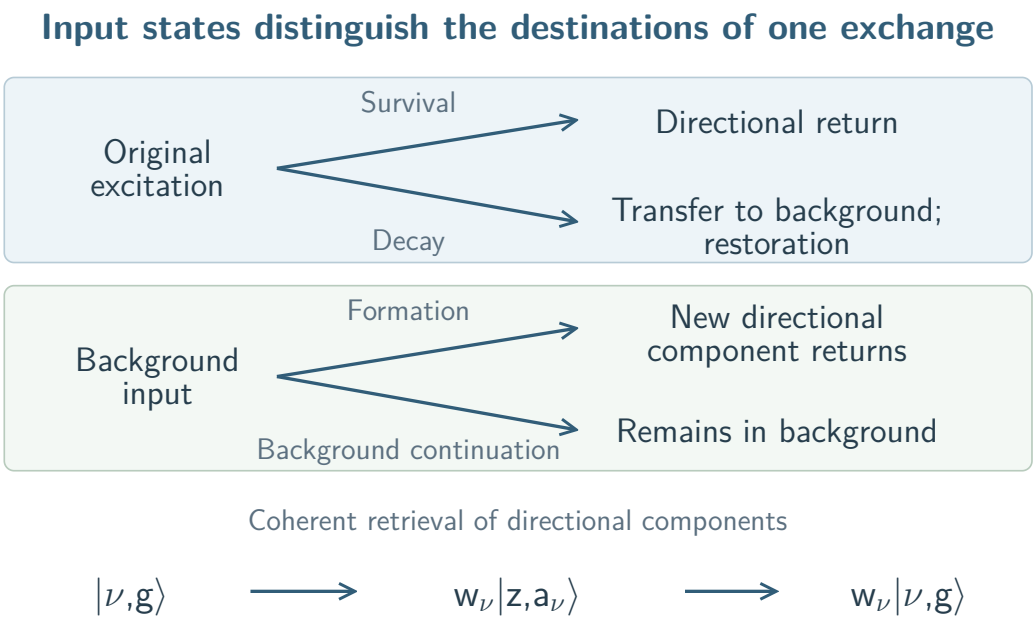


Figure 2-5. Continuation and decay of an original excitation, and formation and continuation in the background, are distinguished by their inputs and output destinations. All directional-return components shown undergo a coherent round trip through zero. Arrows denote process components; actually distinguishing the four record types also requires input labels and final pointers.

subspace coupled to the common input, is $X_A = |e_0\rangle\langle\sigma_0| + \text{h.c.}$. Its eigenpatterns $\Phi_\pm = (e_0 \pm \sigma_0)/\sqrt{2}$ have background and directional components of equal amplitude. Feedback sends both input branches to the same Φ_+, transferring their distinction to records.

Specifically, use a zero-energy pointer and apply a Hadamard, controlled $e^{i\alpha P_A} e^{-i\alpha X_A}$, and another Hadamard. For outcome 1, apply background–direction sign feedback. Finally, apply $e^{-i\alpha X_A}$, where $P_A = X_A^2$. Multiplying the two pointer outputs $(I \pm e^{i\alpha P_A} e^{-i\alpha X_A})/2$ by the feedback gives, on the bright plane,

$$\begin{aligned} A_0|_{\mathcal{B}} &= e^{-i\alpha} P_+ + \cos\alpha\, e^{2i\alpha} P_-, \\ A_1|_{\mathcal{B}} &= -i \sin\alpha\, |\Phi_+\rangle\langle\Phi_-|, \qquad P_\pm = |\Phi_\pm\rangle\langle\Phi_\pm|. \end{aligned} \tag{2.27}$$

At the first full-exchange angle $\alpha = \pi/2$, retaining every pointer and thermal label, any bright-plane input satisfies

$$\boxed{x\Phi_+ + y\Phi_- \longmapsto -i\Phi_+ \otimes (x|0\rangle + y|1\rangle)_D \otimes b_s.} \tag{2.28}$$

Source and records factorize, with all joint information retained. Starting from e_0, the two outcomes each have share one half; failures of the initial thermal preparation are retained separately. One-cycle delivery is not guaranteed for components of the three-dimensional source outside the bright plane; their formation is treated in Section 2.3.7.

For a nonzero spectral response $L(1/2, \theta) \neq 0$, next apply its polar-decomposition unitary on the selected conjugate-direction block and the identity on the background. Writing the bare spectral phase as $\delta_\theta = \arg L(1/2, \theta)$ gives

$$\sigma_{\delta_\theta} = \frac{e^{i\delta_\theta}\psi_\theta + e^{-i\delta_\theta}\psi_{\bar\theta}}{\sqrt{2}}, \qquad \Phi_{\delta_\theta} = \frac{e_0 + \sigma_{\delta_\theta}}{\sqrt{2}}. \tag{2.29}$$

When adopting $R = L/\zeta(1/2)$, perform the background–direction sign action of Section 8.1.2 to change the phase to the convention L. This formation construction uses a common exchange angle, label alignment, and positive-branch feedback; the chosen control realizes the angle. Control strength and total elapsed duration enter the process, including the feedback energy supply for an always-on exchange term.

Group shares of this pattern follow directly from interference. Take the species-layer reference point $x_0 = 1$. Let $\mathcal{D}_j$ be all deeper lifts of the point pair modulo p, P_j the group projection, and $\bar{d}_j$ the normalized equal-phase vector within the group. For an even character faithful and non-self-conjugate on the quotient, $m = (p-1)/2 \geq 3$. Fixing the phase order, uniform lifting gives

$$\begin{aligned} \langle \bar{d}_j | e_0 \rangle &= \frac{1}{\sqrt{m}}, \qquad \langle \bar{d}_j | \psi_\theta \rangle = \frac{e^{2\pi i j/m}}{\sqrt{m}}, \\ \langle \bar{d}_j | \Phi_\delta \rangle &= \frac{1 + \sqrt{2}\cos(\delta + 2\pi j/m)}{\sqrt{2m}}. \end{aligned} \tag{2.30}$$

The complete group projection selects the equal-phase direction within the group from this pattern, hence

$$p_j(\delta) = \langle \Phi_\delta | P_j | \Phi_\delta \rangle = |\langle \bar{d}_j | \Phi_\delta \rangle|^2 = \frac{[1 + \sqrt{2}\cos(\delta + 2\pi j/m)]^2}{2m}, \qquad \sum_j p_j = 1. \tag{2.31}$$

Normalization follows from $\sum_j \cos(\delta + 2\pi j/m) = 0$ and $\sum_j \cos^2(\delta + 2\pi j/m) = m/2$. Cross terms between background and directions produce the hierarchy among groups.

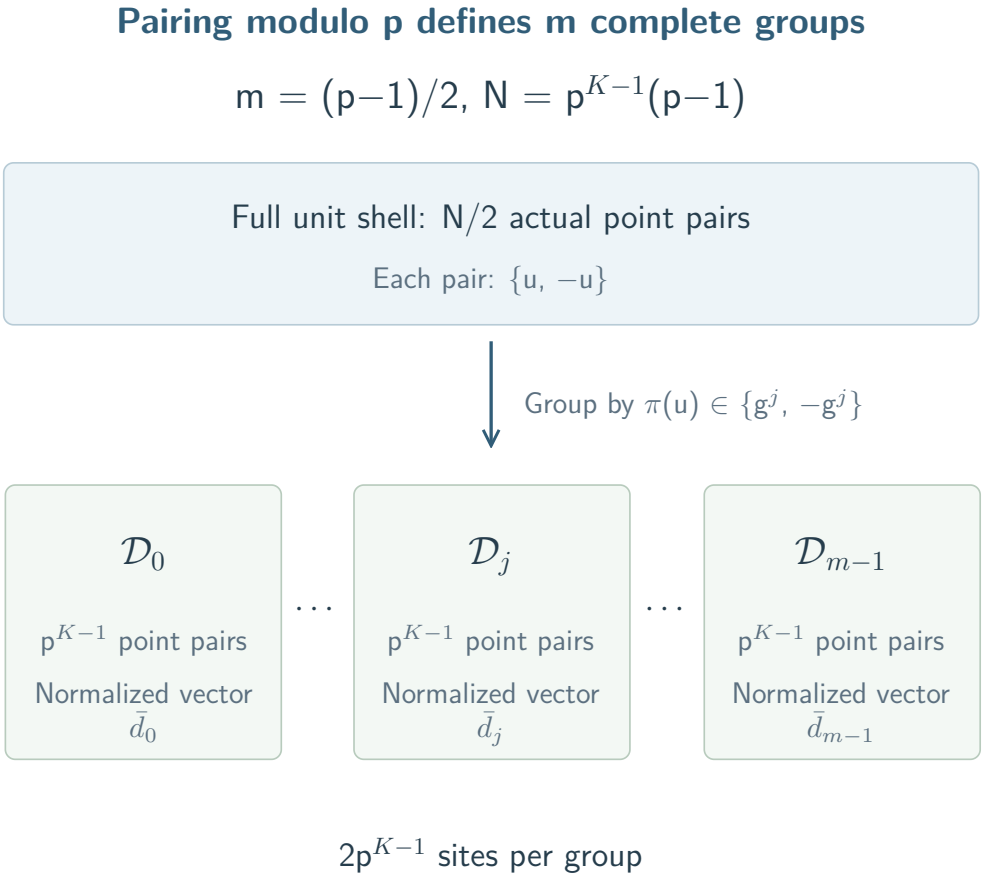


Figure 2-6. Each pair of positions at the species layer corresponds to a complete group containing p^{K-1} deeper observation units. Its normalized equal-amplitude, equal-phase vector is denoted by $\overline{d}_j$. These m complete groups are counted separately from the N/2 point pairs in the full unit shell.

Uniform lifting of the species layer makes the result valid for every K. Lower-order characters, nonuniform deeper preparations, and self-conjugate directions require calculations with their own patterns. Formation has now supplied p_j; Section 2.4.4 calculates how it becomes a coefficient of persistent action.

2.3.3 Formation-Phase Corrections

A formation pattern can acquire a further phase modification during a complete spectral excursion and return. Two quantities must be calculated together: the action produced by spectral auxiliaries and the phase left after they return. Let U_θ be the spectral unitary completed with failure outputs, r its input, and x the specified single-address output. Its effect is

$$Q_\theta = U_\theta^\dagger P_x U_\theta, \qquad w_{\mathrm{sp}} = \langle r|Q_\theta|r\rangle. \tag{2.32}$$

If the instrument delivers R_θ/C with equal amplitudes to N_θ addresses, then $w_{\mathrm{sp}} = |R_\theta|^2/(C^2 N_\theta)$. The actual preparation and outputs determine capacity and address count. The $p = 7$ eight-path instrument of Section 7.1.3 gives $C = 1$, whose single-address effect is precisely Q_θ.

Use two independent spectral auxiliaries with effects Q_1, Q_2, together with one coherent relay through e_0 to a complete observation group. The self-adjoint relay action B commutes with both effects, and the selected independent relay preparation gives $\langle B\rangle = \langle e_0|\bar{d}_j\rangle = 1/\sqrt{m}$. Set $A_0 = Q_1$ and $A_1 = Q_1 Q_2 B$, with a relative minus sign in the relay coupling. For the complete auxiliary preparation r_M,

$$A = A_0 - A_1, \qquad \langle A_0\rangle = w_{\mathrm{sp}}, \qquad \langle A_1\rangle = \frac{w_{\mathrm{sp}}^2}{\sqrt{m}}, \qquad a = \langle A\rangle = w_{\mathrm{sp}} - \frac{w_{\mathrm{sp}}^2}{\sqrt{m}}. \tag{2.33}$$

The squared term comes from two independent auxiliaries; reusing correlated auxiliaries requires recalculating their joint moment. To convert this mean into phase, the auxiliaries must participate in the outward evolution and return to a common input while preserving the action within the formation space.

On the formation space $\mathcal{H}_F = \mathrm{span}\{e_0, \psi_\theta, \psi_{\bar{\theta}}\}$, let $Z_F = P_\theta - P_{\bar{\theta}}$ and embed the common auxiliary input as $\mathcal{J}_F\psi = \psi \otimes r_M$. The two outward stages are generated by $G_0 = Z_F \otimes A_0$ and $G_1 = -Z_F \otimes A_1$. The return retraces the actual subspace path, using simultaneous arms with a relative minus sign to cancel action leaving that subspace while retaining its internal phase. The first stage leaves $s w_{\mathrm{sp}} Z_F$ and the second $-s w_{\mathrm{sp}}^2 Z_F/\sqrt{m}$. They commute within the input space, giving

$$\mathcal{R}_s(\psi \otimes r_M) = e^{isaZ_F}\psi \otimes r_M, \qquad Z_F = P_\theta - P_{\bar{\theta}}. \tag{2.34}$$

The background component e_0 is unchanged, while the two directions acquire opposite phases. The formed source becomes $\Phi_s := \Phi_{\delta_\theta + sa}$, with group shares $p_j(s) := p_j(\delta_\theta + sa)$. Reflections along the path are determined by the input preparation and act on the entire space to be formed and its controllable auxiliaries. The return generator, ordered composition, and actual round-trip duration are derived in Supplementary Proof S2.

Here s is the actual net action. Subsequent evaluation identifies this return with the formation process described by the unit-spectral kernel. For nonzero a, the kernel comparison in Supplementary Proof S2 then fixes $s = 1$. This physical identification constrains the accumulated action; control strength, completed-event count, and full round-trip duration are accounted for separately in the actual process.

2.3.4 Formation Records and Coherence

Formation can leave different records while producing the same source. What matters is whether those distinctions remain correlated with subsequent logical states. For an accepted bright background input, both full-feedback outcomes produce the positive source and execute the spectral modification only once. Completion records no longer participate in later action, and unaccepted resources no longer couple to the source.

Let h label accepted fine preparation histories, b_h their amplitudes, and $d = 0, 1$ the full-feedback outcome. Let $\mathcal{J}_{\text{in}}$ be the isometry combining the fixed background input and retained logic, and $\mathcal{J}_{F,s}$ the common delivery. Finitely many independent preparation attempts satisfy

$$K_{h,d}\mathcal{J}_{\text{in}} = -\frac{ib_h}{\sqrt{2}}\mathcal{J}_{F,s}, \qquad Q_M = \sum_{h \text{ within the first } M \text{ attempts, accepted}} |b_h|^2 \leq 1. \tag{2.35}$$

The two feedback outcomes together assign weight $|b_h|^2$ to each accepted history; unaccepted outcomes and the waiting tail have total share $1 - Q_M$. For single-attempt acceptance probability q_F, $Q_1 = q_F$. Only a separately labeled input branch requires multiplication by its additional weight. Waiting changes the delivery probability and time without repeatedly accumulating the formation angle on a completed source. Under incomplete feedback, negative-branch occupation is multiplied each round by $\cos^2 \alpha$ and must still be treated as input undergoing formation.

Common return has a general criterion. Let $\mathcal{J}, \mathcal{J}_{\text{out}}$ be fixed input and output isometries and V a fixed logical unitary. The return subchannel equals $q\,\mathrm{Ad}_V$ on this space if and only if every actual fine history satisfies

$$K_h\mathcal{J} = c_h\mathcal{J}_{\text{out}}V, \qquad \sum_h |c_h|^2 = q. \tag{2.36}$$

Sufficiency follows by summation. For $q > 0$, necessity follows from the rank-one Choi matrix of the channel: all Kraus vectors must be collinear. When $q = 0$, every $K_h\mathcal{J}$ is zero. With nonzero return, arbitrary input superpositions and coherence with spectator systems return under the same V, a stronger statement than restoration of one pure state.

The common-return criterion retains reversible working auxiliaries, permanent records, and incomplete flags. Reversible auxiliaries participate in later retrieval; permanent records that never recouple preserve path distinctions. Section 2.4.2 explains how environment overlaps enter output shares. Specific return examples appear in Supplementary Proofs S6.1, S4.4, and S14.

2.3.5 Maintenance of a Stable State

To provide a source of mass, the formation pattern must continue to participate in actual exchange. Target branches match different position groups of the source and thereby acquire different interaction strengths. Source, target, and auxiliaries still participating in return form the persistent carrier. The isometry $\mathcal{J}_s$ embeds the target together with the formed source Φ_s, spectral auxiliaries, nodes, and necessary records into the joint space.

The image of this isometric embedding is called the code subspace; inside and outside the code refer to this image and its orthogonal complement. The given preparation determines the projection needed to preserve the carrier. Let $\mathcal{U}_s$ be a reference unitary preparing the fixed source and auxiliaries from a simple input $\mathcal{J}_{\text{in}}$. Define

$$\mathcal{J}_s = \mathcal{U}_s\mathcal{J}_{\text{in}}, \qquad P_s = \mathcal{J}_s\mathcal{J}_s^\dagger, \qquad S_s = 2P_s - I = \mathcal{U}_s(2P_{\text{in}} - I)\mathcal{U}_s^\dagger. \tag{2.37}$$

The operators P_s, S_s are fixed during the persistent stage. Let the actual self-adjoint exchange $X = X^\dagger$ include spectral access, nodes, original address matching, and all auxiliaries and failure outputs. Use two simultaneous arms, each of strength $g/2$:

$$\widehat{H}_X = \frac{g}{2}(X + S_s X S_s), \qquad h_X = g\mathcal{J}_s^\dagger X \mathcal{J}_s. \tag{2.38}$$

Since $S_s\mathcal{J}_s = \mathcal{J}_s$, we have $(X + S_s X S_s)\mathcal{J}_s = 2P_s X \mathcal{J}_s$, and hence

$$\boxed{\widehat{H}_X \mathcal{J}_s = \mathcal{J}_s h_X, \qquad e^{-it\widehat{H}_X}\mathcal{J}_s = \mathcal{J}_s e^{-ith_X}.} \tag{2.39}$$

This intertwining connects formation to maintenance. Formation supplies the joint pattern and its preparation reflection; the matrix elements left by the original exchange within that pattern become the persistent generator. The two arms cancel amplitudes leaving the carrier, while the internal action continues to accumulate phase. The same pattern can therefore retain its identity during persistent exchange. This relation holds for all times, reducing the mass calculation to actual matrix elements of X in the formation preparation.

The reference unitary can be synthesized from existing operations. Starting at the known background input, perform $e^{-i\pi X_A/4}$ and multiply the direction by i to obtain Φ_+, followed by the spectral polar-decomposition unitary and complete round trip. Inverse preparation, input marking, and preparation then implement S_s. This prepares the same fixed state without reversing random feedback records or requiring actual formation to stop at half exchange. Formation auxiliaries and the common reference must be accessible; permanent records remain factors on which the action is the identity, or block labels.

This balance has a direct structural constraint. For the complete carrier P and reflection $S = 2P - I$, $(I - P)(aX + bSXS)P = (a - b)(I - P)XP$. If the original exchange transfers between blocks, preservation at all times requires equal-strength coherent arms; common calibration still determines their overall strength. Both arms must reach the same complete output. Independent records or finite serial pulses generally alter the cancellation. Changing the formation angle changes the pattern, preparation reflection, and background readout. With original address matching fixed, formation shares consequently alter maintenance frequencies through actual matrix elements.

When several actions share this preparation, their sum retains the intertwining. This connects matter maintenance and the background gradient to one carrier: after writing into the background, transporting both actions and their reflections by the same writing unitary preserves their common intertwining. Passing both actions through balanced exchange with the available output column then makes background occupation constrain the matter phase and gradient response by the same availability factor. Chapter 4 derives the force from this structure. The operator proof is in Supplementary Proof S7. Section 2.3.7 connects the same maintenance action to unready inputs to calculate convergence after formation.

2.3.6 Survival and Re-formation

Persistent evolution must also undergo thermal processing. The original fixed restoration removes system coherence between background and directions. Here reversible auxiliaries and thermal outcomes are retained, and the actual deformation

is compensated separately for each outcome, allowing successful returns to continue along the same carrier. This compensation introduces no new mass phase.

Let M_η be the thermal outcome map including preparation weights, defined on the logical space to be returned. If $G_\eta = M_\eta^\dagger M_\eta > 0$, write its polar decomposition as $M_\eta = W_\eta \sqrt{G_\eta}$ and let $\mathcal{J}_T$ denote the target isometry for the carrier and restored auxiliaries. Choose the actual return filter as

$$0 < r_\eta \le \lambda_{\min}(G_\eta), \qquad F_\eta = \sqrt{r_\eta}\, \mathcal{J}_T G_\eta^{-1/2} W_\eta^\dagger, \\ \boxed{F_\eta M_\eta = \sqrt{r_\eta}\, \mathcal{J}_T, \qquad F_\eta^\dagger F_\eta \le I.} \tag{2.40}$$

Set $F_\eta = 0$ outside the image. The inequality $r_\eta G_\eta^{-1} \le I$ implies contraction; adding the orthogonal failure output $\sqrt{I - F_\eta^\dagger F_\eta}$ completes the instrument. Conversely, exact return of all unknown inputs requires $r_\eta I \le G_\eta$. An outcome with a zero eigenvalue cannot return the whole space with positive weight. Every outcome must retain thermal labels usable for control, and failures count as exits.

The logical operator associated with each fine survival record is therefore $\sqrt{r_\eta} I$, with total $q_T = \sum_\eta r_\eta$; all remaining outputs have total effect $(1-q_T)I$. The common per-cycle survival probability comes from this actual instrument, not temperature alone. Transport, filtering, restoration, and recording all contribute to the cycle period.

Let $\mathcal{L}_f$ be the actual trace-nonincreasing family-loading map, with output in the target logical space and $\ell_f(\rho) = \operatorname{Tr} \mathcal{L}_f(\rho)$. After loading, embed by $\mathcal{J}_s$. Each round evolves under $U = e^{-it_H h}$ for an active window t_H, followed by thermal processing and the remaining waiting time. When formation-completion records factorize from subsequent auxiliaries, the output surviving n consecutive cycles is

$$\rho_{\text{live}}^{(n)} = Q_M q_T^n \mathcal{J}_s U^n \mathcal{L}_f(\rho) U^{\dagger n} \mathcal{J}_s^\dagger. \tag{2.41}$$

For normalized input, all destinations satisfy

$$(1 - Q_M) + Q_M(1 - \ell_f) + Q_M(1 - q_T^n)\ell_f + Q_M q_T^n \ell_f = 1. \tag{2.42}$$

The four terms are source-delivery failure, family-loading failure, cumulative first exits, and continued survival. The complete output retains every record. The background after an exit may form new excitations; these are counted separately from uninterrupted survival of the original excitation. Actual decay products, generation lifetimes, and stability of the lowest generation still require their respective exit channels.

Supplementary Proof S13.5 connects reversible neutrino path auxiliaries and thermal records to this survival process. Here n counts from actual loading completion in each history. Different histories have preparation durations T_h; combining them at one absolute time requires summing according to their actual completed cycles. Delivery probability and the waiting distribution change the sample and starting times, while the persistent generator remains $h_X = g \mathcal{J}_s^\dagger X \mathcal{J}_s$. The next subsection inserts repair during operation to examine formation of the persistent carrier from unready inputs.

2.3.7 Convergence of Formation

The preceding delivery specifies an exact carrier. For unready inputs, we must also show that the same operating process gradually establishes it. We insert source and auxiliary repair between active intervals while tracking thermal exit. Restrict the source domain to $\mathcal{S} = \mathrm{span}\{e_0, \psi_\theta, \psi_{\bar{\theta}}\}$ and use the product-form isometric embedding $\mathcal{J}_s\psi = \Phi_s \otimes r \otimes \psi$. Here r contains only auxiliaries that can be independently re-prepared. Target, position, orientation, background gradient, and permanent records remain in ψ. Write $P_s = \mathcal{J}_s\mathcal{J}_s^\dagger$ and $Q_s = I - P_s$. This construction does not replace degrees of freedom carrying permanent information in a nonproduct family encoding.

Each round first applies the complete generator, then pauses activity for repair. Once a fresh auxiliary is independently prepared, an actual exchange replaces the old one, transferring it and its correlations to outgoing records. The source undergoes $\mathcal{T}_s = \mathrm{Ad}_{D_s}\,\mathcal{T}\,\mathrm{Ad}_{D_s^\dagger}$. Here $D_s = e^{isaZ_F} D_L$ is the existing round-trip unitary on the formation space, $D_s\Phi_+ = \Phi_s$, and $\mathcal{T}$ includes all full-feedback outcomes. Returning to the bare axis, applying feedback, and returning again directs repair toward the same pattern without adding the formation angle again.

Let $\mathcal{E}$ be successful repair and $q_R > 0$ the readiness probability of fresh resources; on failure the carrier is bypassed. Retaining outgoing old auxiliaries and flags, the reduced cycle satisfies

$$\begin{aligned} \mathcal{R} &= q_R\mathcal{E} + (1-q_R)\mathrm{Id}, & \kappa &= 1 - q_R/4 < 1, \\ \mathcal{F} &= \mathcal{R}\circ\mathrm{Ad}_{\widehat{U}}, & \mathcal{F}^*(Q_s) &\le \kappa Q_s, \\ \mathcal{F}(\mathcal{J}_s\sigma\mathcal{J}_s^\dagger) &= \mathcal{J}_s U\sigma U^\dagger\mathcal{J}_s^\dagger, & U &= e^{-it_H h}. \end{aligned} \tag{2.43}$$

Here $\widehat{U} = e^{-it_H\widehat{H}}$. Full feedback leaves at most three quarters of the preparation deficit of the three-dimensional source; auxiliary replacement removes unready auxiliaries. The active stage commutes with P_s, yielding κ. Every repair outcome preserves the preparation of an already formed carrier, while the target continues under U. Feedback matrices and proof are given in Supplementary Proof S3.

Thermal return must also cover transient inputs. Apply the transport, exchange, filtering, and retrieval of Section 2.3.6 at the target thermal port, act as the identity on the source, auxiliaries, and logical degrees of freedom not participating in this action, and compensate free unitary evolution on the entire transient domain. Every fine survival action is then $\sqrt{r_\eta}I$ throughout the domain, with $0 < q_T \le 1$. For any allowed normalized input ρ, the surviving output and first-exit probability are

$$\boxed{\rho_{\mathrm{live},n} = q_T^n\mathcal{F}^n(\rho), \qquad \Pr(\text{first exit in cycle } n) = q_T^{n-1}(1-q_T).} \tag{2.44}$$

This counts from the operating input, with no additional delivery probability Q_M. Let $\varepsilon_0 = \mathrm{Tr}(Q_s\rho)$. Supplementary Proof S3 establishes an input-dependent logical state σ_∞ such that

$$\boxed{\left\| \frac{\rho_{\mathrm{live},n}}{q_T^n} - \mathcal{J}_s U^n \sigma_\infty U^{\dagger n}\mathcal{J}_s^\dagger \right\|_1 \le 2\sqrt{\varepsilon_0}\kappa^{n/2} + \varepsilon_0\kappa^n.} \tag{2.45}$$

The surviving component thus approaches a persistent orbit of the same generator. Its limiting logical state σ_∞ depends jointly on input and repair. The parameter

q_R controls formation convergence, q_T the survival envelope, and h phase evolution. A finite lifetime reduces the samples that complete formation; frequencies within the carrier remain unchanged at the scalar thermal-return interface used here.

The selected serial cycle period is $\tau = t_H + T_R + T_X$, where T_R is the repair duration and T_X the thermal-processing duration. Auxiliary replacement, reuse, and complete timing are described in Supplementary Proof S3.3.

Matter, motion, and background are active and paused together within this serial cycle, sharing the active fraction of Section 2.4.6. The product-form isometric embedding and scalar thermal return connect formation, persistence, and exit in one cycle. We now calculate its accumulated phase.

2.4 Elementary-State Mass

A pattern can preserve its form while its phase relative to a common reference changes with time. Rest mass is read from the rate of this relative phase against actual elapsed duration. When a symmetric pair of frequency branches is used, the half-gap is read after identifying their symmetry center. Propagation, node exchange, and return together form this persistent process, through which the excitation's spectral structure enters actual phase evolution. We first derive the mass generator during activity, then include return, thermal processing, and waiting in the total duration to obtain the actual mass reading.

2.4.1 Spectral Response and Exchange Strength

The zero element $z = |0\rangle$ connects shell-average components and directions, but is neither the unit-shell average state e_0 nor the whole-space reference b. Take $N = |S_0|$ and $P_z = |z\rangle\langle z|$, with additive translation $(\tau_a f)(x) = f(x + a)$. From $[P_z, \tau_a] = |z\rangle\langle a| - |-a\rangle\langle z|$ we obtain the directional vertex

$$J_\chi = \frac{1}{\sqrt{N}} \sum_{a \in S_0} \chi(\pi(a))[P_z, \tau_a] = |z\rangle\langle \psi_{\bar{\chi}}| - \chi(-1)|\psi_\chi\rangle\langle z|. \tag{2.46}$$

Each component can be balanced by the environment as in Section 2.3.1. The operator J_χ organizes the order of access; in the non-self-conjugate case it cannot directly serve as a self-adjoint Hamiltonian. The actual exchange pairs each joint matrix element with its adjoint.

Denote the spectral variable by s_{sp}. In its convergence domain, the unnormalized spectral transfer $\widetilde{\mathcal{B}}(s_{\mathrm{sp}}) = \sum_{n \geq 1} n^{-s_{\mathrm{sp}}} T_n$ gives $L(s_{\mathrm{sp}}, \chi)$ along a direction, then continues to the selected regular point s_0. This variable is distinct from the formation round-trip action s. Write $L_\chi = L(s_0, \chi)$ and $R_\chi = L_\chi / \zeta(s_0)$. Every multiplicative pullback preserves the whole-space pattern b, whose unnormalized response is $\zeta(s_0)$, supplying the denominator of the spectral comparison. The unit-shell average state e_0 instead removes multiples of p and has response $(1 - p^{-s_0})\zeta(s_0)$. Spectral reference, zero endpoint, and actual environment preparation thus have distinct roles. An actual instrument must also provide its capacity and complete outputs. If the selected spectral unitary U_χ has matrix element $r_\chi = L_\chi / c_L$ at the specified success port, then

$$K_\chi = \langle \mathrm{suc}|U_\chi|r\rangle, \qquad K_\chi^\dagger K_\chi + \sum_b F_{\chi b}^\dagger F_{\chi b} = I. \tag{2.47}$$

On the accessed eigendirection, the success column produces r_χ. Capacity must at least make every accessed success column contractive; an instrument handling several directions must normalize all of them simultaneously. The quantity $|L_\chi| > 1$ can be a response coefficient but cannot be a passive one-shot success amplitude. We retain c_L and the failure columns until their conditional cancellation in the background calibration using a common source.

2.4.2 Composition of Propagation Paths

Character multiplication specifies label changes by modular addition of CRT coordinates. Actual paths additionally specify how zero, directions, and auxiliaries connect in sequence. For example, $J_{\chi_2} J_{\chi_1} |z\rangle$ returns to zero only when $\chi_2 = \bar{\chi}_1$. To obtain $\chi_1 \to \chi_1\chi_2$ requires the corresponding absorption–formation matrix elements and environment balance; multiplying labels cannot replace exchange.

For the selected conjugate double excursion, the spectral response acts only on directions and zero serves only as a relay. Applying the vertices of Section 2.4.1 successively to the input gives

$$\begin{aligned} J_\theta |z\rangle &= -\theta(-1)|\psi_\theta\rangle, \\ J_{\bar\theta} \widetilde{\mathcal{B}} J_\theta |z\rangle &= -\theta(-1) L_\theta |z\rangle, \qquad L_\theta L_{\bar\theta} = |L_\theta|^2. \end{aligned} \tag{2.48}$$

After the first return, conjugate access is repeated. The two orientation signs multiply to one, and conjugate responses at a real spectral point give the final squared modulus. This first evaluates the algebraic product of spectral accesses; the complete instrument then supplies its actual normalization.

Using two explicit success accesses while retaining every failure output gives the actual joint success amplitude and effect:

$$r_\theta r_{\bar\theta} = \frac{|L_\theta|^2}{c_L^2}, \qquad \langle Q_{\mathrm{path}} \rangle = \frac{|L_\theta|^4}{c_L^4}, \qquad Q_{\mathrm{path}} = U_{\mathrm{path}}^\dagger P_{\mathrm{suc}} U_{\mathrm{path}}. \tag{2.49}$$

This is the specific origin of the fourth spectral power in the effect. A single excursion, coherent composition of an entire path, and independent acceptance at successive stages can give different results; closure alone does not fix the power. Noncommuting paths are composed by full matrix multiplication, and inserting intermediate bases gives their history amplitudes.

For paths from the same pure preparation reaching the same system final state, write their amplitudes as $a_{\mathcal{P}}$ and complete environment final states as $e_{\mathcal{P}}$. Then

$$w_f = \sum_{\mathcal{P},\mathcal{P}'} a_{\mathcal{P}} \overline{a_{\mathcal{P}'}} \langle e_{\mathcal{P}'} | e_{\mathcal{P}} \rangle. \tag{2.50}$$

Identical final states require amplitude addition, orthogonal final states require addition of squared moduli, and partial overlaps retain cross terms; see [9] for the two-path visibility–distinguishability relation. When stages produce actual records, each subsequent stage starts from the preceding actual output. Using outcome maps of quantum instruments [10], let $\mathcal{I}_{r_k}^{(k)}$ be the completely positive, trace-nonincreasing map for outcome r_k at stage k, including every degree of freedom participating later. The joint share of a recorded history is

$$p(r_1, \ldots, r_m) = \mathrm{Tr}\left[\mathcal{I}_{r_m}^{(m)} \circ \cdots \circ \mathcal{I}_{r_1}^{(1)}(\rho_0)\right]. \tag{2.51}$$

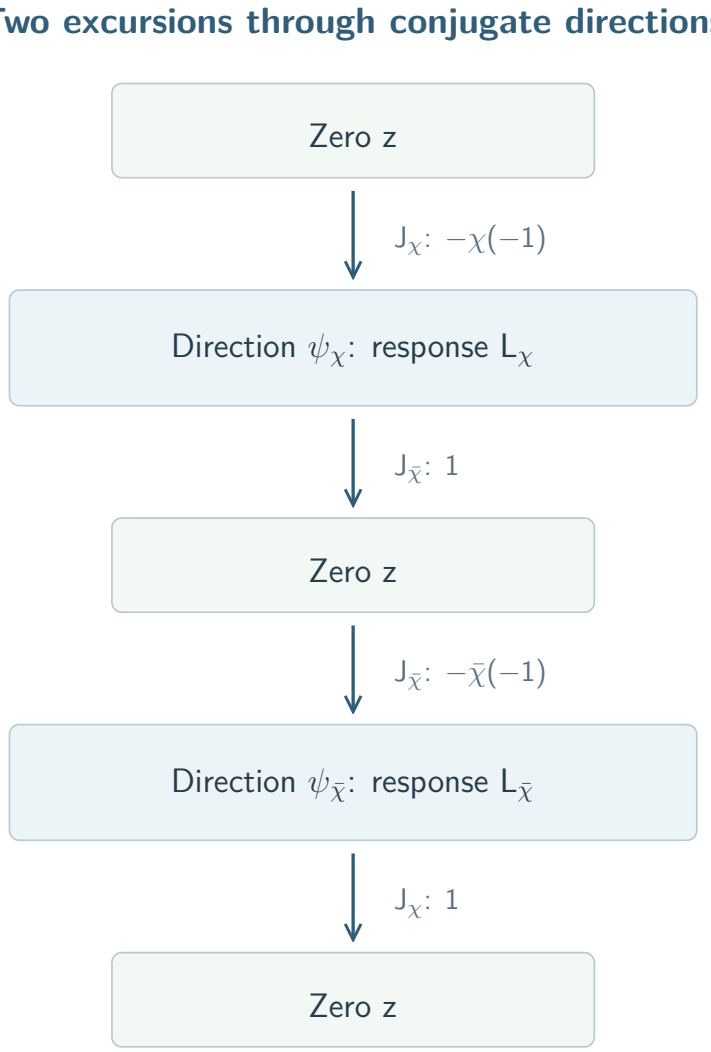


Figure 2-7. The selected double excursion visits χ and $\bar{\chi}$ in sequence. The two formation signs multiply to one; at a real spectral point the responses multiply to $|L(s_0,\chi)|^2$. The diagram gives the order of spectral-path evaluation, not a direct unitary amplitude for the complete cycle.

For two stages this is $p(r_1)p(r_2 \mid r_1)$. It factors further into independent shares only when actual re-preparation or other conditions remove the dependence. Components of mixed preparations are combined with their weights. These rules apply to both open and closed paths.

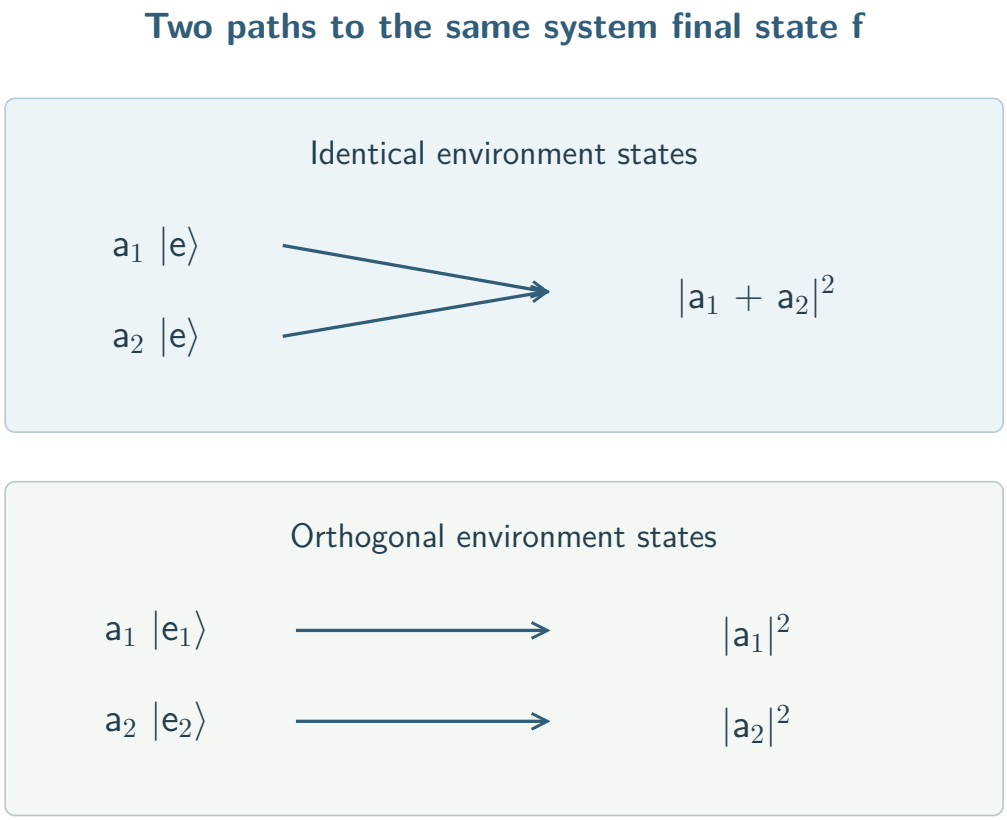


Figure 2-8. Two paths reach the same system final state, but their composition still depends on the complete environment final states. Components with identical environment states add as amplitudes; outcomes with orthogonal environment states add as shares. The amplitudes shown arise from the same pure preparation and actual joint evolution.

The resulting path quantity then enters a self-adjoint action. For example, a specified effect Q_{path} can act on the carrier through conditional exchange between two orientations. The complete spectral auxiliaries must remain involved in maintenance, so evaluating $\mathcal{J}^\dagger X \mathcal{J}$ yields the stated weight. If success has already been measured and only the normalized successful state is delivered, the preceding success probability generally affects yield alone; that is a different process.

2.4.3 Node Normalization Factors

Node factors come from overlaps among input, actual connection, and complete output. On a symmetric orbit of d points, the normalized equal-phase vector $|\eta\rangle = d^{-1/2}\sum_x |x\rangle$ has overlap $1/\sqrt{d}$ with one specified endpoint and event share $1/d$. If both entrance and exit each involve an overlap, both factors must be retained.

A binary coherent node can directly represent $1/\sqrt{2}$. Take $F_N = |1\rangle\langle 0|$, node preparation $n = \cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle$, and actual spectral unitary U. The self-adjoint exchange is then

$$B_N = F_N \otimes U + F_N^\dagger \otimes U^\dagger, \qquad \langle n, r|B_N|n, r\rangle = \frac{\mathrm{Re}\langle r|U|r\rangle}{\sqrt{2}}. \tag{2.52}$$

Once the node preparation and coherent port are fixed, the coefficient follows from the matrix element. The common internal reference determines phase orientation; a real part cannot arbitrarily be replaced by a modulus. If subsequent action uses an endpoint effect, compress that effect rather than applying an amplitude node factor again.

Distinguishable outputs are likewise counted through the complete operator. For d colors, let each color retain the same transfer K, $E = K^\dagger K$, and $dE \le I$, and set $\Omega_d = \sum_{c=1}^{d} |c, \bar{c}\rangle$. Two orthogonal failure-output subspaces give the isometry

$$\begin{aligned} V_{\mathrm{in}}\psi &= (\Omega_d \otimes K\psi \otimes r) \oplus \sqrt{I - dE}\,\psi \oplus 0, \\ V_{\mathrm{out}}\psi &= (\Omega_d \otimes K\psi \otimes U^\dagger r) \oplus 0 \oplus \sqrt{I - dE}\,\psi, \end{aligned} \qquad V_{\mathrm{out}}^\dagger V_{\mathrm{in}} = d\langle r|U|r\rangle E. \tag{2.53}$$

The color count and spectral response thus jointly enter actual return. Replacing each color's transfer by $K/\sqrt{d}$, or allowing coherent overlap at failure outputs, changes the result. Having d colors therefore fixes a multiplier only together with per-color strength and failure outputs.

The geometric factor for a relay to a complete species-layer group is

$$|\bar{d}_j\rangle = \frac{1}{\sqrt{2p^{K-1}}} \sum_{\pi(u)\in\{a_j, -a_j\}} |u\rangle, \qquad \langle \bar{d}_j|e_0\rangle = \frac{1}{\sqrt{m}}. \tag{2.54}$$

It holds at any depth because the port reads all deeper lifts. A single deeper point pair instead gives $\sqrt{2/|S_0|}$. Repeated application of the same projection satisfies $P^2 = P$ and does not automatically add a normalization factor each time. Every new multiplicative factor must correspond to a new access, relay, or preparation.

2.4.4 Generational Factors

Fix the formation source Φ_s and use the original same-group matching between source and target:

$$Q_{TS} = \sum_j P_j^T \otimes P_j^S, \qquad \langle \Phi_s|Q_{TS}|\Phi_s\rangle_S = \sum_j p_j(s) P_j^T. \tag{2.55}$$

Each group coefficient follows from actual source occupation, without assigning a coupling separately to each generation. Complete target groups retain both parity orientations. When they carry motion, their matching to source groups is connected

through auxiliary labels and actual routing; retained labels cannot be replaced by position projections altered by translation.

Place group matching, path effects, and node exchange in the same complete preparation. For the even-conjugate family considered here, the actual exchange before the maintenance frequency-shift insertion is $X_f^{(0)} = Q_{TS}Q_{\mathrm{path}}B_N Y$: group matching reads the source, the complete path effect reads spectral auxiliaries, the node uses a binary preparation without an additional spectral insertion, and Y acts on retained orientation partners. The factors act on specified independent auxiliary units and commuting labels. The exchange is therefore self-adjoint, and its compression can be evaluated factor by factor.

Write the common instrument capacity as Λ and unnormalized path weight as $\mathcal{P}_f$; conjugate double access uses $\mathcal{P}_f = |L_\theta|^4$. Path, node, and source respectively give $\mathcal{P}_f/\Lambda$, $1/\sqrt{2}$, and group shares $p_j(s)$. Substitution into the persistent generator of Section 2.3.5, after extracting the orientation factor Y, gives the rest-frame mass matrix in the active window:

$$M_f^{(0)} = \frac{g\mathcal{P}_f}{\sqrt{2}\Lambda} \sum_j p_j(s) P_j^T. \tag{2.56}$$

Here Λ collects the actual capacities; with only the preceding double spectral access, for example, it contains c_L^4. Renaming does not remove this factor: it must be retained together with the same background readout. Auxiliary units added by new nodes must enter the new $\mathcal{J}_s$ and reflection. Protection omitting those auxiliary units cannot be reused.

The reference group is fixed by x_0. Choose group zero with $p_0 > 0$. The background branch also traverses the same source and actually matches this group, so its matrix element contains p_0. The reference group need not be maximal or fully transmitted. Section 2.5 calibrates both actual branch frequencies together, leaving the relative generational factor p_j/p_0; its denominator was not inserted into the target action in advance.

For a family with a nondiagonal chain-matrix output, retain the full matrix and permanent records before taking eigenvalues. Three nonzero formation shares do not prove three independent persistent spectral lines. The common carrier of Section 2.3 and the actual mass matrix jointly determine whether those lines occur.

2.4.5 Phase Corrections and Frequency Shifts

Formation modification changes the source angle, while a spectral insertion during maintenance can also change the rest-frame generator. Retain the actual self-adjoint action A of Section 2.3, with preparation mean a. In the specified connection commuting with maintenance exchange X, insertion of $I - A$ gives

$$h_0 = g\mathcal{J}_s^\dagger X \mathcal{J}_s, \qquad h_{\mathrm{corr}} = g\mathcal{J}_s^\dagger X(I-A)\mathcal{J}_s, \qquad C_{X,A} = \mathcal{J}_s^\dagger X A \mathcal{J}_s - a\mathcal{J}_s^\dagger X \mathcal{J}_s. \tag{2.57}$$

Direct expansion yields

$$\boxed{h_{\mathrm{corr}} = (1-a)h_0 - gC_{X,A}.} \tag{2.58}$$

Equation (2.58) gives the necessary and sufficient criterion for an exact proportional correction: $C_{X,A} = 0$. This construction uses independent complete modification auxiliaries, so $M_f = (1-a)M_f^{(0)}$, while the motion term retains its original action. For correlated preparations, the additional contribution is $-gC_{X,A}$ in the same equation.

Within-cycle correlations, accumulated formation action, and cycle-period corrections affect this result separately. Fixing marginal means or refreshing auxiliaries each round does not fix the within-cycle $C_{X,A}$. A different dwell protocol that lengthens the cycle yields a corresponding frequency factor such as $1/(1+a)$. Section 2.3.3 explains the unit-spectral-kernel identification $s = 1$ used for the formation angle.

2.4.6 Mass and Motion

Phase is read against a common reference internal to state space. The overall phase of a single eigenstate cancels in the projector $|\psi\rangle\langle\psi|$, so position occupation does not display this rotation. Determining relative phase requires actual nonzero coherence between two components and separate readings of its two quadratures in repeated trials.

Let the eigenvalues of the actual persistent generator be E_a and the active window t_H. Combine repeated preparation repair, thermal processing, return, filtering, and waiting that equalizes branch durations into T_{proc}. Then

$$\tau = t_H + T_{\mathrm{proc}}, \qquad \nu_{\mathrm{cyc}} = \tau^{-1}, \qquad f_{\mathrm{act}} = t_H/\tau. \tag{2.59}$$

Here f_{act} is the fraction of a cycle during which phase accumulates. Formation and initial-loading time are counted only as often as they actually occur; every recurring process enters τ. At the survival interface of Section 2.3.6, the coherence mode and its readable frequency are

$$\lambda_{ab} = q_T e^{-i(E_a-E_b)t_H}, \qquad \omega_{ab}^{\mathrm{obs}} = f_{\mathrm{act}}(E_a - E_b), \qquad \gamma_{\mathrm{exit}} = -\frac{\log q_T}{\tau} \quad (0 < q_T \le 1). \tag{2.60}$$

The argument and modulus of the same survival root determine phase and exit rates, respectively. Maintenance accumulates phase among survivors, while thermal excursions determine how much of the branch remains; both are divided by the complete cycle period. Initial coherence $c_0 \neq 0$ becomes $c_n = \lambda_{ab}^n c_0 = q_T^n e^{-in(E_a-E_b)t_H} c_0$ after n successive rounds. Reading real and imaginary parts, continuously unwrapping the phase change relative to the initial state, negating it, and dividing by $n\tau$ yields $f_{\mathrm{act}}(E_a - E_b)$. Reading logarithmic decay of its modulus yields γ_{exit}. With fixed per-cycle phase and survival probability, adding compensated common waiting rescales both rates by the same period factor. Neither the endpoint principal argument alone nor division by only the phase-active window gives the complete-cycle reading.

For formation during operation in Section 2.3.7, Supplementary Proof S3 also bounds the remaining transient spectral radius by $q_T\sqrt{\kappa}$, below the modulus q_T of the roots within the code. Formation transients therefore decay further relative to the survival envelope, and long-time visible frequencies still come from the same h. A particular line requires nonzero overlaps with preparation and observation. Separation of survival envelope and phase slope depends on the specified thermal

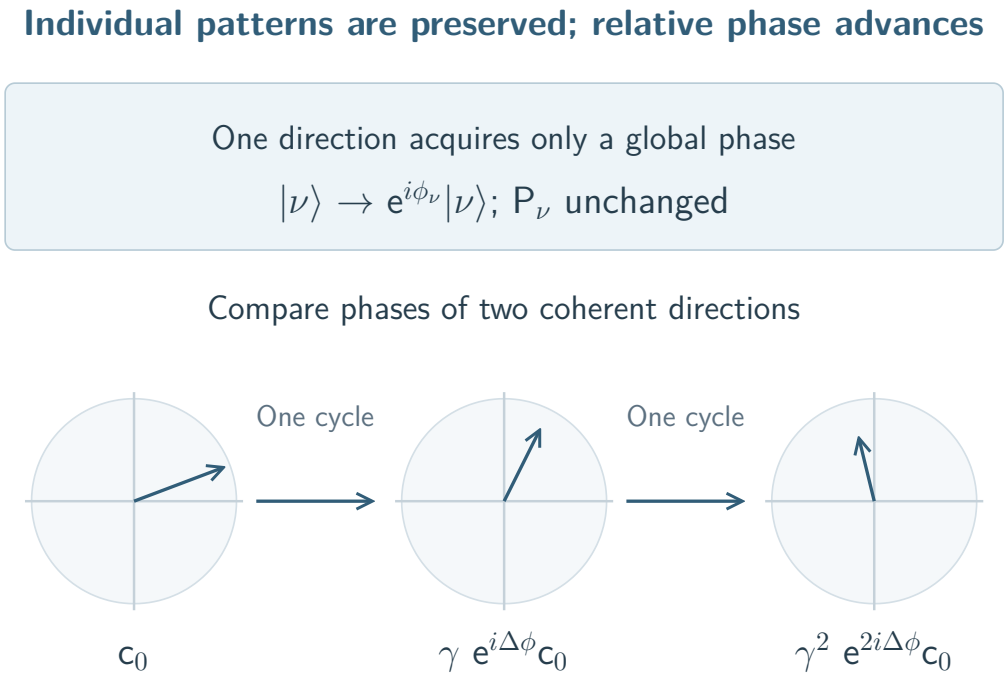


Figure 2-9. The overall phase of a single direction does not change its pattern. Coherence retained between two directions rotates from cycle to cycle and may decay simultaneously. Arrows represent a coherence matrix element, rotating by the same angle and shrinking by the same positive factor each round; they do not depict motion of a stable state in observational space.

interface and need not hold for a general thermal channel. Two-branch interference directly reads a frequency difference: branches $\pm\mu_j$ give beat frequency $2f_{\mathrm{act}}\mu_j$. Assigning the single-branch mass $f_{\mathrm{act}}\mu_j$ requires an established symmetry center or zero-mass reference.

As the carrier propagates between positions, internal maintenance continues to act on the same object. We use a joint preparation in which two opposite exchanges share a coherent port and both endpoints undergo the same simultaneous maintenance. On this carrier, the family mass matrix M_f and position propagation give

$$h_f(k) = M_f \otimes Y + gI \otimes Z_{\mathrm{mov}} \sin(k\ell_x), \qquad Y^2 = Z_{\mathrm{mov}}^2 = I, \quad \{Y, Z_{\mathrm{mov}}\} = 0. \tag{2.61}$$

The operators Y, Z_{mov} act on the two-state space of motion orientations and are defined separately from Z_F, which acts on formation directions. Let $\mu_j > 0$ be a positive eigenvalue of M_f. Squaring cancels cross terms, so

$$E_{j,\pm}^{\mathrm{obs}}(k) = \pm f_{\mathrm{act}} \sqrt{\mu_j^2 + g^2 \sin^2(k\ell_x)}, \qquad c_{\mathrm{obs}} = f_{\mathrm{act}} g \ell_x, \qquad M_{\mathrm{in},j}^{\mathrm{obs}} c_{\mathrm{obs}}^2 = f_{\mathrm{act}} \mu_j. \tag{2.62}$$

The last expression follows from the curvature of the positive branch at the origin, using $\hbar = 1$. On a finite ring, this uses the continuous symbol or a long-wave-packet curvature reading. One parameter controls both the rest-frame half-gap and inertia at the origin. This does not identify the full lattice dispersion with a Lorentz relation at all momenta or exclude other low-energy valleys in the complete position spectrum.

This driven generator must also be distinguished from the shell energy H_K of the original thermal exchange. Energy conservation at a bare joint endpoint does not imply work-free new control or $h_f = H_K$. Supplementary Proof S1.5 retains the constraint on conserved quantities under the same complete unitary, environment energy, and strict contraction; its premises do not extend to the modified joint process here.

2.4.7 General Mass Formula

For fixed preparation, coupling structure, and complete schedule, elementary-state masses are first determined by the actual matrix:

$$\boxed{g\mathcal{J}_s^\dagger X_f \mathcal{J}_s = M_f \otimes Y, \qquad m_{f,j}^{\mathrm{obs}} = f_{\mathrm{act}} |\mu_{f,j}|.} \tag{2.63}$$

Here X_f includes actual rest-frame paths, nodes, modification, target matching, and motion orientation. The right-hand side extracts the orientation factor Y. The quantity $\mu_{f,j}$ is a nonzero eigenvalue of the family matrix M_f and corresponds to symmetric branches of the persistent generator. When paths factorize, modification correlations vanish, and target matching is diagonal, the even-conjugate family gives

$$m_{f,j}^{\mathrm{obs}} = \frac{f_{\mathrm{act}} g}{\Lambda} \frac{\mathcal{P}_f(1-a)}{\sqrt{2}} p_j(s). \tag{2.64}$$

The actual neutral chain retains a nondiagonal matrix; other families use their respective spectral paths and ports. Before the common background supplies the absolute reading, g, Λ, and the active fraction remain in the formula. Source-completion

probability, loading share, and uninterrupted-survival share already enter their respective complete outputs and are not multiplied into this fixed generator again.

Paths, nodes, and group shares have now become coefficients of the persistent generator through exchange. The same parameter gives rest-frame phase and inertia at the origin. The next section calibrates the common factor $f_{\mathrm{act}} g/\Lambda$ against the background from the same source; finite feedback tails, link corrections, and nondiagonal matrices remain within their respective family actions.

2.5 Common Energy Scale

Exchange strengths and persistent frequencies of different stable states must be compared against a common background. Processes can involve different mode ranges and weights, but participating and background responses must use the same scale. This section uses a common source and schedule to compare their squared responses, retaining actual participation ranges and weights.

2.5.1 Participation Weights

Common calibration first requires identifying participating modes and their weights. The total participating mode quantity differs from a normalized pattern. For whole-space participation, for example, b is one coherent direction with nonzero amplitude at all d=p^K positions. Position modes provide an orthogonal expansion basis, while the pattern specifies relations among their components. The projection counting participating positions and that selecting the collective pattern are therefore respectively

$$P_{\mathrm{all}} = \sum_{x \in X_{p,K}} |x\rangle\langle x| = I_{\mathcal{H}}, \qquad P_b = |b\rangle\langle b|, \qquad \operatorname{rank} P_{\mathrm{all}} = d, \quad \operatorname{rank} P_b = 1. \tag{2.65}$$

A normalized pattern can be composed of d positions. The rank of the pattern and dimension of its participation range are counted separately.

First define the common participation quantity in the specified mode space. Let the participating subspace be spanned by normalized orthogonal vectors u_α indexed by $\mathcal{S}$, with orthogonal projection $P_{\mathcal{S}}$. Represent the unnormalized evaluation weights by a dimensionless positive operator Γ, and set

$$P_{\mathcal{S}} = \sum_{\alpha \in \mathcal{S}} |u_\alpha\rangle\langle u_\alpha|, \qquad w_\alpha = \langle u_\alpha|\Gamma|u_\alpha\rangle \geq 0, \qquad W_{\mathcal{S}} = \operatorname{Tr}(P_{\mathcal{S}}\Gamma P_{\mathcal{S}}) = \sum_{\alpha \in \mathcal{S}} w_\alpha. \tag{2.66}$$

Here $W_{\mathcal{S}}$ is the total spectral weight of selected participating modes. In a general participation basis, w_α are diagonal weights; in an eigenbasis of the compressed operator $P_{\mathcal{S}}\Gamma P_{\mathcal{S}}$, they are its nonnegative eigenvalues. With $P_{\mathcal{S}}$ and Γ fixed, a change of orthonormal basis preserves the trace. An orthogonal decomposition of the participating subspace gives additive weights. For Γ=I, every orthogonal mode has unit weight and $W_{\mathcal{S}} = \operatorname{rank} P_{\mathcal{S}}$. Whole-ring position participation therefore gives d=p^K and unit-shell position participation gives p^{K-1}(p−1), the counts for these respective ranges under unit weighting.

Each process must specify its participating modes and justify the weights used. A projection only marks the summation range; formation and exchange operators

must also establish that the process actually involves it. Repeated paths leading to the same indistinguishable final state combine amplitudes first, as in Section 2.4.2. Path names alone cannot increase the number of orthogonal modes.

For $W_{\mathcal{S}} > 0$, the corresponding normalized evaluation weight is

$$\sigma_{\mathcal{S}} = \frac{P_{\mathcal{S}} \Gamma P_{\mathcal{S}}}{W_{\mathcal{S}}}, \qquad \operatorname{Tr} \sigma_{\mathcal{S}} = 1, \qquad \langle u_\alpha | \sigma_{\mathcal{S}} | u_\alpha \rangle = \frac{w_\alpha}{W_{\mathcal{S}}}. \tag{2.67}$$

This separates total quantity from relative shares. Mathematically, $\sigma_{\mathcal{S}}$ is a density operator; identifying its shares as actual observation probabilities additionally requires a specific preparation and readout. Overall scaling of Γ cancels under normalization: for c>0, replacing Γ by cΓ preserves $\sigma_{\mathcal{S}}$ but changes $W_{\mathcal{S}}$ to $cW_{\mathcal{S}}$. If the subsequent energy scale uses $W_{\mathcal{S}}$, its specified absolute weights must be retained rather than changed by arbitrary renormalization.

Weights also do not replace coherent preparation. Both Γ_1=I and $\Gamma_2 = dP_b$ have unit diagonal entries in the site basis and trace d, yet normalization gives I/d and P_b, respectively. Section 2.2.4 described their coherence difference; collective formation must still satisfy Section 2.5.4.

Thermal weighting further requires specifying the spectrum and energy origin. The weight sum on the BC logarithmic energy spectrum indexed by positive integers generally differs from Gibbs weighting of the finite shell generator. This chapter uses the former for spectral evaluation and the latter for actual thermal exchange. Supplementary Proof S5 compares them and treats finite cutoffs. Once the participation projection and absolute weights are fixed, write their total as W and calculate the common response.

2.5.2 Background Response

Participation weights specify mode contributions; background calibration must also specify each mode's response. Take W>1 and a common energy calibration E_0>0. We first calculate squared responses on the process and background sides, then connect them to actual exchange strength.

The logarithmic spectrum indexed by positive integers supplies the spectral reference. For the complete initial spectral segment 1,…,N, the level count and highest energy obey the exact relation

$$\mathcal{N}(\epsilon) = \#\{n \geq 1 : \log n \leq \epsilon\} = \lfloor e^\epsilon \rfloor, \qquad \max_{1 \leq n \leq N} \log n = \log N. \tag{2.68}$$

Here N is the integer number of modes in the initial segment. For general positive participation weights, we adopt the common response scale $\Omega(W) = \log W$ and sum squared mode responses with positive weights. This collective-response prescription extends the initial-segment relation; a general weighted participation quantity W does not itself determine the highest energy. Actual frequency is read from phase and elapsed duration after connecting the response to exchange.

A participation projection and positive weight operator express the common response. Set $A_{\mathrm{coll}} = \Omega(W) P_{\mathcal{S}}$, assigning the same response coefficient throughout the chosen participating subspace. Summing squared responses with positive weights gives, on the process side,

$$\mathcal{Q}_{\mathrm{proc}} = \operatorname{Tr}(\Gamma A_{\mathrm{coll}}^\dagger A_{\mathrm{coll}}) = \Omega(W)^2 \operatorname{Tr}(\Gamma P_{\mathcal{S}}) = W \log^2 W. \tag{2.69}$$

The last step uses $P_{\mathcal{S}}^2 = P_{\mathcal{S}}$ and cyclicity of the trace, also for Γ not commuting with $P_{\mathcal{S}}$. The quantity $\mathcal{Q}_{\text{proc}}$ is this response's unnormalized square. For different response coefficients on orthogonal modes, the general expression is $\sum_\alpha w_\alpha |a_\alpha|^2$; the common-coefficient formula is a special case.

The background side uses adjacent gaps of the same logarithmic spectrum indexed by positive integers. Let $\Delta_n = \log(n+1) - \log n$. Each link has two orientations, each assigned an orthogonal response coordinate. Define $g_{\text{bg}}(n,+) = \Delta_n$ and $g_{\text{bg}}(n,-) = -\Delta_n$ on these coordinates. Their squared norm defines the intrinsic metric constant C*, the background squared-response constant of the complete logarithmic spectrum:

$$\Delta_n = \log\left(1 + \frac{1}{n}\right), \qquad g_{\text{bg}} \in \ell^2(\mathbb{N}^* \times \{+,-\}), \qquad C^* := \|g_{\text{bg}}\|^2 = 2\sum_{n=1}^{\infty} \Delta_n^2. \quad (2.70)$$

Since $0 < \Delta_n < 1/n$, the square sum is finite and positive. The factor 2 counts both orientations of each link: each is squared before summation. The vector g_{bg} describes the gradient response of the spectral reference and is distinct from the constant-function pattern b in the finite state space; its norm cannot be conflated with normalization of b.

The background reference uses the complete logarithmic spectrum. A different choice of finite open chain 1,…,J, J≥2, has J−1 links. Its normalization and truncation error satisfy

$$C_J^* = 2\sum_{n=1}^{J-1} \Delta_n^2, \qquad 0 < C^* - C_J^* < 2\sum_{n=J}^{\infty} \frac{1}{n^2} < \frac{2}{J-1}. \quad (2.71)$$

The final bound follows from a tail integral. For the open chain of nonzero address representatives 1,…,p^K-1, take J=p^K-1 and denominator C_J^*, without adding the zero residue class or a closing link. Background cutoff J and participation-weight cutoff L are specified separately.

Using E/E_0 as the common amplitude with which an excitation drives the background gives response $(E/E_0)g_{\text{bg}}$ and squared quantity $(\text{E}/\text{E}_0)^2\text{C}^*$. Matching the squared quantities on the process and background sides for the same response yields

$$\left(\frac{E(W)}{E_0}\right)^2 C^* = W \log^2 W, \qquad E(W) = E_0\sqrt{\frac{W}{C^*}}\log W, \qquad W > 1. \quad (2.72)$$

Taking the positive root for $E \geq 0$ determines this energy scale; E_0 is supplied by physical calibration. A finite background follows the same derivation, replacing only C^* by C_J^*: $E_J(W) = E_0\sqrt{W/C_J^*}\log W$. At fixed W, E_0, $E_J(W)/E(W) = \sqrt{C^*/C_J^*}$ quantifies the effect of background truncation on the reading.

2.5.3 Background Calibration from a Common Source

Exact mass ratios originate in factors actually shared by matter and background. The formation pattern gives different directions different matching shares, while common preparation, drive, and active time multiply both matter and reference

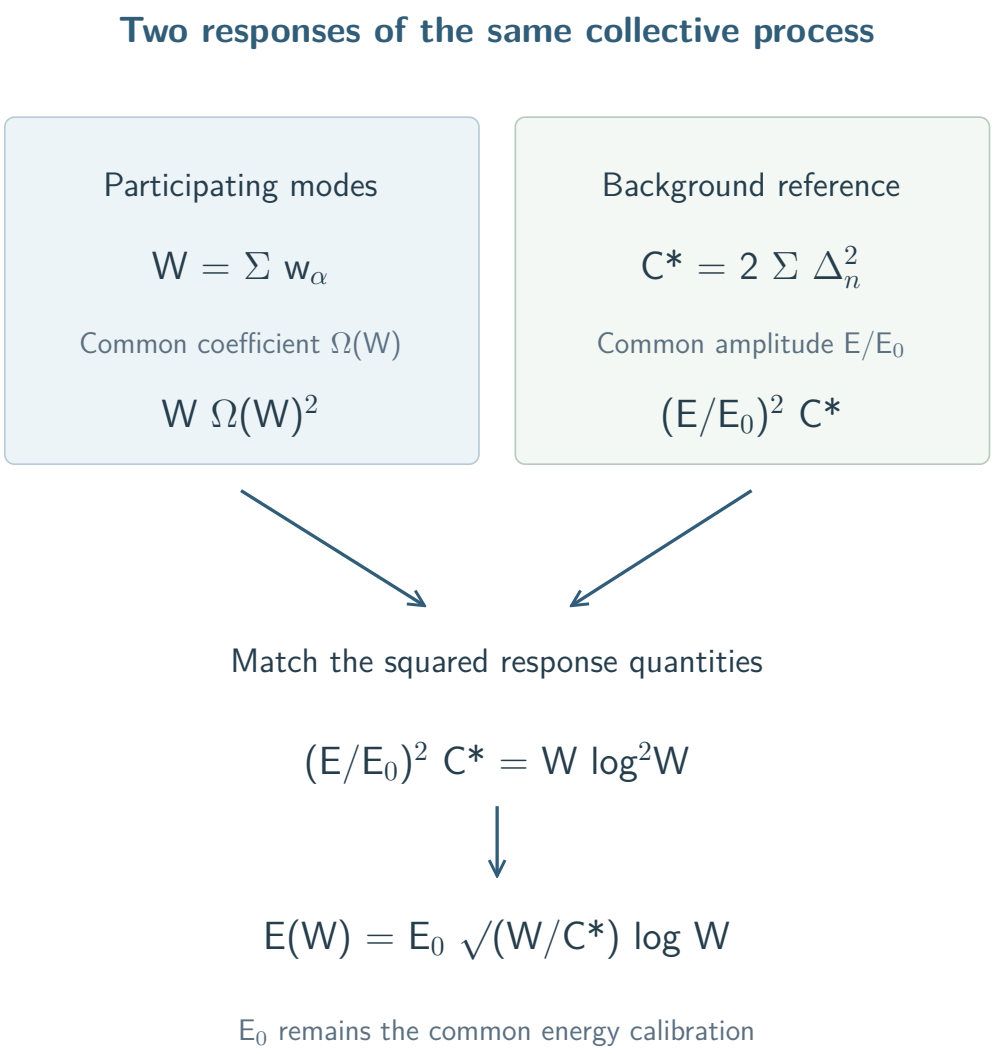


Figure 2-10. The process side uses a common coefficient and sums squares with positive weights; the background side uses the response of bidirectional logarithmic gaps. Matching the squared quantities gives E(W), retaining both the common energy calibration E_0 and background range.

branches. Comparing them cancels common factors and retains directional differences. To realize this relation, place the formed source, auxiliaries, orientations, and background gradient in one isometry $\mathcal{J}_s$. Matter retains group matching Q_{TS} from Section 2.4. Fix the background target to reference group zero, so its common-source matching becomes P_0^S. The background bypasses matter-specific spectral paths and nodes while retaining the same capacity and basic strength.

To specify how capacity enters, take $\Lambda \geq 1$ and set $c = (\Lambda - 1)/2$. Prepare $r_\Lambda = (1, \sqrt{c}, \sqrt{c})^T/\sqrt{\Lambda}$ and $Z_\Lambda = \mathrm{diag}(1, 1, -1)$, so $\langle Z_\Lambda \rangle = 1/\Lambda$. On an independent auxiliary, choose a self-adjoint unitary background-tail action $U_t = U_t^\dagger$, $U_t^2 = I$, and prepare its normalized positive branch $U_t|+_t\rangle = |+_t\rangle$; a binary exchange $U_t = \sigma_x$ is one example.

Represent the background gradient by a self-adjoint contraction block $G_r = \sqrt{b_r}\sigma_x$ between entrance and exit, with $G = \bigoplus_r G_r$ and $\sum_r b_r = C_{\mathrm{bg}}$. For the bidirectional links above, take $r = (n, \pm)$ and $b_r = \Delta_n^2$. The complete background gives $C_{\mathrm{bg}} = C^*$ and a finite open chain gives C_J^*. Actual exchange in each block starts at its entrance preparation. The isometry j_G embeds the gradient space, including both endpoints, into the complete auxiliaries, with the selected self-adjoint dilation satisfying $j_G^\dagger U_G j_G = G$. For independent auxiliary preparations, the explicit exchange is

$$X_B = P_0^S Z_\Lambda U_t U_G, \qquad X_M = Q_{TS} Q_{\mathrm{path}} B_N (I - A) U_t Y. \tag{2.73}$$

Unwritten factors carry the identity. The capacity factor for matter is already included in the path effect, so Z_Λ is not applied again to the matter branch. Background and matter share one source. Applying reflection protection from Section 2.3 on the common preparation code gives the intertwining relations

$$h_B = \frac{g p_0}{\Lambda} G, \qquad h_M = M_f \otimes Y, \qquad M_f = \frac{g \mathcal{P}_f (1 - a)}{\sqrt{2}\Lambda} \sum_j p_j P_j. \tag{2.74}$$

Here $1/\sqrt{2}$ uses the node's own reading, while the path uses its complete effect. In this example, B_N may be the binary node without an additional spectral insertion. Both simultaneous actions preserve the same source pattern. Separate knowledge of their expectations does not ensure this sharing; both actual exchanges must use the same complete preparation and protection.

Starting from the gradient entrance above, block r has actual exit probability $\sin^2[(g p_0/\Lambda)\sqrt{b_r}\, t]$. Summing short-time squared-response coefficients over entrances gives $(g p_0/\Lambda)^2 C_{\mathrm{bg}}$. With the complete cycle of Section 2.4.6, matter and background pause in the same processing window, the background remains coherent across windows, and it continues to final readout after target exit. The coefficient read against complete physical time therefore carries the common active fraction f_{act}. Matching to the process side of the preceding subsection yields

$$\left(\frac{f_{\mathrm{act}} g p_0}{\Lambda E_0}\right)^2 C_{\mathrm{bg}} = W \log^2 W, \qquad \boxed{\frac{f_{\mathrm{act}} g p_0}{\Lambda} = v_E, \quad v_E = E_0 \sqrt{\frac{W}{C_{\mathrm{bg}}}} \log W.} \tag{2.75}$$

The positive branch and positive strength select the positive root. Substituting the already calculated matter action gives

$$\boxed{m_{f,j}^{\mathrm{obs}} = v_E \frac{\mathcal{P}_f(1-a)}{\sqrt{2}} \frac{p_j(s)}{p_0(s)}.} \tag{2.76}$$

Every factor in Eq. (2.76) now has a process origin. Paths and nodes determine the response specific to matter, while p_j/p_0 compares shares of the same formation pattern in the matter direction and background reference direction. Common capacity, basic strength, and active fraction cancel in this comparison, leaving an exact dimensionless ratio; the reference group need only satisfy $p_0 > 0$. These common quantities still determine absolute frequencies, propagation speed, and completion time, and must remain in subsequent motion and metric-strength calculations.

The quantity v_E calibrates the background-response energy scale; the cycle rate $\nu_{\mathrm{cyc}} = 1/\tau$ is determined by the actual schedule. Multiplying per-cycle phase $\mu_j t_H$ by that rate still gives $\nu_{\mathrm{cyc}}\mu_j t_H = f_{\mathrm{act}}\mu_j$. Common calibration does not require $\tau = 1/v_E$.

For collective participation to enter a persistent frequency, the actual output of the participation process must connect to the same coherent entrance. Let the complete unitary preparation retain all address references, spectral records, and auxiliaries, with success column v_W. Use the same capacity Λ_{col} at every port, satisfying $\|v_W\|^2 = W/\Lambda_{\mathrm{col}}^2$. Construct simultaneous reflection-protected exchange from ready state z_0 and this complete output. Its action on the entire success code is

$$X_{v_W} = |1, v_W\rangle\langle z_0| + |z_0\rangle\langle 1, v_W|, \qquad m_W = g_{\mathrm{col}}(\log W)\|v_W\| = \frac{g_{\mathrm{col}}}{\Lambda_{\mathrm{col}}}\sqrt{W}\log W. \tag{2.77}$$

Participating directions couple through one entrance. The normalized bright exit is $v_W/\|v_W\|$, so the entrance-to-bright-exit matrix element is $\|v_W\|$. This gives the dynamical origin of $\sqrt{W}$, and the original logarithmic action supplies $\log W$. The quantity m_W is the rest-frame half-gap of this common carrier; with shared motion it becomes the mass parameter in the inertial term. Participation counts enter maintenance through this bright-mode frequency only when connected as actual output columns with reversible exchange preserved on the complete auxiliaries.

Vacuum and matter ports share this capacity, the actual strength g_{col}, and the background-gradient action. The background frequency of the positive-eigenvalue vacuum bright state (the positive bright state) therefore contains the common factor $g_{\mathrm{col}}/\Lambda_{\mathrm{col}}$. One squared matching gives

$$\boxed{g_{\mathrm{eff}} := \frac{g_{\mathrm{col}}}{\Lambda_{\mathrm{col}}} = \frac{E_0}{\sqrt{C_{\mathrm{bg}}}}, \qquad m_W = E_0\sqrt{\frac{W}{C_{\mathrm{bg}}}}\log W.} \tag{2.78}$$

The common capacity cancels in this single matching. The collective-port quantity g_{eff} and elementary-port quantity gp_0/Λ above are defined by their respective actual actions and cannot directly be identified as one bare control coefficient. Supplementary Proof S6 proves the complete-output and calibration relations. Chapter 6 connects metric return and the residual vacuum output to this common carrier.

This common-source calibration can connect to actual background occupation by matter. Use the same formation source, complete matching, and common binary tail; transport the original matter and gradient arms together with the occupation

preparation, then protect their balanced action on the complete available output. Let E be the complete matching effect, $H_s = g_M E/4$ the original action within the code, and $c_B = g_B q_0/\Lambda$ the background coefficient. Take $0 < s \leq 1$ and $0 \leq e < 1$ and define $r_e = \sqrt{s(1-e)}$; $s = N/\Lambda_B^2$ is the actual occupation-preparation share. The common positive bright branch of this explicit connection is maintained for a duration t_H within the complete cycle period τ, giving

$$\boxed{M_e = \frac{t_H}{\tau}\frac{r_e g_M e}{2}, \qquad V_e = \frac{t_H}{\tau}\,2 r_e c_B, \qquad n_e = \frac{Ne}{2}, \qquad n_e = \frac{2Nc_B}{g_M}\frac{M_e}{V_e}.} \tag{2.79}$$

The same matching e determines both matter phase and occupation share, while available amplitude r_e and holding ratio t_H/τ multiply matter and background gradient alike. Division cancels these two common factors exactly, relating resource occupation to M_e/V_e. The inverse background frequency thus comes from the ratio of two responses of the same carrier: it measures the occupation associated with a given matter phase relative to the background maintenance rate.

The quantity V_e is the coefficient of the entire original gradient operator; the frequency of each gradient spectral line is obtained by multiplying this coefficient by the corresponding eigenvalue. The cycle rate is $1/\tau$, and the relation does not require $V_e\tau = 1$. The full construction, connection of formation input to occupation, and survival records are in Supplementary Proofs S7.1–S7.3. Spatial propagation transports the resource auxiliary system carrying the occupation state, hereafter called the resource packet, while the original gradient action continues on its actual carrier.

These comparisons hold along the same history in which matter and background share accumulated action and total elapsed duration. Combining histories of different durations at one time requires the full phase distribution. Gradient entrances use the actual reference preparation; Section 4.5.5 treats calibration after resource gating. Neutrino transport stages, relays, repair, and thermal processing follow the same active-window schedule; Supplementary Proofs S13.3–S13.5 give the local extension that keeps the background running after exit. Chapter 8 specifies the paths.

2.5.4 Mass of Collective States

The shells of a collective state participate with shares h_j in one pattern b with definite relative phases. Chapter 9 forms this pattern by energy-conserving exchange with coherent reference b_R and connects it to protected persistent action. Its joint output has the same state of the remaining degrees of freedom for all shell components.

Restoration with identical occupations but orthogonal shell records produces only a mixed state. Fixed shell restoration in Supplementary Proof S1 gives $X + y\sigma_g^{\mathrm{sh}}$, where X is supported in the direction space orthogonal to b, and $0 \leq y \leq 1$. Hence

$$\langle b|(X + y\sigma_g^{\mathrm{sh}})|b\rangle = y\sum_{j=0}^{K} g_j h_j < 1, \qquad K \geq 1. \tag{2.80}$$

Every shell has $h_j < 1$, so this restoration does not prepare the pure pattern P_b. For a normalized joint formation component $\Psi = \sum_j |e_j\rangle \otimes \xi_j$, the system matrix element is $\langle \xi_\ell | \xi_j \rangle$. Producing P_b requires exactly

$$\langle \xi_\ell | \xi_j \rangle = \sqrt{h_j h_\ell} \quad \text{for all } j, \ell; \qquad \text{equivalently,} \quad \xi_j = \sqrt{h_j}\, \xi, \quad \|\xi\| = 1. \tag{2.81}$$

The off-diagonal relations saturate Cauchy–Schwarz pairwise and fix relative phases, so every $\xi_j/\sqrt{h_j}$ is the same unit vector and the joint state is $b \otimes \xi$. The reference exchange of Section 9.1.3 produces this common final state; Section 9.3.2 then gives its persistent return. For the collective pattern, every multiplicative pullback gives $T_n b = b$, so the finite positive-weight response satisfies $\widetilde{\mathcal{B}}_L(\beta) b = (\sum_{n=1}^{L} n^{-\beta}) b$. Denote this response coefficient by $Z_L(\beta)$ and adopt the collective-participation identification $W_{\mathrm{em}} = Z_L(\beta)$. For L≥2 and β>0, $W_{\mathrm{em}} > 1$. This identification uses the operation's total response, not the rank of the one-dimensional pattern projection. Supplementary Proof S5 details the distinctions among the logarithmic spectrum, shell spectrum, and truncations. With the fixed common response and background matching, the characteristic energy is

$$E_{\mathrm{em},L}(\beta) = E_0 \sqrt{\frac{Z_L(\beta)}{C^*}} \log Z_L(\beta). \tag{2.82}$$

For fixed β>1, the tail bound in Supplementary Proof S5 controls Z_L. Taking L→ ∞ then gives $E_{\mathrm{em}}(\beta) = E_0 \sqrt{\zeta(\beta)/C^*} \log \zeta(\beta)$, using positive-weight convergence and continuity of the energy-scale function. Here β specifies thermal weighting: varying β describes different participation weights, not elapsed physical time, and does not change the absence of a pole at β=1 for fixed finite L.

Chapter 9 connects collective participation to the persistent generator through orthogonal response ports, complete protection, and common background calibration. Using the rest-frequency reading of Section 2.4.6 and natural units $\hbar$=c=1, collective mass is read from a nonzero coherence mode against a zero-mass reference. Let $\tau_{\mathrm{em}} > 0$ be the complete cycle period and $\widetilde{\varphi}_{\mathrm{em}}$ the continuously unwrapped per-cycle phase relative to that reference. Frequency and energy are connected by

$$M_{\mathrm{em}} = -\frac{\widetilde{\varphi}_{\mathrm{em}}}{\tau_{\mathrm{em}}}. \qquad M_{\mathrm{em}} = E(W_{\mathrm{em}}) \quad \Longleftrightarrow \quad -\widetilde{\varphi}_{\mathrm{em}} = \tau_{\mathrm{em}} E(W_{\mathrm{em}}). \tag{2.83}$$

Phase here is read against a calibrated zero-mass reference, and actual operations determine the complete period. Write $B_{\mathrm{em}} = -\widetilde{\varphi}_{\mathrm{em}} > 0$. Sharing $v_E = E(W_{\mathrm{ref}})$ with elementary states then gives

$$\frac{M_{\mathrm{em}}}{v_E} = \sqrt{\frac{W_{\mathrm{em}}}{W_{\mathrm{ref}}}} \frac{\log W_{\mathrm{em}}}{\log W_{\mathrm{ref}}}, \qquad B_{\mathrm{em}} = \tau_{\mathrm{em}} v_E \frac{E(W_{\mathrm{em}})}{E(W_{\mathrm{ref}})}. \tag{2.84}$$

The common E_0 and background square constant cancel in the ratio, while each period is still timed from its own operations. Per-cycle phase magnitude equals the energy ratio only under the additional condition $\tau_{\mathrm{em}} v_E = 1$. Sections 9.1 and 9.3 provide the full collective-formation, return, and frequency construction.

Background exchange produces a coherent pattern, and actual matching turns group shares into coefficients of the persistent generator. Return preserves the carrier, exit determines its survival envelope, and relative phase gives the rest frequency. The same generator controls inertia at the origin, while squared matching to the same-source background fixes the common mass scale. Collective states use this scale as well, with intershell coherence and complete return realized in Chapter 9. The next chapter treats joint exchange between maintained carriers.

§3 Resonance Forces

Stable states persist through exchange with the background and can also send excitations to other stable states through a common environment. When the emitting and absorbing endpoints have compatible spectral channels, a component released by one can enter the formation or return process of the other, allowing the two carriers to affect each other through actual exchange. This exchange interaction, carried by common spectral channels, is called a resonance force.

An exchange changes the carrier states and may also leave a record in the environment. Its strength must be determined by comparing the input and output of the same joint process: endpoint overlap determines whether the excitation can enter, propagation and return determine how amplitudes combine, and the actual records and elapsed duration determine the event share and rate of change that are read out. With a calibrated momentum reading, the rate of momentum change caused by exchange is the corresponding mechanical force. The participation of elementary and emergent states follows Chapter 2; the responses of both are calculated using these physical relations.

3.1 Resonant Exchange

3.1.1 Transfer of a Directional Excitation

First consider the case in which the emitting and absorbing endpoints can access the same nontrivial direction χ. This direction is represented by the phase pattern ψ_χ on the unit shell, with projection $P_\chi = |\psi_\chi\rangle\langle\psi_\chi|$. Using the shell-preserving response defined at a regular spectral point s_* in Chapter 2, write $R_\chi = R_\chi(s_*)$. Then

$$P_{\chi'} T_{\mathrm{el}}(s_*) P_\chi = \delta_{\chi'\chi} R_\chi P_\chi. \tag{3.1}$$

This selection rule follows from character orthogonality: after traversing this response segment, a component entering direction χ remains in that direction, multiplied by the corresponding complex coefficient. We say that the emitting and absorbing endpoints have a common spectral channel when both have a nonzero component in this direction. This means that their patterns can connect; how much amplitude actually leaves the emitting endpoint, and whether it can reach the absorbing endpoint, must still be determined by the exchanges at both endpoints and the intervening process.

The squared spectral response also has a definite but limited meaning. For a normalized prepared state ρ, selecting this direction with P_χ gives

$$\mathrm{Tr}\Big(\rho P_\chi T_{\mathrm{el}}^\dagger T_{\mathrm{el}} P_\chi\Big) = |R_\chi|^2 \,\mathrm{Tr}(\rho P_\chi). \tag{3.2}$$

Here $\mathrm{Tr}(\rho P_\chi)$ is the input share in the selected direction. The squared response of the normalized conditional input is $|R_\chi|^2$, while the unnormalized spectral response gives $|L(s_*, \chi)|^2$; the actual probability and frequency shift are further determined by the joint exchange and its elapsed duration.

Specify subsystems A and B, an environment C, and their actual interactions within the whole. Use the direction–shell-average exchange of 2.3.1 as the vertex at each endpoint: C receives the component released by A and exchanges with B after

propagation. This tensor decomposition is a condition of the joint construction. We first calculate transfer between the selected endpoints, then incorporate the complete carriers comprising formation source, target, and auxiliaries in 3.1.2. In the actual temporal order, the joint unitary action during transfer is

$$\mathbb{U}_{\mathrm{ex}} = U_{BC}\,(I_{AB} \otimes U_C)\,U_{AC}, \qquad \mathcal{A}_{fi} = \langle f|\mathbb{U}_{\mathrm{ex}}|i\rangle. \tag{3.3}$$

The local operators of A and B act on different factors, but the two complete exchanges share C, so U_{AC} and U_{BC} generally cannot be interchanged. The amplitudes, phases, and records retained by C at the intermediate stage determine what the second vertex actually receives. To discuss return, append the specified release and retrieval actions after absorption; the transfer stage itself occupies only part of the complete cycle.

Let A initially carry the directional excitation, let B absorb from its shell-average endpoint, and let environment C receive and convey the excitation between them. This process can be restricted to a minimal closed transfer subspace. Choose shell-average endpoints e_a and e_b for A and B, and orthogonal states u and v_χ for C, and define three normalized joint vectors:

$$|i\rangle = |\psi_\chi\rangle_A|e_b\rangle_B|u\rangle_C, \quad |m\rangle = |e_a\rangle_A|e_b\rangle_B|v_\chi\rangle_C, \quad |f\rangle = |e_a\rangle_A|\psi_\chi\rangle_B|u\rangle_C. \tag{3.4}$$

These states are balanced at both vertices in total energy and total unit-group charge. Denote the energies of the directional states at the two endpoints by $E_{A\chi}$ and $E_{B\chi}$, and the environment energies by ϵ_u and ϵ_v. The required energy condition is $E_{A\chi} - E_a = \epsilon_v - \epsilon_u = E_{B\chi} - E_b$. The energies used here are those specified by the actual joint generator; with the shell generator H_K of the original exchange in 2.3.1, these are the corresponding shell energy differences. If the mass frequencies of Chapter 2 are further used as these exchange energies, Eq. (S13) in Supplementary Proof S1.5 and the subsequent compatibility conditions between phase and conserved energy must also hold, and the balance must be checked with the corresponding environment energies. A shared character label gives only directional matching.

In this closed three-state subspace, the first vertex performs a unitary rotation between i and m through angle α; the second rotates between m and f through angle γ; and intermediate propagation gives m a relative phase ϑ. The remaining endpoint is unchanged during each rotation, and the two forward-transfer coefficients are chosen as sin α and sin γ. Applying the actions in order gives

$$\begin{aligned} \mathbb{U}_{\mathrm{ex}}|i\rangle &= \cos\alpha|i\rangle + e^{i\vartheta}\sin\alpha\cos\gamma|m\rangle + e^{i\vartheta}\sin\alpha\sin\gamma|f\rangle, \\ \mathcal{A}_{fi} &= e^{i\vartheta}\sin\alpha\sin\gamma, \\ P_{i\to f} &= \sin^2\alpha\sin^2\gamma. \end{aligned} \tag{3.5}$$

The squared moduli of the three orthogonal outcomes sum to 1, and the completed-transfer amplitude is the product of the two vertex amplitudes and the propagation phase. When several paths lead to the same complete final state, propagation phases enter the interference. Actual exchange angles are determined by the vertex actions, while energy balance and the relative phases at successive contacts ensure resonance. The source, auxiliaries, and retrieval of the persistent carriers are incorporated in the next subsection.

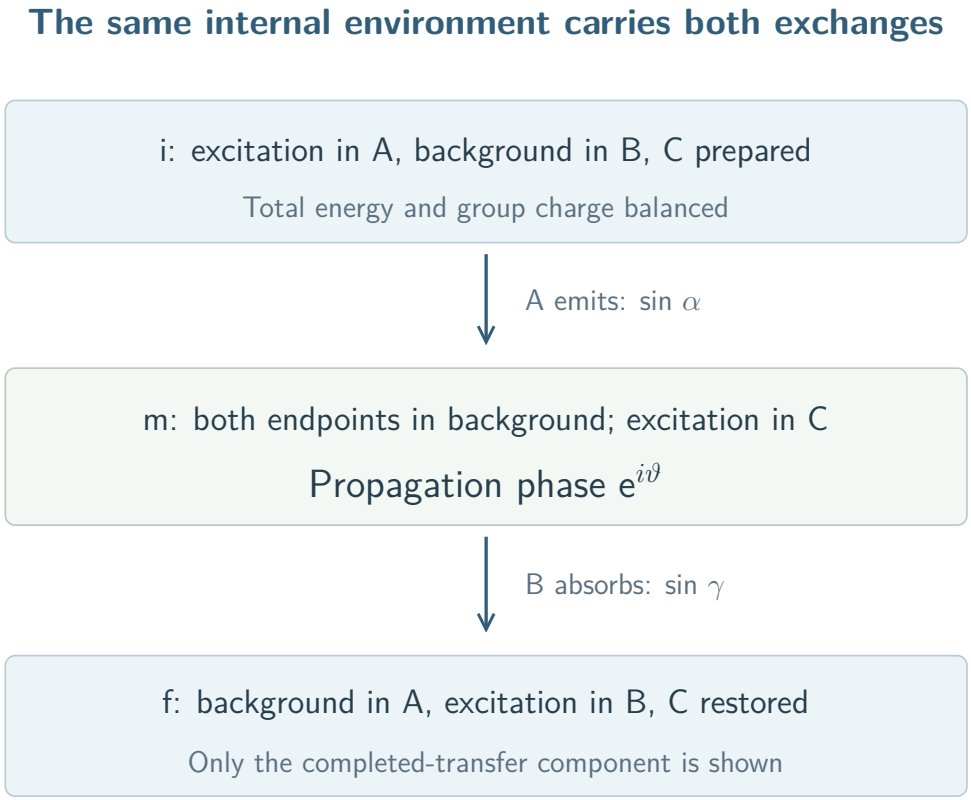


Figure 3-1 An excitation is transferred from A to B through the same C. The completed-transfer component in the three-state model has two vertex amplitudes and one propagation phase; the complete unitary output also includes components that were not emitted or not absorbed. Arrows denote the evolution of components, not measurements already performed at intermediate stages.

Group-charge conservation can be checked within the same joint process. Let $D_A(u)$, $D_B(u)$, and $D_C(u)$ be the unit-group actions on the respective subsystems, and let q denote the character eigenvalue function of each state under these actions. If every exchange and the intermediate propagation commute with the total group action, then every nonzero matrix element between input and output components of definite group charge satisfies

$$\begin{aligned} \mathbb{U}_{\mathrm{ex}}, D_A(u) \otimes D_B(u) \otimes D_C(u)t \quad &= 0 \quad (\forall u), \\ \mathcal{A}_{fi} \neq 0 \implies \prod_{S=A,B,C} q_S^{\mathrm{out}} &= \prod_{S=A,B,C} q_S^{\mathrm{in}}. \end{aligned} \tag{3.6}$$

To prove this, simply move the total group action through the joint unitary operator: the same matrix element is multiplied by the total character of the input and that of the output, so it must vanish when these differ. The notation χ and its conjugate is interchanged when multiplicative pullback is replaced by its inverse action; the eigenvalue function q used here preserves the convention actually adopted. If the input and output states of C belong to the same definite-charge sector, the charge changes at the two endpoints cancel; if C remains in another sector, it carries the remaining charge. Even if a reference containing coherence across charges returns to the same state, one cannot directly cancel a definite charge for it in this component formula. Total energy is balanced in the same way when it is conserved by the joint generator. Direction preservation along the propagation segment and joint conservation in the complete exchange thus have distinct, well-defined scopes.

3.1.2 Exchange between Stable States

Two persistent carriers interact through their target logical states, while their formation sources, auxiliaries, and old records remain part of the complete objects. Using the product-form isometric embedding of 2.3.5, first prepare the two endpoints separately: $\mathcal{J}_\alpha\psi = \omega_\alpha \otimes \psi$ and $\alpha = A, B$, where ω_α contains the source and auxiliaries already prepared at that endpoint, and old records no longer participate in this interaction. For each target choose actually accessible orthogonal modes $|0\rangle_\alpha, |1\rangle_\alpha$ and write $\mathcal{L}_\alpha = \mathrm{span}\{|0\rangle_\alpha, |1\rangle_\alpha\}$. Logical inputs are restricted to $\mathcal{L}_A \otimes \mathcal{L}_B$ and may be entangled with spectator records; 0 does not denote exit into the background.

Relay C has endpoints u, v and is prepared in u at entry. The endpoint projections $P_\alpha = \mathcal{J}_\alpha\mathcal{J}_\alpha^\dagger$ act as the identity on C, allowing it to store the logical state temporarily.

First specify endpoints capable of contacting the target, rather than only the fixed source. Let B_α be an actual self-adjoint action on the local source and auxiliaries, with $\beta_\alpha = \langle\omega_\alpha|\mathsf{B}_\alpha|\omega_\alpha\rangle > 0$. Set the following target transitions to zero outside the selected two levels, and define

$$\begin{aligned} L_A &= |0\rangle_A\langle 1| \otimes |v\rangle_C\langle u|, \quad L_B = |1\rangle_B\langle 0| \otimes |u\rangle_C\langle v|, \\ X_\alpha &= i\mathsf{B}_\alpha \otimes (L_\alpha - L_\alpha^\dagger), \quad \mathcal{J} = \mathcal{J}_A \otimes \mathcal{J}_B \otimes I_C. \end{aligned} \tag{3.7}$$

The other carrier and its auxiliaries, omitted from these expressions, carry the identity. This explicitly selects ports between the carriers: the minimal example takes $\mathsf{B}_\alpha = I$, and a spectral auxiliary can also enter as one actual action. If an

endpoint acts only on the fixed source and as the identity on the target, its compression has no off-diagonal matrix element L_α and cannot implement the transfer below. The two selected modes must also remain closed during exchange; an abstract change of basis alone cannot establish that an actual port is available.

Reuse the reflections $S_\alpha = 2P_\alpha - I$ synthesized from the respective preparations in 2.3.5, with both arms acting simultaneously at each endpoint. Termwise compression gives

$$\begin{aligned} \widehat{H}_\alpha &= \frac{g_\alpha}{2}(X_\alpha + S_\alpha X_\alpha S_\alpha), \\ \widehat{H}_\alpha \mathcal{J} &= \mathcal{J} h_\alpha, \qquad h_\alpha = i\Omega_\alpha(L_\alpha - L_\alpha^\dagger), \qquad \Omega_\alpha = g_\alpha \beta_\alpha. \end{aligned} \tag{3.8}$$

As before, the two arms cancel components outside the local preparation space, while $\mathcal{J}_\alpha^\dagger \mathsf{B}_\alpha \mathcal{J}_\alpha$ gives β_α on the source and auxiliary factors. The preparation at the other endpoint is unaffected, so both terms preserve the common carrier. Sharing C does not make the two terms commute; they are composed below in the actual order.

Write $R_\alpha(x) = \exp[x(L_\alpha - L_\alpha^\dagger)]$, so that $x = \Omega_\alpha t$. First perform the full transfer $T = R_A(\pi/2)$ at endpoint A, temporarily storing A's logical state in C; then perform a partial exchange between C and B through angle γ; finally perform the actual inverse retrieval at the same A endpoint. For C entering in u, the three stages act as

$$\begin{aligned} W &= T^\dagger R_B(\gamma) T, \\ W(\psi_{AB} \otimes u) &= (V_\gamma \psi_{AB}) \otimes u, \\ V_\gamma|00\rangle &= |00\rangle, \qquad V_\gamma|11\rangle = |11\rangle, \\ V_\gamma|10\rangle &= \cos\gamma|10\rangle + \sin\gamma|01\rangle, \\ V_\gamma|01\rangle &= \cos\gamma|01\rangle - \sin\gamma|10\rangle. \end{aligned} \tag{3.9}$$

Check each of the four logical inputs: the first stage sends $|a\rangle_A u_C$ to $|0\rangle_A|a\rangle_C$, the middle stage acts only on B and C, and the final stage completely retrieves both components of C, so C returns to u for every input. T retains the signs of a real rotation at other inputs. If only emission and absorption are retained, return of the mediator for $|10\rangle$ does not guarantee its return for $|01\rangle$; the third stage completes retrieval over the entire logical space.

Composing the exact intertwining relation for each stage, with $\widehat{W}$ denoting the three actual protected pulses, yields the identity on the complete carrier:

$$\boxed{\widehat{W}\,\mathcal{J}(\psi_{AB} \otimes u) = \mathcal{J}(V_\gamma \psi_{AB} \otimes u).} \tag{3.10}$$

Thus the same relation holds for arbitrary mixed inputs and their entanglement with spectator records, without postselecting successful transfers. The sources and auxiliaries at both endpoints retain their original preparations and the mediator returns, while the target states may change and become entangled across the endpoints. For example, $|10\rangle$ becomes the coherent superposition of the two terms shown in the equation; a change of logical state does not mean that the complete carrier exits.

This example is calculated in the interaction picture at actual resonance. If the two modes are eigenstates of the local free generator, energy and charge balance within the code require

$$E_{A1} - E_{A0} = E_{B1} - E_{B0} = \varepsilon_v - \varepsilon_u = \Delta,$$
$$q_{A1} q_{A0}^{-1} = q_{B1} q_{B0}^{-1} = q_v q_u^{-1}. \tag{3.11}$$

The charge equalities apply when the actual endpoints have definite group charges; otherwise covariance must be checked using matrix elements of the full representation. The energy condition within the code makes V_γ commute with the sum of the free energies at the two endpoints. Sufficient conditions for the complete two-arm action to conserve the same joint energy also include $[H_{\mathrm{free}}, X_\alpha] = [H_{\mathrm{free}}, S_\alpha] = 0$; energy conservation at the bare endpoint does not automatically ensure it for the reflected arm. If the actual controls do not satisfy these conditions, the energy supplied by the drive must be included, and shell energy cannot be directly identified with mass frequency.

When the maintenance action is retained and the above resonance conditions hold, free phases can be restored over the entire actual elapsed duration. If maintenance is paused during exchange, the pause still enters the frequency as prescribed in 2.4.6. For a forward middle stage $\gamma \geq 0$, the three-stage duration is $T_{\mathrm{ex}} = \pi/(2\Omega_A) + \gamma/\Omega_B + \pi/(2\Omega_A) + T_{\mathrm{ctrl}}$, with additional propagation, preparation, and recording time included in T_{ctrl}. The final stage uses the actual inverse generator; if a different retrieval pulse is used, its action and duration must be recalculated.

For the specified input $|10\rangle$, the middle-stage strength and readable transfer give

$$\boxed{\gamma = g_B \beta_B T_B, \qquad P_{10\to 01} = \sin^2\gamma, \qquad \Delta\langle E_B\rangle = \Delta \sin^2\gamma.} \tag{3.12}$$

If the local B_B uses the single-address effect of 7.1.3 and retains its specified auxiliary preparation, then $\beta_B = |R_\theta|^2/6$. This incorporates the spectral share into an actual exchange angle; the event probability remains $\sin^2\gamma$ above, with interaction strength and duration retained. For a general port, β_B is obtained from the actual auxiliaries; a common character label cannot replace that calculation, nor can this example be taken as the unique realization of all force vertices.

When both endpoints use the same pattern Φ, they may also share one source. Take the joint embedding $\mathcal{J}_{\mathrm{sh}}\psi = \Phi_S \otimes r_A \otimes r_B \otimes \psi_{ABC}$, let each endpoint act only on the common source, its local auxiliaries, and its original logical port, and protect it with $P_{\alpha,\mathrm{sh}} = P_\Phi \otimes P_{r_\alpha}$. The same two-arm cancellation gives $\widehat{H}_\alpha \mathcal{J}_{\mathrm{sh}} = \mathcal{J}_{\mathrm{sh}} h_\alpha$, so the stagewise retrieval above remains valid and the source is counted only once. Local auxiliaries and complete records are retained; this identification is not made for different patterns or nonproduct permanent records.

No unretrieved path record is left while the relay temporarily stores the state. If the output is subsequently read, or thermal processing is added, retain these outcomes as a complete instrument. In particular, when each endpoint separately uses the thermal processing of 2.3.6, which is scalar on each survival history, let q_A, q_B denote the survival shares at the endpoints and U_0 the free evolution retained over the actual elapsed duration. Summing all survival records gives

$$\mathcal{I}_{\mathrm{live}}(\rho) = q_A q_B \, \mathcal{J}_{AB} U_0 V_\gamma \rho V_\gamma^\dagger U_0^\dagger \mathcal{J}_{AB}^\dagger. \tag{3.13}$$

The shares in which only A exits, only B exits, and both exit are respectively $(1-q_A)q_B$, $q_A(1-q_B)$, and $(1-q_A)(1-q_B)$; together with the preceding expression they form the complete output. Two independent thermal processes are used here, with the same survival effect on every logical state employed; for a shared source, the

operator associated with each thermal outcome acts as the identity on that source and both sets of auxiliaries. The same complete schedule is executed whichever endpoint exits; state-dependent exit or exit through a common environment must be calculated separately. Under the specified port, conservation, and record construction, exchange, retrieval, and subsequent survival of the two complete carriers are thereby connected into one set of finite processes.

3.1.3 Physical Readings of Exchange

The observed outcomes of exchange are determined by the actual pointers and their joint output. Include the actual pointers and other auxiliaries in environment $\mathcal{E}$, let $\mathbb{U}$ denote the joint evolution including the necessary recording actions, and let η be its preparation. The joint final state yields both the reduced map on the two endpoints and the probability of event y:

$$\Omega^{\text{out}} = \mathbb{U}(\rho_{AB} \otimes \eta)\mathbb{U}^\dagger, \quad \Phi_{AB}(\rho_{AB}) = \text{Tr}_{\mathcal{E}}\, \Omega^{\text{out}}, \quad q_y = \text{Tr}[(I_{AB} \otimes Q_y)\Omega^{\text{out}}]. \tag{3.14}$$

Following the complete-record rule of Chapter 2, matrix elements of the actual joint evolution between the environment preparation and output basis give the Kraus actions for the outcomes [11][12]. Summing unread environment components within a single pointer class gives

$$\begin{aligned} \mathcal{I}_y(\rho) &= \sum_a K_{ya}\rho K_{ya}^\dagger, \\ E_y &= \sum_a K_{ya}^\dagger K_{ya}, \\ q_y(\rho) &= \text{Tr}(\rho E_y), \\ \sum_y E_y &= I_{AB}. \end{aligned} \tag{3.15}$$

Here a combines unread preparation and output indices without specifying additional observable events; completeness follows from joint unitarity and normalized preparation. The correspondence between spectral response and coupling must be realized in these actual branches and their effects.

Q_y is an actual pointer projection satisfying the record conditions established in 2.3. The preceding final-state share becomes a recorded event probability only once the corresponding output distinction has been realized. When several paths enter the same event, combine them using the environment-overlap rule of 2.4.2; the return of C to the same prepared state means only that C retains no such distinction, while other parts of the environment may still contain records.

To quantify the effect on B, fix the input, environment preparation, and observation duration, and let Φ_0 be the baseline map with the exchange under discussion switched off. For an already calibrated observable O_B, the mean change caused by exchange is

$$\Delta\langle O_B\rangle = \text{Tr}[(I_A \otimes O_B)\{\Phi_{AB}(\rho_{AB}) - \Phi_0(\rho_{AB})\}]\,. \tag{3.16}$$

Choosing a position, coherence, or momentum readout for O_B measures a distribution change, phase response, or momentum transfer, respectively. Only when

momentum has been calibrated, with T denoting the actual elapsed duration of this comparison, does $\Delta\langle p_B\rangle/T$ give the average mechanical force due to the exchange. The event probability q_y, the record rate per unit time, and this momentum response have different units and cannot share a numerical strength without a specified correspondence. Subsequent evaluation of resonance forces follows the established exchange process to determine its channels, response coefficients, and actual readings separately.

3.2 Structure of Resonance Channels

The preceding section connected transfer between directional endpoints to exchange and retrieval of selected complete carriers. We now fix the corresponding process and distinguish channels by changes in endpoint species: the propagation segment preserves the selected direction, while the endpoint transition is jointly balanced by the system and environment. Comparing the labels of the input and output states identifies which part of the phase structure participates in exchange, after which the corresponding actual action can be determined.

3.2.1 Channel Labels

Using the species-layer decomposition of 2.2, write p−1 as a product of pairwise coprime prime powers. Denote the order of the ith factor by n_i and the group of species labels by Γ:

$$p-1=\prod_{i=1}^{r} n_i, \qquad n_i=q_i^{b_i}, \qquad r=\omega(p-1), \qquad \Gamma=\prod_{i=1}^{r}\mathbb{Z}/n_i\mathbb{Z}. \tag{3.17}$$

The q_i are distinct, and ω counts the number of distinct prime factors. Once this CRT identification is fixed, the label $\ell=(\ell_1,\ldots,\ell_r)$ specifies the character $\chi\ell=\prod_i \chi\ell_i^{(i)}$, and character multiplication is coordinatewise addition modulo n_i. The odd prime p has not yet been assigned a value, and no specific particle or force names are assigned to the factors here.

Use the unit-shell average state e_0 as the trivial label $|0\rangle$, and denote the remaining directions pulled back from the species layer by $|\ell\rangle$. These vectors are normalized as in 2.1 and span the species-layer subspace:

$$\begin{gathered}\mathcal{H}_{\mathrm{sp}}=\operatorname{span}\{|\boldsymbol{\ell}\rangle:\boldsymbol{\ell}\in\Gamma\}\simeq\bigotimes_{i=1}^{r}\mathbb{C}^{n_i},\\ |\boldsymbol{\ell}\rangle\longleftrightarrow|\ell_1\rangle\otimes\cdots\otimes|\ell_r\rangle,\\ \dim\mathcal{H}_{\mathrm{sp}}=p-1.\end{gathered} \tag{3.18}$$

This decomposes one layer of species degrees of freedom. The complete state space also contains other shells and directions of higher conductor; the dimension of the address space at depth K remains p^K. The trivial direction e_0 must be included in this tensor space, since transformations within a factor can mix nontrivial directions with it. Whether each nontrivial direction is a stable state under the specified response is still determined by the nonzero-eigenvalue condition of 2.1.

Fix the convention for the unit-group action so that the eigenvalue function of $|\ell\rangle$ is denoted by $\chi\ell$. When an endpoint changes from ℓ to ℓ', the transition carries character $\chi\ell'\chi\ell^{-1}$, with label $\delta{=}\ell'-\ell$. By the joint covariance of 3.1, a nonzero exchange matrix element between definite-charge input and output components satisfies

$$\mathcal{A}_{fi} \neq 0 \quad \Longrightarrow \quad \Delta\ell_{A,i} + \Delta\ell_{B,i} + \Delta\ell_{C,i} = 0 \pmod{n_i}, \qquad i = 1, \ldots, r. \tag{3.19}$$

Here we first consider processes whose environment charges also belong to this species-layer character group; if the environment carries charge of higher conductor, balance must be imposed in the corresponding full group. When the input and output charges of C coincide, the label changes of A and B cancel coordinate by coordinate; when C retains a charge change, it carries the corresponding difference. Thus a coordinate at one endpoint may change, while conservation constrains the combined change of all participants.

The coordinates involved in the transition form its support $S(\delta) = \{i : \delta_i \neq 0\}$. If S is empty, the species label is unchanged, but the process may still change amplitude, phase, or records. If S contains one coordinate, the transition proceeds along one CRT factor; if S contains several, it changes several species components simultaneously. This classification covers arbitrary r and does not restrict multicoordinate changes to the case in which every coordinate is nonzero.

Joint conservation alone permits multicoordinate transitions. For example, the matrix unit $F\ell,\delta{=}|\ell{+}\delta\rangle\langle\ell|$ carries charge $\chi\delta$. If an environment operator $B_{-}\delta$ carries its inverse charge, a charge-conserving Hermitian action on the joint space can be formed:

$$\begin{aligned} D(u)F_{\boldsymbol{\ell},\boldsymbol{\delta}}D(u)^\dagger &= \chi_{\boldsymbol{\delta}}(u)F_{\boldsymbol{\ell},\boldsymbol{\delta}}, \\ D_C(u)B_{-\boldsymbol{\delta}}D_C(u)^\dagger &= \chi_{\boldsymbol{\delta}}(u)^{-1}B_{-\boldsymbol{\delta}}, \\ V_{\boldsymbol{\delta}} &= gF_{\boldsymbol{\ell},\boldsymbol{\delta}} \otimes B_{-\boldsymbol{\delta}} + \overline{g}F^\dagger_{\boldsymbol{\ell},\boldsymbol{\delta}} \otimes B^\dagger_{-\boldsymbol{\delta}}. \end{aligned} \tag{3.20}$$

In each term the system and environment charges cancel, so $V\delta$ commutes with the joint unit-group action without requiring δ to have only one nonzero coordinate. This construction checks charge conservation; its use as an actual vertex additionally requires the corresponding environment endpoints and energy balance. Single-factor channels must therefore be specified explicitly in the effective action.

3.2.2 Single-Factor Transitions

Following the single-factor step rule for formation paths, we organize elementary channels through effective transitions that act on only one CRT factor at a time and as the identity on all others. The matrix units of the ith family are

$$X_i(a,b) = I_1 \otimes \cdots \otimes |a\rangle_i\langle b| \otimes \cdots \otimes I_r, \qquad a, b \in \mathbb{Z}/n_i\mathbb{Z}. \tag{3.21}$$

For $a{\neq}b$, this step carries charge whose only nonzero coordinate is $a{-}b$ in position i; for $a{=}b$, it is a zero-charge diagonal action on that factor. This specifies both which labels change and the identity action on spectator factors. If a transition coefficient depends on labels of other factors, that conditional control must be retained; a single-coordinate label difference alone does not justify the above form.

These matrix units describe effective transitions within the species layer. The actual exchange vertices defined in 2.3 may also pass through the zero element and the environment, and the transfer of 3.1 also has intermediate states. Only after composing these steps and calculating the relevant matrix elements can one determine whether an effective channel is realized, together with its coefficient and phase. The trivial species-layer direction e_0 and the terminal zero element e_K belong to different shells and cannot be interchanged at this step.

Local matrix units on different factors commute. Every label difference also has a unique coordinate decomposition, so whenever all required local steps are available, a multicoordinate transition can be organized into a factor-by-factor path:

$$\boldsymbol{\delta} = \sum_{i \in S(\boldsymbol{\delta})} \delta_i \mathbf{e}_i, \qquad \chi_{\boldsymbol{\delta}} = \prod_{i \in S(\boldsymbol{\delta})} \chi_{\delta_i \mathbf{e}_i}, \qquad [X_i(a,b), X_j(c,d)] = 0 \quad (i \neq j). \tag{3.22}$$

Here e_i is the unit vector of the label coordinates, not a shell-average state. The single-factor rule gives r families of local channels, and multicoordinate paths compose some of these families. Although the label decomposition is unique, the actual order is set by the process; complete exchanges sharing an environment need not commute, and their amplitudes also include intermediate states and records. Likewise, $\mathrm{R}\chi$ generally depends on the full character: the CRT decomposition does not imply factorwise multiplication of Dirichlet L-values.

3.2.3 Factor-Preserving Actions

Single-factor transitions determine the scope of actions within each factor. Extending these actions to arbitrary unitaries within each factor gives the factor-preserving unitary group, which organizes actions on the species layer that preserve the tensor-factor decomposition. Whether it also preserves a chosen spectral response must be checked by commutation with that response. This group is defined as the image of the tensor-product action:

$$\mathcal{G}_{\mathrm{fact}} = \{A_1 \otimes \cdots \otimes A_r : A_i \in U(n_i)\} \subset U(\mathcal{H}_{\mathrm{sp}}). \tag{3.23}$$

A scalar phase can be factored out of each A_i, leaving a determinant-one part; multiplying the scalars leaves only one overall phase. Hence the map $\Phi{:}(\mathrm{z},\mathrm{S}_1,\ldots,\mathrm{S}_r)\mapsto \mathrm{z}(\mathrm{S}_1 \otimes \cdots \otimes \mathrm{S}_r)$ from $\mathrm{U}(1)\times \prod_i \mathrm{SU}(\mathrm{n}_i)$ onto this group is surjective. If its image is the identity, fixing eigenvectors of the other factors and comparing any two eigenvalues of the ith factor shows that S_i must be scalar. The kernel is therefore completely characterized by

$$S_i = \xi_i I_i, \qquad \xi_i^{n_i} = 1, \qquad z\prod_{i=1}^{r} \xi_i = 1; \qquad \ker \Phi \simeq \prod_{i=1}^{r} \mu_{n_i} \simeq \mathbb{Z}/(p-1)\mathbb{Z}. \tag{3.24}$$

Here μ_n is the group of nth roots of unity. The final isomorphism uses the pairwise coprimality of the n_i; it also specifies the embedding of the central subgroup being quotiented out: each tuple ξ_i determines $\mathrm{z}=(\prod_i \xi_i)^{-1}$. The resulting global group and Lie algebra are

$$\mathcal{G}_{\text{fact}} \simeq \frac{U(1) \times \prod_{i=1}^{r} SU(n_i)}{\mathbb{Z}_{p-1}}, \qquad \mathfrak{g}_{\text{fact}} \simeq \mathfrak{u}(1) \oplus \bigoplus_{i=1}^{r} \mathfrak{su}(n_i). \tag{3.25}$$

Each $\mathrm{SU(n}_i\mathrm{)}$ factor is generated by i times the traceless Hermitian actions within that factor, including off-diagonal transitions and relative diagonal phases; it has n_i^2-1 real generator directions. Scalar actions from all factors coincide as the same central U(1) on the tensor space. The Lie algebra thus has r+1 factors and total dimension $1+\sum_i(\mathrm{n}_i^2-1)$. This counts the structure of factor-preserving transformations on the species layer.

The central phase multiplies every vector in $\mathcal{H}_{\text{sp}}$ by z, and every density operator satisfies $(\mathrm{zI})\rho(\mathrm{zI})^\dagger=\rho$; reading only the state within this space reveals no distinguishable change. It becomes a relative phase only if the actual process preserves coherence with another reference sector and applies the phase to the species layer alone. For example, take a reference subspace $\mathcal{R} \subset \mathcal{H}_{\text{sp}}^{\perp}$ within the whole system, with projections P_{sp} and $P_{\mathcal{R}}$. On their direct sum,

$$\widetilde{U}_z = zP_{\text{sp}} + P_{\mathcal{R}}, \qquad P_{\text{sp}}\widetilde{U}_z\rho\widetilde{U}_z^\dagger P_{\mathcal{R}} = zP_{\text{sp}}\rho P_{\mathcal{R}}. \tag{3.26}$$

The corresponding observation must read this coherence block; if the preparation has no coherence across sectors, or recording has removed it, the action remains indistinguishable. The reference here is internal to the whole, and its preparation and phase action must be specified by the actual process. The central U(1) therefore does not correspond to a label difference with every coordinate nonzero: a scalar action preserves all labels, whereas a multicoordinate transition changes them.

3.2.4 Symmetries of a Fixed Response

The factor-preserving group also cannot be directly identified with the symmetry group of fixed dynamics. In the chosen character basis, let T_{sp} be the restriction of the shell-preserving response to $\mathcal{H}_{\text{sp}}$, and denote all its eigenvalues by $\lambda\ell$. For nontrivial labels, $\lambda\ell=\mathrm{R}\chi\ell$; the shell-preserving response used here is projected onto direction space $\mathcal{M}$ on both sides, so $\lambda_0=0$. This zero follows from the projection defining the directional response; the background in the complete cycle is still treated through the actual exchange. For an arbitrary unitary A, preservation of this fixed response requires

$$T_{\text{sp}}|\boldsymbol{\ell}\rangle = \lambda_{\boldsymbol{\ell}}|\boldsymbol{\ell}\rangle, \qquad [A, T_{\text{sp}}] = 0 \quad \Longleftrightarrow \quad (\lambda_{\boldsymbol{\ell}'} - \lambda_{\boldsymbol{\ell}})\langle\boldsymbol{\ell}'|A|\boldsymbol{\ell}\rangle = 0 \quad (\forall \boldsymbol{\ell}, \boldsymbol{\ell}'). \tag{3.27}$$

Thus only directions with identical complex eigenvalues can mix under this symmetry; equality of eigenvalue moduli alone is insufficient. Arbitrary unitary transformations are possible within each degenerate spectral block, while a simple spectrum leaves only phase transformations of the individual directions. A symmetry of the complete cycle must also be compatible with the actual exchanges, environment preparation, and readout; changing basis simultaneously for all states, operators, and pointers is a change of representation of the same process.

Under the specified single-factor effective step rule, this section obtains r families of local channels, together with r SU factors and one central U(1) from the tensor structure of the species layer. Identifying these r+1 structural factors with

r+1 independent elementary resonance forces further requires the corresponding exchanges, readable responses, and their independence to be established individually; in particular, the central phase needs a specified relative reference. Specific force names and stable-state identities are fixed when channels are selected. Here we first calculate each channel's coupling from its actual endpoints, participating directions, and environment output.

3.3 Couplings of Elementary Resonance Forces

After an exchange, environment records give the share of the selected outcome, while system readings give the change experienced by the stable state. Comparing the strengths of different resonance channels requires comparing responses of the same kind under common preparation and reference conventions. We first consider the direct action at an effective vertex, before incorporating repeated exchanges that leave and return, and call the corresponding dimensionless response coefficient the elementary coupling or bare coupling.

Using the notation of 3.2, δ denotes the species-label difference between input and output endpoints, while χ denotes the participating stable-state direction. The former determines the charge that the environment must balance in the exchange; the latter determines the spectral response used. Even processes carrying the same transfer charge may act on different stable states, so $\mathrm{R}\chi$ in the coupling must be evaluated for the direction that actually participates.

Fix the regular spectral point s_* used in 2.1, retain the full background spectral reference $\zeta_* = \zeta(s_*) \neq 0$, and abbreviate $L_\chi = L(s_*, \chi)$. By the directional response of 3.1, the squared response for input ρ and direction projection $\mathrm{P}\chi$ is $|R_\chi|^2 \operatorname{Tr}(\rho P_\chi)$. We now calculate how the actual preparation and endpoint matching determine this share.

3.3.1 Participating Labels

Once the directional response is fixed, the actual matching share remains to be calculated: an equal-weight preparation supplies orthogonal alternatives, of which the vertex accepts only one normalized direction, so the dimension of the alternative space is needed. To organize candidate exchanges in the species layer, write $G = (\mathbb{Z}/p\mathbb{Z})^\times$ and $m = (p-1)/2$, and arrange the alternatives by relative unit-group action u, negation-pair group j, and conjugate orientation o. First stipulate that these three labels vary independently. Then

$$\Lambda_\chi = G \times \{0, \dots, m-1\} \times \{+, -\}, \qquad m = \frac{p-1}{2}. \tag{3.28}$$

Here j groups the species layer by $\{a, -a\}$; in deeper reductions it represents the entire group of lifts described in Supplementary Proof S4.2–Supplementary Proof S4.3, not the total number of point pairs at depth K. The label u describes the relative action, while j describes the group in which preparation or matching occurs. If the same address determines both in the actual process, the actually allowed subset of labels must be used, without multiplying their counts independently again.

We classify by the value of $\chi(\mathrm{u})$, retaining orientation when χ differs from its conjugate and combining the two orientations into one class when χ is real. Let

$\eta\chi$(o) take values + and − in the former case and remain 0 in the latter. The classification map is

$$r_\chi(u,j,o)=(\chi(u),\eta_\chi(o)),\qquad \eta_\chi(o)=\begin{cases}o, & \chi\neq\bar\chi,\\ 0, & \chi=\bar\chi.\end{cases}\tag{3.29}$$

This map first gives an algebraic classification. Realizing it as a record requires the corresponding internal reference and pointer; applying u to a pure character state produces only a common phase and does not by itself read χ(u). The following count also does not assume that these phases can be perfectly distinguished in a single readout.

Fix a class in the image of this map. For its phase value $\chi(\mathrm{u}_0)$, the full preimage is $\mathrm{u}_0\ker\chi$, giving $|\ker\chi|$ choices of u; j has m choices; and orientation has cχ preimages, where cχ=1+[χ=$\bar\chi$]. With dχ=ordχ, the kernel–image relation for a group homomorphism gives the number of labels in this class:

$$N_\chi=|\ker\chi|\,m\,c_\chi=\frac{(p-1)^2}{2d_\chi}\,c_\chi,\qquad d_\chi=\operatorname{ord}\chi,\qquad c_\chi=1+\mathbf{1}_{\{\chi=\bar\chi\}}.\tag{3.30}$$

Here Nχ counts label preimages within the same accepted class, with its three factors coming from relative action, point-pair group, and orientation. To use this count for an actual matching share, the preparation or environment must contain corresponding independent alternatives carrying these labels. In particular, the factor 2 in the self-conjugate case requires two actual orientation alternatives; if forward and reverse are merely two descriptions of the same complete state, they are counted once. If they are two coherent histories leading to the same joint final state, amplitudes are combined first, and their strength is determined jointly by phase and environment overlap.

3.3.2 Endpoint Matching

Suppose that all labels in the selected class are indeed carried by Nχ orthogonal preparation alternatives, with label space $\mathcal{L}\chi$. Take the equal-weight mixed preparation $\sigma\chi$=I$\mathcal{L}\chi$/Nχ and suppose this vertex matches only one normalized direction vχ, with normalized matched output fχ. Following the spectral-response assignment, specify the unnormalized action of this vertex as Aχ=L$\chi|\mathrm{f}\chi\rangle\langle\mathrm{v}\chi|$. Any normalized input ξ first enters the matched direction with overlap $\langle\mathrm{v}\chi|\xi\rangle$ and then acquires the spectral factor Lχ; averaging over the prepared state gives

$$\begin{aligned}A_\chi|\xi\rangle&=L_\chi\langle v_\chi|\xi\rangle|f_\chi\rangle,\\ P_{v_\chi}&=|v_\chi\rangle\langle v_\chi|,\\ A_\chi^\dagger A_\chi&=|L_\chi|^2P_{v_\chi},\\ \mathrm{Tr}(\sigma_\chi A_\chi^\dagger A_\chi)&=\frac{|L_\chi|^2}{N_\chi}.\end{aligned}\tag{3.31}$$

The last step uses only Tr(Pvχ)=1. Thus the denominator comes from the share of an equal-weight preparation lying in one matched direction. Dividing this mean response by the specified squared background spectral response gives the bare coupling:

$$\alpha_\chi^{(0)} := \frac{\mathrm{Tr}(\sigma_\chi A_\chi^\dagger A_\chi)}{|\zeta_*|^2} = \frac{|R_\chi|^2}{N_\chi} = \frac{2d_\chi}{(p-1)^2 c_\chi}\left|\frac{L(s_*,\chi)}{\zeta(s_*)}\right|^2. \tag{3.32}$$

The superscript (0) indicates that repeated exchange has not yet been included. Equal-weight labels can be realized by thermal preparation in the equal-energy block $H_L = \epsilon I$, whose conditional Gibbs state is I/N_χ [13]. For a general normalized preparation σ_χ, retaining a common spectral amplitude on the actual matching subspace, namely $A_\chi^\dagger A_\chi = |L_\chi|^2 P_{\mathrm{match}}$, gives

$$\alpha_\chi^{(0)} = |R_\chi|^2 \,\mathrm{Tr}(\sigma_\chi P_{\mathrm{match}}); \qquad \sigma_\chi = I/N_\chi,\ \mathrm{rank}\, P_{\mathrm{match}} = q \quad\Longrightarrow\quad \alpha_\chi^{(0)} = \frac{q|R_\chi|^2}{N_\chi}. \tag{3.33}$$

Thus the denominator counts the share of the equal-weight input lying in the matching subspace; summation of orthogonal outcomes and amplitude combination of coherent paths are treated separately as in 2.4.2.

3.3.3 Actual Events

The bare coupling is a squared-response ratio relative to a reference; actual events must still be specified by a normalized instrument. For a single matched direction, choose $\kappa{>}0$ such that $\kappa^2|\mathrm{R}\chi|^2 \leq 1$ and set, on the same label space,

$$M_\chi = \kappa R_\chi |f_\chi\rangle\langle v_\chi|, \qquad M_\varnothing = (I - \kappa^2 |R_\chi|^2 P_{v_\chi})^{1/2}, \qquad M_\chi^\dagger M_\chi + M_\varnothing^\dagger M_\varnothing = I. \tag{3.34}$$

The record χ and remaining outcomes $\varnothing$ complete the output; κ calibrates the interaction amplitude, and the event share of the equal-weight input is $\kappa^2\alpha_\chi^{(0)}$. For a given exchange, checking whether its actual effect equals $\kappa^2|R_\chi|^2 P_{\mathrm{match}}$ determines whether it realizes this response. The criterion for all inputs and its short proof are given in Supplementary Proof S5.1.

For the transfer from a unit-shell direction to the terminal zero element already specified in Supplementary Proof S5.1, this connection can be expressed directly as cycle data. In this section θ denotes the exchange angle, to distinguish it from coupling α. Let $\mathrm{aK}\chi$ be the corresponding preparation share, $\theta\mathrm{K}\chi$ the exchange angle, and $\mathrm{tu,K}\chi$ the environment upper-state weight. Then

$$\epsilon_{K\chi} = a_{K\chi} t_{u,K\chi} \sin^2\theta_{K\chi}; \qquad a_{K\chi}\sin^2\theta_{K\chi} = \alpha_\chi^{(0)} \quad\Longrightarrow\quad \epsilon_{K\chi} = t_{u,K\chi}\alpha_\chi^{(0)}. \tag{3.35}$$

This realization carries the bare coupling through a combination of preparation share and exchange angle, while the actual event probability also contains the thermal environment weight. When several directions share one preparation, their $\mathrm{aK}\chi$ obey a sum constraint. Supplementary Proof S5.1 provides a common construction when $\mathrm{S}{=}\sum \chi\alpha\chi^{(0)}{<}1$: the background share is strictly positive, the exchange angles take nonendpoint values, and the spectral evaluations of all directions are realized by the same normalized instrument. Specific values are checked within this range.

The persistent port of 3.1.2 places the single-address mean $\beta_B = \alpha_{\mathrm{EM}}^{(0)}$ in angle $\gamma = g_B T_B \beta_B$, giving probability $\sin^2\gamma$; the one-shot acceptance instrument of this section places the spectral response in the amplitude, giving $\kappa^2\alpha^{(0)}$ for an equal-weight input. The former scales as $(\alpha^{(0)})^2$ at short times, whereas the latter scales linearly with $\alpha^{(0)}$. These use different actual endpoints and show that the location at which a spectral share enters an operator determines its reading.

3.3.4 Comparison of Couplings

With the same spectral parameter and reference, equal-weight preparations, and one matched direction each, the ratio of two nonzero elementary couplings is

$$\frac{\alpha_{\chi}^{(0)}}{\alpha_{\psi}^{(0)}} = \left|\frac{L(s_*,\chi)}{L(s_*,\psi)}\right|^2 \frac{N_{\psi}}{N_{\chi}}. \tag{3.36}$$

Relative strength is determined jointly by spectral response and matching share. Character order determines the kernel size and thus enters the participation count; the L-value also depends on the particular phases of that character at each integer, so ordering by character order cannot replace calculation of response magnitudes. If two processes have different thermal environment weights or matching ranks, these factors must also be retained when comparing actual recorded shares.

A response bound must also specify the range of the spectral parameter. For real s>1, absolute convergence of the Dirichlet series and the triangle inequality give

$$|L(s,\chi)| \le \sum_{p\nmid n} n^{-s} = (1-p^{-s})\zeta(s), \qquad |R_{\chi}(s)| \le 1-p^{-s} \quad (s>1). \tag{3.37}$$

When evaluation at s_0=1/2 uses analytic continuation, this proof no longer guarantees the above maximal-response bound. In that case $|\zeta(s_0)|^2$ serves as the specified spectral reference, while preparation and the instrument control the range of actual probabilities. Each elementary coupling is thereby connected to a definite direction, participating labels, and matching endpoint. If repeated exchange changes accessible paths or final-state records, the response must be recombined from those actual actions.

3.4 Virtual Processes and Effective Couplings

When a stable state participates in an exchange, the excitation may reach the selected endpoint directly, or first enter an intermediate direction and then return to the original direction or reach the same endpoint. Such round trips change both the amplitude reaching the endpoint and the preparation share. For processes that preserve the spectral vertex, virtual corrections change the matching share of the same direction, replacing the participation count in the bare-coupling denominator with an effective matching denominator. If the spectral vertex or matched direction also changes with the process, those changes must be included in the complete response.

Here intermediate round trips that are not read as independently terminating events are called virtual processes. Direct and round-trip paths are combined using the environment-overlap formula of 2.4.2, and the resulting complete output becomes the input to the next round. Virtual processes thus change both endpoint response and subsequent preparation; these changes are calculated separately below.

3.4.1 Effective Matching Share

First consider a case admitting an exact effective-denominator form: intermediate actions change the preparation at the matching endpoint while leaving the response Aχ and matched direction vχ specified in 3.3 unchanged. Let $\Gamma\chi \ge 0$ be a nonzero

unnormalized preparation on this label space, with $\sigma\chi$=$\Gamma\chi$/Zχ and Zχ=Tr$\Gamma\chi$; write hχ=$\langle$vχ|$\Gamma\chi$|v$\chi\rangle$>0 and wχ=|Rχ|^2>0. Since A$\chi^\dagger$Aχ=|Lχ|^{2}Pvχ,

$$\begin{aligned} \alpha_\chi^{\text{eff}} &:= \frac{\mathrm{Tr}(\sigma_\chi A_\chi^\dagger A_\chi)}{|\zeta_*|^2} = w_\chi \frac{h_\chi}{Z_\chi} = \frac{w_\chi}{N_\chi^{\text{eff}}}, \\ N_\chi^{\text{eff}} &:= \frac{Z_\chi}{h_\chi} \geq 1, \\ \Delta N_\chi &:= N_\chi^{\text{eff}} - N_\chi, \\ \alpha_\chi^{\text{eff}} &= \frac{w_\chi}{N_\chi + \Delta N_\chi}. \end{aligned} \tag{3.38}$$

The inequality follows from 0≤Pvχ ≤I, which gives hχ ≤Zχ. The equal-weight preparation can be represented by $\Gamma\chi$=I, giving Zχ=Nχ and hχ=1 and recovering the bare coupling. If hχ=0, the preparation has no matching share and the coupling vanishes; the effective denominator can only be understood as an infinite limit. Multiplying $\Gamma\chi$ by an overall positive scalar does not change the ratio.

Thus the bare value Nχ may come from a structural label count, whereas Nχ^{eff} is the ratio of total preparation weight to matching weight and is generally no longer an integer. An intermediate process that increases the matching share decreases the effective denominator; one that relatively increases other preparation shares increases it. If the actual correction also changes Aχ, the matched direction, or coherent paths, first recalculate Tr($\sigma\chi$A$\chi^\dagger$Aχ), without presuming that all changes enter the denominator.

The link between this response and event probability still uses the actual calibration of 3.3. For example, if the conditional operator of the selected outcome is Mχ=κAχ/ζ∗, with κ fixed and satisfying M$\chi^\dagger$Mχ ≤I, then pχ($\sigma\chi$)=$\kappa^2\alpha\chi^{eff}$. Preparation dependence, the calibration factor, and the duration of each round are all retained explicitly. To turn a spectral-response ratio into a stationary record rate, one must additionally use the actual stationary input and divide by τ.

Now calculate the first-order correction near the equal-weight preparation. Let a real parameter λ control the added action, and suppose $\Gamma\chi(\lambda)$=I+λGχ+O(λ^2), where Gχ is self-adjoint and $\Gamma\chi$ remains positive in a neighborhood. Writing $\gamma\chi$=TrGχ and gχ=$\langle$vχ|Gχ|v$\chi\rangle$, direct expansion of the above ratio gives

$$\begin{aligned} N_\chi^{\text{eff}}(\lambda) &= \frac{N_\chi + \lambda\gamma_\chi + O(\lambda^2)}{1 + \lambda g_\chi + O(\lambda^2)} = N_\chi + \lambda(\gamma_\chi - N_\chi g_\chi) + O(\lambda^2), \\ \alpha_\chi^{\text{eff}}(\lambda) &= \alpha_\chi^{(0)} \left[1 + \lambda \left(g_\chi - \frac{\gamma_\chi}{N_\chi}\right) + O(\lambda^2)\right]. \end{aligned} \tag{3.39}$$

The quantity $\gamma\chi$ measures the change in total weight, while gχ measures the change at the matching endpoint; their relative sizes determine the sign of the correction. Even for an exactly linear preparation $\Gamma\chi$=I+λGχ, the effective denominator remains (Nχ+$\lambda\gamma\chi$)/(1+λgχ) and does not automatically truncate to a linear expression.

3.4.2 Virtual-Process Corrections

Finding Gχ requires tracing how the added action changes total preparation weight and endpoint matching weight. Structural labels first constrain the directions

through which these contributions can pass: if the system operator $\mathrm{X}\chi$ of a step carries definite character charge χ under the unit-group action, then composing it twice in the same direction gives

$$D(u)X_\chi D(u)^\dagger = \chi(u)X_\chi \quad \Longrightarrow \quad D(u)X_\chi^2 D(u)^\dagger = \chi(u)^2 X_\chi^2. \tag{3.40}$$

The environment must still balance this system charge in the joint exchange. Thus a nonzero matrix element sends system label η to $\eta\chi^2$; the composite operator may also vanish through endpoint mismatch. For nontrivial χ, $\chi^2{=}1$ if and only if ord $\chi{=}2$, and $\chi^2{=}\bar{\chi}$ if and only if ord $\chi{=}3$. Other squares lie in other nontrivial directions. The class whose same-direction square is the trivial character is called Type II; the remaining class is called Type I. This classification first specifies the destination of the composite charge.

Since G is cyclic of even order, the nontrivial quadratic character $\chi\mathrm{q}$ is unique. But any χ multiplied by $\bar{\chi}$ also gives 1, so this does not imply a unique neutral round trip. For order 3, $\{1, \chi, \overline{\chi}\}$ is closed under label multiplication; higher-order characters also have finite multiplicative closures and can form longer return paths. Which paths occur depends on the actual vertices, shells, energy balance, and environment output.

For Type I, the first-order effective-denominator shift equals $\mathrm{V}_0{-}2\mathrm{w}\chi$ provided the actual correction satisfies $\gamma\chi{-}\mathrm{N}\chi\mathrm{g}\chi{=}\mathrm{V}_0{-}2\mathrm{w}\chi$. If the total-weight term and endpoint-enhancement term are further specified separately as $\gamma\chi{=}\mathrm{V}_0$ and $\mathrm{g}\chi{=}2\mathrm{w}\chi/\mathrm{N}\chi$, then

$$\gamma_\chi = V_0, \qquad g_\chi = \frac{2w_\chi}{N_\chi} \quad \Longrightarrow \quad N_\chi^{\mathrm{eff}} = N_\chi + \lambda(V_0 - 2w_\chi) + O(\lambda^2). \tag{3.41}$$

V_0 must be calculated from the actual preparation and operator of the selected background fluctuation. If a connected fluctuation is defined as $\langle \mathrm{B}^\dagger\mathrm{B}\rangle{-}|\langle \mathrm{B}\rangle|^2$, B and the state in which the expectation is taken must be specified; identifying it with $|\mathrm{R}\chi\mathrm{q}|^2$ remains an additional correspondence. The coefficient 2 must likewise come from two actual contributions and their coherence relation, and cannot be inferred merely from passage through two vertices.

Equation (3.41) determines a first-order shift. To use this shift in a finite corrected denominator, write $\Delta N_{\chi,\mathrm{I}} = V_0 - 2w_\chi$ and take, in the range $N_\chi + \Delta N_{\chi,\mathrm{I}} \geq 1$,

$$\Delta N_{\chi,\mathrm{I}} = V_0 - 2w_\chi, \qquad N_{\chi,\mathrm{I}}^{\mathrm{fin}} = N_\chi + \Delta N_{\chi,\mathrm{I}}, \qquad \alpha_{\chi,\mathrm{I}}^{\mathrm{fin}} = \frac{w_\chi}{N_\chi + V_0 - 2w_\chi}. \tag{3.42}$$

The superscript fin denotes the finite-correction prescription used here. It retains the above first-order coefficient; the first-order expansion has not determined higher-order changes in the actual preparation. The connected background contribution increases the effective denominator, while the two endpoint enhancements decrease it; their net effect determines whether the coupling is enhanced or reduced relative to its bare value. The common spectral factor w_χ remains unchanged at this step.

The same applies to cross-direction actions in Type II. Only if the actual routing calculation gives first-order coefficients $\epsilon\chi\psi$ and a self-conjugate-alternative contribution $\mathrm{c}\chi\mathrm{w}\chi$ can one write

$$N_\chi^{\mathrm{eff}} = N_\chi + \lambda \left(\sum_\psi \epsilon_{\chi\psi} w_\psi + c_\chi w_\chi \right) + O(\lambda^2), \qquad w_\psi = |R_\psi|^2. \tag{3.43}$$

The sum covers the directions that actually participate. Only mutually orthogonal alternatives with equal response weights directly give integer multiplicities; coherently combined paths require their cross terms to be retained. Each correction must be realized as a change in total preparation weight or matching weight. Under the same finite-correction prescription, denote the sum of these contributions by $\Delta N_{\chi,\mathrm{II}}$ and require the resulting denominator to be at least one. Then

$$\begin{aligned} \Delta N_{\chi,\mathrm{II}} &= \sum_\psi \epsilon_{\chi\psi} w_\psi + c_\chi w_\chi, \\ N_{\chi,\mathrm{II}}^{\mathrm{fin}} &= N_\chi + \Delta N_{\chi,\mathrm{II}}, \\ \alpha_{\chi,\mathrm{II}}^{\mathrm{fin}} &= \frac{w_\chi}{N_\chi + \sum_\psi \epsilon_{\chi\psi} w_\psi + c_\chi w_\chi}. \end{aligned} \tag{3.44}$$

Both classes of formulas express corrections as the matching share at the same endpoint. Type I retains the connected background contribution and endpoint enhancement in the given direction; Type II additionally includes redistribution across directions obtained from the actual routing. Signs and multiplicities of correction terms are determined by the corresponding actions and complete output, with the conditions for adopting a finite denominator kept distinct from the first-order expansion.

3.4.3 Composition of Repeated Processes

Higher-order responses from repeated round trips can be obtained directly from the actual intermediate actions. First consider a coherence-preserving process: paths of different lengths are indeed indistinguishable in the same final readout, and their propagation phases and durations are included in their amplitudes. Let b be the amplitude vector entering the intermediate space, C the linear action from that space to the selected endpoint, and B the amplitude action for one further round trip within it. Denote the total amplitude combining all intermediate paths by a_{int}. Splitting according to whether another round trip occurs gives $a_{\mathrm{int}} = b + Ba_{\mathrm{int}}$. When the spectral radius satisfies spr(B)<1, this relation has a unique solution, hence

$$a_{\mathrm{int}} = (I - B)^{-1} b, \qquad C a_{\mathrm{int}} = \sum_{n=0}^{\infty} C B^n b, \qquad \mathcal{A}_{\mathrm{eff}} = \mathcal{A}_{\mathrm{dir}} + C(I - B)^{-1} b. \tag{3.45}$$

In finite dimensions, the spectral-radius condition ensures that B^n tends to zero; multiplying the finite sum by I−B gives I−B^{n+1}, and taking the limit yields the formula. Here a_{int} is the total path amplitude under the common terminal readout, not a separately normalized intermediate or stationary state. C, B, and b contain the actual matrix elements and propagation factors. The coupling correction is obtained only after substituting the effective amplitude into a response comparison with the

same preparation and spectral reference; if different durations have left orthogonal time records, one must instead sum recorded outcomes.

A scalar alternating series is only one special case: the corrected amplitude becomes its first term divided by 1+r only if the accessible intermediate mode is actually multiplied by −r at each repetition and |r|<1. Orientation reversal alone does not fix this sign. For example, on a conjugate pair let S exchange the two orientations, with S^2=I. If B=rS, then

$$(I - rS)^{-1} = \frac{I + rS}{1 - r^2} = \frac{P_+}{1 - r} + \frac{P_-}{1 + r}, \qquad P_\pm = \frac{I \pm S}{2}, \qquad |r| < 1. \tag{3.46}$$

The factor 1/(1+r) remains alone only when the component actually entered and read out lies in the antisymmetric mode. If both modes participate, both terms must be retained. Label closure for a character of order 3 does not by itself determine which mode is entered, the magnitude of the repetition ratio r, or whether the environment retains a distinction.

When the environment has recorded the intermediate outcomes, repeated processes are instead composed through completely positive actions. Let $\mathcal{Q}$ denote the conditional action in which the exchange has not yet terminated, and $\mathcal{J}\chi$ the action terminating with the selected outcome; all terminating outcomes together with $\mathcal{Q}$ form a complete trace-preserving one-step action. If spr($\mathcal{Q}$)<1, the probability of eventually obtaining a χ-class outcome from normalized initial state ρ is

$$p_\chi^{\mathrm{end}} = \sum_{n=0}^{\infty} \operatorname{Tr} \mathcal{J}_\chi \mathcal{Q}^n(\rho) = \operatorname{Tr}\big[\mathcal{J}_\chi (I - \mathcal{Q})^{-1}(\rho)\big] . \tag{3.47}$$

The terms correspond to different termination rounds and are individually nonnegative; summing over every terminating outcome gives 1 under the above termination condition. This probability calculation and coherent amplitude summation describe different recording arrangements. To calculate strength per unit time, the actual duration of each path must also be included; a termination probability cannot be directly treated as a count rate.

To compare different corrections, fix the preparation procedure, spectral reference, and outcome read. Let $\{\mathcal{I}_r\}$ be the one-round instrument including all intermediate actions, $\mathcal{F}$ the complete cycle, and $\mathcal{I}\chi$ the instrument for the selected terminal-record class. Each round uses the same fresh environment preparation and has actual period τ. Then

$$\sum_r \mathcal{I}_r = \mathcal{F}, \qquad p_\chi(\rho) = \operatorname{Tr} \mathcal{I}_\chi(\rho), \qquad \rho_* = \mathcal{F}(\rho_*), \qquad \mathscr{R}_{\chi,*} = \frac{p_\chi(\rho_*)}{\tau}. \tag{3.48}$$

Under the positive-exchange and restoration conditions in §4.4 of Ref. [2], iteration of the original thermal cycle converges to $\rho*$, and the expected cumulative event count divided by total duration tends to $\mathcal{R}\chi,*$ as the number of cycles grows. This result for the expected count rate does not require events in successive rounds to be independent (Hu, 2026, §§4.4 and 6.6) [2]. Stationary occupation is determined jointly by preparation and exchange, so the stationary record rate generally also contains stable-state occupation and thermal environment weights. Couplings must

be compared using the same input and calibration; if stationary inputs are used instead, their dependence must be retained as well.

The pχ above is the share of rounds producing the selected terminal outcome and lies between 0 and 1. If several χ-class events are actually recorded within one round, their mean number is $\sum$h nχ(h)Pr(h), where h denotes a distinguishable complete record history. This may exceed 1 and differs from the probability of at least one occurrence in the round. Coherent paths that leave no distinction are combined at the amplitude level and are not counted among these additive histories.

3.4.4 Corrected Couplings

Combining coherent paths and recorded histories determines how repeated processes enter the actual response. If these actions preserve the spectral vertex, matched direction, and comparison reference, their changes to the same effective denominator can be examined further. Let δN_n be the added denominator contribution from repetition n, with first term $\Delta N_\chi^{\mathrm{fin}}$ chosen as the finite-correction shift of Type I or Type II above. For a fixed real number r, when the actual denominator increments satisfy the following recurrence and $|r| < 1$, their sum and the corresponding coupling are

$$\begin{aligned} \delta N_{n+1} = -r\delta N_n, \qquad \delta N_0 = \Delta N_\chi^{\mathrm{fin}}, \qquad r \in \mathbb{R}, \quad |r| < 1, \\ N_\chi^{\mathrm{full}} = N_\chi + \sum_{n=0}^{\infty} \delta N_n = N_\chi + \frac{\Delta N_\chi^{\mathrm{fin}}}{1+r}, \\ \alpha_\chi^{\mathrm{full}} = \frac{w_\chi}{N_\chi^{\mathrm{full}}} = \frac{w_\chi}{N_\chi + \Delta N_\chi^{\mathrm{fin}}/(1+r)}. \end{aligned} \tag{3.49}$$

Induction on the recurrence gives $\delta N_n = (-r)^n \Delta N_\chi^{\mathrm{fin}}$. The sum of the first $J+1$ terms is $\Delta N_\chi^{\mathrm{fin}}[1-(-r)^{J+1}]/(1+r)$; under $|r| < 1$ the remainder tends to zero, yielding the effective denominator. Dividing the same spectral response w_χ by this denominator gives the corrected coupling. The resulting N_χ^{full} must still be at least one and compatible with probability normalization of the actual instrument; when $r = 0$, this formula recovers the preceding finite-correction prescription.

This resummation acts on the increments of the denominator, leaving the bare denominator N_χ unchanged. The denominator recurrence must be established separately from the corresponding process and cannot be replaced directly by a geometric amplitude series: taking a coherent amplitude's squared modulus generates cross terms, and preparation normalization also generates higher-order terms. The sum is exact once the recurrence is given; the finite prescription for its first term and the repetition ratio retain their respective physical-correspondence conditions. Here full denotes the complete sum of this specified correction sequence.

Exact normalization may also arise from ratio preservation in a recorded process. Independently prepare a state on a fixed output auxiliary space. Let the continuation branch leave that auxiliary system untouched and apply two types of filtering only at the first terminal event. If the two endpoint effects have the same ratio for every input, every segment of continuation evolution applies the same linear map to them, preserving the ratio segment by segment. Summing histories that have already been distinguished also preserves it. Retries therefore change the total delivery probability and waiting time, while the two conditional mixture shares after delivery remain

unchanged, as does the matching denominator of the output state. Supplementary Proof S8 gives the complete proof and the realization used in Chapter 7.

This reveals another structural stability of effective coupling: repeated processes may change when delivery occurs and how much is delivered while preserving how the delivered directions are mixed. The denominator recurrence and this fixed-ratio mechanism are each used according to the actual process; they connect the composition laws of the complete output back to calculable matching shares.

3.5 Emergent Resonance Forces

An emergent state describes a collective excitation through a pattern involving the entire space. In resonance, exchange may occur between two collective excitations or connect a collective pattern with an elementary direction. Calculating this interaction requires identifying the actual ports through which the collective pattern participates, the distinguishable outcomes left by exchange, and their total response relative to the same background reference.

3.5.1 Ports of the Collective Pattern

First consider the collective pattern itself. In 2.1, b is invariant under every multiplicative pullback and has full spectral response ζ(s)b, while an elementary direction has response L(s,χ)$\psi\chi$. Thus, at a spectral point where $\zeta*$=ζ(s$*$) is finite and nonzero and the corresponding L-values are well defined, their spectral factors relative to the collective reference are respectively 1 and Rχ=L(s$*$,χ)/$\zeta*$. These factors describe responses to a specified spectral action. Actual exchange also contains port overlaps, exchange angles, propagation phases, and environment shares, which must be calculated with the process.

The absence of a nontrivial unit-group character in the collective pattern does not remove these exchange factors. For example, using the zero-element vertex of 2.4.1, take a normalized nontrivial unit-shell direction $\psi\chi$ and let eK be the normalized zero-element state. Then

$$\begin{aligned} V_{0,\chi} &= |e_K\rangle\langle\psi_{\bar{\chi}}| - \chi(-1)|\psi_\chi\rangle\langle e_K|, \\ V_{0,\chi} b &= -\chi(-1) p^{-K/2} \psi_\chi, \\ |\langle\psi_\chi|V_{0,\chi}|b\rangle|^2 &= p^{-K}. \end{aligned} \tag{3.50}$$

The first term vanishes because $\psi\bar{\chi}$ is orthogonal to b, while the second extracts the zero-element component of b. This calculates a vertex matrix element; the complete joint exchange must still satisfy energy and charge balance and include the actual interaction duration. If this vertex enters the selected channel, its p^{-K} share must remain in that channel's weight. Moreover, b contains coherent components of different shell energies, so resonance conditions in the actual cycle are still determined by phase and exchange as in 3.1; the trivial character alone does not establish zero-frequency resonance.

3.5.2 Affine Channels

The collective pattern may participate in exchange through different ports, requiring a set of operation labels to organize these paths. Addition and unit multiplication

supply such candidates: at the species layer, composing translations x↦x+b with scalings x↦ax gives the affine group. Whether different operations produce distinguishable exchange outputs also depends on the actual input and environment. To distinguish it from the unit group G used above, denote this group by Gaff:

$$\begin{aligned} G_{\text{aff}} = \text{Aff}(\mathbb{F}_p) &= \{(a,b) : a \in \mathbb{F}_p^{\times},\ b \in \mathbb{F}_p\}, \\ (a,b)(a',b') &= (aa',\ b+ab'), \\ |G_{\text{aff}}| &= p(p-1). \end{aligned} \tag{3.51}$$

Here b as the second coordinate of a group element denotes a translation amount; the pattern b written on its own still denotes the constant function on the full space. For K≥2, affine operations admit a nonredundant identification with unit addresses at depth two. Let U_2=($\mathbb{Z}/p^2\mathbb{Z}$)× and let [a] be the multiplicatively compatible Teichmüller representative. Specify the following action on U_2:

$$F_{a,b}(x) = [a]x + pb \pmod{p^2}, \qquad F_{a,b} \circ F_{a',b'} = F_{aa',\, b+ab'}. \tag{3.52}$$

If $F_{a,\beta}$ fixes a unit address, reduction modulo p first gives a=1, and reduction modulo p^2 then gives b=0. Thus no nonidentity operation fixes that address; since both the operation count and address count are p(p−1), the action is free and transitive. Choosing reference address x_0=1 associates each operation with a unique $F_{a,\beta}(x_0)$, and every other address occurs exactly once. This is an affine action on the set U_2; the multiplicative group of U_2 itself is abelian, and the two group structures are used separately. If the original state space has depth K greater than two, using these labels additionally requires specifying reduction to U_2 and the preparation of deeper addresses; all positions of the entire unit shell cannot be counted as this label set.

The same set of operations can produce very different results on different inputs. These address permutations all preserve the uniform vector on U_2, so every operation sends the same uniform input to the same vector. Conversely, applying these permutations to a chosen point state produces mutually orthogonal address states. The first case shows that operation count does not equal output count; the second provides a preparation realizing all address distinctions. The actual channels of an emergent state must be determined from its own joint exchange.

3.5.3 Combination of Complete Outputs

Suppose the same normalized matched input v produces complete output F_g along candidate path g, where F_g already includes both system and environment and is individually normalized, and the path amplitude is c_g. For the selected exchange outcome, the intensity of the complete response is

$$\left\| \sum_g c_g F_g \right\|^2 = \sum_{g,h} \overline{c_g} c_h \langle F_g | F_h \rangle. \tag{3.53}$$

The Gram matrix of the complete outputs determines the total response of this exchange; subsequent return also depends on the output states. Retaining a separate record for each path and retaining only a common record for a group of paths can give the same outcome effect while preserving different coherence within the group.

Supplementary Proof S4.4 shows how this difference changes the input occupation at the same return vertex.

In the additive case, fix an exchange class, a matched input v, and mutually orthogonal actual complete outputs F_g. Suppose every channel has the same spectral factor r, with the remaining input, propagation, and exchange factors included in c_g. The channel response normalized by the collective spectral reference is $B_{g,r} = rc_g|F_g\rangle\langle v|$. The response operator of each output satisfies $B_{g,r}^\dagger B_{g,r} = |r|^2|c_g|^2|v\rangle\langle v|$; taking the trace against input preparation ρ and summing all actual orthogonal outputs gives

$$\begin{aligned} B_{g,r} &= rc_g|F_g\rangle\langle v|, \\ W_{\text{aff}} &:= \sum_g |c_g|^2, \\ h_v(\rho) &:= \langle v|\rho|v\rangle, \\ \alpha_r(\rho) &:= \sum_g \mathrm{Tr}(B_{g,r}\rho B_{g,r}^\dagger) = |r|^2 W_{\text{aff}} h_v(\rho). \end{aligned} \tag{3.54}$$

Waff is the total channel weight of this process. For matched input ρ=|v⟩⟨v|, $h_v = 1$. Only if all p(p−1) labels are actually realized and every channel has $|c_g|^2 = 1$ in the same response units does Waff=p(p−1). If g is merely a preparation label randomly chosen with probability a_g, then c_g already contains $\sqrt{a_g}$; when every other per-channel response is one and $\sum_g a_g = 1$, the total weight is one. Adding alternative labels to a normalized preparation does not automatically increase total response.

For an interaction between emergent states, take r=1 above if the unnormalized spectral response of each channel is $\zeta_* c_g$. For an interaction between an emergent state and elementary direction χ, take r=Rχ if the per-channel response is $L(s_*, \chi)c_g$. Only when these two processes have the same total channel weight and are each compared at their matched input are their couplings respectively

$$\alpha_{\text{emg}} = W_{\text{aff}}, \qquad \alpha_{\text{mix}}(\chi) = W_{\text{aff}}|R_\chi|^2. \tag{3.55}$$

These formulas make explicit the roles of the common weight and the spectral factor. Having collective states at both endpoints does not alone fix the spectral factor of an exchange vertex: if the actual exchange passes through elementary direction χ, its response must still be retained. If a mixed process adds a zero-element input, directional filtering, or a different environment preparation, these factors must enter its own c_g before Waff is calculated. For example, entry from b into fχ through a zero-element vertex carries amplitude $\mathrm{p}^{-K/2}$; a comparison conditional on the selected zero-element component must separately state the probability of that condition.

To use the response for a one-round event, normalize with a common coefficient κ as in 3.3.3 and complete the output with the no-exchange outcome. Let $P_v = |v\rangle\langle v|$. The total-effect condition in this section is $\kappa^2|r|^2 W_{\text{aff}} P_v \le I$, and the event share is $p_r(\rho) = \kappa^2 \alpha_r(\rho)$. If the two exchange classes are alternative outcomes in the same round, their effects must be added before checking this bound. Evaluating the share at the actual stationary input and dividing by the complete cycle period gives the event count rate.

3.5.4 Node Weights

The channel sum reduces total response to the individual path coefficients c_g. When layer labels are used to assign node weights, one must first count how often each node occurs in the specified layer product and then connect that product to actual channel weights. Representations of the affine group in characteristic p organize these multiplicities; their layer chain counts labels, while actual quantum propagation is still determined by joint operators on the complex state space. In characteristic p, the group order contains the factor p, so a representation need not split into a direct sum of simple modules. Let $A = \mathbb{F}_p[G_{\mathrm{aff}}]$, let $U = \mathbb{F}_p^{\times}$ be the scaling subgroup, and induce from its trivial module to obtain P_0. Denoting unit translation by t, we may write

$$\begin{gathered} P_0 = A \otimes_{\mathbb{F}_p[U]} \mathbf{1} \simeq \mathbb{F}_p[t]/((t-1)^p), \\ z = t - 1, \\ \operatorname{rad}^k P_0 = z^k P_0, \\ \dim_{\mathbb{F}_p} z^k P_0 = p - k \\ (0 \le k \le p). \end{gathered} \tag{3.56}$$

The element t satisfies $t^p{=}1$, and in characteristic p, $t^p{-}1{=}(t{-}1)^p$. Every submodule must first be translation-invariant and hence an ideal of this truncated polynomial ring; these ideals are precisely (z^k), and all are scaling-invariant. They form the unique successive inclusion chain, with one-dimensional successive quotients. This is the Loewy-layer structure used here. [14] Since the order p−1 of U is invertible in $\mathbb{F}$p, its trivial module is projective; induction preserves projectivity, and P_0 has the trivial module as its unique simple top, so it is the projective cover of the trivial module.

Scaling by a sends t to t^a, with $t^a{-}1{\equiv}a(t{-}1) \bmod (t{-}1)^2$. It therefore acts on the kth successive quotient by multiplication by a^k:

$$z^k P_0 / z^{k+1} P_0 \simeq \chi_{\mathrm{nat}}^k, \qquad \chi_{\mathrm{nat}}(a) = a, \qquad k = 0, \dots, p-1. \tag{3.57}$$

Thus every nontrivial character occurs exactly once in the intermediate layers, while the trivial character occurs at both ends. This chain has p layers and P_0 has dimension p; the earlier p(p−1) counts all affine operations, a different quantity. To associate these layer labels with complex spectral directions, first choose a label convention from $\mathbb{F}$p× to complex roots of unity, consistent with the orientation of the pullback action used above. This associates character labels; it does not turn the nonsplit chain over $\mathbb{F}$p into a quantum propagation chain in the complex state space. A complex representation decomposes using an invariant inner product, and actual transitions between layers must still be supplied by joint operators.

The layer structure determines the multiplicities of a given node prescription rigorously. Retain the decomposition $p-1 = \prod_i n_i$ of 2.2, where the n_i are pairwise coprime prime powers and characters are labeled by coordinates $\ell_i(\chi) \in \mathbb{Z}/n_i\mathbb{Z}$. Now explicitly specify an evaluation prescription: assign a positive factor t_i when coordinate i is zero and one otherwise; take the product of these factors for each nontrivial layer, then multiply over all nontrivial layers. Denoting the total factor by $\mathcal{L}$p gives

$$\mathcal{T}_\chi = \prod_{i:\ell_i(\chi)=0} t_i, \qquad \mathcal{L}_p := \prod_{\chi\neq 1} \mathcal{T}_\chi = \prod_i t_i^{(p-1)/n_i-1}. \tag{3.58}$$

The proof is a count: fixing coordinate i to zero leaves $(p-1)/n_i$ choices for the other coordinates; after removing the fully trivial label, the remaining $(p-1)/n_i-1$ nontrivial layers each contribute one factor t_i. Multiplying the contributions of all factors gives the formula. Conjugating all labels negates every coordinate and leaves the zero-coordinate counts unchanged.

This theorem determines multiplicities; the physical values of the node factors must be established as in Supplementary Proof S4.2 and Supplementary Proof S5.1. The factor $n_i^{-1/2}$ is the single-point amplitude of a uniform vector; its square is the corresponding probability. The total share of n_i orthogonal, equally weighted outcomes is instead obtained by addition. Connecting these quantities to the same layer weight t_i requires specifying the input, endpoint, spectral reference, and whether the final quantity is an amplitude or a share. If consecutive layers represent successive recorded exchanges, their joint probability must be obtained by sequentially composing conditional instruments; only when the actual composition yields scalar factors may it be replaced by the product of those factors.

Using this layer product as the denominator of emergent channel weight corresponds to the assignment

$$W_{\mathrm{aff}} = \frac{p(p-1)}{\mathcal{L}_p} \quad\Longrightarrow\quad \alpha_{\mathrm{emg}} = \frac{p(p-1)}{\mathcal{L}_p}, \qquad \alpha_{\mathrm{mix}}(\chi) = \frac{p(p-1)}{\mathcal{L}_p}|R_\chi|^2. \tag{3.59}$$

For example, this total weight is realized if all affine channels are orthogonal, have equal responses, and actually each take $|c_g|^2 = 1/\mathcal{L}_p$. But the Loewy chain alone does not prove that actual exchange uses these c_g. Choosing P_0 as the emergent evaluation object, prescribing multiplication of node weights along its layers, and placing the resulting product in the channel denominator are all conditions retained in this evaluation correspondence. The integer number of group elements remains $\mathrm{p}(\mathrm{p}-1)$; it is the response weight that changes. General positive factors t_i do not guarantee $\mathcal{L}\mathrm{p}>1$, so the correction may enhance or suppress response and cannot universally be interpreted as removal of channels.

Comparisons of responses at different spectral parameters s must retain these preparation and weight conditions. Only with matched inputs, fixed c_g, and its fixed evaluation correspondence can all variation of the mixed response be assigned to $\mathrm{R}\chi(\mathrm{s})$, while the purely emergent response remains the same Waff. The spectral parameter s, inverse environment temperature β, and actual time are distinct variables. Interpreting parameter variation as time evolution further requires a relation between that parameter and time, together with calculation of changes in preparation and exchange. Finite thermal preparations always use finite partition sums; the pole of the infinite ζ partition sum at β=1 does not directly imply divergence of a one-event probability.

The physical process of a resonance force consists of actual exchange between two carriers: an excitation passes through compatible endpoints and a common environment, allowing a change in one carrier to enter the state evolution of the other. In the constructed protection and retrieval process, the sources and auxiliaries retain their original preparations and the mediator returns, while the two targets

may still undergo transfer and become entangled. Exchange strength is calculated along the same process: matching shares at elementary vertices give Eq. (3.33); repeated round trips modify the response according to coherence and record relations of the complete output; and collective exchange sums to Eq. (3.54) when outputs are orthogonal. Actual instruments realize these responses as event shares, while momentum changes with the same input, reference, and elapsed duration give the corresponding mechanical readings.

Stable states also alter the background with which they continually exchange. If that change propagates through the background, it can further affect the maintenance and motion of other stable states. This leads to metric forces.

§4 Metric Forces

Matter changes the local state of the background; this change propagates through the background and alters the maintenance of other stable states. **The resulting rate of momentum change is called a metric force.**

Stable states maintain themselves through persistent exchange with the background. This exchange both leaves background occupation and relies on the remaining available background to continue accumulating phase, giving the matter source and maintenance response a common dynamical origin. Metric action connects different stable states through this shared background; its universal part comes from a common modulation of the complete maintenance generator by background changes. This chapter develops the connection using a fixed source at rest and a passively moving carrier, with the source, background resource units, and test carrier sharing actual positions and with time and distance calibrated by the same process.

4.1 Matter Sources

A matter source must be produced by the stable state's own action. Connecting the persistent generator of the complete carrier to the local background makes paths, nodes, and formation shares determine the writing through the same operator; source strength is thereby connected to maintenance phase. We first describe a source process that retains fresh outputs by refreshing pointers, then connect the common matching of Chapter 2 into a reversible occupation round trip. Both obtain the source from the actual joint action, with definite forms for the background and return histories left behind.

Denote the isometric embedding of the protected carrier by J_f, with complete action satisfying $\widehat{H}_f J_f = J_f \mathsf{H}_f$, where both generators are measured in frequency units. During a readout window with displacement paused, prepare a fresh pointer in $|0\rangle_B$, implement $U_{\mathrm{read}} = \exp(-it_{\mathrm{read}} \widehat{H}_f \otimes \sigma_y^B)$ for effective readout duration t_{read}, then read the pointer in the σ_x^B basis. Adding and subtracting the amplitudes of its two components gives the operators within the code:

$$\begin{aligned} k_\pm &= \frac{\cos(t_{\mathrm{read}}\mathsf{H}_f) \pm \sin(t_{\mathrm{read}}\mathsf{H}_f)}{\sqrt{2}}, \\ E_\pm = k_\pm^\dagger k_\pm &= \frac{I \pm \sin(2t_{\mathrm{read}}\mathsf{H}_f)}{2}, \qquad E_+ + E_- = I. \end{aligned} \tag{4.1}$$

Completeness shows that every outcome is included. The readout preserves rest-frame eigenbranches and their degenerate subspaces; after tracing out the pointer, coherence between distinct eigenvalues generally decays. The field source below is a fixed rest-frame eigenbranch and is not replaced by an arbitrary unknown superposition.

A fixed positive source can be produced by an already formed carrier. Using the common return column of 2.5.3 and Supplementary Proof S6, with input z_0, normalized bright output e, and target orientations $Q = Q^\dagger$ and $Q^2 = I$, the complete rest-frame vertex is $F_W = \mu_W \sigma_x \otimes Q$, where $\mu_W = (g_{\mathrm{col}}/\Lambda)\sqrt{W}\log W > 0$ is the half-gap during loading. Pause displacement and isolate resource gating; after

a quarter pulse, apply phase i at the output to obtain

$$\boxed{U_+ = D_i e^{-i\pi F_W/(4\mu_W)}, \qquad \mathcal{J}_+\psi = U_+(z_0 \otimes \psi) = \frac{z_0 \otimes \psi + e \otimes Q\psi}{\sqrt{2}}, \qquad \mathcal{J}_+^\dagger \mathcal{J}_+ = I, \qquad F_W \mathcal{J}_+ = \mu_W \mathcal{J}_+.} \tag{4.2}$$

Target information is retained in correlations between port and orientation, and the isometry also preserves arbitrary entanglement with subsystems not participating in this action. The definite preparation already formed in Chapter 2 has an actual inverse procedure; in the same ready layout, first invert back to the preparation input and then execute U_+, so every accepted history still yields the same $\mathcal{J}_+$. The total acceptance probability remains the original Q_M, without an additional positive-branch share. Different layouts require an actual input conversion and inclusion of its duration; the complete proof of delivery is given in Supplementary Proof S9.1.

Source writing is a function of the complete generator, so the operator associated with each outcome is a scalar multiple of the identity on this positive-source code subspace and preserves target logic. The same-vertex procedure below also includes the common maintenance phase without changing these outcome probabilities. The object prepared here is a source at rest; displacement generally couples positive and negative carriers, so rest-frame preservation does not directly establish preservation of the same code in motion. The loading half-gap and the mass read over the complete operating cycle are also read separately.

Place an identical local two-state background resource unit at each actual propagation position, with $N_i = |1\rangle\langle 1|$ projecting onto its unavailable state. With input-independent probability ε_i, perform the entire readout segment, load outcomes $+, -$ into endpoints 1 and 0 of a fresh resource unit, respectively, and swap it into the background position. If the segment is not performed, retain the old resource unit; old outputs and outcome records remain within the joint process. For eigenvalue λ_i of source generator H_f, let

$$\xi_i = \langle N_i \rangle - \frac{1}{2}, \qquad b_i = \frac{1}{2}\sin(2t_{\mathrm{read}}\lambda_i), \qquad \boxed{\Delta\xi_i = -\varepsilon_i \xi_i + \varepsilon_i b_i.} \tag{4.3}$$

The executed branch replaces resource unit occupation with the reading E_+, while the bypass retains the old occupation, so this update includes both writing and replenishment. The half-occupied reference is realized by unbiased supply with equal probabilities for the two states and boundary preparation; it is not a constant subtracted afterward from an arbitrary background. A weak source gives $b_i = t_{\mathrm{read}}\lambda_i + O((t_{\mathrm{read}}\lambda_i)^3)$ without removing the replenishment term.

The local source thus has a definite physical meaning: the stable state's action changes the unavailable share of background resource units, while replenishment in the same update limits its accumulation. When $0 < t_{\mathrm{read}}\lambda_i \ll 1$, source writing increases the unavailable share and reduces the available share relative to the unbiased reference; background exchange then conveys this local change to other positions.

The matter source can also be realized directly by the same complete matching of 2.5.3. Write occupation using an already formed carrier: the matter action is still given by $H_s = g_M E/4$, and the written occupation share is also determined by E, rather than by a separately specified source coefficient. After entry into the common positive bright carrier, the mean resource occupation is $n_e = Ne/2$, and

the available bright-channel amplitude r_e relative to zero blockage $r_0 = \sqrt{s}$ satisfies

$$\boxed{\frac{r_e^2}{r_0^2} = 1 - e = 1 - \frac{2n_e}{N}.} \tag{4.4}$$

Formation preparation can connect directly to this occupation without a separately specified post-formation input. Denote the actual formation acceptance column by K_5 and its pre-formation acceptance effect by $D_F = K_5^\dagger K_5$. In the connection where the same formation column supplies subsequent matching, the post-formation effect is $E_{\mathrm{RO}} = K_5 K_5^\dagger$ and the complete matching is $E = q_0 E_{\mathrm{RO}}$. Hence $K_5^\dagger E_{\mathrm{RO}} K_5 = D_F^2$: matching once more after formation makes the occupation effect on the pre-formation input contain D_F^2. Available, occupied, old-failure, and unformed outputs together complete the identity, showing that this occupation indeed comes from the same formation–exchange process.

The original formation reflection preserves the matter action over the entire pattern; its available component carries the same matter phase into background exchange, and the positive bright-channel amplitude is determined by the norm of this actual available column. Once the positive bright carrier is actually loaded, equal halves at the two outer ports also give the resource-occupation relation in Eq. (4.4). The pre- and post-formation domains, complete normalization, and common intertwining are given in Supplementary Proof S7.1.

The same matching determines both source phase and occupation share, connecting persistent matter matching to a change in background availability. We first calculate field establishment and motion using continual refreshing with fresh pointers; the realization in which the same resource packet returns along its original path is developed in 4.5.7, where complete histories determine both resource occupation at the end of a round and accumulation along the path.

4.2 Background Propagation

Exchange between background resource units conveys the local bias already produced. Propagation paths are determined by resource unit pairs that can actually exchange, while time and distance are calibrated by the corresponding process readings.

Observational geometry and reading structure follow the observation construction in *Dynamics of the Bost–Connes State Space* [2]: the complex matrix algebra of nonzero observation units and its positivity give observational spacetime of signature $(1, 3)$, while cycle duration and logarithmic coordinates of integer representatives provide time and position readings, respectively. The original proofs are in §§5.1–5.2, §5.4, and §§2.6 and 5.5 of that paper. Specific propagation further requires common positions, adjacency, and lengths for resource units and the test carrier. Here we fix actual position projections P_i, neighboring displacements, and supply boundaries, with pointwise correspondence between the two. Using the position ring already realized in Chapter 2, calibrate adjacent length as ℓ_x; periodic wraparound is also an actual edge. The background is transported along this adjacency, with distant exchange against unbiased resource units through supply branches.

After source updating in each round, choose a complete swap of neighboring identical resource units or an idle step with specified probabilities. Let R be the first-moment transport matrix. Writing $D_\varepsilon = \mathrm{diag}(\varepsilon_i)$, the round recurrence, with

refreshing followed by transport, and its stationary solution are

$$\boxed{\begin{aligned}\xi' &= R(I - D_\varepsilon)\xi + RD_\varepsilon b,\\ \xi_* &= [I - R(I - D_\varepsilon)]^{-1} RD_\varepsilon b.\end{aligned}} \tag{4.5}$$

This is a finite mixing process: each connected component can reach an unbiased supply endpoint, and an idle step has positive probability. The resulting contraction makes the inverse exist, and the initial bias converges to the unique solution over successive rounds. The full configuration may still have correlations; closure of first moments follows from the chosen swaps and repreparations and does not require the biased stationary state to be a product.

When every internal edge and supply branch has common selection probability c, $R = I - cL$, where L is the graph Laplacian including the supply boundary. Let $\xi^+ = (I - D_\varepsilon)\xi + D_\varepsilon b$ denote the stationary bias after refreshing and before transport. Then

$$\boxed{cL\xi^+ = D_\varepsilon(b - \xi), \qquad \xi = R\xi^+.} \tag{4.6}$$

The first relation follows directly from $\xi^+ - \xi = D_\varepsilon(b - \xi)$. Interior regions without matter refreshing satisfy $L\xi^+ = 0$; the post-transport ξ is also harmonic away from refreshing sites and their neighbors. With only conservative transport outside the source, there is no additional bulk decay term; refreshing throughout the bulk changes the equation and field profile.

Actual adjacency determines where this change propagates, while boundary supply and source replenishment determine its stationary distribution. The three spatial axes specify observational geometry, and the position ring retains its actual circular adjacency. Spectral addresses n and their logarithmic distances are used separately from propagation positions i; using the former for this process additionally requires specifying a correspondence between them.

The position kernel and coherent memory of the original joint thermal cycle are given in Supplementary Proofs S9.2–S9.3; this section uses the explicitly specified resource-unit swaps and supply boundaries above.

The resulting $\xi_{*,i}$ describes background occupation bias relative to the unbiased reference at every position in the presence of the source. The action felt by the test carrier must still be determined from the actual role of these resource units in its maintenance process.

4.3 Maintenance Response

Local background changes enter matter through the original maintenance action. Chapter 2 has already placed matter phase and background gradient in the same carrier; when local resources constrain this common exchange, both responses acquire the same availability factor. Spatial differences caused by the source thus change the local rhythm of stable states relative to the shared background, and position gradients become impulses through actual motion. We first formulate a complete cycle using single-resource-unit gating, then explain how collective configurations produce finite common modulation. The source remains a fixed rest-frame eigenbranch, and the test carrier writes no additional source record during this round.

Let $\mathbf{n} = (n_i)$ denote the resource configuration in a round. Single-resource-unit availability response is $h_i(\mathbf{n}) = 1 - n_i$. If π_b is the stationary configuration

distribution produced by all source updates and transport, then

$$\bar{h}_i = \sum_{\mathbf{n}} \pi_b(\mathbf{n}) h_i(\mathbf{n}) = \frac{1}{2} - \xi_{*,i}, \qquad h_0 = \frac{1}{2}. \tag{4.7}$$

This holds exactly for any admissible finite bias. Multiple-resource-unit response is generally a square-root function whose mean must be calculated from the full configuration distribution, not replaced by the square root of mean occupation. We fix a single resource unit here to retain a complete calculation of the same process.

This available share enters maintenance through an actual response port. Use mutually orthogonal normalized input $|z_i\rangle$ and output direction $|a_i\rangle$ of the background port, and let $r_v > 0$ be the spectral-response coefficient fixed at full availability. An unavailable single resource unit switches off this response. For a fixed configuration, the port action and preparation are

$$\begin{gathered} X_i(\mathbf{n}) = r_v(1-n_i)(|a_i\rangle\langle z_i| + |z_i\rangle\langle a_i|), \\ |\chi_i(\mathbf{n})\rangle = \begin{cases} (|z_i\rangle + |a_i\rangle)/\sqrt{2}, & n_i = 0, \\ |z_i\rangle, & n_i = 1, \end{cases} \qquad X_i(\mathbf{n})|\chi_i(\mathbf{n})\rangle = r_v h_i(\mathbf{n})|\chi_i(\mathbf{n})\rangle. \end{gathered} \tag{4.8}$$

The final equality follows directly from exchange between the two endpoints. When unavailable, the port uses a zero-response input without normalizing a zero vector. This realizes control of an actual interaction edge by the same resource unit, making h_i an interaction coefficient.

On the matter side, reuse the two-arm protection of Chapter 2, denoting its dimensionless action by $\widehat{\mathsf{F}}_t$ with $\widehat{\mathsf{F}}_t J_t = J_t \mathsf{F}_t$. Multiply both arms by the local X_i and prepare working array $|\Xi_{\mathbf{n}}\rangle = \bigotimes_i |\chi_i(\mathbf{n})\rangle$ at all positions. Take joint embedding $\mathcal{J}_{\mathbf{n}}(\psi \otimes |i\rangle) = (J_t\psi \otimes |i\rangle) \otimes |\Xi_{\mathbf{n}}\rangle$. This array depends on resource configuration, independently of the position occupied by the test carrier.

Let g_T be the bare maintenance strength and ν the displacement strength. Acting as the identity on other working factors, the actual maintenance action and compression relations for the two terms are

$$\begin{gathered} \widehat{H}_H(\mathbf{n}) = g_T \sum_i (\widehat{\mathsf{F}}_t \otimes P_i) \otimes X_i(\mathbf{n}), \\ \widehat{H}_H(\mathbf{n})\mathcal{J}_{\mathbf{n}} = \mathcal{J}_{\mathbf{n}}(g_T r_v \mathsf{F}_t \otimes h(\mathbf{n})), \qquad h(\mathbf{n}) = \sum_i h_i(\mathbf{n}) P_i, \\ \widehat{H}_x \mathcal{J}_{\mathbf{n}} = \mathcal{J}_{\mathbf{n}}(\nu Z \otimes D_x), \qquad D_x = \frac{S^\dagger - S}{2i}. \end{gathered} \tag{4.9}$$

For input $\psi \otimes |i\rangle$, P_i in the first term selects the local response; Eq. (4.8) and matter-side intertwining then give coefficient $g_T r_v h_i$. The original displacement S changes only position and acts as the identity on the working array, so it still preserves the same $\mathcal{J}_{\mathbf{n}}$. This explains the preparation of an all-position array: if only a position-dependent working state were attached, the original bare displacement would generally fail to preserve that carrier.

Both terms can therefore act simultaneously on the same object. Let the test action have the pair of eigenvalues $\pm f_t$, with the original motion orientation satisfying $Z^2 = I$ and $\{Z, \mathsf{F}_t\} = 0$. Denote the accumulated coefficients during the active

window by $a_H = \int g_T(t) r_v \, dt$ and $a_x = \int \nu(t) \, dt$. If the two terms are constant or share an envelope, the activity and accumulated generator are

$$\boxed{U_{\mathbf{n}} = e^{-iK_{\mathbf{n}}}, \qquad K_{\mathbf{n}} = a_H \mathsf{F}_t \otimes h(\mathbf{n}) + a_x Z \otimes D_x.} \tag{4.10}$$

Different temporal schedules require the actual time-ordered propagator. In each round, freeze the configuration after resource updating, prepare the array, run this action, then reverse the working preparation. Preparation and its inverse do not change logical position; the common period τ includes all array preparation, source readout, transport, and waiting.

Let $P(\mathbf{n}'|\mathbf{n}; b)$ be the transition probability of the full resource configuration. At the cycle stage where the joint state is block diagonal in the resource-configuration basis, one round acts on any classically correlated resource–carrier blocks $\rho_{\mathbf{n}}$ as

$$\boxed{\begin{aligned} \widetilde{\rho}_{\mathbf{n}'} &= \sum_{\mathbf{n}} P(\mathbf{n}'|\mathbf{n}; b) \rho_{\mathbf{n}}, \\ \rho'_{\mathbf{n}'} &= U_{\mathbf{n}'} \widetilde{\rho}_{\mathbf{n}'} U_{\mathbf{n}'}^{\dagger}. \end{aligned}} \tag{4.11}$$

Its Kraus operators are $\sqrt{P(\mathbf{n}'|\mathbf{n}; b)} |\mathbf{n}'\rangle\langle\mathbf{n}| \otimes U_{\mathbf{n}'}$. Each probability column sums to 1, so the process preserves trace and retains all resource outcomes; it does not require decorrelating the carrier from the background anew in every round.

Equation (4.11) places source updating and background transport from the preceding sections in the same round as maintenance and motion. Each resource outcome corresponds to an actual action $U_{\mathbf{n}}$; correlations generated between background and carrier enter the next round, allowing changes in carrier motion to be calculated directly.

Multiple-resource-unit configurations can modulate frequency through the same port structure. Suppose N_C configurations are coherently produced with equal weights from one input, and occupation projection O blocks only the configuration matching the incoming resource packet label. The Gram operator of the complete available column is $\beta_C^2(I - O/N_C)$, where $\beta_C = \sqrt{W_C}/\Lambda_C$. Its positive square root is the bright-channel coupling operator, giving $\beta_C(I - \delta_C O)$ and $\delta_C = 1 - \sqrt{1 - 1/N_C}$. Matching first changes the norm of the output column and thereby the persistent frequency.

When the original matter and background gradient share this bright carrier and motion transports the entire loaded working system, this bright-channel coupling operator modulates the complete generator in common. Forward and inverse loading return the auxiliaries while retaining the holding phase, as proved in Supplementary Proof S6.5. The coupling operator and loading duration both enter the background calibration of this realization.

4.4 Motion and Force

4.4.1 Momentum Change

Within the active window of each fixed configuration, first consider how the carrier moves between positions. Let $u \in [0, 1]$ be normalized elapsed time in the window, and denote the position blocks of the carrier by $\rho_{ij}(u)$ and $p_i = \operatorname{Tr} \rho_{ii}$. From $\partial_u \rho = -i[K_{\mathbf{n}}, \rho]$,

$$J_i(u) = a_x \operatorname{Re} \operatorname{Tr}(Z \rho_{i,i+1}(u)), \qquad \partial_u p_i = J_{i-1} - J_i. \tag{4.12}$$

The local maintenance term cancels in the position trace, while neighboring displacement gives incoming and outgoing probability currents, conserving total probability around the ring. Background action changes coherence between orientation and position, thereby changing subsequent probability flow; background-occupation differences alone do not determine the carrier's actual motion.

Take the finite-difference momentum operator $\Pi_x = I \otimes D_x/\ell_x$ on the same position ring. Resource updating does not change test-carrier momentum, and the displacement term also commutes with Π_x. Integrating the Heisenberg equation for each configuration gives the exact finite-round relation

$$\boxed{\begin{aligned}\Delta\langle\Pi_x\rangle &= \sum_{\mathbf{n}} \mathrm{Tr}(\widetilde{\rho}_{\mathbf{n}}\,\mathcal{I}_{\mathbf{n}}),\\ \mathcal{I}_{\mathbf{n}} &= \int_0^1 e^{iuK_{\mathbf{n}}}\, ia_H[\mathsf{F}_t \otimes h(\mathbf{n}), \Pi_x] e^{-iuK_{\mathbf{n}}}\, du.\end{aligned}} \tag{4.13}$$

The proof uses only $\frac{d}{du}(e^{iuK}\Pi_x e^{-iuK}) = e^{iuK} i[K,\Pi_x] e^{-iuK}$. It does not assume that finite position and momentum satisfy the canonical commutation relation of infinite space, and it includes the periodic wraparound edge. The mean metric force per round is therefore $\overline{F}_x = \Delta\langle\Pi_x\rangle/\tau$. Spatial background differences generate impulse through the commutator of maintenance with momentum, completing the connection from local background change to actual force; a uniform maintenance term contributes zero here.

4.4.2 Potential Energy and Acceleration

The background changes maintenance frequency, appearing in slow motion as position-dependent potential energy. Since resources still fluctuate each round, the range in which a mean action is valid must first be determined from the complete channel.

Fix a finite mixing resource process with stationary distribution π_b and per-round accumulation $K_{\mathbf{n}} = \eta B_{\mathbf{n}}$, where $B_{\mathbf{n}}$ is bounded and η sufficiently small. Write $a_H = \eta A_H$ and $a_x = \eta A_x$, with mean action $\bar{B} = A_H \mathsf{F}_t \otimes \sum_i \bar{h}_i P_i + A_x Z \otimes D_x$. Starting from a stationary product input, the complete channel retains correlations generated subsequently; finite mixing makes their correction to the reduced motion per round $O(\eta^2)$, so within $N\eta \le T$,

$$\boxed{\sup_{N\eta\le T} \left\| \rho^{(N)} - e^{-iN\eta\bar{B}} \rho^{(0)} e^{iN\eta\bar{B}} \right\|_1 \le C_T \eta.} \tag{4.14}$$

The constant depends on the mixing rate and operator bounds of the fixed resource process, and actual elapsed duration is $N\tau$; see Supplementary Proof S9.8. Within this window, converting to rates using the complete cycle period gives the mean generator and the source-induced change:

$$\begin{aligned} H_{\mathrm{av}} &= \frac{a_H}{\tau}\mathsf{F}_t \otimes \sum_i \bar{h}_i P_i + \frac{a_x}{\tau} Z \otimes D_x,\\ \boxed{\delta H} &= \boxed{\frac{a_H}{\tau}\mathsf{F}_t \otimes \sum_i (\bar{h}_i - h_0) P_i.} \end{aligned} \tag{4.15}$$

We now calculate the implications for motion. Using the symmetric eigenbranches of Chapter 2, take $f_t > 0$ and write $\mathsf{F}_t = f_t\beta_t$ on these two branches, with $\beta_t^2 = I$ and $\{\beta_t, Z\} = 0$. The uniform baseline has rest-frame half-gap $m_t = a_H f_t h_0/\tau$ and long-wavelength speed $c_0 = a_x\ell_x/\tau$. Let $\Phi(x) = [\bar{h}(x) - h_0]/h_0$. When the background varies slowly on the wave-packet scale, first freeze its local value to obtain the matrix symbol

$$\begin{aligned} H_{\mathrm{loc}}(x,k) &= m_t[1+\Phi(x)]\beta_t + \frac{a_x}{\tau} Z \sin(k\ell_x), \\ H_{\mathrm{loc}}(x,k)^2 &= \left\{ m_t^2[1+\Phi(x)]^2 + \frac{a_x^2}{\tau^2}\sin^2(k\ell_x) \right\} I. \end{aligned} \tag{4.16}$$

Cross terms cancel by anticommutation. This is the local symbol with the background frozen; the square of the full nonuniform operator also contains spatial-gradient terms, whose corrections are treated separately under slow variation. Taking $|\Phi| \ll 1$ and $|k\ell_x| \ll 1$ with $c_0|k| \ll m_t$, the positive-energy branch expands as

$$\begin{aligned} E_+(x,k) &= \sqrt{m_t^2[1+\Phi(x)]^2 + \frac{a_x^2}{\tau^2}\sin^2(k\ell_x)} \\ &= m_t + m_t\Phi(x) + \frac{c_0^2k^2}{2m_t} + O\!\left(|\Phi|\frac{c_0^2k^2}{m_t} + \frac{c_0^4k^4}{m_t^3} + \frac{c_0^2\ell_x^2k^4}{m_t} \right). \end{aligned} \tag{4.17}$$

The first term is the baseline rest energy, the second the background-induced energy change, and the third the kinetic energy $k^2/(2M_{\mathrm{in},t})$. This identifies both $M_{\mathrm{in},t} = m_t/c_0^2$ and

$$\boxed{V_t(i) = m_t\Phi_i, \qquad \Phi_i = \frac{\bar{h}_i - h_0}{h_0} = -2\xi_{*,i}.} \tag{4.18}$$

For a single resource unit, the final equality is an identity for the mean response; using it as a motion potential retains the averaging, positive-energy, and slow-variation conditions above. In the long-wavelength reading $\hbar = 1$, let X, k denote the center coordinate and wave number of a localized wave packet. The leading equations of motion are $\dot{X} = \partial_k E_+ = c_0^2 k/m_t$ and $\dot{k} = -\partial_X E_+ = -m_t\partial_X\Phi$. Hence

$$F_{t,x} = -m_t\partial_x\Phi, \qquad \ddot{X} = -c_0^2\partial_x\Phi. \tag{4.19}$$

The same maintenance action determines both the inertial parameter in kinetic energy and the coefficient of the background potential, so test mass cancels from the leading acceleration. Higher orders retain corrections from local inertia, lattice dispersion, and spatial gradients; the exact finite-round momentum change is still given by Eq. (4.13).

To compare source and test carrier, first use the same fixed frequency scale $\nu_* > 0$ to write $\widehat{H}_s = \nu_*\widehat{\mathsf{F}}_s$, and let $f_s = \lambda_s/\nu_*$ and $\theta = \nu_* t_{\mathrm{read}}$. Then $t_{\mathrm{read}}\lambda_s = \theta f_s$ and the source reading of 4.1 is unchanged; f_s and the test coefficient f_t of 4.3 use the same dimensionless normalization. All passive-source strength formulas below retain this θ. The source itself uses the same background clock and has leading zero-signal mass $m_s = a_H f_s h_0/\tau$. Let $\mathcal{Q} = [I - R(I - D_\varepsilon)]^{-1}RD_\varepsilon$ and substitute the weak source $b_s = \theta f_s + O((\theta f_s)^3)$ to obtain

$$\boxed{V_t(i) = -m_t\frac{\theta\tau}{a_H h_0^2}\sum_s \mathcal{Q}_{is}m_s + \text{controlled corrections.}} \tag{4.20}$$

Source and test carrier share the baseline mass calibration; coherent readings of the source itself must retain its actual instrument and cannot use the passive test's configuration kernel instead. Corrections include source-sine nonlinearity, finite accumulation and averaging error, and weak-field, low-momentum, and slow-variation errors. For a fixed source and process, both the field's spatial structure and strength come from the same return kernel $\mathcal{Q}$.

4.4.3 Continuous Survival and Energy Balance

Continuous survival of matter and the stationary distribution of the background field describe different objects. The channel above preserves source and test carrier within the protected study window; if the actual cycle also performs thermal processing, the complete process must specify how the original carrier is stored during that processing, when it returns, and when it exits.

One existing pattern-preserving realization uses an auxiliary copy with the same spectrum and group representation as the system. A full swap first stores the unknown carrier and its correlations in the copy, while the original thermal processing acts on a known system input; after successful filtering, the inverse swap returns the carrier. The full swap commutes with the sum of the two bare energies. Activity is switched off during thermal processing, and the bare free phases of system and copy are actually compensated. If the success outcomes at the fixed input act on the carrier as $\sqrt{r_\eta} I$, with total survival probability $q_T = \sum_\eta r_\eta$, then the protected activity U satisfies

$$\mathcal{Q}_T(\rho) = q_T U \rho U^\dagger, \qquad \operatorname{Tr} \mathcal{Q}_T(\rho) + \operatorname{Tr} \mathcal{X}_T(\rho) = \operatorname{Tr} \rho. \tag{4.21}$$

The exit map $\mathcal{X}_T$ retains failed outputs and the stored unknown state. Envelope and phase evolution of the successful component are thereby separated: survival probability changes the envelope, while frequency must also be divided by the complete cycle period including storage, thermal exchange, filtering, compensation, and recording.

Energy changes during activity can also be balanced within the same space. For an active window at fixed configuration, denote the bare energy by H_0 and specify a self-adjoint generator H_{on} on the full space that preserves the protected code, with actual activity generator $K_{\mathbf{n}}/t_H$ within the code. Let $D = H_{\mathrm{on}} - H_0$, without assuming that H_0 preserves the same code subspace. Taking positive work to mean energy supplied to the system by control, rectangular switching gives

$$\begin{gathered} \rho' = e^{-it_H H_{\mathrm{on}}} \rho e^{it_H H_{\mathrm{on}}}, \qquad W_{\mathrm{on}} = \operatorname{Tr}(D\rho), \quad W_{\mathrm{off}} = -\operatorname{Tr}(D\rho'), \\ \boxed{W_{\mathrm{on}} + W_{\mathrm{off}} = \operatorname{Tr}[H_0(\rho' - \rho)].} \end{gathered} \tag{4.22}$$

Total energy is conserved within the active window, namely $\operatorname{Tr}[H_{\mathrm{on}}(\rho' - \rho)] = 0$; substituting $H_{\mathrm{on}} = H_0 + D$ gives the identity. Finite switching uses the actual time-ordered propagator and includes $\int \operatorname{Tr}(\rho_t \dot{H}_t) dt$; protection throughout switching must still be checked against the generators used. The original thermal exchange continues to balance the total bare energies of system, copy, and bath, including the energy carried away by exit outputs. Control work, thermal exchange, and exit thus have separate, explicit accounts.

Spatial background differences change carrier momentum through the same maintenance action; within the controlled weak-field, low-speed window, they appear as

a potential-energy gradient and give a leading acceleration independent of test mass. Thermal processing determines continued survival and complete timing; greater equilibrium occupation alone does not establish directed motion.

4.5 Strength Calibration

Absolute strength compares two outputs of the same process: the mass phase accumulated by the source itself and the position-dependent phase acquired by the test carrier through the altered background. Their relation is determined jointly by source writing, spatial arrival and return, and local maintenance. Only after converting both phases into mass and potential energy using their total elapsed durations, and obtaining impulse from the same motion process, can actual force be compared with source and test masses. Common background calibration constrains the activity ratios at source and test endpoints, while the actual operation sequence determines the time needed to complete those activities. The spatial kernel and clock must therefore be obtained within the same outgoing-and-return process.

4.5.1 Actual Field Strength

The amplitude of a metric force is determined jointly by how much bias the source writes, how the background transports and replenishes it, and how the carrier responds. Equation (4.20) already combines these actions in one kernel. For a localized single source at s, let $\mathcal{G} = L^{-1}$ and take source-site refreshing probability $0 < \varepsilon < 1$. Let $j_s = \varepsilon(b_s - \xi_s)$ denote the net bias written per round. From the refreshing and transport relations in Eq. (4.6), stationarity gives

$$\begin{aligned} &cL\xi^+ = j_s e_s, \qquad \xi = \xi^+ - j_s e_s, \qquad \xi^+ = \frac{j_s}{c}\mathcal{G}_{\cdot s}, \\ &\xi_s = \left(\frac{\mathcal{G}_{ss}}{c} - 1\right) j_s, \qquad \left[1 - \varepsilon + \frac{\varepsilon \mathcal{G}_{ss}}{c}\right] j_s = \varepsilon b_s. \end{aligned} \tag{4.23}$$

Here e_s is the unit column vector at the source position. The final equation substitutes the bias propagated back to the source into the same writing rule. Solving for j_s and using $\xi_i = \xi_i^+$ at $i \neq s$ gives

$$\xi_{*,i} = \frac{\varepsilon \mathcal{G}_{is}}{c(1-\varepsilon) + \varepsilon \mathcal{G}_{ss}}\, b_s, \qquad i \neq s. \tag{4.24}$$

This denominator results from joint balance between source and background: the source writes bias each round, while propagation returns part of it to the source, reducing the difference still available to be written next round. Eliminating the unknown source-site value makes the same return kernel determine net writing and the distant field profile. A weak source only linearizes the writing relation; return feedback remains. Reading this profile as strength in three-dimensional space additionally requires the corresponding actual adjacency and spatial window.

The three-dimensional field profile depends on actual spatial propagation. If positions are realized as an equal-spacing three-dimensional nearest-neighbor grid with equal exchange weights in all six directions, its operator has smooth long-wavelength limit $-\ell_x^2 \nabla^2$. Fourier transformation directly gives the Green kernel of

the infinite unit lattice:

$$\mathcal{G}(\mathbf{j}) = \int_{[-\pi,\pi]^3} \frac{e^{i\mathbf{q}\cdot\mathbf{j}}}{2\sum_{a=1}^{3}(1-\cos q_a)} \frac{d^3q}{(2\pi)^3} = \frac{1}{4\pi|\mathbf{j}|} + O(|\mathbf{j}|^{-2}). \tag{4.25}$$

The denominator near the origin is $|\mathbf{q}|^2+O(|\mathbf{q}|^4)$; isolating this low-wave-number part gives the leading term. In a finite box, every missing neighboring edge is connected to unbiased supply, and its kernel tends to this kernel as the boundary recedes. Restoring $r = \ell_x|\mathbf{j}|$, the response in the far region outside the source therefore has shape $1/r$. This uses three-dimensional adjacency and its spatial calibration; the principal unit-position ring alone does not yield this conclusion.

On the same three-dimensional positions, if test-carrier motion also satisfies the potential-readout conditions of Eqs. (4.18)–(4.19), substituting the far-region kernel into Eq. (4.20) gives the leading radial action

$$\boxed{\begin{gathered} V_t(r) = -G_{\mathrm{dyn}}\frac{m_s m_t}{r}, \qquad F_{t,r} = -G_{\mathrm{dyn}}\frac{m_s m_t}{r^2}, \\ G_{\mathrm{dyn}} = \frac{\theta\tau\varepsilon\ell_x}{4\pi a_H h_0^2[c(1-\varepsilon)+\varepsilon\mathcal{G}_{ss}]}. \end{gathered}} \tag{4.26}$$

Here G_{dyn} is the leading strength of the present source, transport, and passive-response process. Until the three-dimensional spatial conditions are adopted, the original position ring continues to use the finite kernel of Eq. (4.20), without radial substitution. This establishes both the spatial conditions for field shape and the operational origin of strength; conversion to common physical units further requires the background scale.

4.5.2 Background Scale

The common scale comes from the squared background spectral response. Fix odd prime p and depth K, let $d = p^K$ and $M = d-1$, take spectral function $h(n) = \log n$ on nonzero integer representatives, and denote spectral-address projections by P_n. The increment between adjacent spectral addresses is $\Delta_n = \log(1+1/n)$. Assigning the squared gradient of every spectral edge to both endpoints gives

$$\begin{gathered} B_K = \sum_{n=1}^{M-1} \Delta_n^2 (P_n + P_{n+1}), \qquad B_K P_0 = 0, \\ \boxed{\operatorname{Tr} B_K = C_M^* = 2\sum_{n=1}^{M-1} \log^2(1+1/n).} \end{gathered} \tag{4.27}$$

Every edge appears twice, directly giving the trace. Since $\Delta_n < 1/n$, the total tends to a finite value C^* as the spectral cutoff increases, with $0 < C^* - C_M^* < 2/(M-1)$. Spectral addresses, edges, and physical resource positions are counted separately here.

Here $h(n)$ measures spectral response, whereas the matter-induced potential Φ_i is determined by the actual background distribution. They are connected through common frequency calibration, without setting Φ_i equal to the logarithm of a position label. A normalized state gives relative response shares; unnormalized participation weights record the total response.

For a normalized state ρ supplied by the cycle, the mean squared spectral gradient is

$$Q_{\mathrm{met}}(\rho) = \mathrm{Tr}(\rho B_K), \qquad Q_{\mathrm{met}}(I/d) = \frac{C_M^*}{d}, \qquad Q_{\mathrm{met}}(I_{\neq 0}/M) = \frac{C_M^*}{M}. \tag{4.28}$$

The finite total squared gradient is distributed among all participating positions in an equal-weight state, so its mean decreases as the number of positions grows. The normalized trace already performs one average and is not divided again by the mode count. This quantity gives the squared response of the specified state; it does not alone determine the actual field amplitude.

Using common background matching, let positive operator Γ record unnormalized response weights and let $P_{\mathcal{S}}$ select the participation range. Take $W = \mathrm{Tr}(P_{\mathcal{S}}\Gamma P_{\mathcal{S}}) > 1$ and common coefficient $\log W$. Then

$$\boxed{E(W)^2 C_{\mathrm{bg}} = E_0^2 W \log^2 W, \qquad E(W) = E_0 \sqrt{\frac{W}{C_{\mathrm{bg}}}} \log W.} \tag{4.29}$$

The quantity C_{bg} uses the background squared response actually selected: C^* for the full spectrum and C_M^* for a finite open chain. The physical energy unit is supplied by E_0. This equation retains the physical identification through squared-response matching; a normalized state alone is insufficient to recover W.

Define the inverse square of the common metric energy scale as the strength scale:

$$\boxed{G_E(W) = \frac{1}{E(W)^2} = \frac{C_{\mathrm{bg}}}{E_0^2 W \log^2 W}.} \tag{4.30}$$

When every orthogonal mode has unit response weight and the entire space participates, $W = d = p^K$. If $C_{\mathrm{bg}} = C_M^*$ is also used, $E_0^2 G_E(d) = Q_{\mathrm{met}}(I/d)/\log^2 d$. Here $1/d$ comes from the specified state average, whereas the squared logarithm comes from common energy-scale matching; they play different roles.

This distinction directly affects strength when structural weights change. Retain the positive factor $\mathcal{L}_p$ of Chapter 3. If the metric process uses effective participation weight $W_{\mathrm{met}} = d\mathcal{L}_p > 1$, then with the same background and units,

$$\boxed{\frac{G_E(d\mathcal{L}_p)}{G_E(d)} = \frac{1}{\mathcal{L}_p}\left[\frac{\log d}{\log(d\mathcal{L}_p)}\right]^2.} \tag{4.31}$$

The same weight enters both participation and its logarithm, so multiplying only by $1/\mathcal{L}_p$ omits part of the contribution. For example, with $\Gamma = \mathcal{L}_p I$, the normalized state remains I/d and its Q_{met} is unchanged, while absolute participation and energy scale change. The Loewy-layer structure supplies the algebraic hierarchy used; assigning metric participation weights this way remains an explicitly adopted physical correspondence.

4.5.3 Strength Matching

Compare this spectral scale with the actual strength in Eq. (4.26). Here take a finite open-chain background $C_{\mathrm{bg}} = C_M^*$ and metric-end scale $G_E = G_E(W_m)$.

Write $r_v = \sqrt{W_v}\log W_v$ and $r_m = \sqrt{W_m}\log W_m$, where W_v, W_m are the prescribed participation weights at the maintenance and metric endpoints, respectively.

Let $a_B = \int_{\text{background active window}} g(t)\,dt$ denote accumulated background activity and $g(t)$ the common bare drive strength. When maintenance and background share the same active window, $a_H = r_v a_B$. At the common weak-accumulation level, the squared background phase from the same actual readout is $C_{\text{ph}} = C_M^* h_0^2 + O(a_B^2)$, and the period is calibrated by $\tau = a_B\sqrt{C_{\text{ph}}}/E_0$. Hence $m_f = v_{\text{spec}} f$ and $v_{\text{spec}} = E_0 r_v/\sqrt{C_M^*}$. Further adopting event scale $\ell_x = 1/v_{\text{spec}}$ and substituting into the actual strength gives its ratio to the spectral scale:

$$\frac{G_{\text{dyn}}}{G_E} = \frac{\theta\varepsilon r_m^2}{4\pi h_0 r_v^2[c(1-\varepsilon)+\varepsilon\mathcal{G}_{ss}]}, \qquad h_0 = \frac{1}{2}. \tag{4.32}$$

This compares their common leading order, with source and test masses using the same background clock. It shows how actual source readout, transport, and return enter strength; a common period and common units do not remove these ratios. Adopting $G_{\text{dyn}} = G_E$ as a physical calibration requires the selected process to satisfy this correspondence, rather than assigning the same field strength to every allowed exchange.

4.5.4 Static Response at a Common Vertex

Denote the source and test angles in this section by $z_s = a_B r_s$ and $z_t = a_B r_t$. Under the normalization $a = r_v a_B$ below, $f_s = r_s/r_v$ and $f_t = r_t/r_v$ are dimensionless phase coefficients; when produced through the complete common spectral port of S6, $r_s = r(W_s)$ and $r_t = r(W_t)$. Other families must use their own actual phase coefficients; the dimensionful eigenvalue λ_i of 4.1 cannot be substituted directly here.

When the same vertex supplies both maintenance phase and source readout, field strength and spectral scale can be compared within common preparation, readout, and timing. Take a static process with common refreshing at source and test endpoints, conservative transport outside the source, and unbiased supply at the boundary. Test-end refreshing is retained in the coherent return, giving a different return kernel from the passive test of Eq. (4.11); strength must be calculated independently rather than transferred from Eq. (4.32).

Take contact points s, t, common refreshing probability ε, and probability c for each exchange edge and supply branch. A fresh resource unit remains coherent until the common-vertex action and coherent readout are complete. The two contact refreshes are mutually exclusive, $0 < 2\varepsilon < 1$, and transport retains at least one-half idle probability, namely $c|\mathcal{E}| \le 1/2$, where $\mathcal{E}$ includes supply branches. The bias of the fresh resource unit at the source is $b_s = \sin(af_s)/2$, where $a = r_v a_B$ and a_B is the effective accumulation at the common background endpoint; after connecting the common capacity of S6, it is calculated from $a_B = \int g_{\text{col}}(t)/\Lambda\,dt$. At the test endpoint, take the coherence slope at the origin of its own accumulation, where $b_t = 0$, while retaining unbiased refreshing.

Let P_S project onto the two contact points, and set $\alpha = c(1-\varepsilon)/\varepsilon$. Stationarity still gives $\xi^+ = (I - \varepsilon P_S)\xi + \varepsilon P_S b$ and $\xi = (I - cL)\xi^+$. Outside the source, $L\xi^+ = 0$; substituting at the contacts into the latter equation eliminates ξ and gives $(\alpha L + P_S)\xi^+ = P_S b$. Write $\xi^+ = \mathcal{G}_{\cdot S}\mathbf{J}$ and denote $u = \mathcal{G}_{ss}$, $v = \mathcal{G}_{tt}$, and $w = \mathcal{G}_{st}$. Solving

over the whole space reduces to

$$\begin{pmatrix} \alpha+u & w \\ w & \alpha+v \end{pmatrix} \begin{pmatrix} J_s \\ J_t \end{pmatrix} = \begin{pmatrix} b_s \\ 0 \end{pmatrix}, \qquad \Delta = (\alpha+u)(\alpha+v) - w^2 > 0. \tag{4.33}$$

Here $\mathbf{J}$ is the net bias written per round at a contact divided by c, namely $\mathbf{j} = c\mathbf{J}$; it is not a probability. Positive definiteness of the matrix follows from $\mathcal{G} > 0$ and $\alpha > 0$. Direct inversion gives $J_s = (\alpha+v)b_s/\Delta$ and $J_t = -wb_s/\Delta$, hence $\xi_t^+ = wJ_s + vJ_t = \alpha w b_s/\Delta$. Define the cross-response and its far-region coefficient by

$$K_{ts} = \frac{\alpha w}{(\alpha+u)(\alpha+v) - w^2}, \qquad F = \frac{\alpha}{(\alpha+u)(\alpha+v)}, \qquad \xi_t^+ = K_{ts} b_s. \tag{4.34}$$

The test phase must also be read using this instrument. Branches in which the test endpoint is not selected use the response of the old resource unit $1 - n_t$; when it is selected, the fresh resource unit gives the unbiased mean $1/2$ during the coherent process. The normalized slope of the actual coherence root at the origin of test accumulation is therefore

$$\begin{gathered} H_{t|s} = (1-\varepsilon)\left(\frac{1}{2} - \xi_t\right) + \frac{\varepsilon}{2} = \frac{1}{2} - \xi_t^+, \\ \boxed{\Phi_t = \frac{H_{t|s} - 1/2}{1/2} = -2\xi_t^+ = -K_{ts}\sin(af_s).} \end{gathered} \tag{4.35}$$

This uses $\xi_t^+ = (1-\varepsilon)\xi_t$ when $b_t = 0$. Test-end refreshing thus enters both the stationary kernel and phase readout; removing that update changes the result. At zero source, the same reading gives source mass $m_s = af_s/(2\tau)$. For a weak source, in the three-dimensional far region and the range $w^2 \ll (\alpha+u)(\alpha+v)$, substitute $w \simeq \ell_x/(4\pi r)$ and read the actual leading coefficient $G^{(0)} = \tau \ell_x F/(2\pi)$ from test energy shift $m_t \Phi_t$.

For this active two-contact process, thermal survival and replenishment of a positive source can be composed in their complete temporal order. Each round performs source activity, scalar thermal survival, and end-of-round replenishment; the thermal window acts as the identity on resources, test carrier, and their correlations, while complete preparation of a vacant site produces the same positive-source carrier before the next round. Let $f = Q_M$ be the actual per-round replenishment probability including every readiness step, q_T the survival probability of the thermal process transported to the positive-source code subspace, and π_j the probability of presence before activity in round $j+1$. Taking $(1-f)q_T < 1$, roundwise counting gives

$$\boxed{\begin{gathered} \pi_{j+1} = f + (1-f)q_T\pi_j, \qquad \pi_* = \frac{f}{1-(1-f)q_T}, \\ M_s = \pi_* E_s, \qquad E_s = \frac{z_s}{2\tau}. \end{gathered}} \tag{4.36}$$

Here E_s is the individual phase mass verified at the same port, and M_s is the mean counting mass of individuals actually present. If additional input loading before formation has not been included, f must be replaced by the total acceptance probability summed over all accepted histories. The complete joint instrument for source absence and presence and proof for correlated inputs are given in Supplementary Proof S9.4.

Source occupation multiplies both the stationary background response and counting mass, so this common factor cancels in strength per unit mass for a weak source and weak test. Thermal exit still determines individual lifetime, and replenishment still has an actual cost; neither disappears from the complete process. Starting from a vacant site also gives an establishment delay, and finite-angle response must be calculated using the complete instrument.

With persistent replenishment, the stationary weak-test potential is $\Phi_\infty = -\pi_* K_{ts} \sin z_s$, where $z_s = a f_s$. Substituting $M_s = \pi_* E_s$ above and the same reading $E_s = z_s/(2\tau)$ gives $\Phi_\infty = -2\tau K_{ts} M_s \operatorname{sinc} z_s$. Thus, after normalization by actual counting mass, the weak-source leading coefficient remains $\tau \ell_x F/(2\pi)$. Changing the replenishment schedule changes the complete cycle period, so the old clock cannot continue to be used after this cancellation. Complete finite-source and field-establishment results are given in 4.5.6.

The same test resource unit also connects source writing to background calibration. Let $H_{t|s} = 1/2 - \pi_* K_{ts} b_s$ be the available share after contact. The actual gradient exchange has short-time squared response $C_{\mathrm{ex}} = C_M^* H_{t|s}$, while the squared weak-angle phase rate is $C_{\mathrm{ph}} = C_M^* H_{t|s}^2$. Their baselines are $C_M^*/2$ and $C_M^*/4$ respectively, but their first-order changes at the unbiased reference point are both $-C_M^* \pi_* K_{ts} b_s$. The source-induced availability deficit therefore enters both exchange response and maintenance phase; the complete signal and proof of this connection are given in Supplementary Proof S9.5.

4.5.5 Complete Timing and Strength

Matter formation, positive-source loading, and thermal replenishment all take actual time. Actual strength therefore depends jointly on the source-to-test return response and the total elapsed duration of a round. We now recalibrate the background within the same gated operating procedure; the bare coefficient for the ungated half-gap in 2.5.3 cannot be additionally applied here.

The same instrument also performs background calibration. Let $\mathcal{R}$ be the full-configuration transport matrix and E_b unbiased refreshing at contact b. Retaining the actual coherence between positive and negative test eigenbranches, the zero-source matrix is

$$T_t(z) = \mathcal{R}\left[(1-2\varepsilon) D_t(z) + \varepsilon \sum_{b=s,t} \mathsf{E}_b D_t(z) \mathsf{E}_b\right], \qquad D_t(z) = \operatorname{diag} e^{-2iz(1-n_t)}, \quad z = a f_t. \tag{4.37}$$

Full-configuration reversal $\mathcal{J}$ commutes with transport and unbiased refreshing and sends $1 - n_t$ to n_t, hence $\mathcal{J}\overline{T_t(z)}\mathcal{J} = e^{2iz} T_t(z)$. The unique simple root continued from the origin satisfies $e^{iz}\lambda_t(z) \in \mathbb{R}$; on a connected branch where it is nonzero and visible and its centered root is positive, $\arg \lambda_t = -z$, with normalized frequency $-\arg \lambda_t/(2z) = 1/2$. Applying the same reading to every background-gradient branch gives $C_{\mathrm{ph}} = C_M^*/4$.

Write $C = C_M^*$ and $\Omega = 2E_0/\sqrt{C}$; the common background reading gives $a_B = \Omega\tau$. In a serial schedule with common bare rate g_{col}, A_M sums the actual dimensionless cost of inactive steps, including inversion of the formation reference, positive-source loading, thermal transport, replenishment, transport, and completion of failure branches; t_w is waiting time that does not scale with the bare rate.

The active-window cost is Λa_B, so

$$\boxed{\begin{aligned} \tau &= \frac{\Lambda a_B + A_M}{g_{\mathrm{col}}} + t_w, \qquad \rho_{\mathrm{clk}} = \frac{g_{\mathrm{col}}}{\Omega}, \quad \delta = \Omega t_w, \\ a_B &= \frac{A_M + \rho_{\mathrm{clk}}\delta}{\rho_{\mathrm{clk}} - \Lambda}, \qquad \rho_{\mathrm{clk}} > \Lambda. \end{aligned}} \tag{4.38}$$

This uses a serial procedure in which A_M is independent of the accumulation being solved for; if costs depend on that accumulation, retain the implicit relation in the first line. Idle steps also occupy a complete execution window, and the resulting schedule uses the common phase branch of 4.5.6.

Let $F_{\mathrm{ret}} = F$ denote the far-region coefficient of Eq. (4.34), and set $r_m = \sqrt{W_{\mathrm{met}}}\log W_{\mathrm{met}}$ and $r_v = \sqrt{W_v}\log W_v$. With the same three-dimensional spatial correspondence, a stationary weak source and weak test, and a controlled far region, substituting $\ell_x = 2/(\Omega r_v)$ into the actual response gives

$$\boxed{\begin{aligned} G^{(0)}_{\mathrm{response}} &= \frac{\tau \ell_x}{2\pi} F_{\mathrm{ret}}, \qquad E_{\mathrm{met}} = \frac{E_0 r_m}{\sqrt{C}}, \\ \mathfrak{C}_0 := G^{(0)}_{\mathrm{response}} E^2_{\mathrm{met}} &= \kappa_0 \frac{A_M + \rho_{\mathrm{clk}}\delta}{\rho_{\mathrm{clk}} - \Lambda}, \qquad \kappa_0 = \frac{r_m^2 F_{\mathrm{ret}}}{4\pi r_v}. \end{aligned}} \tag{4.39}$$

The return coefficient comes from actual contact and propagation, and accumulated action comes from the complete schedule. Once this process is specified, the above equation determines its leading strength. Stationary occupation has canceled through the same counting mass, while preparation and replenishment costs remain in A_M; see Supplementary Proofs S9.5–S9.6.

The condition for equality between actual response and the inverse spectral square is therefore

$$\boxed{G^{(0)}_{\mathrm{response}} = E^{-2}_{\mathrm{met}} \quad \Longleftrightarrow \quad \rho_{\mathrm{clk}} - \Lambda = \kappa_0(A_M + \rho_{\mathrm{clk}}\delta).} \tag{4.40}$$

This paper adopts one physical calibration, identifying leading response strength with the inverse square of the spectral energy scale; Eq. (4.40) constrains actual driving, accumulation, and the complete-cycle timing under that calibration. It does not follow from the common background clock alone. For dynamically controlled actions that can be scaled together, the required action and schedule are given in Supplementary Proof S9.6; for an independently fixed process, Eq. (4.39) directly gives strength. Subsequent evaluations use this same calibration, with finite-source and propagation corrections calculated from the complete outputs below.

4.5.6 Finite Response

Finite source angle, propagation buildup, and test readout all come from the same complete process. Let $\mathbb{T}(z_t; b_s)$ retain the source-presence flag, all resource configurations, and test coherence, and let v_0 be the specified joint input. Composing two-contact activity, thermal exit, and end-of-round replenishment in order gives the finite signal and source-induced phase:

$$\begin{aligned} C_N(z_t; b_s) &= \mathbf{1}^{\mathsf{T}} \mathbb{T}(z_t; b_s)^N v_0, \\ \varphi_N = \mathrm{Arg}_{\mathrm{unfold}} \frac{C_N(z_t; b_s)}{C_N(z_t; 0)}, &\qquad \Phi_N = -\frac{\varphi_N}{N z_t}, \quad b_s = \frac{\sin z_s}{2}. \end{aligned} \tag{4.41}$$

The potential reading requires established common baseline masses for source and test, nonzero signals, and a phase unwrapped along the actual continuous branch. At finite test angle, Φ_N generally depends on the test port; only the weak-test limit gives a common response. The complete operator and proof are given in Supplementary Proofs S9.4 and S9.7.

Starting with an unbiased resource preparation, write $\mathsf{A} = (I - \varepsilon P_S)(I - cL)$. The finite buildup kernel of the original active process is

$$K_N = \frac{\varepsilon}{N} \sum_{l=0}^{N-1} (N-l)(\mathsf{A}^l)_{ts}, \qquad K_N \nearrow K_{ts}. \tag{4.42}$$

When source occupation starts at equilibrium, the weak-test potential is $-\pi_* K_N \sin z_s$; starting from a vacant site additionally requires the buildup delay caused by replenishment histories. A sufficiently long kernel response does not mean infinite individual survival, nor guarantee that rereading mass over the same long window preserves the original phase identification.

A common phase reading also requires signals to remain nonzero and visible throughout the observation window. Supplementary Proof S9.7 gives a sufficient finite-window bound; a specific process uses K_N to assess field buildup and C_N to assess phase and visibility.

In the stationary weak-test reading, let $\eta = w^2/[(\alpha+u)(\alpha+v)] < 1$. Retaining the source sine and finite return gives the potential in the three-dimensional bulk:

$$\boxed{\Phi_\infty(r) = -\frac{G^{(0)}_{\text{response}} M_s}{r} \frac{\operatorname{sinc} z_s}{1-\eta}, \qquad \operatorname{sinc} z_s = \frac{\sin z_s}{z_s}.} \tag{4.43}$$

Use continuous extension at zero source angle and account separately for lattice and boundary errors. Actual force requires a positional difference or derivative of this potential. If source mass, source angle, period, and contact parameters do not vary with test position, u, v are approximately constant in the bulk, and the spatial approximation $w \simeq \ell_x/(4\pi r)$ also controls derivatives, the leading radial force is

$$\boxed{\mathcal{F}_r = -\frac{G^{(0)}_{\text{response}} M_s m_t}{r^2} \operatorname{sinc} z_s \frac{1+\eta}{(1-\eta)^2}.} \tag{4.44}$$

Return corrections to potential and force therefore differ. Finite buildup, finite test angle, and general geometry are calculated in Supplementary Proof S9.7; these corrections describe actual outputs beyond the leading calibration, and their magnitudes are compared separately with experimental uncertainties.

4.5.7 Calibration and Strength of the Common Round Trip

The round trip of the same resource packet uses the occupation preparation of 4.1, with source-end matching effect E determining both matter phase and packet occupation. Unlike refreshing that removes the old resource unit, this process retains the same packet and its transport records and reverses their order on return; without remote readout, the packet can be restored while holding along the path still leaves a phase. We calculate visits, reception, impulse, and clock consecutively for this process.

For the same resource packet returning along its original path, actual paths give a finite visit kernel. Let R=I−cL be the mean transport matrix of the actual edge-swap and idle-step selection rule, where L is the Laplacian of the undirected position graph used. Freeze and read after every complete edge swap; after m outgoing steps, return in reverse order using the same record. The position sequence is $\mathrm{x}_1,\ldots,x_m,x_{m-1},\ldots,\mathrm{x}_0$, with each endpoint read once and intermediate positions twice. The mean visit counts for unit source column e_{x_0} are

$$\boxed{u^{(m)} = (I+R)\sum_{j=0}^{m-1} R^j e_{x_0}, \qquad \mathbf{1}^{\mathsf{T}} u^{(m)} = 2m.} \tag{4.45}$$

This visit kernel and the refreshing stationary kernel of 4.2 describe their respective processes. A single terminal readout reads only $R^m e_{x_0}$; accumulated readout along the path produces the geometric sum above. On a finite connected graph with idle steps, its nonuniform part has an exact Green decomposition and finite remainder; see Supplementary Proof S10.2. The original position ring's L^+ gives a periodic quadratic profile; neither the configuration space of 6.2 nor the calibration graph of the gradient spectrum automatically becomes a three-dimensional position graph.

During these frozen windows, the same resource packet controls the common final segment of the receiving matter. Let K_t be the joint action of the target's complete matter and original background gradient, and O_B the incoming resource packet's occupation projection; local swap-in, holding, and return implement $K_t(I-O_B)$. The target code preserves its pattern exactly, but finite reception generally entangles it with the packet. For resource-packet occupation probability $q_s = se/2$ of a fixed source branch, time integral b of the target holding envelope in each window, and time integral h_t of the additional ungated holding envelope, the actual coherence for spectral difference $\Delta\kappa$ between two values of K_t at the same position x is

$$\mathcal{C}_{m,x} = e^{-i(h_t+2mb)\Delta\kappa}\left[1-q_s+q_s\,\mathbb{E}_\eta e^{ib\Delta\kappa N_x(\eta)}\right]. \tag{4.46}$$

Here N_x is the visit count of the same outgoing-and-return history, and $\mathbb{E}_\eta$ sums actual exchange-selection records. This is not a channel power for independent fresh resource packets, and mean visits cannot be placed inside the exponential in advance. Within the accumulated weak-readout regime, the complete output gives

$$\boxed{\Delta H_t^{(1)} = H_t^{(0)}\Phi_m(X), \quad H_t^{(0)} = \frac{h_t+2mb}{\tau}K_t, \quad \Phi_m(x) = -\frac{q_s b}{h_t+2mb}u_x^{(m)}.} \tag{4.47}$$

The same Φ_m multiplies the target's entire zero-field action, so matter phase and original background gradient acquire the same leading relative change. This explicitly connects the common background to a universal response: for the adopted common coupling structure, different stable states read local changes through their complete maintenance action. At finite angles, the complete output also gives visibility changes, gradient occupation probabilities, and source residuals; these and phase follow from the same history formula, as shown in Supplementary Proof S10.1.

This round trip can be calculated directly for a target in a position superposition. Initially the target is independent of source and network, with internal network correlations retained; target displacement is paused during frozen readout. The leading impulse of the complete position-controlled action is

$$\boxed{\Delta\langle\Pi\rangle^{(1)} = -iq_s b\langle K_t[u^{(m)}(X),\Pi]\rangle.} \tag{4.48}$$

This is a momentum difference of the complete output, not a total derivative obtained by moving the target and recomputing stationarity. It reduces to a potential gradient only in the original long-wavelength, slowly varying window; when the original gradient input has zero mean, the leading mean force on a fixed matter branch is $-M_t\nabla\Phi_m$. Further identifying this phase mass with inertial mass requires checking the complete zero-field process formed by this round trip and actual motion, without borrowing free curvature from another schedule. A fully reflection-symmetric isolated source may change its surrounding background while having zero net self-impulse; its background response need not vanish.

The common round trip also constrains accumulation at source and target through the same background scale. Place both endpoints themselves in the same zero-blockage reference $r_0 = \sqrt{s}$, with common original gradient coefficient c_B and complete period τ; also take an unoccupied baseline for the target's incoming resource packet. The calibrated gradient coefficient V_0 requires

$$V_0 = 2r_0c_B\frac{t_H}{\tau} = 2r_0c_B\frac{h_t + 2mb}{\tau}. \tag{4.49}$$

Preparing only the incoming resource packet in the unoccupied state does not remove blockage of the target itself. Under this chapter's connection in which all target maintenance is resource-gated, run the actual pause/compensation procedure outside reception windows, namely $h_t = 0$; repeating the same equal-window module along the path gives $2mb = t_H$. Unequal windows generally fix only total time integral of the reception envelope, while the actual envelope remains determined by the controls used.

Source holding and target readout use constant unit normalized envelopes and are performed serially; outgoing and return transport, resource-packet swapping, and thermal processing have total fixed positive cost C_m. The complete period is $\tau = 2t_H + C_m$. Common calibration fixes holding ratio $\alpha = V_0/(2r_0c_B) = t_H/\tau$, giving the unique positive solution $t_H = \alpha C_m/(1-2\alpha)$ and $b = t_H/(2m)$ when $0 < \alpha < 1/2$. Initial loading is timed as startup, while operations executed each round enter C_m; the duration of each operation in the specified sequence and nonunit envelopes are given in Supplementary Proof S10.3.

Substitution into actual reception gives $\Phi_m(x) = -q_s u_x^{(m)}/(2m)$. Take $0 < e < 1$ and $r_s = \sqrt{s(1-e)}$, and read the source phase already accumulated before reception as $M_s = r_s g_M e t_H/(2\tau)$ using the same τ. The finite first-order potential difference between two nonsource positions satisfies

$$\boxed{\frac{\Phi_m(x) - \Phi_m(y)}{M_s} = -\frac{sr_0c_B}{mr_s g_M V_0}\left(u_x^{(m)} - u_y^{(m)}\right).} \tag{4.50}$$

The common factor from accumulated holding cancels from this complete first-order ratio. With a controlled finite Green tail, weak matching $r_s \to r_0$, and $g_B = g_M$, the coefficient of the Green difference at nonsource positions is $\mathcal{A}^{(0)}_{\mathrm{Green}} = 2sq_0/(mc\Lambda V_0)$. In a window where the actual three-dimensional position graph gives $L^+_{xs} \simeq \ell_x/(4\pi r)$, the corresponding leading strength is $G^{(0)} = \mathcal{A}^{(0)}_{\mathrm{Green}}\ell_x/(4\pi)$. It uses this process's exchange-selection rule, cycle length, complete-cycle timing, and source phase mass. The inertial reading in the same procedure, its correspondence with the spectral scale, and source residuals left by finite readout are given in Supplementary Proof S10. The physical calibration of 4.5.5 must be established separately for this procedure.

Metric force thus develops along one physical chain: matter matching changes background availability; actual exchange propagates that change; and local complete maintenance produces phase gradients and impulses. The common carrier fixes ratios between responses, while the return kernel and complete-cycle timing determine actual strength; this paper adopts the physical calibration to the inverse spectral square for the process used.

§5 State Space and p-Channels

The preceding chapters retained odd prime p as a parameter while discussing stable patterns, address point-pair groupings, and exchange channels. Evaluating specific matter requires choosing one channel and relating directions, representations, and actual branches. This chapter determines which structural conditions select $p = 7$ and which definite objects this choice supplies for subsequent mass and coupling calculations.

For a fixed odd prime p, the state spaces and constructions at all finite depths K are collectively called a p-channel. The parameter p determines the arithmetic structure of the species layer, while K determines address-resolution depth; exchanges, phases, and preparations of the actual cycle must still be specified. Channel selection therefore first constrains structure, with physical identities subsequently tested on the corresponding carriers and actions.

5.1 Structure of p-Channels

The directional decomposition of Chapter 2 and CRT channels of Chapter 3 supply the structural quantities needed to compare different p. The species-layer unit group is cyclic of order $p-1$; after excluding the trivial character, its direction-label count and number of negation-pair groups are respectively

$$\boxed{\#\big(\widehat{G}_{p,1} \setminus \{1\}\big) = p-2, \qquad m = \#\big(G_{p,1}/\{\pm 1\}\big) = \frac{p-1}{2}.} \tag{5.1}$$

Both counts are independent of depth K. A deeper unit shell has $(p-1)p^{K-1}$ addresses, with each species point-pair group lifting to p^{K-1} deeper point pairs; the complete direction space, including higher-conductor directions and directions on other shells, has dimension $p^K - K - 1$.

Choose a generator of the unit group and label characters by ℓ. Conjugation, negation parity, and order satisfy

$$\overline{\chi_\ell} = \chi_{-\ell}, \qquad \chi_\ell(-1) = (-1)^\ell, \qquad \operatorname{ord}\chi_\ell = \frac{p-1}{\gcd(\ell, p-1)}. \tag{5.2}$$

There is exactly one nontrivial self-conjugate direction χ_m, while the others form $(p-3)/2$ pairs; the negation parity of χ_m is $(-1)^m$. The symmetric whole-group formation of 2.3.2 uses a faithful nonreal character of quotient group $G_{p,1}/\{\pm 1\}$, whose order is m. Thus

$$m \geq 3 \quad \Longleftrightarrow \quad p \geq 7 \qquad (p \text{ is an odd prime}). \tag{5.3}$$

This excludes smaller channels but still permits larger odd primes. Further decomposing $p-1 = \prod_i n_i$ into pairwise coprime prime powers gives the complete species space and factor-preserving group of Chapter 3:

$$\mathcal{H}_{\mathrm{sp}} \simeq \bigotimes_{i=1}^{r} \mathbb{C}^{n_i}, \qquad \mathcal{G}_{\mathrm{fact}} \simeq \frac{U(1) \times \prod_{i=1}^{r} SU(n_i)}{\mathbb{Z}_{p-1}}, \qquad \mathfrak{g}_{\mathrm{fact}} \simeq \mathfrak{u}(1) \oplus \bigoplus_{i=1}^{r} \mathfrak{su}(n_i). \tag{5.4}$$

The number of structural factors is $1+\omega(p-1)$ and the real Lie-algebra dimension is $1+\sum_i(n_i^2-1)$. The decomposition uses whole prime powers; for example, factor 4 corresponds to SU(4).

Channels can therefore be compared as follows. The label count includes only nontrivial species characters. The group-before-central-quotient column lists U(1) and SU factors before taking the central quotient, without identifying the factor-preserving group in advance with the symmetry group of a fixed cycle.

Table 5-1 Species-layer structures of different p-channels. The factor-preserving group in every row includes the specified central quotient.

p	p−1	Point-pair groups m	Nontrivial labels	Group before central quotient	Structural factors
3	2	1	1	U(1) × SU(2)	2
5	2^2	2	3	U(1) × SU(4)	2
7	2 × 3	3	5	U(1) × SU(2) × SU(3)	3
11	2 × 5	5	9	U(1) × SU(2) × SU(5)	3
13	2^2 × 3	6	11	U(1) × SU(4) × SU(3)	3
17	2^4	8	15	U(1) × SU(16)	2
31	2 × 3 × 5	15	29	U(1) × SU(2) × SU(3) × SU(5)	4

The group in every row must be quotiented by the center $\mathbb{Z}_{p-1}$ specified above. For example, p=7, 11, and 13 all have three structural factors, but their SU-factor dimensions differ; having three factors alone does not select a unique channel. The point-pair group count and nontrivial-label count each determine p and satisfy p−2=2m−1, so they are not independent arithmetic information. If associated with different observables, their independence comes from the separate grounds for the two physical correspondences.

Comparing channels also requires distinguishing different action types within the same label set. Following 3.2, labels with exactly one nonzero CRT coordinate describe single-factor steps, while labels with several nonzero coordinates have composite support. Choosing zero or nonzero for each coordinate decomposes the complete label count over nonempty supports:

$$\begin{aligned} p-2=\prod_{i=1}^{r} n_i-1=\sum_{\varnothing\neq S\subseteq\{1,\dots,r\}}\prod_{i\in S}(n_i-1),\\ \#\{\text{single-factor nontrivial labels}\}=\sum_{i=1}^{r}(n_i-1). \end{aligned} \tag{5.5}$$

This identity follows by expanding $\prod_i[1+(n_i-1)]$. For p=7, factors 2 and 3 give 1+2=3 single-factor nontrivial labels and 1×2=2 two-factor labels, totaling 5. The latter two have order 6, while single-factor labels have order 2 or 3. A classification of single-factor steps therefore applies only to the corresponding label set and cannot be extended unchecked to all elementary directions.

For every odd prime, nonzero negation pairs give two-dimensional complex observation units; the positive-cone and unit conditions used in 4.2 give signature (1,3), as proved in §§5.1–5.2 of Ref. [2]. The scaling-invariant direction over the entire space is always $\mathbb{C}$b, and logarithmic position readings have the same structure; see §5.5 of that paper. These shared structures permit comparison of different p-channels but neither select p nor exhaust collective-matter identities.

Channels differ in their species-layer point-pair grouping, nontrivial phase directions, conjugation and negation structures, and composition of coprime factors. We now relate these features to the matter structure adopted.

5.2 Channel Selection and Physical Correspondence

We first give the structural conditions selecting p=7, then explain the species, complete matter representations, and interaction correspondences within that channel.

5.2.1 Selection Conditions for Single-Factor Channels

Adopt the single-factor step rule of 3.2. Write p−1 as a product of pairwise coprime prime powers n_i, and let $\mathcal{L}$ contain all labels in the full character group with exactly one nonzero CRT coordinate. The complete single-factor label set is required here. Among possible relations for a repeated nontrivial label χ in this set, two are particularly simple: its square equals the trivial label, or its square equals its conjugate. Denoting these by A and B respectively,

$$\begin{aligned} \text{A}: &\quad \chi^2 = 1 &&\iff \quad \operatorname{ord}\chi = 2, \\ \text{B}: &\quad \chi^2 = \bar{\chi} &&\iff \quad \operatorname{ord}\chi = 3. \end{aligned} \tag{5.6}$$

The first equivalence follows directly from nontriviality of χ; the second uses $\bar{\chi}{=}\chi^{-1}$ to rewrite the relation as $\chi^3{=}1$ and then excludes the trivial label. If every label in $\mathcal{L}$ is required to belong to A or B, every nontrivial character in each factor must have order two or three. Every cyclic factor has a character of order exactly n_i, forcing n_i to be 2 or 3. Conversely, if the factors are only 2 or 3, all their nontrivial labels satisfy the corresponding relations. Hence

$$\big[\forall \chi \in \mathcal{L},\ \text{A or B}\big] \iff \big[n_i \in \{2,3\} \text{ for all } i\big] \iff p-1 \mid 6 \iff p \in \{3,7\}. \tag{5.7}$$

This uses the fact that p is an odd prime, so p−1 is even, and no factor repeats in the coprime prime-power decomposition. A-class labels exist in every odd-prime channel because its unit-group order is even. B-class labels exist exactly when 3 divides p−1. Requiring both A and B therefore excludes p=3 and gives

$$\boxed{\begin{array}{l} \mathcal{L} \text{ is the complete set of single-factor nontrivial labels,} \\ \big[\text{A, B exhaust } \mathcal{L} \text{ and both types occur}\big] \quad \iff \quad p = 7. \end{array}} \tag{5.8}$$

This gives the complete arithmetic content of the selection conditions. It expresses a requirement on elementary steps: the square of every single-factor label falls into one of the two relations, with both relations present. Finite reduction and the cycle axioms alone do not enforce this requirement; if only some labels of

a larger channel are opened, the proof also does not select a unique p. Subsequent evaluations adopt it as a channel-selection condition.

Label multiplication must also connect to actual processes. For example, the excursion operator of 2.4.2 satisfies $J_{\chi_2}J_{\chi_1}|z\rangle = -\chi_1(-1)\delta_{\chi_2,\bar{\chi}_1}|z\rangle$ at the zero-element endpoint. Although an order-three χ has $\chi^2{=}\bar{\chi}$ at the label level, $J_\chi J_\chi|z\rangle$ at that endpoint vanishes. The square relations therefore classify labels only; nonzero amplitude of actual compositions and return to the original pattern still depend on vertices and endpoints.

Correspondence with physical generations provides another check. The known classification of elementary fermions contains three generations (see the PDG electroweak review) [15]. If all m whole groups of 2.3.2 are identified with all generation branches, with each group corresponding to an actually maintainable branch, m=3 gives (p−1)/2=3 and also selects p=7. Thus the two classes of label relations and the identification of all generation branches are compatible at p=7. If a larger channel forms only three visible branches, the three-generation count no longer determines p by itself.

5.2.2 Matter Species

At p=7, use 3 as a generator of the unit group modulo 7 and write characters as χ_ℓ. Their CRT coordinates and orders are

$$\chi_\ell(3^a) = e^{2\pi i\ell a/6}, \qquad \ell \longleftrightarrow (\ell \bmod 2, \ell \bmod 3), \qquad \operatorname{ord}\chi_\ell = \frac{6}{\gcd(\ell,6)}. \tag{5.9}$$

The five nontrivial labels comprise one real direction and two complex-conjugate pairs. Table 5-2 fixes the matter names used in subsequent evaluations: species labels record phase structure, whereas color, orientation, and charge belong to the complete matter representation; formation and maintenance processes give the mass branches. This section lists the representation relations needed for that correspondence; masses and couplings are calculated in later chapters.

Table 5-2 Species labels and candidate matter correspondences at p=7.

Direction	CRT coordinates	Order	Conjugate direction	Candidate matter name
χ_3	(1,0)	2	Itself	Neutrino-type
χ_2	(0,2)	3	χ_4	Charged lepton
χ_4	(0,1)	3	χ_2	Corresponding antilepton
χ_1	(1,1)	6	χ_5	Up-type quark
χ_5	(1,2)	6	χ_1	Down-type quark

Species conjugation and charge conjugation act on different information. In the table, χ_1, χ_5 are complex conjugates, whereas up- and down-type quarks carry charges $+2e/3$ and $-e/3$ respectively [17]. When the color carrier and its conjugate have been realized in the complete state, a compatible arrangement of branches is

$$\begin{aligned} u &\leftrightarrow (\chi_1, \mathbf{3}), \qquad \bar{u} \leftrightarrow (\chi_5, \bar{\mathbf{3}}), \\ d &\leftrightarrow (\chi_5, \mathbf{3}), \qquad \bar{d} \leftrightarrow (\chi_1, \bar{\mathbf{3}}). \end{aligned} \tag{5.10}$$

Thus d and $\bar{u}$ share χ_5 but are distinguished by their color representations. This shows how labels accommodate physical names; color carriers, embeddings, charges, and exchanges must still be realized by the complete representation.

The set $\mathcal{L}$ consists of χ_3, χ_2, and χ_4; the remaining two directions have two-factor support. Once names within conjugate pairs are chosen, label multiplication also connects the orientations of the two pairs:

$$\bar{\chi}_2 = \chi_4, \qquad \bar{\chi}_1 = \chi_5, \qquad \chi_3\chi_2 = \chi_5, \qquad \chi_3\chi_4 = \chi_1. \tag{5.11}$$

These multiplicative relations fix combinations of species phases; their masses are evaluated through the actual cycles of Chapter 2.

Neutrality is likewise determined on the complete state. Let Ψ_3 be a definite-charge state, C the conjugation operation, and Q a self-adjoint charge operator. Then

$$C\Psi_3 = e^{i\alpha}\Psi_3, \qquad Q\Psi_3 = q\Psi_3, \qquad CQC^{-1} = -Q, \qquad \Longrightarrow \qquad q = 0. \tag{5.12}$$

Conjugation reverses charge, while the same ray retains its original charge, hence $q = -q = 0$. Self-conjugacy of χ_3 supplies species structure for a neutral correspondence; Majorana identification additionally requires the complete fermionic degree of freedom to coincide with its conjugate [16]. The two charged-lepton branches are exchanged by actual conjugation and carry opposite nonzero charges.

The three sets of mass formulas use specified preparations and endpoints. At $p = 7$, the in-phase point-pair endpoint of 2.3.2 directly receives nonreal even direction χ_2, χ_4, while odd direction χ_1, χ_3, χ_5 has zero amplitude at that endpoint. The quark and neutrino branches of Chapter 8 use real-direction connections, parity-closed words, and indirect-chain endpoints as specified there; the whole-group formula for even directions cannot be applied directly. Their nonzero responses come from their respective complete processes.

5.2.3 Channel Structure of Forces

Matter correspondences must use the same representations as interactions. The CRT decomposition at p=7 gives factors of dimensions two and three; following 3.2, the factor-preserving action is

$$\mathcal{H}_{\text{sp}} \simeq \mathbb{C}^2 \otimes \mathbb{C}^3, \qquad \mathcal{G}_{\text{fact}} \simeq \frac{U(1) \times SU(2) \times SU(3)}{\mathbb{Z}_6}. \tag{5.13}$$

The nonzero steps of the two- and three-dimensional factors are χ_3 and χ_2, χ_4 respectively; together with diagonal actions, they give 3 and 8 traceless self-adjoint directions, supplying the structural origins of weak and color interactions.

The complete matter representation must also accommodate color-singlet leptons. The natural action $I_2 \otimes SU(3)$ on species space gives

$$\mathcal{H}_{\text{sp}}|_{SU(3)} \simeq \mathbf{3} \oplus \mathbf{3}, \qquad (\mathcal{H}_{\text{sp}})^{SU(3)} = \{0\}. \tag{5.14}$$

The central element $e^{2\pi i/3} I_3$ multiplies the entire space by a nontrivial phase, so no nonzero vector is fixed by every color action. This natural representation therefore cannot directly serve as a complete matter representation containing both leptons and quarks.

The charge generator must likewise be determined in the complete representation [15]. The generator of the central $U(1)$ takes the same value on all six directions. If charge is required to commute with every color action in the natural color representation above, irreducibility gives, block by block,

$$[Q, I_2 \otimes U] = 0 \quad (\forall U \in SU(3)) \quad \Longrightarrow \quad Q = Q_2 \otimes I_3. \tag{5.15}$$

Self-adjoint Q_2 can give at most two charge values, each at least triply degenerate. Thus the lepton, quark, and antiparticle names used in Table 5-2 require a corresponding complete matter representation, whose descent through the central quotient must be checked. Negation parity, real or complex species, and step labels play their respective roles in subsequent endpoint selection; weak symmetry breaking and color-singlet formation are determined by their actual actions.

Collective exchange and metric response retain the process definitions of the preceding chapters. Nuclear interaction uses the collective exchange and response combination of 3.5, with the trivial unit-shell direction e_0 and full-space constant direction b serving their respective endpoints. The metric process uses coefficient G_E obtained in 4.5 for the chosen participation range and energy scale; its connection to gravitational coupling continues to be tested through that section's mass source, spatial calibration, and actual return response.

Under the stated single-factor selection conditions, the channel is fixed as $p = 7$. Its three species-layer point-pair groups and the matter correspondences in Table 5-2 supply the objects for subsequent evaluation: Chapter 6 establishes the common energy scale, and Chapters 7 and 8 calculate couplings and masses, respectively.

§6 Energy Scales

Different processes have widely separated scales, yet must be compared using the same background and energy unit. In the $p = 7$-channel selected in Chapter 5, electroweak processes use the remaining participation of the unit shell, metric processes use the weighted full space, and nucleon processes use a collective spectral preparation. All three face the same question: which components actually participate, and how are their squared responses matched to the same background?

Adopt the matching relation $E(W) = E_0\sqrt{W/C^*}\log W$ of 2.5, retaining the common physical energy calibration E_0 and the full logarithmic-spectrum background constant C^*. Each section calculates a different W and then gives the energy scale, comparison quantities, and conditions of applicability.

This chapter retains the coherent return and background matching of Chapter 2, using $r(W) = \sqrt{W}\log W$ as the dimensionless response coefficient of the common port. For rest-frame branches, common switching histories, and inactive segments that act as the identity or are compensated, complete timing gives $\nu_W = \bar{g}_{\text{eff}} r(W)$, where $\bar{g}_{\text{eff}} = T^{-1}\int_0^T g_{\text{eff}}(t)\,dt$ includes actual capacity and active and paused durations. Vacuum squared-response matching fixes $\bar{g}_{\text{eff}} = E_0/\sqrt{C_{\text{bg}}}$, hence $\nu_W = E(W)$. Section 6.2 constructs the metric port from actual return, and 9.3.2 supplies the collective spectral port; their preparation and action conditions remain attached to this ungated interface. The resource-gated process of Chapter 4 performs a new common calibration using the complete phase of 4.5.5, with the same spectral energy scale but without applying both bare-drive normalizations together.

6.1 Electroweak Scale

Here the electroweak scale supplies the common energy scale for elementary-state mass readings. It depends on unit addresses distinguishable by the process and the weight on those addresses still participating in the common background response. The calculation must determine resolution depth, bare participation, and effective participation after exchange in sequence, then substitute into the same energy-scale formula.

6.1.1 Resolution Depth of the Electroweak Process

Let k denote the resolution depth used by this process, retaining K for the complete construction depth, with K≥k. For every odd prime p, the unit group at depth k has a Teichmüller decomposition, and the filtration of its principal-unit part is

$$\begin{aligned}(\mathbb{Z}/p^k\mathbb{Z})^\times &\simeq \mu_{p-1}\times U_1,\\ U_j &= 1+p^j\mathbb{Z}/p^k\mathbb{Z},\\ U_k &= \{1\},\\ U_j/U_{j+1} &\simeq (\mathbb{F}_p,+)\\ &(1\le j<k).\end{aligned} \tag{6.1}$$

The map $1+\mathrm{p}^j\mathrm{a}\mapsto \mathrm{a} \bmod \mathrm{p}$ gives the final isomorphism: multiplication of two principal units corresponds modulo p^{j+1} to addition of a, and its kernel is exactly

the next layer. There are therefore k−1 p-valued layer directions, with p^{k-1} principal units in total. For a Teichmüller unit ξ, use the power action $u \mapsto u^{\xi}$ on principal units, whose action on each layer satisfies

$$(1 + p^j a)^{\xi} \equiv 1 + p^j \xi a \pmod{p^{j+1}}. \tag{6.2}$$

In this power action, ξ is a finite-precision representative of the corresponding p-adic unit. Since 2j≥j+1, quadratic and higher terms of the binomial expansion vanish in this quotient layer. All layers therefore have the same $\mathbb{F}_p$-linear action, but these finite-field layers do not yet constitute a complex color carrier or automatically give a continuous unitary symmetry mixing the layers.

Adopt the following depth-matching condition: identify the three component labels of the actual color carrier one-to-one with the principal-unit layer labels used, with one component carried by each layer, all three components included, and no extra layer. This correspondence and its actual representation must still be realized as in 5.2. Given this prescription, k−1=3, so the process uses k=4. This choice restricts electroweak participation, while the complete state space may have any K≥4.

At resolution depth k, deeper addresses in the same residue class modulo p^k are combined with equal amplitude and phase. Representing this preparation in a deeper K-layer state space, define for every unit address a

$$\begin{aligned} |a; k\rangle &= p^{-(K-k)/2} \sum_{x \equiv a \pmod{p^k}} |x\rangle, \\ P_k &= \sum_{a \in (\mathbb{Z}/p^k\mathbb{Z})^\times} |a; k\rangle\langle a; k|, \\ \operatorname{rank} P_k &= \varphi(p^k) = (p-1)p^{k-1}. \end{aligned} \tag{6.3}$$

Each sum contains p^{K-k} orthogonal point basis states, and its prefactor normalizes the vector; distinct a have disjoint supports and are therefore orthogonal. The projection P_k selects one coherent direction in each address fiber, excluding other phase arrangements within the fiber. With this participation range and one unit weight per mode, bare participation is

$$\boxed{N_{\mathrm{EW}}^{(0)} = \operatorname{Tr} P_4 = \varphi(7^4) = 6 \times 7^3 = 2058.} \tag{6.4}$$

Four-layer resolution and equal-amplitude lifting form the participation scheme selected here, and 2058 is its orthogonal-mode count. Equal amplitudes refer to each address fiber internally and do not prescribe equal-amplitude matching of the weak endpoint over all labels. The complete K-layer unit shell contains $6\times7^{K-1}$ points; a process participating in all these independent directions uses that count instead. Section 5.2 fixes p, and this section's depth and lifting scheme further fix the participation range of the electroweak process.

6.1.2 Effective Participation

Weak exchange releases and replenishes some participating modes. The same process forms an occupied preparation and leaves a remainder available for continued

background response. In the independent-label preparation of 3.3, χ_3 is a real character of order two, its kernel has 3 elements, and there are 3 species-layer point-pair groups. Combining two orientation alternatives carried by distinct preparations or environments into one class gives c_3=2. The bare matching class has label count

$$N_W = |\ker \chi_3|\, m\, c_3 = 3 \times 3 \times 2 = 18. \tag{6.5}$$

Three independent registers carry these 18 labels. This section uses a role embedding in the same participation space P_4 compatible with joint energy and group charge: initial occupation projection P_B has rank 18, and release projection $P_r \le P_B$ rank 3; two replenishment projections P_2, P_3 have ranks 1 and 2, are mutually orthogonal, and are orthogonal to P_B. All 21 roles lie within P_4. This embedding, unit occupation, and 3/1/2 routing belong to the selected participation scheme.

Let $w_\ell = |R_\ell|^2 \in [0,1]$. Each role accesses its corresponding fixed spectral probe χ_1, χ_2, χ_3, using the complete success and failure outputs of the unit-capacity spectral instrument. A success record controls only an occupation-bit flip, without moving the address: the three release roles leave occupation upon spectral acceptance and remain occupied upon failure; the one and two replenishment roles load upon their respective spectral acceptance and remain empty upon failure. The other 15 roles remain occupied throughout, and all original spectral-failure records are retained. With an orthogonal address reference retained for the uniform address preparation, the occupied output is proportional to

$$\boxed{B_W = P_B - w_1 P_r + w_2 P_2 + w_3 P_3, \qquad 0 \le B_W \le P_4.} \tag{6.6}$$

The block occupation weights are respectively 1, $1 - w_1$, w_2, and w_3, so positivity and the upper bound hold blockwise, with total occupation $N_{\mathrm{occ}} = \operatorname{Tr} B_W = 18 - 3w_1 + w_2 + 2w_3$. Let Z_0 be the common normalization of the complete address preparation. The unnormalized reduced address outputs for occupation and the remainder within the participation space are B_W/Z_0 and $(P_4 - B_W)/Z_0$ respectively; complete spectral records remain in the respective branches, and reference-background addresses are also retained. Total and matching weights thus come from the same actual output.

Fix normalized matching endpoint $v \in \operatorname{Ran} P_B$, unchanged by replenishment modes. After loading, read the weak response with independent or refreshed spectral auxiliaries, and let $\eta = \langle v|P_r|v\rangle$. Equation (3.38) directly gives

$$\begin{aligned} h_W &= \langle v|B_W|v\rangle = 1 - w_1\eta, \\ N_W^{\mathrm{eff}} &= \frac{N_{\mathrm{occ}}}{1 - w_1\eta}, \qquad \alpha_W = \frac{w_3(1 - w_1\eta)}{N_{\mathrm{occ}}}. \end{aligned} \tag{6.7}$$

The effective denominator requires $h_W > 0$; the response vanishes when $h_W = 0$. When $w_1 > 0$, the effective matching denominator equals the occupation trace if and only if $P_r v = 0$. One specific coupling structure uses orientation reversal modulo 2 and places the three color-release roles in the intermediate block of a self-loop: $P_r = I_u \otimes |j_0\rangle\langle j_0| \otimes |-o_0\rangle\langle -o_0|$. Endpoint $v = \xi \otimes |o_0\rangle$ is therefore orthogonal to the release space, while ξ may be arbitrarily superposed over kernel and point-pair-group labels. Definite orientation is a sufficient realization of matching preservation here; the general criterion remains $P_r v = 0$, with orientation reversal and release location specified by the adopted actual coupling structure.

In this matching-preserving process, release reduces total occupation without weakening endpoint weight, giving

$$\boxed{N_W^{\mathrm{eff}} = N_{\mathrm{occ}} = 18 - 3w_1 + w_2 + 2w_3,} \qquad w_\ell = \left| \frac{L(\frac{1}{2}, \chi_\ell)}{\zeta(\frac{1}{2})} \right|^2 . \tag{6.8}$$

Equation (6.8) is a finite result under the complete spectral control above, independent of a first-order truncation. Changing the matching endpoint or release location requires returning to Eq. (6.7) and calculating matching weight as well. For example, equal-amplitude matching over all 18 labels gives $\eta = 1/6$, so the same denominator cannot be retained. The evaluation point $s_0 = 1/2$ remains an analytic spectral reference and does not imply a normalizable infinite positive-integer Gibbs distribution there.

Using the $\chi_\ell(3^a) = \exp(2\pi i \ell a/6)$ convention of 5.2, direct evaluation gives

$$\begin{aligned} w_1 &= 0.344759924154\ldots, \\ w_2 &= 0.047561935882\ldots, \\ w_3 &= 0.616447967850\ldots, \\ N_W^{\mathrm{eff}} &= 18.246178099119\ldots. \end{aligned} \tag{6.9}$$

These numbers can be recalculated from a finite Hurwitz ζ combination: summing over a=1,…,6 gives $L(s, \chi_\ell) = 7^{-s} \sum_a \chi_\ell(a) \zeta(s, a/7)$, evaluated at s=1/2. Conjugate directions have identical squared moduli. Unrounded responses are used in calculation, and only final results are rounded to the stated significant figures.

The unoccupied branch of the same preparation supplies the background response. Since $0 \le B_W \le P_4$ has been constructed, the remaining participation operator and total weight are

$$\Gamma_{\mathrm{vac}} = P_4 - B_W \ge 0, \qquad W_{\mathrm{vac}} = \operatorname{Tr} \Gamma_{\mathrm{vac}} = 2058 - \operatorname{Tr} B_W. \tag{6.10}$$

This remainder also supplies the actual common-background amplitude. Retaining the address reference and all spectral records, denote the remainder's purification column produced by the complete occupation procedure by u_{vac}; its squared norm equals the trace of the reduced output, namely W_{vac}/Z_0. With capacity $\Lambda_{\mathrm{col}} \ge \sqrt{Z_0}$ common to all ports used, applying a flag rotation of acceptance amplitude $\sqrt{Z_0}/\Lambda_{\mathrm{col}}$ in the remainder branch gives

$$\|u_{\mathrm{vac}}\|^2 = \frac{W_{\mathrm{vac}}}{Z_0}, \qquad v_{\mathrm{vac}} = \frac{\sqrt{Z_0}}{\Lambda_{\mathrm{col}}} u_{\mathrm{vac}}, \qquad \|v_{\mathrm{vac}}\|^2 = \frac{W_{\mathrm{vac}}}{\Lambda_{\mathrm{col}}^2}. \tag{6.11}$$

All other outputs and rejection flags are retained. Actual use of this amplitude requires coherent control of the complete preparation and its inverse; the reduced mixed state after a partial trace alone cannot produce the common input. Thus vacuum participation and the common background column required by 2.5.3 arise from the same occupation process; see Supplementary Proof S6.3.

Under matching preservation, $\operatorname{Tr} B_W = N_W^{\mathrm{eff}}$ is realized by the above output, and weak coupling and the background remainder satisfy $\alpha_W(2058 - W_{\mathrm{vac}}) = w_3$. Total occupation may exceed 18 because three actual replenishment modes are added; no mode in the 18-dimensional subspace is assigned occupation greater than one. The remainder after subtraction is

$$W_{\mathrm{vac}} = 2058 - 18.246178099119\ldots = 2039.753821900881\ldots. \tag{6.12}$$

A noninteger value represents partial occupation of modes. Occupation-loading success probability is N_{occ}/Z_0, so conditional weak response still differs from the all-trial event rate: when the actual signal effect is $\kappa_W^2 w_3 P_v$, matching preservation gives event probability $\kappa_W^2 w_3/Z_0$ per complete attempt. A fixed attempt duration must include address preparation, loading failure, spectral auxiliaries, readout, and reset. If coherent paths or subsequent exchanges alter occupation and matching, the same complete output must be recalculated.

6.1.3 Evaluation of the Electroweak Scale

Sum the squared bidirectional logarithmic intervals of the complete background using Eq. (2.70), then apply squared-response matching from Eq. (2.72) to obtain the common energy scale:

$$\begin{gathered} C^* = 2\sum_{n=1}^{\infty} \log^2\!\left(1+\frac{1}{n}\right) = 1.954378366537873\ldots, \\ E(W) = E_0\sqrt{\frac{W}{C^*}}\log W \\ (W>1). \end{gathered} \tag{6.13}$$

This section retains common response coefficient $\log W$ and the full logarithmic-spectrum background reference; if finite background C_J^* is used, the denominator must be changed accordingly, and the two normalizations cannot be mixed within one numerical comparison. Substitute the W_{vac} obtained here and denote the resulting candidate electroweak scale by v_{EW}. This gives

$$\boxed{\frac{v_{\mathrm{EW}}}{E_0} = \sqrt{\frac{W_{\mathrm{vac}}}{C^*}}\log W_{\mathrm{vac}} = 246.191373933421\ldots.} \tag{6.14}$$

Without occupation subtraction, $\mathrm{E}(2058)/\mathrm{E}_0$=247.579034659055…; subtracting only the Born-level label count 18 gives $\mathrm{E}(2040)/\mathrm{E}_0$=246.210128943076…; including the finite correction above gives electroweak scale $E(W_{\mathrm{vac}})/E_0 = 246.191373933421\ldots$. All three steps use the same background squared-response constant and energy calibration, isolating the contributions of participation subtraction and finite correction.

With physical calibration $E_0 = 1$ GeV, one obtains $v_{\mathrm{EW}} \simeq 246.1914$ GeV. This calibration connects dimensionless response to actual time and energy. Chapter 2 gives the common-background relation $f_{\mathrm{act}} g p_0/\Lambda = v_{\mathrm{EW}}$, with complete timing as in 2.4.6; below we abbreviate $v = v_{\mathrm{EW}}$.

For an independent numerical comparison, the PDG electroweak review gives Fermi constant $G_F = 1.1663785(6) \times 10^{-5}\mathrm{GeV}^{-2}$. Its conventional Fermi-scale definition $v_F = (\sqrt{2}G_F)^{-1/2}$ gives $v_F \approx 246.21967$GeV (see §10.3.2) [15]. With E_0=1 GeV and the above prescription, the difference is

$$v_{\mathrm{EW}} - v_F \simeq -0.02830\ \mathrm{GeV}, \qquad \frac{v_{\mathrm{EW}} - v_F}{v_F} \simeq -1.1493 \times 10^{-4} = -0.01149\%. \tag{6.15}$$

The quantity G_F defines only the external comparison; E_0 and correction coefficients are specified under the preceding conditions. The bare count 18 gives a relative deviation of approximately -0.00388%; the finite correction lowers the result and does not improve this comparison. No theoretical error bound has yet been established for the participation identification and correction prescription, so numerical proximity does not mean attainment of the experimental precision of G_F or completion of the weak-interaction correspondence.

Sensitivity of the energy scale to participation follows from its derivative. Fix E_0, C*, and bare participation, and increase estimated occupation by a small amount δN_W. Then

$$\frac{\delta v_{\mathrm{EW}}}{v_{\mathrm{EW}}} = -\left(\frac{1}{2} + \frac{1}{\log W_{\mathrm{vac}}}\right)\frac{\delta N_W}{W_{\mathrm{vac}}} + O((\delta N_W/W_{\mathrm{vac}})^2). \tag{6.16}$$

The current values give $\delta v_{\mathrm{EW}} \simeq -0.07619\,\delta N_W$ GeV. Changes in participation can thus be converted directly into changes in energy scale; physical estimation error and numerical convergence accuracy are calculated separately. Subsequent masses and couplings share this scale.

6.2 Planck Scale

Participation in metric action covers the entire state space. Unlike the preceding section, which resolves one set of unit-shell modes, this includes all valuation shells and the zero element, weighted by actual responses of the metric process. The common energy-scale formula established in 4.5 calculates the scale associated with this full-space participation; identifying it as the Planck scale additionally retains the physical correspondences for mass sources, spatial readings, and gravitational coupling.

6.2.1 Full-Space Participation and Construction Depth

First fix odd prime p and finite depth K. If every orthogonal mode carries one unit of weight in the same response units, both participation projection and weight operator are the identity. Using this chapter's intrinsic metric constant C* for the full logarithmic spectrum gives bare participation and energy scale

$$W_{\mathrm{met}}^{(0)} = \operatorname{Tr} I_{\mathcal{H}} = p^K, \qquad M_{\mathrm{met}}^{(0)}(p,K) = E_0\sqrt{\frac{p^K}{C^*}}\,K\log p. \tag{6.17}$$

Here M_{met} denotes the metric scale in energy units; it may be read as mass in natural units $\hbar$=c=1. Although the zero element is absent from the nonzero-address open chain of 4.5, it retains one unit of weight in this full-space participation range. The support of the gradient background and the range of participating modes are specified separately; the open-chain edge count cannot replace p^K.

The augmentation ideal of the affine group algebra supplies a configuration space whose dimension must be separately related to address depth. Let $G_{\mathrm{aff}} = \mathrm{Aff}(\mathbb{F}_p)$, with p(p−1) elements x↦ax+b, a≠0. The augmentation map on the group algebra sums the coefficients of a formal linear combination; its kernel I_{aug} satisfies

$$
\begin{aligned}
A &= \mathbb{F}_p[G_{\mathrm{aff}}],\\
\varepsilon\left(\sum_g a_g g\right) &= \sum_g a_g,\\
I_{\mathrm{aug}} &= \ker \varepsilon,\\
r &:= \dim_{\mathbb{F}_p} I_{\mathrm{aug}} = p(p-1)-1.
\end{aligned}
\tag{6.18}
$$

The augmentation map sends every group element to 1, and rank–nullity gives r. The vectors $g-1$ after excluding the identity element form a basis, giving p^r configurations. The comparison using a common source below supplies their actual formation and complete encoding. This paper takes this configuration code as the participating object in the metric process and carries it in the smallest p-adic register capable of distinguishing every configuration: capacity $p^K \geq p^r$ gives minimum $K = r$, which is 41 at $p = 7$. This depth belongs to the configuration code used and does not bound the maximum depth of the entire background.

These configurations can be established explicitly by comparing outgoing and return transport of the same source. Take the translation and scaling generators of the original affine transport, with one p-valued program coordinate in the source for each group element. Comparing two transports of the same source reads coordinate differences; all edge differences remove only the common constant coordinate. The actually distinguishable settings and their dual are therefore

$$
\boxed{Q = \mathbb{F}_p^{G_{\mathrm{aff}}}/\mathbb{F}_p\mathbf{j}, \qquad Q^* \simeq I_{\mathrm{aug}}, \qquad N_C = |Q| = p^{p(p-1)-1}.}
\tag{6.19}
$$

Under the specified coherent uniform source preparation, write comparison outcomes into fresh probe records that no longer couple back, forming edge-difference modes round by round. Connectivity of the graph generated by the two transports ensures that after finitely many rounds every direction in the Fourier basis labeled by I_{aug} has nonzero occupation probability; continued updating approaches uniformity. Complete encoding uses the distinguishable configuration outputs below. Proofs concerning the multicoordinate source, probe preparation, and destination of records are given in Supplementary Proofs S6.4.1–S6.4.2.

Configuration counts must become event shares through actual matching. A complete edge-difference query encodes different quotient classes as orthogonal outputs; the receiver decodes inversely according to its own setting and detects the ready output, giving

$$
\begin{aligned}
P_{\mathrm{eq}} &= \sum_{q\in Q} |q,q\rangle\langle q,q|, \qquad p_{\mathrm{match}} = \mathrm{Tr}(\rho_{SR}P_{\mathrm{eq}}),\\
\rho_{SR} &= \rho_S \otimes \rho_R, \quad \langle q|\rho_S|q\rangle = N_C^{-1} \quad \Longrightarrow \quad p_{\mathrm{match}} = N_C^{-1}.
\end{aligned}
\tag{6.20}
$$

Independent uniform preparation gives a one-shot configuration-matching share; correlated inputs use the full matching trace in the first line. The distinction between a finite echo and complete encoding, and the corresponding convergence, are given in Supplementary Proof S6.4.3.

6.2.2 Layer Structure and Participation-Weight Corrections

Full-space participation uses the projective cover P_0 and layer weights constructed in 3.5.4. Its radical filtration has p layers: the first and last are trivial characters,

and each nontrivial character occurs once in between. At p=7, the labels used in this chapter give S_0, S_1, S_2, S_3, S_4, S_5, S_0 from top to bottom. Here P_0 has dimension 7; it and the preceding 41-dimensional comparison space I_{aug} serve layer counting and configuration comparison, respectively. The node prescription below connects layer labels to complex spectral responses.

Now assign weights by the node prescription of 3.5: use the corresponding positive factor when a CRT coordinate is zero and one when nonzero, then multiply over every nontrivial layer. For p−1=2×3, specify node factors $t_2=1/\sqrt{2}$ and $t_3=3/|\zeta(1/2)|$. The five layer weights are

$$(\mathcal{T}_1,\mathcal{T}_2,\mathcal{T}_3,\mathcal{T}_4,\mathcal{T}_5)=\left(1,\frac{1}{\sqrt{2}},\frac{3}{|\zeta(1/2)|},\frac{1}{\sqrt{2}},1\right). \tag{6.21}$$

Labels 2 and 4 have zero mod-2 coordinate, while label 3 has zero mod-3 coordinate, so t_2 occurs twice and t_3 once. Their product is

$$\boxed{\mathcal{L}_7=\prod_{j=1}^{5}\mathcal{T}_j=\frac{3}{2|\zeta(1/2)|}=1.027147854134904784\ldots.} \tag{6.22}$$

The layer structure specifies where these nodes occur. For their product to become an actual participation weight, one must also determine input, return, and reference readout within the same process. Here we use return from a common source: the signal retains one coherent endpoint of the binary node, while the reference retains one point of the ternary orbit, with common normalized preparation, actual input coefficient, and spectral-background control.

Specifically, write $Z=|\zeta(1/2)|$, with input-address dimension d. Common isometric preparation C and retained projections Q_s,Q_r relative to an internal group reference satisfy $C^\dagger Q_s C=I_d/2$ and $C^\dagger Q_r C=I_d/3$. Simultaneous reflection makes the actual inputs $F_s=\gamma Q_s C$ and $F_r=\gamma Q_r C$. Denote the normalized real spectral part protected by zero-element reflection by $\Delta_0>0$, with $\Delta_0 z=z$. The unnormalized spectrum gives signal background $B_s=c_B Z\Delta_0$, and the normalized spectrum gives reference background $B_r=c_B\Delta_0$. From the same input,

$$\begin{aligned}
\mathcal{S}_s&=F_s^\dagger B_s^{-1}F_s=\frac{|\gamma|^2}{2c_B Z}I_d,\\
\mathcal{S}_r&=F_r^\dagger B_r^{-1}F_r=c_{\mathrm{ref}}I_d,\qquad c_{\mathrm{ref}}=\frac{|\gamma|^2}{3c_B},\\
\frac{\mathcal{S}_s}{c_{\mathrm{ref}}}&=\frac{3}{2Z}I_d=\mathcal{L}_7 I_d.
\end{aligned} \tag{6.23}$$

Thus amplitudes at the two endpoints enter the same outgoing and return trip, the ternary factor comes from the ratio of a complete orbit to a single-point reference, and the spectral factor enters the inverse background response. Their respective locations in the action are fixed. Common input and background control cancel in the ratio, and the ternary count is not multiplied in again as extra source strength. Supplementary Proof S6.2 gives the complete operators and actual return reading under finite driving.

Using the same single-point reference response as the common participation unit and deploying the same procedure at all $d=7^K$ addresses gives

$$W_{\mathrm{met}} = \operatorname{Tr}\Gamma_{\mathrm{met}} = 7^K \mathcal{L}_7, \qquad \Gamma_{\mathrm{met}} = \mathcal{L}_7 I_{\mathcal{H}}, \qquad \frac{\Gamma_{\mathrm{met}}}{W_{\mathrm{met}}} = \frac{I_{\mathcal{H}}}{7^K}. \tag{6.24}$$

Here Γ_{met} is an unnormalized weight operator in fixed response units and need not be bounded by the identity. The factor $\mathcal{L}_7 > 1$ neither increases state-space dimension nor gives a one-event probability exceeding one. Normalization still yields a uniform state, leaving the uniform gradient mean of 4.5 unchanged; the energy-scale change comes from each address's actual return response relative to the same reference. Choosing P_0, identifying layer nodes with the common-source procedure above, full-address participation, and the reference unit remain adopted physical conditions. Layers share one reference and common controls and cannot be normalized separately layer by layer; if an independently fixed unit c_{unit} is used instead, the participation operator must also retain factor $c_{\mathrm{ref}}/c_{\mathrm{unit}}$.

6.2.3 Evaluation of the Metric Energy Scale

Return response first gives participation weight, while persistent mass comes from the actual return amplitude entering the common carrier. Write the baseline signal-end action of the preceding subsection as $H = \left(\begin{smallmatrix} 0 & F^\dagger \\ F & B \end{smallmatrix}\right)$, where $F^\dagger F = J I_d$ and $BF = bF$, giving $W_{\mathrm{met}} = dJ/(bc_{\mathrm{ref}})$. Before multiplying in logarithmic driving, let $s = \sqrt{b^2 + 4J}$. After the actual half-return π/s, compensation of the common phase gives amplitude $2F/s$ at the virtual endpoint.

Prepare all inputs with a normalized internal address reference, retain outer-port residuals, and apply a flag rotation of common capacity. The complete success column and its norm are

$$v_{\mathrm{met}} = \frac{\operatorname{vec} F}{\Lambda_{\mathrm{col}}\sqrt{bc_{\mathrm{ref}}}}, \qquad \|v_{\mathrm{met}}\|^2 = \frac{W_{\mathrm{met}}}{\Lambda_{\mathrm{col}}^2}, \qquad \operatorname{vec} F = \sum_a F|a\rangle \otimes |\bar{a}\rangle. \tag{6.25}$$

Here $|\bar{a}\rangle$ is the conjugate reference for internal addresses. Common capacity must satisfy $\Lambda_{\mathrm{col}} \geq s\sqrt{d}/(2\sqrt{bc_{\mathrm{ref}}})$ and also cover the vacuum endpoint of 6.1.2. The simultaneous-reflection exchange of 2.5.3 connects this complete column to a common input; after applying drive $\log W_{\mathrm{met}}$, the actual rest-frame half-gap is $(g_{\mathrm{col}}/\Lambda_{\mathrm{col}})\sqrt{W_{\mathrm{met}}}\log W_{\mathrm{met}}$. All addresses are coherently connected within this new persistent carrier, so the square-root enhancement does not rely on directly tracing degenerate return frequencies. For the common-source configurations of 6.2.1, if each gives response κ under the same reference, the same complete-column construction gives $W_C = N_C\kappa$. Configuration count and response unit are determined by their respective preparation and return; see Supplementary Proof S6.5.

The vacuum remainder column and metric return column use the same capacity, drive history, and background action, and one squared-response matching fixes effective strength. Supplementary Proof S6.3 proves this connection and includes the total preparation and control duration. The actual force process after gating is recalibrated once as in 4.5.5; here we retain the common spectral scale. Taking full background $C_{\mathrm{bg}} = C^*$, obtain the metric energy scale from Eq. (2.78), then take its inverse square according to Eq. (4.30):

$$\boxed{\begin{aligned} M_{\mathrm{met}}(K) &= E_0\sqrt{\frac{7^K\mathcal{L}_7}{C^*}}\,(K\log 7+\log\mathcal{L}_7)\,, \\ G_E(K) &= \frac{C^*}{E_0^2\,7^K\mathcal{L}_7\,(K\log 7+\log\mathcal{L}_7)^2} = \frac{1}{M_{\mathrm{met}}(K)^2}. \end{aligned}} \tag{6.26}$$

The layer correction enters both the square root and logarithm. Relative to the bare value at the same K, the exact ratio of energy scales is

$$\frac{M_{\mathrm{met}}(K)}{M^{(0)}_{\mathrm{met}}(7,K)} = \sqrt{\mathcal{L}_7}\left(1+\frac{\log\mathcal{L}_7}{K\log 7}\right). \tag{6.27}$$

Now adopt the minimum depth K=41 for the above configurations and calculate with unrounded C* and $\mathcal{L}_7$. The bare mode count and effective weight are

$$7^{41} = 4.456764032636\ldots\times 10^{34}, \qquad W_{\mathrm{met}} = 7^{41}\mathcal{L}_7 = 4.577755612508\ldots\times 10^{34}. \tag{6.28}$$

These give, respectively,

$$\begin{aligned} \frac{M^{(0)}_{\mathrm{met}}(7,41)}{E_0} &= 1.204792171318\ldots\times 10^{19}, \\ \frac{M_{\mathrm{met}}(41)}{E_0} &= 1.221446368809\ldots\times 10^{19}, \\ E_0^2 G_E(41) &= 6.702721805350\ldots\times 10^{-39}. \end{aligned} \tag{6.29}$$

With the same physical calibration E_0=1 GeV as in the preceding section, the corrected scale is approximately 1.2214464×10^{19} GeV, and G_E approximately 6.70272×10^{-39} GeV^{-2}. The layer correction increases the energy scale by about 1.3823%. Replacing the full background with the nonzero open-chain background at the same depth, the bound of 4.5 gives $0 < C^* - C^*_{7^K-1} < 2/(7^K-2)$, below 4.49×10^{-35} at K=41. This truncation difference is far smaller than the numerical deviation below.

The comparison uses the nonreduced Planck scale, defined in $\hbar$=c=1 units by $M_P = G_N^{-1/2}$; the reduced scale is $M_P/\sqrt{8\pi}$, and the two cannot be interchanged. CODATA 2022 gives Planck-mass energy equivalent $1.220890(14)\times10^{19}$ GeV, using gravitational constant $G_N = 6.67430(15)\times10^{-11}\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$ (see page 1 of the constants table) [18]. Calculating the comparison center with this G_N and exact unit conversions gives, under the conditions of this section,

$$\begin{aligned} \frac{M^{(0)}_{\mathrm{met}} - M_P}{M_P} &\simeq -1.3185\%, \\ \frac{M_{\mathrm{met}} - M_P}{M_P} &\simeq +0.04556\%, \\ \frac{G_E - G_N}{G_N} &\simeq -0.09106\% \\ (\hbar = c = 1). \end{aligned} \tag{6.30}$$

The final quantity first converts G_N to the same natural units; the equivalent SI reading is $G_E \approx 6.66822\times10^{-11}\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$. Conversion uses $G_{\mathrm{SI}} = \hbar c^5/E_J^2$,

where E_J is the metric energy scale in joules. Experimental constants serve as external comparisons; $\mathcal{L}_7$ is determined by the reference procedure above, and E_0 retains the physical calibration. Since M_P is defined by G_N, comparing the energy scale and gravitational constant gives two forms of the same test, not independent observational constraints.

The layer correction improves the central-value comparison, but the deviation still exceeds the experimental relative uncertainty of G_N, and no combined theoretical error bound has yet been established for layer weights, depth, and the common energy scale. Section 4.5.5 derives the leading field strength of the active process from positive-source preparation, actual return, and complete timing, and states the calibration condition equating it with the inverse spectral square here. When participation changes, both the square root and logarithm must be updated.

Relative to the electroweak scale, the common E_0 and C^* cancel, giving a hierarchy relation independent of the absolute unit:

$$\frac{M_{\mathrm{met}}(41)}{v_{\mathrm{EW}}} = \sqrt{\frac{7^{41}\mathcal{L}_7}{W_{\mathrm{vac}}}}\frac{\log(7^{41}\mathcal{L}_7)}{\log W_{\mathrm{vac}}} = 4.961369479745\ldots \times 10^{16}. \tag{6.31}$$

This hierarchy mainly comes from the enormous difference between the two participation ranges: electroweak evaluation uses effective weights on four-layer unit addresses, whereas metric evaluation in this example uses full-space weights at 41 layers. The factor $\mathcal{L}_7$ supplies only a finite correction. Holding the layer factor fixed while increasing depth by one gives exactly

$$\frac{M_{\mathrm{met}}(K+1)}{M_{\mathrm{met}}(K)} = \sqrt{7}\left(1 + \frac{\log 7}{K\log 7 + \log \mathcal{L}_7}\right), \qquad \frac{M_{\mathrm{met}}(42)}{M_{\mathrm{met}}(41)} = 2.710260172825\ldots. \tag{6.32}$$

A one-layer depth change is much larger than the layer-weight correction here, showing the dominant role of participation range in the energy-scale hierarchy. The next scale concerns the full-space constant-function pattern b, whose collective participation is determined by spectral preparation.

6.3 Nucleon Scale

The nucleon scale is established using the collective-pattern mass reading of 2.5. The spectral weight of collective participation determines its energy scale, while internal color structure restricts allowed representations. We first find the number of fundamental color representations in the minimal color singlet, then adopt the corresponding spectral preparation, and finally substitute the sum of positive weights into the common energy-scale formula.

6.3.1 Minimal Color Singlet

Suppose the actual color carrier has been realized as $V_c = \mathbb{C}^{d_c}$ $(d_c \geq 2)$ according to 5.2, with color acting in the fundamental representation of $SU(d_c)$. Consider the tensor product $V_c^{\otimes n}$ of n such representations and require the color part of the collective state to be invariant under every color action. Here n counts fundamental-color-representation tensor factors; their interpretation as a particular internal projection

of the collective state still requires actual formation and readout. Tensor notation alone does not establish a bound state of several independent particles.

The counting condition for a color singlet follows directly. Central element $\exp(2\pi i/d_c)I$ acts on n tensor factors as $\exp(2\pi in/d_c)$, so a nonzero invariant vector requires d_c to divide n. Conversely, the fully antisymmetric tensor in d_c fundamental representations is multiplied by det g=1 under g; taking products of these tensors in groups gives an invariant vector in every $n = d_c r$. Hence

$$(V_c^{\otimes n})^{SU(d_c)} \neq \{0\} \quad \Longleftrightarrow \quad d_c \mid n, \qquad n \geq 1. \tag{6.33}$$

At the minimum $n = d_c$, this invariant subspace is exactly one-dimensional. Diagonal color transformations require every color label to occur once, so an invariant tensor can contain only permutations of all labels. Applying an SU(2) mixing of any two colors further requires the coefficient to change sign when those labels are exchanged. All coefficients are thus fixed by one coefficient, yielding the fully antisymmetric tensor. With unit normalization,

$$|\varepsilon_{d_c}\rangle = \frac{1}{\sqrt{d_c!}} \sum_{\sigma \in S_{d_c}} \operatorname{sgn}(\sigma) |\sigma(1), \ldots, \sigma(d_c)\rangle, \qquad g^{\otimes d_c} |\varepsilon_{d_c}\rangle = |\varepsilon_{d_c}\rangle. \tag{6.34}$$

In the candidate matter correspondence at p=7, $d_c = 3$, so the minimal nonempty color singlet has n=3 and color part ϵ^{abc}. This does not require three distinct species labels; color indices are carried separately from species, spatial, and spin indices. Identifying specific matter additionally requires specifying the remaining parts and checking statistics and maintenance conditions for the complete state.

This minimality applies only to tensor products of the same fundamental color representation. If conjugate representations are included, $V_c \otimes \overline{V}_c$ already contains a singlet corresponding to the identity matrix, a different internal representation class. For the three-dimensional fundamental representation alone, allowed n are 3, 6, 9, …; n=4 and 5 fail the necessary color-singlet condition above. If higher excitations use those integers as labels, the degrees of freedom being counted must be specified separately.

6.3.2 Nucleon Spectral Preparation

Color structure fixes the minimal internal representation but has not yet fixed collective participation weight. Following 2.5, use the logarithmic spectrum indexed by positive integers $H_{\log}|j\rangle = \log j |j\rangle$ as the evaluation reference, fixing the energy zero at j=1. Denote spectral cutoff by L, separately from address depth K. The corresponding positive weights, normalized shares, and infinite sum are

$$Z_L(\beta) = \sum_{j=1}^{L} j^{-\beta}, \qquad w_{j,L}(\beta) = \frac{j^{-\beta}}{Z_L(\beta)}, \qquad Z_L(\beta) \longrightarrow \zeta(\beta) \quad (\beta > 1). \tag{6.35}$$

The complete logarithmic spectrum indexed by positive integers and its ζ partition function for β>1 are standard structures in BC quantum statistics [19]. Here β is the dimensionless parameter of this spectral preparation. The shell generator H_K has its own shell energies and degeneracies, so the two partition sums cannot

be equated merely by using the same temperature-parameter name. The full-space constant-function pattern b satisfies $T_j b = b$ for every multiplicative pullback; consequently the unnormalized response $\sum_j j^{-\beta} T_j$ gives $Z_L(\beta)$ on b. Only upon adopting the participation correspondence of 2.5 do we take $W_B = Z_L(\beta)$ and, for the infinite spectral reference, $W_B = \zeta(\beta)$. The trivial unit-shell direction excludes multiples of p and has a response different from this full integer sum.

Averaging over these normalized spectral shares and differentiating Z_L directly gives the mean and variance of logarithmic energy:

$$\langle H_{\log} \rangle_{\beta,L} = -\partial_\beta \log Z_L, \qquad V_L(\beta) := \mathrm{Var}_{\beta,L}(H_{\log}) = \partial_\beta^2 \log Z_L. \tag{6.36}$$

The first expression sums weighted log j, and another derivative gives $\langle H_{\log}^2 \rangle - \langle H_{\log} \rangle^2$. For β>1, logarithmic moments of every order converge locally uniformly, so the infinite sum can be differentiated in the same way. If $E_0 H_{\log}$ is taken as physical energy, its variance has an additional factor E_0^2; below we use dimensionless V. The Laurent expansion of ζ near β=1 gives

$$V(\beta) = \partial_\beta^2 \log \zeta(\beta) = \frac{1}{(\beta-1)^2} - 2\gamma_1 - \gamma^2 + O(\beta - 1) \quad (\beta \to 1^+). \tag{6.37}$$

Here γ is Euler's constant and γ_1 is defined by $\zeta(1+\mathrm{u}) = 1/\mathrm{u} + \gamma - \gamma_1 \mathrm{u} + \mathrm{O}(\mathrm{u}^2)$. Expanding log ζ as $-\log \mathrm{u} + \gamma\mathrm{u} + (-\gamma_1 - \gamma^2/2)\mathrm{u}^2 + \mathrm{O}(\mathrm{u}^3)$ and differentiating twice gives the formula. For fixed finite L, Z_L is finite at β=1 and has no such pole.

The collective spectral preparation adopts this parameter condition: for collective evaluation with n fundamental-color-representation factors, choose β by setting the leading term of the infinite-spectrum variance equal to n. This connects the number of fundamental color factors to spectral preparation and is an evaluation rule independent of the color-singlet condition. On the β>1 side, it gives

$$\boxed{\frac{1}{(\beta-1)^2} = n \quad \Longrightarrow \quad \beta(n) = 1 + \frac{1}{\sqrt{n}}, \qquad \beta_B = 1 + \frac{1}{\sqrt{3}}.} \tag{6.38}$$

This equation follows neither from Gibbs shares nor from the color-singlet requirement alone, and it is not equivalent to the exact condition V(β)=n. Here $\beta_B - 1 \approx 0.5774$, outside a limit arbitrarily close to the pole. We therefore use it as the prescribed leading-term rule and separately examine its difference from the exact variance.

6.3.3 Nucleon Scale and Approximation Range

For the minimal nonempty color singlet made solely of fundamental color representations, n=3, with the above β prescription and complete spectral participation, the unrounded parameter and total weight are

$$\beta_B = 1.577350269189\ldots, \qquad W_B = \zeta(\beta_B) = 2.349636114511\ldots. \tag{6.39}$$

Continue using the same intrinsic metric constant C* and common energy calibration E_0, substitute into the collective energy-scale formula of 2.5, and denote the resulting candidate nucleon scale by M_B:

$$\boxed{\frac{M_B}{E_0} = \sqrt{\frac{\zeta(1+1/\sqrt{3})}{C^*}}\log\zeta(1+1/\sqrt{3}) = 0.936669312626\ldots.} \tag{6.40}$$

With E_0=1 GeV, this gives $M_B \approx 936.6693$MeV, the overall scale of collective participation. Sections 9.1 and 9.3 give collective-pattern formation, complete return, and calibration using a common source, making the rest-frame phase of the specified carrier satisfy $-\tau_B M_B$; τ_B counts the actual total elapsed duration. This conditional realization connects the spectral evaluation here to mass while retaining its port, protection, and calibration assumptions.

Using proton mass for this comparison, CODATA 2022 gives mass energy equivalent $m_p c^2 = 938.27208943(29)$MeV (see page 3 of the constants table) [18]. Under the above conditions, the difference is

$$M_B - m_p \simeq -1.60278\ \text{MeV}, \qquad \frac{M_B - m_p}{m_p} \simeq -0.17082\% \qquad (c=1). \tag{6.41}$$

This number is determined by spectral-preparation parameter β_B, participation weight, background, and physical calibration E_0, with proton mass serving as an independent comparison. Splitting between nucleon species and other internal readings depend on the corresponding matter structures. Numerical precision and the accuracy of the participation and β prescriptions must be evaluated separately.

At the same β_B, the exact variance is V(β_B)=2.860214657918…, approximately 4.6595% below 3. This confirms that the prescription used is the leading-term one; the deviation cannot directly serve as a mass error bar. Comparisons with alternative prescriptions and finite spectral cutoffs are given in Supplementary Proof S11.

Within the allowed family containing only fundamental color representations and retaining the above β prescription and common calibration, the energy scale increases with factor count n, so n=3 gives the lowest scale. Subsequent stability is determined by the actual return and exit processes of Chapter 9.

Electroweak, metric, and nucleon scales thus share the same background-matching function. Their differences come respectively from remaining unit-shell weight, full-space participation with node weights, and collective spectral preparation, rather than separate energy units for each class. Their participation ranges, spectral preparations, and physical correspondences determine applicability; subsequent couplings and masses are calculated using these scales.

§7 Force Coupling Constants

Exchange between stable states depends both on their respective spectral responses and on the share of the input preparation that can enter the common endpoint. When the background and environment participate, intermediate processes also modify the matching share at the endpoint. The coupling strength must therefore be read from the relationship among the spectral response, the actual preparation, and the complete output.

Chapter 3 established the operator relations among these quantities. This chapter adopts the $p = 7$ channel and physical identifications of Chapter 5 and evaluates them using the common energy scale of Chapter 6. The spectral response is denoted by R_ℓ, and the response ratio obtained through endpoint matching by α_i. Converting the latter into an event probability or a record rate per unit time still requires the normalization of the actual instrument and the elapsed duration of the cycle.

7.1 Vertex Spectral Response

The same stable state can participate in exchange through different endpoints. Its spectral response describes the propagation coefficient in the selected direction; endpoint selection and the preparation share determine how much of it enters the actual process. We first determine this common spectral factor R_ℓ. In this chapter, the spectral parameter s_* retained in Chapter 3 is set to $s_0 = 1/2$ following Supplementary Proof S5.1.

7.1.1 Spectral Weights

When a spectral weight enters a coherent amplitude, the square root of the probability weight is used. Taking the logarithmic spectrum $H_{\log}|n\rangle = \log n|n\rangle$ as a reference, the normalized thermal preparation at a given β, within a finite cutoff L, and its standard purification are

$$\rho_{\beta,L} = \frac{1}{Z_L(\beta)} \sum_{n=1}^{L} n^{-\beta} |n\rangle\langle n|, \qquad |\Omega_{\beta,L}\rangle = \frac{1}{\sqrt{Z_L(\beta)}} \sum_{n=1}^{L} n^{-\beta/2} |n\rangle \otimes |n\rangle. \tag{7.1}$$

Taking the partial trace over the second factor recovers $\rho_{\beta,L}$. Thus the probability weight $n^{-\beta}$ corresponds to the unnormalized amplitude modulus $n^{-\beta/2}$. This relation fixes the modulus; the relative phases also depend on the auxiliary preparation and actual paths. Here β belongs to the logarithmic-spectrum preparation and is defined separately from the thermal parameter of the shell generator H_K.

This chapter adopts the exponent $s_0 = 1/2$ corresponding to the critical parameter $\beta = 1$. This vertex spectral prescription is used together with the relative phases, reference, and actual exchange. The preparation at finite L remains normalizable, whereas the partition sum over the infinite logarithmic spectrum diverges at this parameter.

Accordingly, this chapter retains the following vertex spectral prescription and denotes its squared response by w_ℓ:

$$\boxed{s_0 = \frac{1}{2}, \qquad R_\ell = \frac{L(s_0, \chi_\ell)}{\zeta(s_0)}, \qquad w_\ell = |R_\ell|^2.} \tag{7.2}$$

Here $\zeta(s_0)$ is the common analytic spectral reference: $\zeta(1/2) < 0$, which differs from the partition sum of a finite positive thermal preparation. Equation (7.2) gives the directional response to be realized. The actual success block must still be constructed using a complete spectral instrument, with all other outputs retained.

7.1.2 Directional Spectral Values

Fix the generator 3, label the six characters by $\chi_\ell(3^a) = \exp(2\pi i \ell a/6)$, and extend them by zero on multiples of 7. The generator gives the residue classes (1,3,2,6,4,5) in order. For a nontrivial character, the sum over a period vanishes, so its Dirichlet series converges for real s>0, absolutely for s>1 and conditionally for 0<s≤1. For the principal character and the background ζ, analytic continuation is used at s_0. All values can be calculated using the finite Hurwitz formula (DLMF 25.15.3) [20]:

$$L(s, \chi_\ell) = 7^{-s} \sum_{a=1}^{6} \chi_\ell(a) \zeta\left(s, \frac{a}{7}\right), \qquad \zeta(1/2) = -1.4603545088095868\ldots. \tag{7.3}$$

Table 7-1 Responses at the same spectral point; ℓ=0 is the trivial direction of the unit shell, and ℓ=1,…,5 are the nontrivial directions.

ℓ	ord χ_ℓ	$L(1/2, \chi_\ell)$	w_ℓ
0	1	−0.9083923865	0.3869281968
1	6	0.7139433438 + 0.4749021828i	0.3447599242
2	3	0.3100893626 − 0.0726419314i	0.0475619359
3	2	1.1465856669	0.6164479678
4	3	0.3100893626 + 0.0726419314i	0.0475619359
5	6	0.7139433438 − 0.4749021828i	0.3447599242

Conjugacy at a real spectral point gives $L_5 = \bar{L}_1$ and $L_4 = \bar{L}_2$, so the two pairs of conjugate directions have equal w within each pair. The principal character removes multiples of 7. The Euler-factor identity in the region of absolute convergence, followed by analytic continuation, gives

$$L(s, \chi_0) = (1 - 7^{-s})\zeta(s), \qquad R_0 = 1 - \frac{1}{\sqrt{7}} = 0.6220355269907728\ldots. \tag{7.4}$$

R_0 is the resonance ratio, and $w_0 = R_0^2$ is its square; they are not interchangeable in evaluations. The ℓ=0 entry in the table is the value of the full spectral response on the unit-shell average state e_0. In Section 3.2, T_{el} is projected onto the nontrivial

direction space on both sides, so $T_{\mathrm{el}} e_0 = 0$. The constant-function pattern b on the full space, the unit-shell average state e_0, and the terminal zero element e_K are three different objects.

The difference between the analytic spectral ratio and direct truncated normalization can be checked. Let $H_L^{(1/2)} = \sum_{n \le L} n^{-1/2}$ and use the same finite positive-weight sum as the denominator. The numerator for a nontrivial character converges, whereas the principal-character numerator removes multiples of 7. Hence

$$\frac{\sum_{n \le L} \chi_\ell(n) n^{-1/2}}{H_L^{(1/2)}} \longrightarrow \begin{cases} 0, & \ell \neq 0, \\ 6/7, & \ell = 0. \end{cases} \tag{7.5}$$

The principal-character limit follows from $H_L^{(1/2)} \sim 2\sqrt{L}$. Equation (7.5) shows that direct normalization of truncated positive weights yields a different set of responses; Table 7-1 uses the analytic reference in Eq. (7.2). Subsequent evaluations of energy scales, couplings, and masses retain this choice and use unrounded spectral values.

7.1.3 Realizing the Spectral Response

For a spectral response to enter the dynamics, it must occur in an actual action that retains every outcome. We first construct a complete spectral instrument with response R_χ on the nontrivial directions. Formation and force coupling then use their respective complete preparations and endpoints. Let the action on the species layer within the same shell satisfy $D_a \psi_\chi = \chi(a) \psi_\chi$, with uniform lifts of the modulo-7 directions at deeper layers. Subtracting the trivial direction from Eq. (7.3), set

$$\begin{aligned} b_a &= \frac{7^{-1/2} \zeta(1/2, a/7)}{\zeta(1/2)} - \frac{1 - 7^{-1/2}}{6}, \qquad S = \sum_{a=1}^{6} b_a D_a, \\ S\psi_\chi &= R_\chi \psi_\chi \quad (\chi \neq 1), \qquad S e_0 = 0. \end{aligned} \tag{7.6}$$

The sum of a nontrivial character vanishes, so the subtraction leaves its response unchanged; for the trivial direction, the coefficient sum vanishes. This realizes the restriction of the shell-preserving response to the species layer. It is not extended to a claim about the analytic response of every higher-conductor direction.

All coefficients are real. Euler–Maclaurin evaluation of the Hurwitz sums with remainder bounds gives $0.973 < C_0 := \sum_a |b_a| < 0.975 < 1$. This strict upper bound allows two identity paths $d_\pm = \pm(1 - C_0)/2$ to be added so that $\sum_i |d_i| = 1$ and $\sum_i d_i D_i = S$. Using eight orthogonal path labels, prepare

$$|a_{\mathrm{sp}}\rangle = \sum_i \sqrt{|d_i|} |i\rangle, \qquad U_{\mathrm{sp}} = \sum_i \mathrm{sgn}(d_i) D_i \otimes |i\rangle\langle i|, \qquad \boxed{\langle a_{\mathrm{sp}} | U_{\mathrm{sp}} | a_{\mathrm{sp}} \rangle = S.} \tag{7.7}$$

The preparation is normalized and every controlled block is unitary, so Eq. (7.7) defines a complete spectral instrument of unit capacity $C = 1$. Final projection back onto a_{sp} produces the success block S, while the remaining orthogonal outputs sum to the effect $I - S^\dagger S$. The two canceling paths belong to this complete action. Unit capacity is thus realized; adopting this path preparation and coherent readback remains a choice of process.

The same spectral response enters the formation instrument and the label-matching instrument separately, each retaining its own complete channels and records. Following Section 2.3.3, the formation modification reads one modulo-7 address and all its deeper lifts using the actual address projection P_x. Force coupling instead reads the share of an independent label preparation that falls in the matching direction. Consider formation first: the auxiliary state $r_{\mathrm{sp}} = \psi_\theta \otimes a_{\mathrm{sp}}$ has equal amplitudes on the six addresses, so

$$Q = U_{\mathrm{sp}}^{\dagger}(\mathsf{P}_x \otimes P_{a_{\mathrm{sp}}})U_{\mathrm{sp}}, \qquad \boxed{w_{\mathrm{sp}} = \langle r_{\mathrm{sp}}|Q|r_{\mathrm{sp}}\rangle = \frac{|R_\theta|^2}{6}.} \tag{7.8}$$

The factor 1/6 in Eq. (7.8) comes from a single-address projection. Spectral success and the single-address output have already been counted together once, so no further division by the success probability is needed. Formation retains the complete spectral auxiliaries and computes the modification using two independent spectral auxiliary units and a coherent relay. Force coupling uses a separate label-matching instrument, whose preparation and matching direction jointly determine the accepted share.

7.2 Born-Level Couplings

Once the spectral instrument supplies the directional response, the strength of exchange still depends on matching the preparation to the endpoint. An equal-weight input is distributed over several independent labels, whereas the selected endpoint accepts only one normalized direction, producing a matching share. Section 3.3 expressed this share as a trace in label space; here we calculate its denominator for $p = 7$. Born level means that corrections from virtual processes have not yet been included.

A force's transfer label and its participating direction are used separately. For an endpoint transition from ℓ to ℓ', the channel character is $\chi_{\ell'-\ell}$, expressing the species difference that the environment must balance. The vertex spectral factor R_ℓ instead uses the response of the participating stable state. Under the candidate physical identifications in Section 5.2, color channels transfer along the modulo-3 coordinate and weak channels along the modulo-2 coordinate; electromagnetic relative phases are implemented separately through the actual charges and reference. Spectral evaluations use χ_1/χ_5, χ_2/χ_4, and χ_3 as the representative responses for color, electromagnetic, and weak vertices, respectively.

The correspondence between these representative responses and physical interactions follows Section 5.2: the color, weak, and charge readings of matter must be implemented jointly by the complete representation, exchange vertices, and readout. The overall $U(1)$ of the species layer gives a common phase within a fixed sector; the charges of different stable states use their respective charge representations. The following counts are made under this correspondence.

For the species-layer unit group G=$\mathbb{F}_7\times$, take m=3 opposite-point-pair groups. Let $d_\ell = \operatorname{ord}\chi_\ell$. The preimage of a phase class is a coset of $\ker\chi_\ell$, giving $6/d_\ell$ relative group actions. There are m alternatives for the point-pair group; if χ_ℓ is self-conjugate, the classification also includes two orientation alternatives. Taking the three labels to be carried independently as specified in Section 3.3 gives

$$\boxed{N_\ell = |\ker \chi_\ell|\, m\, c_\ell, \qquad |\ker \chi_\ell| = \frac{6}{d_\ell}, \quad m = 3, \quad c_\ell = 1 + \mathbf{1}_{\{\chi_\ell = \bar{\chi}_\ell\}}.} \tag{7.9}$$

The factor 2 for the weak direction corresponds to two orientations carried by different preparations or environmental alternatives. Two descriptions of the same complete state are counted only once; coherent paths leading to the same joint final state require amplitudes to be added first. $m = 3$ counts point-pair groups in the species layer, separately from the address depth K and the total number of nonzero observation units in the full space.

Substitute these N_i into the equal-weight preparation and single-matching-direction construction of Section 3.3. Take $\sigma_i = I/N_i$, with matching vector v_i and normalized output f_i. At the specified spectral point, the unnormalized vertex is $A_i = L(1/2, \chi_{\ell_i})|f_i\rangle\langle v_i|$, giving

$$\mathrm{Tr}(\sigma_i A_i^\dagger A_i) = \frac{|L(1/2, \chi_{\ell_i})|^2}{N_i}, \qquad \alpha_i^{(0)} = \frac{\mathrm{Tr}(\sigma_i A_i^\dagger A_i)}{|\zeta(1/2)|^2} = \frac{w_{\ell_i}}{N_i}. \tag{7.10}$$

Here we use $\mathrm{Tr}\,|v_i\rangle\langle v_i| = 1$. For an arbitrary preparation, replace $1/N_i$ by $\langle v_i|\sigma_i|v_i\rangle$ according to Eq. (3.33); for several matching directions, take the trace against the corresponding projection. The actual event effect still uses the matching conditions in Section 3.3. Table 7-2 lists the three counts and Born responses.

Table 7-2 Label counts and Born-level responses for the three candidate vertex types.

Vertex identification	Representative χ	Order d	Kernel order	c	N	$\alpha^{(0)}$
Color force	χ_1/χ_5	6	1	1	3	0.1149199747
Electromagnetic force	χ_2/χ_4	3	2	1	6	0.0079269893
Weak force	χ_3	2	3	2	18	0.0342471093

The inverses of the three Born-level coupling constants are

$$\begin{aligned} (\alpha_s^{(0)})^{-1} &= 8.701707448621\ldots, \\ (\alpha_{\mathrm{EM}}^{(0)})^{-1} &= 126.151299115580\ldots, \\ (\alpha_W^{(0)})^{-1} &= 29.199544712231\ldots. \end{aligned} \tag{7.11}$$

To convert response ratios into event shares, use the complete instrument of Section 3.3. For example, take $M_i = \kappa_i R_{\ell_i}|f_i\rangle\langle v_i|$ and complete the remaining outputs when $\kappa_i^2 w_{\ell_i} \le 1$. The Born preparation then gives $p_i^{(0)} = \kappa_i^2 \alpha_i^{(0)}$. When several outcome types share one round, the contraction condition applies to the total effect. The spectral response, matching share, and instrument calibration thus jointly determine the actual events.

Here $\alpha_i^{(0)}$ is the Born response under the label preparation. The representative electromagnetic direction gives precisely $\alpha_{\mathrm{EM}}^{(0)} = w_2/6 = w_{\mathrm{sp}}$, equal to the single-address reading at formation. The two arise from label matching and address projection, respectively, and each retains its own complete channels and records. The

persistent port of Section 3.1.2 can feed this value into the exchange angle and then read the exchange probability through its squared sine. Virtual processes use the Born value here as input; the corrected response is denoted by α_i.

7.3 Virtual-Process Corrections

An intermediate process can preserve the vertex spectral response while changing the preparation reaching the endpoint and its matching share. Following Section 3.4, let the unnormalized positive preparation be $\Gamma_i \geq 0$ and the matching weight at the endpoint be $h_i = \langle v_i|\Gamma_i|v_i\rangle > 0$. The effective denominator is then $N_i^{\mathrm{eff}} = \operatorname{Tr}\Gamma_i/h_i$. It compares the total preparation weight with the accepted weight, so the following corrections act on the matching share without requiring a change to R_{ℓ_i}.

7.3.1 Effective-Denominator Corrections

The three representative characters for p=7 satisfy $\chi_1^2{=}\chi_2$, $\chi_2^2{=}\chi_4{=}\bar{\chi}_2$, and $\chi_3^2{=}\chi_0$. If the composition of two actions carrying the same charge is nonzero, its composite charge is χ^2. Thus the color and electromagnetic representatives are Type I, and the weak representative is Type II. The trivial character here only states that the composite charge is zero; by itself, it selects neither a particular background state nor a neutral return path.

For Type I, take the connected background contribution to be $V_0 = w_3$ and the total contribution of the two endpoint enhancements to be $2w_\ell$. If the connected contribution arises from the variance of an actual background operator B, this correspondence requires the background preparation to satisfy $\langle B^\dagger B\rangle - |\langle B\rangle|^2 = w_3$. This defines the connected quantity. Merely naming $R_0 - w_3$ the disconnected part and subtracting it does not determine that variance.

In the first-order expansion of Eq. (3.41), these rules correspond to $\gamma_i - N_i g_i = V_0 - 2w_{\ell_i}$. Using this shift to form the finitely corrected denominator according to Eq. (3.42) then gives

$$\begin{aligned} N_s^{\mathrm{eff}} &= 3 + w_3 - 2w_1 = 2.926928119541\ldots, \\ N_{\mathrm{EM}}^{\mathrm{eff}} &= 6 + w_3 - 2w_2 = 6.521324096085\ldots. \end{aligned} \tag{7.12}$$

The shift for the color direction is approximately -0.07307, increasing the matching share; that for the electromagnetic direction is approximately $+0.52132$, decreasing the matching share. These signs follow from comparing the specific response values and are not predetermined by the Type I label.

For the Type II weak direction, specify three color contributions that reduce the denominator, together with one electromagnetic contribution and two self-conjugate weak contributions that increase it. One contribution w_1 or w_2 here uses the weight of the selected representative direction, or equivalently the normalized average of the conjugate pair. Actually summing two orthogonal conjugate outputs with equal responses would give 2w rather than w. The actual routing, orthogonality, and weights of these three terms must satisfy the conditions in Section 3.4. Character multiplication determines the allowed charge destinations, while the coefficients -3, $+1$, and $+2$ belong to this routing prescription. Substitution of these coefficients and the same set of w into Eq. (3.44) gives

$$\boxed{N_W^{\mathrm{eff}} = 18 - 3w_1 + w_2 + 2w_3 = 18.246178099119\ldots.} \tag{7.13}$$

The complete occupation process in Section 6.1.2 already realizes this finite denominator. For the positive occupation output in Eq. (6.6), when the released subspace is orthogonal to the matching direction, the weight subtracted by release does not diminish the endpoint matching weight. Equation (6.7) then gives Eq. (7.13). The weak coupling and background remainder are determined by the same output. If the actual endpoint overlaps the released subspace, the matching weight and background remainder must still be calculated separately from that same output.

7.3.2 Higher-Order Electromagnetic Corrections

An intermediate electromagnetic excursion passes through the conjugate direction and returns to the original port. When the returning part reenters the same intermediate action, it produces a nested response. Whether the contributions combine coherently depends on whether the complete final state retains an excursion record; the distinction between two orientation labels alone cannot decide this. We give a coherent static realization that retains all spectral auxiliaries and uses explicit port connections. Its insertion order and the actual termination-round count described in Section 3.4.3 are calculated separately.

Denote the Born inputs by $e = \alpha_{\mathrm{EM}}^{(0)} = w_2/6$ and $u = \alpha_W^{(0)} = w_3/18$. Distribute the first-order net shift equally among the six electromagnetic labels. At second order, assign weights along the routing of two weak returns and one conjugate return of fixed orientation, obtaining

$$\begin{aligned} x = 3u - 2e = \frac{w_3 - 2w_2}{6}, \qquad y = e(2u+e), \qquad r = \frac{y}{x}, \\ x = 0.086887349348\ldots, \qquad r = 0.006972132346\ldots. \end{aligned} \tag{7.14}$$

These multiplicities are retained as routing conditions. x is the net response shift, and y is the second-order weight; neither directly defines a continuation or termination probability. To place them in the same persistent action, use two independent electromagnetic spectral auxiliary units and one weak spectral auxiliary unit. All purification auxiliaries are coherently accessible and are prepared only once, before the entire intermediate process begins. Denote the complete spectral action of each unit by U_i and its acceptance projection by Π_i. The acceptance projection checks both spectral success and rank-one matching in label space, with the equal-weight label preparation of Section 7.2. The ready-state expectations of the pulled-back projections $Q_i = U_i^\dagger \Pi_i U_i$ are therefore e, e, u, respectively. The actual intermediate action neither reads these flags nor discards the other spectral outputs.

Let the common ready state be s and the preparation code projection be $P_c = I_{\mathrm{carrier}} \otimes |s\rangle\langle s|$. Following the simultaneous equal-strength arms of Chapter 2, apply reflection averaging to the joint operator:

$$\begin{aligned} &A_0 = 3Q_W - 2Q_{E_a}, \qquad A_1 = 2Q_{E_a}Q_W + Q_{E_a}Q_{E_b}, \\ &S_c = 2P_c - I, \qquad \Phi_c(A) = \tfrac{1}{2}(A + S_c A S_c), \\ &\widehat{X} = \Phi_c(A_0), \quad \widehat{Y} = \Phi_c(A_1), \qquad \widehat{X}P_c = xP_c, \quad \widehat{Y}P_c = yP_c. \end{aligned} \tag{7.15}$$

Projections on different spectral auxiliary units commute. Independent preparation gives $\langle A_0\rangle_s = x$ and $\langle A_1\rangle_s = y$. Moreover, because $\Phi_c(A) = P_cAP_c + (I - P_c)A(I - P_c)$, the code subspace decouples from its complement. The expression above therefore preserves the same preparation for every insertion order and depth, while the output space outside the code remains fully retained.

Between a neutral, equal-energy external port a and virtual port b, use the same $\widehat{X}$ as the round-trip vertex and take the virtual-port self-term to be $-\widehat{X} - \widehat{Y}$. The complete self-adjoint generator and its restriction to the code are

$$\begin{aligned} H_J/g &= (|a\rangle\langle b| + |b\rangle\langle a|) \otimes \widehat{X} - |b\rangle\langle b| \otimes (\widehat{X} + \widehat{Y}), \\ H_J/g|_{\text{code}} &= \begin{pmatrix} 0 & x \\ x & -(x+y) \end{pmatrix}, \qquad g > 0. \end{aligned} \tag{7.16}$$

The matrix within the code omits the identity factor on the spectator carrier. The same-vertex connection, virtual-port self-term, and controllable preparation code are conditions on the action in this realization. Fix the amplitude of the driven external port at a. The zero-frequency virtual-port equation is $0 = xa - (x + y)b$, and the returned contribution is xb. Eliminating the virtual port gives

$$\boxed{D(0) = \frac{x^2}{x+y} = \frac{x}{1+r}, \qquad D(0) = \sum_{n=0}^{\infty} x(-r)^n \quad (|r| < 1).} \tag{7.17}$$

At nonzero frequency $\nu = \omega/g$, the same elimination gives $D(\nu) = x^2/(\nu+x+y)$, with static window $|\nu| \ll x + y$. This is a driven response, with physical self-energy gD. Undriven unitary evolution can still oscillate; waiting a long time does not automatically produce a zero-frequency static state. These results make the all-order coefficients consequences of this joint generator.

Under the effective-path interpretation of Section 3.4.4, identify the returned contribution with the denominator shift for six labels. Then $\delta N_n = 6x(-r)^n$, and the original denominator is

$$\boxed{N_{\text{EM}}^{\text{full}} = 6 + \sum_{j=0}^{\infty} \delta N_j = 6 + \frac{w_3 - 2w_2}{1+r} = 6.517714521921\ldots.} \tag{7.18}$$

The shift from the first $J+1$ terms is $6x[1-(-r)^{J+1}]/(1+r)$, and the remainder tends to zero when $|r| < 1$. Equation (7.18) has the same all-order structure as the static response. Reading this response as an actual coupling additionally requires the same positive preparation to satisfy $\operatorname{Tr}\Gamma/h = 6+6D$. Recorded termination histories still contribute nonnegative probabilities according to Eq. (3.47); the alternating insertion terms are not treated as round-by-round event probabilities.

For the color and weak directions, take $N_s^{\text{full}} = N_s^{\text{eff}}$ and $N_W^{\text{full}} = N_W^{\text{eff}}$. The weak denominator is supplied by the complete occupation process above, while the color direction retains the stated finite-correction prescription. More general round-trip actions must be evaluated from their respective complete outputs. Electromagnetic resummation acts only on the virtual-process shift and leaves the Born-level denominator 6 unchanged.

7.3.3 Corrected Preparations

Coupling normalization compares the total and matching weights of the same output. For the weak process, the complete spectral control in Section 6.1.2 already gives B_W. When matching is preserved, using $B_W/\operatorname{Tr} B_W$ yields Eq. (7.13). The electromagnetic process uses the six-label projection P, a rank-two projection $R_d \le P$ onto two excluded modes, and a background mode P_b orthogonal to them. The normalized matching direction v_m satisfies $Pv_m = v_m$ and $R_d v_m = 0$. Taking the initial preparation $\Omega = (P + w_3 P_b)/(6 + w_3)$, the two complete filtering paths are

$$\begin{aligned} K_+ &= \sqrt{P - w_2 R_d} + P_b, \quad & L_+ &= \sqrt{w_2 R_d}, \\ K_- &= P, & L_- &= P_b, \\ K_\pm^\dagger K_\pm + L_\pm^\dagger L_\pm &= I_{\text{pool}}. \end{aligned} \tag{7.19}$$

Let $B_{\text{EM}} = w_3 P_b - w_2 R_d$. Then $\operatorname{Tr} B_{\text{EM}} = 6x$ and $B_{\text{EM}} v_m = 0$. The retained outputs of the two paths are $(P + B_{\text{EM}})/(6 + w_3)$ and $P/(6 + w_3)$, respectively, and the excluded outcomes are retained by $L_\pm$. Negative corrections appear only in the difference between two positive preparations.

To give a fully normalized conditional realization, take the squared amplitudes of two terminal transfers to correspond to x, y and use a common interaction strength. This is an additional physical identification at the terminal stage. Introduce a neutral, equal-energy three-state record $|0\rangle, |+\rangle, |-\rangle$ independent of the virtual port, and on the invariant preparation code above use

$$\begin{gathered} C_R = \sqrt{x}|+\rangle\langle 0| + \sqrt{y}|-\rangle\langle 0|, \qquad H_R = i g_R (C_R - C_R^\dagger), \\ \theta = g_R t_R \sqrt{x+y}, \quad s_R = \sin^2\theta > 0, \qquad p_+ = \frac{s_R x}{x+y}, \quad p_- = \frac{s_R y}{x+y}, \quad p_0 = 1 - s_R. \end{gathered} \tag{7.20}$$

Starting from record state $|0\rangle$, H_R couples only to the bright direction $(\sqrt{x}|+\rangle + \sqrt{y}|-\rangle)/\sqrt{x+y}$. The matrix exponential directly gives the three probabilities above. The record is generated at the boundary after the entire virtual process. $+$ and $-$ control the two filtering paths, respectively, while 0 produces no delivery. Combining the retained outcomes of the two orthogonal records, the actual unnormalized output is

$$\boxed{\begin{aligned} \Gamma_{\text{keep}} &= \frac{s_R}{6 + w_3}\left(P + \frac{x}{x+y} B_{\text{EM}}\right), \\ \frac{\operatorname{Tr}\Gamma_{\text{keep}}}{\langle v_m|\Gamma_{\text{keep}}|v_m\rangle} &= 6 + \frac{6x^2}{x+y} = N_{\text{EM}}^{\text{full}}. \end{aligned}} \tag{7.21}$$

The expression in parentheses equals $[x(P + B_{\text{EM}}) + yP]/(x+y)$ and is therefore positive. The matching weight is $s_R/(6+w_3)$, and the total weight is $6 + 6x^2/(x+y)$ times that matching weight. Thus the same output supplies the preparation, matching, and $\alpha_{\text{EM}} = w_2/N_{\text{EM}}^{\text{full}}$ together. The probabilities of all retained outcomes, excluded outcomes, and record 0 sum to 1. After connecting the signal effect $\kappa^2 w_2 P_{v_m}$, the signal probability per attempt is $\kappa^2 w_2 s_R/(6 + w_3)$; it equals $\kappa^2 \alpha_{\text{EM}}$ only conditional on retention. When a common survival factor preserves this complete output, it cancels in the conditional ratio. The actual count rate still includes the total elapsed duration of preparation, action, failure, and readout.

The complete filtering process above also permits finite repetition: each round uses the same instrument, history records remain orthogonal and never recouple, and the continuation process does not access the independently prepared auxiliary system before filtering. Only the first terminal outcome triggers filtering, while every other exit and cutoff outcome is retained. Denote the two terminal effects by E_+, E_-. The local condition for preserving the same delivery ratio for every input is

$$\boxed{yE_+ = xE_-, \qquad N_L = 6 + \frac{6x^2}{x+y}.} \tag{7.22}$$

The common continuation process may change the internal state; the relation between the two effects ensures preservation of the ratio. See Supplementary Proof S8 for the proof. The three-state terminal stage in this section actually realizes this relation. It does not read the virtual port and yields the same output even after H_J is removed. Thus identifying this Born normalization with the static return of the preceding section still requires the original virtual process to generate the terminal effects used here. This physical identification is retained explicitly at this point.

The color direction continues to retain the finite prescription in Eq. (7.12). Its compatibility with a positive preparation can be checked using $\sigma_s = (N_s^{\mathrm{full}})^{-1} P_{v_s} + [1 - (N_s^{\mathrm{full}})^{-1}](I - P_{v_s})/2$. This is an existence construction and does not by itself determine the actual formation process. Once the three responses are connected to the fixed instrument of Section 7.2, their conditional event shares are all $p_i = \kappa_i^2 \alpha_i$, with the spectral vertices and calibration conditions retained.

7.4 Numerical Values of the Elementary Resonance Forces

The three elementary resonance forces share the same spectral point and background reference; their differences arise from the participating directions and endpoint matching. The weak denominator uses the matching-preserving occupation output of Section 6.1.2. The electromagnetic denominator uses the conditional realization under the static response and terminal identification described in Section 7.3. The color direction retains the finite-correction prescription. Taking traces with unrounded spectral values under the same spectral vertices gives

$$\boxed{\begin{aligned} \alpha_s &= \frac{w_1}{3 + w_3 - 2w_1}, \\ \alpha_{\mathrm{EM}} &= \frac{w_2}{6 + (w_3 - 2w_2)/(1+r)}, \\ \alpha_W &= \frac{w_3}{18 - 3w_1 + w_2 + 2w_3}. \end{aligned}} \tag{7.23}$$

Table 7-3 The three corrected responses. The displayed digits support reproducibility and do not imply physical predictions of the same precision.

Identification	N^{full}	α	$1/\alpha$
Color force	2.9269281195	0.1177889959	8.4897574065
Electromagnetic force	6.5177145219	0.0072973334	137.0363590341
Weak force	18.2461780991	0.0337850461	29.5988940685

The color response increases because its matching denominator decreases, whereas the weak response decreases because its denominator increases. The inverse electromagnetic coupling constant changes from 126.1512991 at Born level to 137.1122511 after the finite correction, and then to 137.0363590 through the all-order static expression and terminal identification above. These three numbers belong to different levels of the same evaluation chain. If a subsequent mass phase specifies a Born-level input, it continues to use $\alpha^{(0)}$ from Section 7.2.

Comparison with observations first requires the definition and scale of the coupling bearing the same name to be fixed. The electromagnetic response is identified with the low-energy fine-structure constant, whose CODATA 2022 inverse is 137.035999177(21) (CODATA 2022 table of constants) [18]. The difference given by this section's prescription is

$$\alpha_{\mathrm{EM}}^{-1} - \alpha(0)^{-1} = 0.000359857137\ldots, \qquad \frac{\alpha_{\mathrm{EM}}^{-1} - \alpha(0)^{-1}}{\alpha(0)^{-1}} = 2.6260044 \times 10^{-6}. \quad (7.24)$$

This is approximately 2.63 ppm or 0.0002626%, exceeding the measurement uncertainty of the constant. Assessing physical precision also requires the spectral and vertex identifications and the theoretical errors of the finite corrections and recursion.

The usual reference value for the color coupling has an energy scale and a renormalization scheme. PDG 2025 gives the strong-coupling average $\alpha_s(M_Z^2) = 0.1180 \pm 0.0009$ in the M$\bar{S}$ definition (PDG 2025, QCD §9.4.8) [21]. If this section's color response is identified with that quantity, the central values differ by approximately -0.1788%. This comparison is conditional on the physical identification with the M$\bar{S}$ scheme and the M_Z scale; matching the actual cycle to that scheme and scale remains to be established.

The weak term likewise requires a definition. If it is identified with the gauge coupling $\mathrm{g}^2/(4\pi)$, the comparison at a chosen common scheme and scale μ should use

$$\alpha_W(\mu) = \frac{g(\mu)^2}{4\pi} = \frac{\alpha(\mu)}{\sin^2\theta_W(\mu)}, \qquad \alpha_W(\mu)^{-1} = \sin^2\theta_W(\mu)\,\alpha(\mu)^{-1}. \quad (7.25)$$

The PDG electroweak review distinguishes on-shell, effective-angle, M$\bar{S}$, and other definitions (PDG electroweak review) [15]. A precise weak-coupling comparison requires compatible scales and schemes for the electromagnetic coupling and mixing angle, as well as sufficient numerical precision. The framework gives 29.5988940685; physical comparison requires the scale and normalization of the actual response to be established.

External numerical values are used only for comparison and do not enter the spectral weights, label counts, or virtual-process corrections. The common structure of the three results is a directional response multiplied by a matching share. Precise comparison with experimental quantities additionally depends on the scale, scheme, and normalization corresponding to the actual process.

7.5 Emergent Resonance Forces

Exchange involving collective states can reach several distinguishable complete outputs from the same matching input. The total response therefore depends on the outputs actually open and their weights. Matching shares at elementary endpoints and the output sum here are two different stages of exchange. Section 3.5 established the corresponding operator relations. This section evaluates them using the affine labels for $p = 7$ and the layer weights of Chapter 6.

Translations and unit scalings of the species layer form $\mathrm{Aff}(\mathbb{F}_7)$, with 7×6=42 operations. If actual exchange provides 42 mutually orthogonal complete outputs F_g for the same matching input v, with response $B_g = R_1 c_g |F_g\rangle\langle v|$ in each channel, then

$$\alpha_{\mathrm{mix}}(\rho) = \sum_{g \in \mathrm{Aff}(\mathbb{F}_7)} \mathrm{Tr}(B_g \rho B_g^\dagger) = w_1 W_{\mathrm{aff}} \langle v|\rho|v\rangle, \qquad W_{\mathrm{aff}} = \sum_g |c_g|^2. \tag{7.26}$$

This applies the orthogonal-output sum of Section 3.5 to the 42 actual exits in this section. The weight of each output is specified by the layer identification below.

Under the node weights specified in Section 6.2, the Loewy-layer product is $\mathcal{L}_7$=3/[2|ζ(1/2)|]. This section continues to use the weight identification of Section 3.5: each actual affine channel carries $|c_g|^2 = 1/\mathcal{L}_7$, and the matching input is ρ=|v⟩⟨v|. Thus

$$\begin{aligned} W_{\mathrm{aff}} &= \frac{42}{\mathcal{L}_7} = 28|\zeta(1/2)| = 40.889926246668\ldots, \\ \mathcal{L}_7 &= 1.027147854134904784\ldots. \end{aligned} \tag{7.27}$$

The value 40.8899 is the total response weight, while 42 is the number of affine-group elements. The module structure determines the layer labels and multiplicities. This section retains the weight assignments of the respective processes: the metric process uses $\Gamma_{\mathrm{met}} = \mathcal{L}_7 I$, and nuclear exchange uses $|c_g|^2 = 1/\mathcal{L}_7$ per path, giving $7^K \mathcal{L}_7$ and $42/\mathcal{L}_7$, respectively. These are separate process inputs; the module structure itself determines only the algebraic layers being used.

Nucleon exchange uses the following vertex identification: take the spectral factor of χ_1 or its conjugate χ_5 as the elementary directional part, and identify the mixed response above with a candidate nuclear coupling. This gives

$$\boxed{\alpha_{\mathrm{nuclear}} := \frac{42}{\mathcal{L}_7} w_1 = 14.097207871478\ldots.} \tag{7.28}$$

Without the layer weight, the result is 42w$_1$=14.479916814483…. If the same preparation describes a purely emergent–emergent response and every channel has spectral factor 1, the result is $\alpha_{\mathrm{emg}} = W_{\mathrm{aff}}$. Using w$_1$ for nucleon exchange additionally requires the mixed-vertex identification above; it does not follow merely from both endpoints being emergent states. If actual paths include a zero-element entrance, additional filtering, or incomplete matching, the corresponding shares must also be included in c_g or ⟨v|ρ|v⟩.

$\alpha_{\mathrm{nuclear}}$ is a response ratio. After applying the complete-instrument calibration κ of Section 3.3, the total event probability for the matching input is $p = \kappa^2 \alpha_{\mathrm{nuclear}} \leq 1$.

The record rate includes the actual input and total elapsed duration as in Section 3.5.

Numerical comparison of the nuclear coupling requires a specific scattering channel and vertex definition. The Roy–Steiner scattering analysis uses $g^2/(4\pi)$=13.7±0.2 (Roy–Steiner analysis, §3.2) [22]. Under the same πNN vertex identification, this section's value is approximately 2.8993% above its central value. This reference comes from a particular extraction of a scattering coupling. A precise assessment additionally requires the charged or neutral channel, vertex normalization, and scattering quantity being compared to be fixed.

The nuclear coupling is obtained jointly from the directional response, layer weights, and specified nuclear vertex; the observed value is used for comparison. Chapter 9 presents collective formation and maintenance, and Section 10.5 continues to use the same routing and record rules to organize flavor responses. The specific spatial interaction between nucleons is determined by the two endpoint vertices and the propagation kernel. For the distinction between reversible routing and permanent records, see Supplementary Proof S4.4.

7.6 The Metric Force Constant

The metric force is transmitted through the background. Equation (4.26) obtains the actual leading-order strength from source writing, background transport, and the passive carrier's maintenance response. Equation (6.26), following the common-source return, common amplitude, and single background calibration of Section 6.2, gives the metric energy scale from $W_K = 7^K \mathcal{L}_7$. Taking the inverse square of the energy scale according to Eq. (4.30), and retaining the complete background C^* used for the numerical values in this section, we have in units of $\hbar = c = 1$

$$\boxed{G_E(K) := \frac{1}{M_{\mathrm{met}}(K)^2} = \frac{C^*}{E_0^2 W_K (\log W_K)^2}, \qquad W_K = 7^K \mathcal{L}_7.} \tag{7.29}$$

G_E has units of inverse energy squared. At mass scale m, the dimensionless gravitational strength is $G_E m^2$. This section retains the single physical calibration of Section 4.5.5, using the inverse square of the spectral energy scale as the leading Newtonian strength of the process under consideration. Each realization uses its own return kernel and complete-cycle timing; finite-source and propagation corrections are calculated as in Section 4.5.6.

Taking the conditional parameters K=41 and E_0=1 GeV, with the complete background and node weights of Section 6.2, gives

$$G_E = 6.702721805351 \times 10^{-39}\ \mathrm{GeV}^{-2}, \qquad M_{\mathrm{met}} = 1.221446368810 \times 10^{19}\ \mathrm{GeV}. \tag{7.30}$$

Compared in the same units with the CODATA 2022 Newton constant, the relative difference is approximately −0.0910582% (CODATA 2022 table of constants) [18]. The difference is calculated from the SI value of G using exact unit conversion and still exceeds the cited measurement uncertainty. A small relative difference alone therefore does not establish agreement within experimental precision. The Planck energy scale and G_E are related by an inverse square by definition, so their proximity provides only the same single comparison.

Retaining general K also allows the dependence of this result to be checked directly. The coupling ratio between adjacent depths is

$$\frac{G_E(K+1)}{G_E(K)} = \frac{1}{7}\left(\frac{\log W_K}{\log W_K + \log 7}\right)^2, \qquad G_E \propto E_0^{-2}. \tag{7.31}$$

The construction depth K and energy calibration E_0 determine this dependence of the metric force constant. Within their respective required participating spaces, the three elementary couplings follow from species-layer spectral ratios and label denominators, and nuclear exchange from affine-channel weights. None of these formulas uses K=41. The metric depth can therefore be tested as a separate physical input while the other couplings continue to be evaluated from their respective processes.

Table 7-4 places these five evaluations side by side. Its formulas retain the preparation and identification conditions already stated in the respective sections. Experimental comparisons are listed under their respective definitions, with numerical proximity and physical identification assessed separately.

Table 7-4 Summary of coupling evaluations for five forces.

Force	Coupling evaluation	Result in this chapter	Comparison definition and difference
Color force	$\alpha_s = w_1/N_s^{\text{full}}$	$\alpha_s \approx 0.117789$	$M\bar{S}$, $\mu = M_Z$: $\alpha_s = 0.1180 \pm 0.0009$; central-value difference -0.1788%.
Electromagnetic force	$\alpha_{\text{EM}} = w_2/N_{\text{EM}}^{\text{full}}$	$\alpha_{\text{EM}}^{-1} \approx 137.036359$	Low-energy $\alpha(0)^{-1} = 137.035999177(21)$; inverse difference $+2.626$ ppm.
Weak force	$\alpha_W = w_3/N_W^{\text{full}}$	$\alpha_W^{-1} \approx 29.598894$	Precise comparison requires $g^2/(4\pi)$ at a common scale and scheme and its physical matching.
Nuclear force	$\alpha_{\text{nuclear}} = (42/\mathcal{L}_7)w_1$	$\alpha_{\text{nuclear}} \approx 14.097208$	Against the πNN reference 13.7 ± 0.2 cited in Section 7.5, the central-value difference is $+2.8993\%$.
Metric force (gravity)	$G_E = C^*/[E_0^2 W_K (\log W_K)^2]$	$G_E \approx 6.70272 \times 10^{-39}\text{GeV}^{-2}$	For K=41, E_0=1 GeV, and the identification $G_E \leftrightarrow G_N$, the CODATA central-value difference is -0.09106%.

Here $w_\ell = |R_\ell|^2$, N_i^{full} is given in Section 7.3, and $W_K = 7^K\mathcal{L}_7$. The first four entries are dimensionless responses; the last retains units of inverse energy squared. K=41 is used only in the last row. Sources and conditions for the comparisons

appear in Sections 7.4–7.6; the table does not combine data at different scales into a single low-energy limit.

Elementary resonance-force couplings are jointly determined by directional spectral responses and endpoint matching shares. Collective exchange additionally sums distinguishable outputs, while metric action connects the background participation energy scale to the actual field strength. These quantities enter the same dynamical description through their respective preparations, exchanges, and readouts. Chapter 8 continues to use the relevant spectral responses in the mass processes of each family, distinguishing Born-level from corrected inputs according to the actual mode of coupling.

§8 Elementary-State Mass Spectrum

Why do elementary states in the same background exhibit both mass hierarchies within families and scale differences between families? Formation first leaves a coherent pattern, after which excitations flow out, exchange, and return along their respective paths. Branches that persist yield frequencies read against the same phase reference. The common background supplies the energy calibration; paths and nodes determine the coefficients of actual exchange, while projections of the pattern onto the different endpoint groups distinguish some branches within a family. Mass differences arise from combining these actions in the same persistent process.

This chapter adopts the physical identifications for $p = 7$ in Chapter 5, the background energy scale v in Chapter 6, and the spectral responses in Chapter 7. The complete preparations, persistent actions, and calibration using a common source constructed in Chapter 2 connect each family's path response to its mass reading. We determine the paths, nodes, records, and return mechanisms of the respective families below. Finite-link truncation and feedback tails are given with each realization.

8.1 Phases and Generational Factors

The even conjugate directions χ_2 and χ_4 form a common part of several formation paths. The modulus of the same complex spectral value enters the path scale, while its phase changes the pattern formed by their superposition with the background. Three endpoint groups assign different intensities to this pattern, providing one source of mass relations across generations. Phase modification during formation and frequency shifts during maintenance act on the pattern and cycle reading, respectively, and must be calculated against the same phase reference.

8.1.1 Spectral Phase

Fix the generator 3 and character convention $\chi_2(3^a) = e^{2\pi i a/3}$ of Section 7.1, and write $L_2 = L(1/2, \chi_2)$. χ_2 is an even primitive character, with conjugate χ_4. Initially, take the formation angle to be the phase of L_2 relative to the real background. The denominator of the resonance ratio $R_2 = L_2/\zeta(1/2)$ is negative, so its argument differs from $\arg L_2$ by π; the two cannot be interchanged while keeping the background phase unchanged. Finite Hurwitz summation gives

$$L_2 = 0.3100893625983677 - 0.0726419313701779\, i, \qquad \delta_2 := \arg L_2. \tag{8.1}$$

This phase can also be related to finite Gauss and Jacobi sums. We use the standard definitions and product relations over finite fields [23]. Extending the characters by zero at the zero element, define

$$\tau(\chi) = \sum_{a=1}^{6} \chi(a) e^{2\pi i a/7}, \qquad J(\chi, \chi) = \sum_{a \in \mathbb{F}_7} \chi(a)\chi(1-a). \tag{8.2}$$

Let $\omega_3 = e^{2\pi i/3}$. Evaluating the finite Jacobi sum with the fixed character orientation gives

$$J(\chi_2,\chi_2) = 2 + 3\omega_3^2 = -1 - 3\omega_3 = \frac{1 - 3i\sqrt{3}}{2}, \qquad |J| = \sqrt{7}. \tag{8.3}$$

Group the Gauss-sum product by $a + b$. In the nonzero groups, using $a/(a + b)$ as the variable gives $\tau(\chi_2)^2 = J(\chi_2,\chi_2)\tau(\chi_2^2)$; the zero-sum group vanishes because χ_2^2 is nontrivial. We also have $\chi_2^2 = \chi_4$ and $\tau(\chi_2)\tau(\chi_4) = 7$. The central functional equation for an even primitive character gives $L_2 = \tau(\chi_2)\overline{L_2}/\sqrt{7}$ (DLMF 25.15.5) [20]. Therefore

$$e^{2i\delta_2} = \frac{\tau(\chi_2)}{\sqrt{7}}, \qquad e^{6i\delta_2} = \frac{J(\chi_2,\chi_2)}{\sqrt{7}}, \qquad 6\delta_2 \equiv \arg J \pmod{2\pi}. \tag{8.4}$$

The Jacobi sum determines a phase congruence class. The complex spectral value above selects the branch $-\pi/6<\delta_2<0$, which yields the following value. This branch condition is retained when taking one sixth of the argument.

$$\boxed{\delta_2 = -\frac{1}{6}\arctan(3\sqrt{3}) = -0.230111787241405\ldots \text{ rad} = -13.1844342251\ldots^\circ.} \tag{8.5}$$

The spectral value and chosen argument branch jointly fix the relative phase between the background and conjugate directions. This phase enters the formation pattern. The spectral weight $|R_2|^2$ used in Chapter 7 is unaffected by the sign of the real denominator.

8.1.2 Formation Corrections and Generational Factors

On the unit shell, take normalized directions ψ_2 and ψ_4 and the shell-average state e_0. Under the original exchange and full positive-branch feedback of Section 2.3.2, the accepted bright background entrance produces an equal-amplitude source. All feedback outcomes are retained, and the source factorizes from these records. When the background component keeps a real phase and the conjugate directions acquire opposite phases δ, the pattern is

$$\sigma_\delta = \frac{e^{i\delta}\psi_2 + e^{-i\delta}\psi_4}{\sqrt{2}}, \qquad \Phi_\delta = \frac{e_0 + \sigma_\delta}{\sqrt{2}}. \tag{8.6}$$

The normalization factor $1/\sqrt{2}$ of Φ_δ follows from orthonormality. Equal amplitudes of the background and conjugate combination follow from the eigenstructure of the selected exchange and positive-branch feedback. Feedback first produces the positive pattern $\Phi_0 = (e_0 + \sigma_0)/\sqrt{2}$. Keeping the real-background convention of Section 8.1.1, extend the unitary polar factor V_R of R as the identity on the background, then apply $Z_{\mathrm{bd}} = P_{e_0} - P_{\psi_2} - P_{\psi_4}$. On the formation space, this action $D_L = Z_{\mathrm{bd}}V_R$ supplies the phase of L_2, turns the directional combination into σ_{δ_2}, and produces Φ_{δ_2}. This step is performed once after positive-branch feedback. The zero success block of the complete response on e_0 and the identity extension of the unitary polar factor here are defined separately.

Next, following Section 2.3.3, retain two independent spectral auxiliaries, the coherent relay, and the actual outward and return paths. The background remains unchanged and the auxiliaries return to the common entrance, while the two directions acquire opposite additional phases. Taking Z to vanish on e_0 gives

$$Z = |\psi_2\rangle\langle\psi_2| - |\psi_4\rangle\langle\psi_4|, \qquad e^{i\eta Z}\Phi_{\delta_2} = \Phi_{\delta_2+\eta}. \tag{8.7}$$

The angular shift of this return is $\eta = sa$, where $a = w_{\mathrm{sp}} - w_{\mathrm{sp}}^2/\sqrt{3}$ and s is the accumulated action. Using the eight-path spectral instrument of unit capacity constructed in Section 7.1.3, the actual output for one modulo-7 address and all its deeper lifts gives $w_{\mathrm{sp}} = \alpha_0 = |R_2|^2/6$. The unit spectral-kernel identification of Section 2.3.3 fixes $s = 1$. The formation angle for the charged branches is

$$\alpha_0 := \alpha_{\mathrm{EM}}^{(0)} = \frac{|R_2|^2}{6}, \qquad \alpha_{\mathrm{eff}} := \alpha_0\left(1 - \frac{\alpha_0}{\sqrt{3}}\right), \qquad \eta = \alpha_{\mathrm{eff}}, \quad \delta = \delta_2 + \alpha_{\mathrm{eff}}. \tag{8.8}$$

The first term arises from one spectral effect. The second arises from two independent effects and the relative minus sign of the relay, where $1/\sqrt{3}$ is the overlap of an entire endpoint group with the uniform background. The complete auxiliaries, reflections along the paths, and two simultaneous return arms are retained as in Section 2.3.3. Unit capacity has been proved for the chosen instrument, whereas $s = 1$ depends on the physical identification of unit formation; neither implies the other. With another instrument of capacity C, the general angular shift remains $\eta = s[\alpha_0/C^2 - \alpha_0^2/(\sqrt{3}C^4)]$. The Born input α_0 is used here; the corrected α_{EM} of Chapter 7 is used for the complete return response below.

$$\begin{aligned} \alpha_0 &= 0.0079269893137113\ldots, \\ \alpha_{\mathrm{eff}} &= 0.0078907102627129\ldots, \\ \delta &= -0.222221076978692\ldots \text{ rad} = -12.7323298297\ldots^\circ. \end{aligned} \tag{8.9}$$

For p=7, the species-layer point pairs, ordered by the generator 3, are $\{1,6\}$, $\{3,4\}$, and $\{2,5\}$. At depth K, all unit lattice points reducing modulo 7 to a given pair form the entire group $\mathcal{D}_j$ of Section 2.3.2, with normalized in-phase vector $\bar{d}_j$. Since $\chi_2(3^j)$=exp(2πij/3), the coherent preparation above gives

$$\langle \bar{d}_j|\Phi_\delta\rangle = \frac{f_j}{\sqrt{6}}, \qquad f_j = 1 + \sqrt{2}\cos\left(\delta + \frac{2\pi j}{3}\right), \qquad j = 0,1,2. \tag{8.10}$$

The amplitude and phase are constant within each group, so the share of the whole-group projection equals the squared modulus of the vector readout above. Trigonometric identities give $\sum_j f_j = 3$ and $\sum_j f_j^2 = 6$, and the three group probabilities sum to one. Taking the zeroth group as a nonzero reference gives

$$\boxed{p_j = \frac{f_j^2}{6}, \qquad \varrho_j := \frac{p_j}{p_0} = \left(\frac{f_j}{f_0}\right)^2, \qquad \sum_{j=0}^{2} p_j = 1.} \tag{8.11}$$

These shares are supplied by the formation source. They enter actual matrix elements through the same-group matching in Eq. (2.55)–Eq. (2.56), then enter the persistent generator through the common preparation and simultaneous arms of Section 2.3.5. The background branch matches the zeroth group of the same source, so calibration using a common source in Eq. (2.75)–Eq. (2.76) leaves p_j/p_0. Branches that use this process directly take $\mathrm{Gen}_{f,j} = \varrho_j$. Branches involving segmented readout, feedback, or an indirect relay must be calculated from their respective actions.

Table 8-1 Three group readings after the specified equal-amplitude preparation and formation modification. ϱ_j is the relative intensity, and p_j the normalized share.

j	f_j	p_j	ϱ_j
0	2.3794385286	0.9436212852	1
1	0.5802103735	0.05610734626	0.05945960221
2	0.04035109788	0.0002713685167	0.0002875820215

In group 2, the background and conjugate components nearly cancel, producing a very small relative intensity. All three amplitudes here are positive. More general formation phases can make a group's amplitude negative; its intensity is still given by the square.

8.1.3 Cycle Frequency Shifts

Formation modification determines the pattern produced, while maintenance causes that pattern to accumulate phase persistently. Under the persistent repair of Section 2.3.7, the spectral round trip conjugates the entire feedback process, driving the source toward the same Φ_δ. The formation angle does not accumulate from round to round. Insertions during maintenance change the phase accumulated by each branch per round and thereby enter the frequency reading.

By Eq. (2.63), eigenvalues of the actual persistent generator give masses after complete time accounting. Once the pattern enters the persistent action, path spectral responses and node overlaps determine the family coefficient A_f, generational factors $\mathrm{Gen}_{f,j}$ distinguish branches in the same family, and the common background supplies the energy scale v. If different paths are used within one physical family, their A_f must be calculated separately.

Following the complete time accounting and negative-argument convention of Eq. (2.59)–Eq. (2.60), denote the phase reading per round before insertion of the maintenance frequency shift by $B_{f,j}$ and its correction by $\kappa_{f,j}$. The generator eigenvalue during the active time t_H is $\mu^{(0)}_{f,j}$, and the complete cycle period is $\tau'_{f,j}$. The superscript (0) excludes only the maintenance frequency shift; the source already includes formation modification. Dividing the actual accumulated phase by the complete cycle period, calibration using a common source gives

$$M_{f,j} = \frac{B_{f,j} - \kappa_{f,j}}{\tau'_{f,j}}, \qquad B_{f,j} = t_H \mu^{(0)}_{f,j} = \tau'_{f,j} v A_f \,\mathrm{Gen}_{f,j}. \tag{8.12}$$

Here $\tau' v = 1$ is not assumed: v is calibrated by the background response of Chapter 6, and the period is calculated from the actual schedule. By Eq. (2.57)–Eq. (2.58), independent modification auxiliaries give the joint correlation term $C_{X,A} = 0$. Under the same spectral normalization $a = \alpha_{\mathrm{eff}}$, the generator acquires a proportional correction, so $\kappa_{f,j} = B_{f,j}\alpha_{\mathrm{eff}}$. Writing the common frequency-shift factor as $F := 1 - \alpha_{\mathrm{eff}}$, charged branches that use this maintenance insertion and satisfy the factorization and path conditions above have

$$\boxed{M_{f,j} = v A_f \,\mathrm{Gen}_{f,j}(1 - \alpha_{\mathrm{eff}}), \qquad 1 - \alpha_{\mathrm{eff}} = 0.992109289737287\ldots.} \tag{8.13}$$

For branches with the same path coefficient, common period, and relative frequency shift, the overall factor cancels from mass ratios. If the branch identification is $\mathrm{Gen}_{f,j} = \varrho_j$, then

$$\frac{M_{f,j}}{M_{f,0}} = \frac{\mathrm{Gen}_{f,j}}{\mathrm{Gen}_{f,0}} = \varrho_j, \qquad \mathrm{Gen}_{f,0} = 1. \tag{8.14}$$

Formation modification enters the mass baseline through the source pattern, and the maintenance insertion then produces a proportional frequency shift; they act at different stages. Complete time accounting follows Section 2.4.6. Formation and thermal-survival shares control formation convergence and the survival envelope, respectively.

Each family's masses must therefore be calculated from its own complete paths: first determine the spectral quantities, nodes, and branch shares entering the persistent generator, then obtain the masses through calibration using a common source and complete time accounting.

8.2 Charged Leptons

Charged-lepton formation selects the even conjugate directions χ_2 and χ_4. They are in phase under negation but have different phases across the three observation units of the species layer. The same family therefore has both a common formation scale and three distinct group readings. Under the physical identifications of Chapter 5, χ_2 is the representative and χ_4 supplies the conjugate orientation. Under the branch identification of Section 8.1, groups 0, 1, and 2 correspond to τ, μ, and e, respectively.

8.2.1 Conjugate Lepton Paths

Use the double excursion of Section 2.4.2: connect from the zero element to the χ_2 direction, pass through its spectral response, and return; then connect to χ_4 and return again. The intermediate zero element serves only as a connection and inserts no additional spectral response. Since the spectral point used is real, $\mathrm{L}_4{=}\bar{L}_2$. The two excursions preserve the specified conjugate connection, with spectral product $\mathrm{L}_2\mathrm{L}_4$. Applying a closed-path intensity readout to this complete component gives

$$a_{24} = L_2 L_4 = |L_2|^2, \qquad S_\ell = |a_{24}|^2 = |L_2|^4 = 0.0102885039752424\ldots. \tag{8.15}$$

The closed-path readout takes the squared modulus of the complete spectral product of both excursions, yielding the lepton path's spectral intensity. Through the node exchange and same-group matching of Sections 2.4.1–2.4.4, this intensity enters the generator; calibration using a common source in Section 2.5.3 then cancels the common capacity. Re L_2, read by directional diagonal averaging, is a different response.

At the entrance, use the normalized in-phase vector in the binary orientation factor and connect one specified component. Its overlap is $1/\sqrt{2}$. The three group readings were already normalized together in Section 8.1; this entrance factor is counted only once. Combining it with the closed-path spectral quantity in the actual node exchange gives the family coefficient and generational factors:

$$A_\ell = \frac{|L_2|^4}{\sqrt{2}}, \qquad \mathrm{Gen}_{\ell,j} = \varrho_j, \qquad (j = 0, 1, 2). \tag{8.16}$$

The node amplitude overlap $1/\sqrt{2}$ enters A_ℓ, and the relative intensity ϱ_j after same-group matching enters the generational factor; each is counted once. The phase per round enters $\tau' v$ according to Eq. (8.12), where τ' is the complete cycle period. Node and generational factors must be recalculated if the endpoints or the source's relative background amplitude change.

8.2.2 Masses of Three Generations

The three branches use a common period and the frequency-shift factor F of Section 8.1. Multiplying their respective ϱ_j by the common path factor, node factor, and background energy scale gives the three absolute masses. Using $\varrho_0 = 1$, write them as

$$\boxed{m_\tau = \frac{v|L_2|^4}{\sqrt{2}} F, \qquad m_\mu = m_\tau \varrho_1, \qquad m_e = m_\tau \varrho_2, \qquad F = 1 - \alpha_{\mathrm{eff}}.} \tag{8.17}$$

With the E_0=1 GeV calibration of Section 6.1, calculation using unrounded v, L_2, and the three group readings gives

$$\begin{aligned} m_\tau &= 1776.926974\ldots \text{ MeV}, \\ m_\mu &= 105.6553711\ldots \text{ MeV}, \\ m_e &= 0.5110122512\ldots \text{ MeV}. \end{aligned} \tag{8.18}$$

The τ reading sets the reference scale for this preparation. The μ and e masses differ through the intensities of the same coherent pattern at the other two endpoint groups. Group 2 in particular is near cancellation, so a small phase change can substantially change its relative intensity. The significant digits of the electron mass are therefore limited not only by numerical precision but also by the accuracy of the formation phase and relative background amplitude. Comparisons with experiment and the common spectral-shape relation are collected in Section 8.6.

8.3 Down-Type Quarks

Down-type quarks connect from the even direction χ_2 to the real direction χ_3, yielding the representative direction χ_5; their CRT coordinates are $(0, 2)$, $(1, 0)$, and $(1, 2)$, respectively. The same real direction participates differently in the three branches: b retains the linear real response, s uses recorded segmented intensities, and d adds a flip link to that segmented branch. Mass differences therefore involve both path readout and connections between groups.

8.3.1 Down-Type Quark Propagation Paths

Character multiplication first fixes the direction label. The actual transfer still passes through the common zero element z as in Section 2.4.2. Write $F_{K,\chi} = |z\rangle\langle\psi_\chi|$, where K labels the shell used. Absorbing χ_2 first and then forming χ_5 gives

$$\chi_2\chi_3 = \chi_5, \qquad F^{\dagger}_{K,\chi_5} F_{K,\chi_2} = |\psi_5\rangle\langle\psi_2|. \tag{8.19}$$

This connection must be accompanied by compatible environmental inputs and outputs, with the real-direction response retained at the required steps. χ_5 is an odd direction and has zero amplitude at a symmetric endpoint. An even entrance does not imply that the same symmetric endpoint remains in use at the end. A nonzero output in the down-type direction χ_5 must be received by the corresponding antisymmetric endpoint or a complete directional readout.

Specify that the terminal real direction of the b branch remains in the linear response. Because L_3 is positive and real, the spectral quantity retains one factor L_3. The s and d branches instead use recorded segmented readout, so the real direction enters through a squared modulus and retains L_3^2. The preceding even-conjugate closure gives $|\mathrm{L}_2|^4$ in both readouts:

$$S_b = |L_2|^4 L_3, \qquad S_{s,d} = |L_2|^4 L_3^2, \qquad L_3 = L(\tfrac{1}{2}, \chi_3) > 0. \tag{8.20}$$

The difference in powers comes from the location of the two readouts: a linear response retains one real spectral factor, whereas a recorded intensity contains its squared modulus. Without a reset between segments, the joint record must still be calculated by multiplying conditional probabilities. The label "passes through χ_3" alone does not determine which spectral quantity to use.

Nodes are likewise determined by their respective readouts. Denote the full-spectrum reference by $\zeta_0 = \zeta(1/2)$. The b branch uses a single-component projection at the binary entrance, the sum of equal-weight linear responses from three terminal color components, and one full-spectrum background reference. The color components are summed as $\operatorname{Tr} I_3 = 3$, giving node factor $3/(\sqrt{2}|\zeta_0|)$. Each of the two conditional-response segments in the s, d branch retains one factor $|\zeta_0|^{-1}$, without the terminal color sum above. Hence

$$\zeta_0 := \zeta(\tfrac{1}{2}), \qquad \mathcal{T}_b = \frac{3}{\sqrt{2}|\zeta_0|}, \qquad \mathcal{T}_{s,d} = \frac{1}{|\zeta_0|^2}. \tag{8.21}$$

These node factors use the per-color transfer strengths and entrance projection specified in Section 2.4.3, with orthogonal failure outputs completing the exchange. The additional auxiliary units participate in preparation and protection, allowing node responses to enter the persistent generator. Both routings use the full-spectrum reference ζ_0 for spectral normalization. The shell-average state along the path serves as a connection, while the normalization denominator remains ζ_0.

8.3.2 Flip Amplitude

The s and d branches use the same overall spectral scale of the recorded branch, but d additionally reads one flip link. The two relative components of the conjugate directions are supplied by the spectral phase δ_2. Select a link whose two successive overlaps are $\cos\delta_2$ and $\sin\delta_2$. Its unmodified amplitude magnitude is

$$\lambda_0 := |\sin\delta_2 \cos\delta_2| = \frac{1}{2}|\sin 2\delta_2| = 0.222074215171030\ldots. \tag{8.22}$$

This link uses the unmodified spectral phase δ_2; δ in Section 8.1 is used for the complete formation pattern, so the two have different roles. The link's two label

paths satisfy $\chi_3\chi_2{=}\chi_5$ and $\chi_3\chi_4{=}\chi_1$. If the actual correction readout retains mutually exclusive records of both paths and all three colors, with relative weight $\alpha_0/2$ for each contribution, the total coefficient of the first-order probability correction is $3\alpha_0$. The corresponding first-order amplitude expansion is $\lambda_0(1{+}3\alpha_0/2)$. Here this truncated amplitude expression is adopted as the finite evaluation prescription:

$$\boxed{\lambda := \lambda_0\left(1+\frac{3}{2}\alpha_0\right) = 0.224714785066798\ldots, \qquad \lambda^2 = 0.0504967346276171\ldots.} \tag{8.23}$$

The finite amplitude and finite probability prescriptions differ at second order. If $\lambda_0^2(1{+}3\alpha_0)$ is taken directly as the complete probability, the amplitude is 0.224699270177370…. This chapter retains the amplitude prescription above. Its first-order calculation, associated with the two recorded paths, does not also determine the actual return action at second and higher orders. Identifying λ with a particular weak-mixing matrix element additionally requires the corresponding charged-current vertex and normalization convention.

8.3.3 Masses of Three Generations

First combine the coefficients of the two routes: $A_b = S_b\mathcal{T}_b$ and $A_{s,d} = S_{s,d}\mathcal{T}_{s,d}$. Then choose the generational factor of b as 1, that of s as ϱ_1, and that of d as $\varrho_1\lambda^2$. The identification of these three readouts with matter labels is specified with the paths and cannot be determined by character multiplication alone. Multiplication by the common calibration v and frequency shift F gives the three masses:

$$\boxed{m_b = v|L_2|^4 L_3 \frac{3}{\sqrt{2}|\zeta_0|}F, \qquad m_s = v|L_2|^4 L_3^2 \frac{\varrho_1}{|\zeta_0|^2}F, \qquad m_d = m_s\lambda^2.} \tag{8.24}$$

$$\begin{aligned} m_b &= 4185.420021\ldots \text{ MeV},\\ m_s &= 92.10919837\ldots \text{ MeV},\\ m_d &= 4.651213747\ldots \text{ MeV}. \end{aligned} \tag{8.25}$$

The difference between b and s jointly tests linear versus segmented readout, nodes, and group matching, whereas $d/s = \lambda^2$ separately tests the flip link. Comparing absolute values with experiment also requires a common quark-mass renormalization scheme and scale; see Section 8.6.

8.4 Up-Type Quarks

The representative up-type quark direction χ_1 has CRT coordinates $(1,1)$ and satisfies $\chi_4\chi_3 = \chi_1$. Its path contains a closed combination of even, real, and odd directions. The three branches share its spectral and node coefficients: c uses the reference readout, t accumulates closed returns, and u takes a single weak-channel delivery after a flip.

8.4.1 Spectral Factor of the Closed Path

The even and odd directions each undergo one conjugate double excursion, followed by a closed-path intensity readout, leaving a fourth spectral power for each. The real direction uses an intensity and leaves L_3^2. The entrance counts the binary single-component projection once, and both CRT coordinates at the terminal endpoint are nonzero. The actual node connection used in this routing introduces neither an additional background reference nor a terminal color sum. Denote the common path coefficient of the entire up-type family by A_u, where the subscript u here denotes the up-type family. Then

$$\chi_2\chi_3\chi_1 = \chi_0, \qquad S_u = |L_2|^4 L_3^2 |L_1|^4, \qquad A_u = \frac{S_u}{\sqrt{2}}. \tag{8.26}$$

The label product of this path is the trivial character, while the powers of the spectral factors are determined by the actual visit counts and intensity readouts. The conjugate double excursions of the even and odd directions, together with the intensity response of the real direction, give S_u. Each absorption, formation, and environmental balancing step is realized as in Section 2.4.2.

Take c as the reference branch, with $\mathrm{Gen}_c = 1$. The common background calibration and frequency shift give

$$m_c = vA_uF = \frac{v|L_2|^4 L_3^2 |L_1|^4}{\sqrt{2}} F = 1262.842920\ldots \text{ MeV}. \tag{8.27}$$

8.4.2 Top-Quark Mass

The top-quark branch shares the closed path of the c branch. With continuation coefficient $0 \le q < 1$, load the success effect of repeated attempts into one complete carrier, which subsequently accumulates phase persistently. Starting with the initial contribution, the linear return kernel satisfies $\mathcal{R} = 1 + q\mathcal{R}$, so

$$\mathcal{R} = \sum_{n=0}^{\infty} q^n = \frac{1}{1-q}, \qquad q = 1 - \alpha_{\mathrm{EM}} \quad \Longrightarrow \quad \mathcal{R} = \frac{1}{\alpha_{\mathrm{EM}}}. \tag{8.28}$$

The corrected $q = 1-\alpha_{\mathrm{EM}}$ is used here; it is defined separately from the thermal-survival share containing environmental weights. Finite loading retains success, competing failures, and the unfinished tail of $n+1$ attempts. All working registers participate in the subsequent simultaneous two-arm protection. The success effect increases by a factor $\mathcal{R}_n$ relative to the c branch, without changing the other path factors, nodes, or frequency shift.

Once loading is complete, subsequent survival rounds do not repeat the attempts. Both branches share the drive and active window and operate so that the additional working registers do not lengthen the per-round repair or thermal-processing time. They therefore have the same complete cycle period, giving

$$\boxed{\frac{m_t^{(n)}}{m_c} = \mathcal{R}_n = \frac{1-q^{n+1}}{\alpha_{\mathrm{EM}}}, \qquad \frac{1}{\alpha_{\mathrm{EM}}} - \mathcal{R}_n = \frac{q^{n+1}}{\alpha_{\mathrm{EM}}}.} \tag{8.29}$$

The time spent on initial loading belongs to the formation history; this frequency relation applies to survival cycles after loading. A short proof of the complete effect,

protective intertwining, and common time accounting appears in Supplementary Proof S12. With the finite tail controlled, the result approaches the reading of the effective return kernel below.

Under the specified infinite effective return kernel and phase identification, using the corrected α_{EM} of Section 7.4 gives

$$\boxed{m_t = \frac{m_c}{\alpha_{\mathrm{EM}}} = 173.0553957\ldots\ \mathrm{GeV}.} \tag{8.30}$$

8.4.3 Up-Quark Mass

The up-quark branch uses one flip readout and one weak-channel delivery. The flip intensity is λ^2 from Section 8.3, and the delivery response is α_W from Section 7.4. The conditional process used makes the delivery response compatible with the flip record and uses their product as the dimensionless generational factor relative to the c branch:

$$\mathrm{Gen}_u = \lambda^2\alpha_W, \qquad \boxed{m_u = m_c\lambda^2\alpha_W = 2.154453597\ldots\ \mathrm{MeV}.} \tag{8.31}$$

Here α_W is the delivery-response coefficient specified in Section 7.4. Its correspondence with the actual conditional transition probability, preparation share, and cycle phase is determined by the complete delivery process. If these process conditions change, it must be reevaluated using the actual instrument.

The three up-type branches share A_u, with generational factors 1, $1/\alpha_{\mathrm{EM}}$, and $\lambda^2\alpha_W$, respectively. Here Gen_u denotes only the up quark's generational factor. Once the common spectral quantity is fixed, the difference between accumulated returns and additional delivery produces the hierarchy of this family and yields separately testable proportional relations.

8.5 Neutrinos

The neutrino type uses the self-conjugate direction χ_3, whose amplitudes have opposite signs on a pair of points related by negation. The formation seed in this section uses the symmetric direction of an observation unit, with equal amplitudes at the two points. The formation entrance and target direction have different phase structures; their connection must be supplied by the exchange process adopted.

8.5.1 Indirect Neutrino Channels

For a unit lattice point a, define $d_a^+ = (|a\rangle + |-a\rangle)/\sqrt{2}$. Parity immediately gives

$$\langle d_a^+|\psi_3\rangle = 0, \qquad \chi_3(-a) = -\chi_3(a). \tag{8.32}$$

It is this particular symmetric amplitude that vanishes; the lattice-point occupation of χ_3 and its antisymmetric endpoint response can still be nonzero. The neutrino identification below restricts direct preparation to this symmetric entrance, followed by actual exchange into the selected relay path. The fact that χ_3 is a real character alone does not exclude direct excitation of this direction by other preparations.

The indirect path contains two even-conjugate closures, each giving $|L_2|^4$. Two binary nodes each supply an amplitude matrix element $1/\sqrt{2}$ through the actual exchange of Section 2.4.3. The relay uses the participating space P_4 defined in Section 6.1 and prepares the uniform state $e = N^{-1/2}\sum_x |x\rangle$. The outward path matches the specified address $P_x = |x_0\rangle\langle x_0|$, and the return path matches the uniform background $P_e = |e\rangle\langle e|$. Both actions retain their success and failure flags and complete inverses. The dimension and conditional shares along the double-success history are

$$\begin{aligned} N &= \operatorname{rank} P_4 = \varphi(7^4) = 2058, \\ p(r_1) &= \frac{1}{N}, \\ p(r_2 \mid r_1) &= \frac{1}{N}, \\ p(r_1, r_2) &= \frac{1}{N^2}. \end{aligned} \tag{8.33}$$

The two filtering stages compose to $R = P_e P_x$. From $Re = e/N$, the double-success share is $\|Re\|^2 = N^{-2}$, while the relay background is restored. These are two different matches on the same preparation and require no intermediate repreparation; the complete instrument does not thereby become two statistically independent trials. Both flags participate as reversible working registers in the subsequent complete return, whose inverse connection adds no matching probability. Repeating the same projection $P^2 = P$ instead gives only one share. Here N is the bare mode count of the selected four-layer participating space. Neither the weighted background remainder 2039.7538… nor the complete construction depth K is the denominator of this matching. See Supplementary Proof S13.2 for the complete instrument.

Connecting the two spectral responses, two coherent nodes, and two-stage filtering effect to the persistent action of Section 2.3.5 gives

$$A_\nu = |L_2|^8 \frac{1}{2N^2}, \qquad |L_2|^8 = 0.000105853314\ldots. \tag{8.34}$$

N^{-2} is the complete success effect of this relay. Simultaneous protection lets it enter the persistent generator together with the spectral responses and node matrix elements. The actual mass is then read using the total elapsed duration. Relative sizes of the three branches are determined by the relay pattern and differences among the point-pair chains.

8.5.2 Spectral Kernel of the Indirect Channel

First calculate the kernels of the three point-pair chains on the species layer $K = 1$. For $a = 1, \ldots, 6$ and spectral variable $s > 1$, define the residue-class spectral sums. Here s is used separately from the accumulated action in Section 8.1.2:

$$\zeta_a(s) := \sum_{n \equiv a \pmod 7} n^{-s} = 7^{-s}\zeta_{\mathrm{H}}(s, a/7), \qquad \langle 1|\widetilde{\mathcal{B}}(s)|a\rangle = \zeta_a(s). \tag{8.35}$$

The second expression uses $T_n f(x) = f(nx)$, since $\langle 1|T_n|a\rangle$ equals 1 when $n \equiv a \pmod 7$ and 0 otherwise. This is the matrix element from input a to output 1. At deeper layers, the corresponding readout uses equal-amplitude lifts of the species

layer; the modulo-7 sum here is not relabeled as the matrix element of a single deeper lattice point.

The actual neutral spectral transport stage transports the entire three-dimensional odd sector U_- of the species layer. Each unit uses $n^{-1/2}$ along multiplier n. Its top exterior-power amplitude is $\det(n^{-1/2}T_n|_{U_-}) = \chi_3(n)n^{-3/2}$. Different multipliers remain orthogonal working outputs, so the transfer column $w_a = \sum_{n\equiv a\ (7)} \chi_3(n)n^{-3/2}|n\rangle$ at address a gives $\|w_a\|^2 = \zeta_a(3)$. Here w_a is the actual transfer column, defined separately from $\widetilde{\mathcal{B}}(s)$ in the amplitude sum above. The spectral exponent 3 is jointly produced by the adopted whole-sector transport and intensity readout.

The first transport stage writes the point-pair chain label a, and the second transport stage transports its partner $7-a$ along the same label. For the three point pairs $\{1,6\}$, $\{2,5\}$, and $\{3,4\}$, multiply the weights of the two transport stages. To distinguish them from w_ℓ in the coupling chapter, denote the chain weights by h_a:

$$h_a := \zeta_a(3)\zeta_{7-a}(3), \quad a = 1,2,3, \qquad Q_\nu := \mathrm{diag}(h_1,h_2,h_3). \tag{8.36}$$

These series converge absolutely and all weights are positive. The choice of the three-dimensional odd sector is part of this neutral channel's transport rule. Given that rule, the exponent and chain weights follow from Supplementary Proof S13.1. Their values are h_1=0.005349484164…, h_2=0.001123729206…, and h_3=0.000641318453….

The relay uses the three group components $f_j(\delta_*) = 1 + \sqrt{2}\cos(\delta_* + 2\pi j/3)$ of the even-conjugate pattern and connects the three chains in cyclic order. Initially retaining an arbitrary real phase δ_* gives

$$C_\nu(\delta_*) = \begin{pmatrix} f_0 & f_2 & f_1 \\ f_1 & f_0 & f_2 \\ f_2 & f_1 & f_0 \end{pmatrix}, \qquad \boxed{\mathcal{M}_\nu = C_\nu Q_\nu C_\nu^T.} \tag{8.37}$$

Column a of C_ν contains the three output components of chain a. Taking outer products and summing with weights h_a gives the dimensionless positive kernel $\mathcal{M}_\nu$. The complete transfer columns of the two transport stages retain the full weights of all three chains; columns are neither normalized individually nor averaged with equal probabilities across the chains. Chain and multiplier labels are orthogonal in intermediate states and continue to participate in the inverse action and protection. They are not delivered midway as uncontrollable permanent records. The Gram matrix of the complete output therefore gives the expression above while still admitting connection into a coherent persistent action. The pattern phase determines each column's components, and the chain weights determine its interaction strength.

8.5.3 Neutrino Mass Spectrum

From $\sum_j f_j = 3$ and $\sum_j f_j^2 = 6$, the inner product of different cyclic columns is $\sum_{i<j} f_i f_j = [(\sum_j f_j)^2 - \sum_j f_j^2]/2 = 3/2$. Thus

$$C_\nu^T C_\nu = \frac{9}{2} I_3 + \frac{3}{2}\mathbf{1}\mathbf{1}^T =: G_\nu, \qquad \mathbf{1} = (1,1,1)^T. \tag{8.38}$$

The right-hand side does not contain δ_*. Let $X = C_\nu Q_\nu^{1/2}$; then $\mathcal{M}_\nu = XX^T$, while $X^T X = Q_\nu^{1/2} G_\nu Q_\nu^{1/2}$. The two have the same nonzero eigenvalues. Positive

definiteness of G_ν and $h_a > 0$ also ensure that X is invertible. Denoting the three eigenvalues of this dimensionless chain kernel in increasing order by ε_i gives

$$\begin{aligned} \operatorname{spec}(\mathcal{M}_\nu) &= \operatorname{spec}\bigl(Q_\nu^{1/2} G_\nu Q_\nu^{1/2}\bigr)\,, \\ \varepsilon_1 &= 0.003296381026\ldots, \\ \varepsilon_2 &= 0.006475614825\ldots, \\ \varepsilon_3 &= 0.03291519509\ldots. \end{aligned} \tag{8.39}$$

For fixed h_a and this cyclic connection, shifting the formation phase does not change the chain kernel's eigenvalues. The matrix itself and its eigenvectors may still vary with phase, so this result does not also fix mixing angles. Changes to chain weights, relative background amplitude, or the connection require the spectrum to be recalculated.

The actual persistent action retains the entire three-dimensional family space and its orientation partner Y. Its generator within the code is

$$h_\nu(t) = g(t)\frac{p_0}{\Lambda}\frac{\kappa_{\mathrm{port}}|L_2|^8}{2N^2}\,\mathcal{M}_\nu \otimes Y. \tag{8.40}$$

Here p_0 comes from routing through the same reference group as the background, and Λ is the common capacity of both ends. Ports outside the nodes use the equal-amplitude binary flags $|+\rangle$ and $P_q = |0\rangle\langle 0|$, so $\kappa_{\mathrm{port}} = \langle +|P_q|+\rangle = 1/2$. Together with $1/2$ from the product of the two coherent node matrix elements, this port share supplies the two factors of one half in the total coefficient. Port normalization is specified by the actual preparation. Protective intertwining of the complete output code ensures that the matrix above generates all maintenance phases; masses are therefore given by the eigenvalues of $\mathcal{M}_\nu$. See Supplementary Proof S13.3.

One round completes, in order, the two transport stages and relay, maintenance at the output, the complete inverse connection, repair, and thermal processing. Matter and background are active together in the same window; all other processing pauses them together and fills fixed-duration processing intervals. Denote the complete cycle period by T and the accumulated drive at both ends by Θ_ν, Θ_B. The normalization separates as

$$\eta_\nu = \frac{\Theta_\nu}{\Theta_B} = 1, \qquad \kappa_\nu = \eta_\nu \kappa_{\mathrm{port}} = \frac{1}{2}, \qquad \frac{p_0 \Theta_B}{\Lambda T} = v_E. \tag{8.41}$$

The last equality uses the background calibration of Section 2.5.3 for the same complete schedule. Numerical values in this chapter use v for that energy scale. κ_ν is defined separately from the per-round phase correction $\kappa_{f,j}$ of Section 8.1. Both relay stages are already included in N^{-2}; the duration of the inverse connection enters T, with no additional return factor of one half. Loading and thermal survival modify the delivery yield and survival envelope, while continuously surviving states remain maintained by the same generator. For the proof retaining complete records, see Supplementary Proofs S13.4–S13.5. This neutral action contains no charged proportional frequency shift F, yielding

$$M_\nu^{\mathrm{phys}} = v A_\nu \kappa_\nu \mathcal{M}_\nu, \qquad \boxed{m_{\nu_i} = \frac{v|L_2|^8}{4N^2}\,\varepsilon_i, \quad \kappa_\nu = \frac{1}{2}, \quad i = 1,2,3.} \tag{8.42}$$

The chain kernel's three eigenvalues thus become three masses, with the absolute scale fixed by calibration using a common source. Proofs for the three-chain transport, nodes, port flags, and complete return are collected in Supplementary Proof S13. Calculation gives

$$(m_{\nu_1}, m_{\nu_2}, m_{\nu_3}) = (5.070653857,\ 9.961106144,\ 50.63175635)\ldots\ \text{meV},$$
$$\sum_i m_{\nu_i} = 65.66351635\ldots\ \text{meV}. \tag{8.43}$$

Eigenvalues are numbered from smallest to largest, with the physical identification of normal ordering. The three absolute masses and their sum are model results under these conditions. Oscillation experiments directly constrain mass-squared differences; $\sqrt{\Delta m^2_{31}}$ cannot be treated as an already measured third absolute mass.

8.6 Testing the Mass Spectrum

The families share one energy calibration, while their mass hierarchies arise from formation projections, paths and nodes, and return or delivery responses. Below we fix all evaluation prescriptions of this chapter, retaining the previously stated differences between finite-link and feedback realizations. Experimental data serve as comparison references. Charged-lepton masses, quark masses in specified renormalization schemes, and neutrino oscillation quantities are each tested according to their physical definitions.

8.6.1 Charged Elementary-State Masses

The following table uses the PDG 2025 lepton summary [17] and quark summary [17]. The u, d, and s masses are M$\bar{S}$ masses at 2 GeV; c and b use $m_Q(m_Q)$ in the $\overline{\text{MS}}$ scheme. Uncertainties for these five entries are 90% confidence intervals, following that summary. The t entry uses the direct-reconstruction combination. Its mass parameter must be matched through the relevant analysis to rigorously defined pole and M$\bar{S}$ masses (PDG top review §61.1.1) [24]. Deviations in the table use unrounded model values and are given relative to the reference central values.

Table 8-2 Conditional evaluations for nine charged elementary states compared with PDG 2025. All masses are in MeV. Parenthetical uncertainties apply to the final digits; other uncertainties and schemes are specified in the text.

Particle	Model value (MeV)	Reference value (MeV)	Relative deviation
e	0.5110122512	0.51099895000(15)	+0.002603%
μ	105.6553711	105.6583755(23)	−0.002844%
τ	1776.926974	1776.93 ± 0.09	−0.000170%
d	4.651213747	4.70 ± 0.07	−1.038%
s	92.10919837	93.5 ± 0.8	−1.487%
b	4185.420021	4183 ± 7	+0.05785%
u	2.154453597	2.16 ± 0.07	−0.2568%
c	1262.842920	1273.0 ± 4.6	−0.7979%

Particle	Model value (MeV)	Reference value (MeV)	Relative deviation
t	173055.3957	172560 ± 310	+0.2871%

The three lepton values are relatively close to the reference central values, but the electron and muon residuals still exceed experimental uncertainties. The quark entries additionally require matching the model mass operator to the listed schemes and scales. Until that matching is established, the percentages express only differences from the chosen numerical references and cannot serve as rigorous goodness-of-fit measures.

8.6.2 Mass-Spectrum Relations

The charged leptons use the same positive coefficient m_τ/f_0^2. Since f_0, f_1, and f_2 are all positive on the present branch, square-root masses are proportional to f_j. Using $\sum_j f_j = 3$ and $\sum_j f_j^2 = 6$ then gives

$$\boxed{\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{\sum_j f_j^2}{(\sum_j f_j)^2} = \frac{2}{3}.} \tag{8.44}$$

This algebraic form coincides with the charged-lepton mass relation proposed by Koide [25]. The spectral-shape relation is independent of v and the magnitude of the common frequency shift, but depends on equal amplitudes of the background and normalized conjugate combination, the same generation identification, and the amplitude-sign branch. If any f_j is negative, the mass square root must use $|f_j|$, and the denominator can no longer use $(\sum_j f_j)^2$.

Relations between different families also allow common factors to cancel directly. The charged leptons and two reference quark branches give

$$\frac{m_b}{m_\tau} = \frac{3L_3}{|\zeta_0|} = 2.35542601468\ldots, \qquad \frac{m_c}{m_\tau} = L_3^2|L_1|^4 = 0.710689261785\ldots. \tag{8.45}$$

The flip and delivery branches satisfy

$$\begin{aligned} \frac{m_d}{m_s} &= \lambda^2, \\ \frac{m_t}{m_c} &= \frac{1}{\alpha_{\mathrm{EM}}}, \\ \frac{m_u}{m_c} &= \lambda^2 \alpha_W, \end{aligned} \tag{8.46}$$

$$\boxed{\frac{m_t m_u}{m_c^2} = \frac{\lambda^2 \alpha_W}{\alpha_{\mathrm{EM}}} = 0.233788757123\ldots.}$$

These equalities are algebraic results under the stated evaluation conditions, testing the consistency of nodes, generational factors, and responses. For observational comparison, model masses appearing in b/τ and the up-type cross-ratio must first be physically matched to the mass definitions, renormalization schemes, and scales listed in Section 8.6.1, then tested against the corresponding data.

8.6.3 Neutrino Mass-Squared Differences

The three model neutrino masses were listed in Section 8.5. Defining $\Delta m^2_{21} = m^2_{\nu_2} - m^2_{\nu_1}$ and $\Delta m^2_{31} = m^2_{\nu_3} - m^2_{\nu_1}$ for normal ordering gives

$$\begin{aligned} \Delta m^2_{21} &= 7.351210508 \times 10^{-5}\ \mathrm{eV}^2, \\ \Delta m^2_{31} &= 2.537863220 \times 10^{-3}\ \mathrm{eV}^2, \\ \boxed{\frac{\Delta m^2_{31}}{\Delta m^2_{21}} = \frac{\varepsilon_3^2 - \varepsilon_1^2}{\varepsilon_2^2 - \varepsilon_1^2} = 34.52306552\ldots.} \end{aligned} \tag{8.47}$$

To fix the data combination, we use the normal-ordering IC24 set including SK atmospheric data from NuFIT 6.0 (2024) [26]. The 1σ intervals in its parameter table [26] are compared with the current model values below. Best-fit values vary slightly across combinations; the same input set must be retained in each comparison.

Table 8-3 Oscillation quantities for normal ordering. Experimental entries use the NuFIT 6.0 IC24 combination including SK atmospheric data and its 1σ intervals.

Quantity	Model value	Reference value
Δm^2_{21} (10^{-5} eV^2)	7.35121	7.49 ± 0.19
Δm^2_{31} (10^{-3} eV^2)	2.53786	2.513 (+0.021 / −0.019)

The ratio of mass-squared differences cancels common factors such as v, the two matches, and κ_ν, and therefore focuses on the relative spectrum of the chain weights and cyclic matrix. A common activity ratio and port flags change only the overall scale. Noncommon weighting across chains, transport rules, and general finite truncations can change the relative spectrum; see Supplementary Proof S13.6. The absolute values of the two mass-squared differences also test the overall mass scale. Two marginal 1σ intervals alone cannot assign an independent Gaussian error to their ratio; a rigorous joint test requires the correlated likelihood from the same oscillation analysis.

The common background energy scale combines with each family's paths, nodes, and branch actions to give this chapter's mass spectrum. The charged-lepton hierarchy comes from three group projections of the same source. Down-type branches distinguish real-direction readouts and a flip link; up-type branches distinguish a reference, returns, and delivery; neutrinos read the three eigenvalues of an indirect chain kernel. The comparisons above separately test spectral-shape relations, the overall scale, and their correspondence with actual physical masses.

The mass spectrum above covers the selected nine charged elementary states and three neutrino branches. The common nucleon scale was calculated from collective participation in Section 6.3. Baryon formation must additionally explain how color components form a whole stable pattern, how internal resonance maintains it, and why an individual color component cannot independently become the same kind of output. These structures determine how elementary states enter composite matter.

§9 Baryons, Quarks, and Color Confinement

Baryon formation requires internal color components to enter a whole pattern together and to continue maintaining that whole after exchange. Nucleons have both a common collective scale and distinguishable charge, species, and spin readings. To place these properties in one state, we must specify what the full-space mode carries, what the internal representation carries, and how the cycle preserves both simultaneously.

Retaining the $p = 7$ identification of Chapter 5, the complete nucleon object consists of the common mode b and internal color, species, and spin states. Cross-shell coherence specifies the whole pattern, the internal representation distinguishes proton from neutron, and the actual return determines both phase and survival. We use this object to calculate the common scale, internal mass difference, and exchange response, then test whether a colored component can become an independent output.

9.1 Collective Baryon Structure

A nucleon's integrity is first expressed by its participating components jointly producing one maintainable pattern. The full-space vector b of Chapter 2 provides a candidate form of such collective participation. It is invariant under all multiplicative pullbacks and gives every address the same amplitude. These addresses label modes in state space; position in observational spacetime is still determined by the corresponding readout. Participation of every address does not directly specify the nucleon's spatial size.

9.1.1 The Full-Space Collective Pattern

First retain an arbitrary finite depth K≥1. Write $d = p^K$, with normalized shell-average states e_j and shares $h_j = |S_j|/d$. The common pattern and its shell expansion are

$$b = \frac{1}{\sqrt{d}} \sum_{x \in X_K} |x\rangle = \sum_{j=0}^{K} \sqrt{h_j}\, e_j, \qquad h_j = \frac{|S_j|}{d}, \qquad \sum_j h_j = 1. \tag{9.1}$$

For the multiplicative pullback $(T_n f)(x) = f(nx)$, a constant function clearly satisfies $T_n b = b$. Conversely, if f is fixed by every T_n, then in particular $T_p f = f$. Repeating K times gives $f(x) = f(p^K x) = f(0)$. Thus the common fixed space is exactly one-dimensional:

$$\boxed{\bigcap_{n \geq 1} \ker(T_n - I) = \operatorname{span}\{b\}.} \tag{9.2}$$

Invariance under all multiplications includes cross-shell scaling, and T_p alone already determines the unique direction above. Invariance under the unit group alone permits different constant values on different shells; the fixed space remains $\mathcal{C} = \operatorname{span}\{e_0, \ldots, e_K\}$, of dimension K+1. The unit-shell average state e_0 belongs to $\mathcal{C}$ but differs from b. Cross-shell action links the independent shell constants, leaving the same full-space pattern.

T_n describes the arithmetic response structure here. When p divides n, it is generally not unitary. The common fixed-point property of all T_n determines the arithmetic structure of b. Formation and maintenance of this pattern in the actual cycle still require examination of the joint action of formation exchange, coherent retrieval, and background restoration, as well as establishment of the required coherence between shells.

In b, all shell phases are aligned to the same reference. Writing the mode density operator as ρ, the matrix element between shells j and ℓ can be tested through two self-adjoint observables:

$$A_{j\ell} = |e_j\rangle\langle e_\ell| + |e_\ell\rangle\langle e_j|, \qquad B_{j\ell} = i\big(|e_j\rangle\langle e_\ell| - |e_\ell\rangle\langle e_j|\big), \quad j < \ell. \tag{9.3}$$

Their expectations are $2\Re\langle e_j|\rho|e_\ell\rangle$ and $2\Im\langle e_j|\rho|e_\ell\rangle$, respectively. For b, the former is $2\sqrt{h_j h_\ell}$ and the latter zero; for the shell-dephased background σ_h^{sh}, both vanish. The shell shares are identical, but the cross-shell response still distinguishes these preparations. Actually making this distinction requires a joint readout realizing these operators and retention of the phase reference used.

9.1.2 Internal Degrees of Freedom of Collective States

The common pattern b occupies only one ray. A single pure-state ray cannot distinguish proton from neutron; their difference must be represented by internal states and the corresponding observables. The color part may share one singlet, while the remaining species, spin, and associated spatial parts must be organized according to the complete representation.

To clarify this organization, distinguish the mode factor $\mathcal{H}_K$ from the internal factor $\mathcal{H}_{\mathrm{int}}$ in the specified joint carrier, and denote the normalized joint state by Ω_B. This tensor decomposition differs from the direct-sum decomposition into shell-average and direction spaces. It is a condition of the physical identification, requiring the existing color, species, and other degrees of freedom to be implemented in actual carriers and actions. The fixed-vector property of b does not itself construct this decomposition. If the reduced mode state equals P_b exactly, the complete state must have

$$\boxed{\mathrm{Tr}_{\mathrm{int}}\,\Omega_B = P_b \quad\Longleftrightarrow\quad \Omega_B = P_b \otimes \sigma_B, \qquad \sigma_B \geq 0, \quad \mathrm{Tr}\,\sigma_B = 1.} \tag{9.4}$$

Let $Q_b = (I - P_b) \otimes I$. By positivity, $\mathrm{Tr}(Q_b\Omega_B) = \|Q_b\Omega_B^{1/2}\|_{\mathrm{HS}}^2 = 0$ forces Ω_B to be supported on $\mathrm{span}\{b\} \otimes \mathcal{H}_{\mathrm{int}}$, so the normalized state must be $P_b \otimes \sigma_B$. The converse follows by taking the partial trace.

The internal state may be entangled or mixed. If the mode and internal factors are nontrivially entangled, the mode marginal is no longer P_b and must be calculated from the complete joint state.

In this representation, the expectation of a mode observable O is determined by the common pattern, whereas that of an internal observable E is determined by the internal preparation:

$$\mathrm{Tr}[\Omega_B(O \otimes I)] = \langle b|O|b\rangle, \qquad \mathrm{Tr}[\Omega_B(I \otimes E)] = \mathrm{Tr}(\sigma_B E). \tag{9.5}$$

Different internal states σ_A and σ_C can produce different internal responses while preserving the same mode readings. Proton and neutron labels must therefore be

accompanied by concrete internal states and distinguishable readouts. Error-free distinction in a single measurement requires orthogonal supports, and the actual instrument must give definite and different outcomes on those supports. For necessity and sufficiency over the complete measurement class, see Supplementary Proof S4.1. The one-dimensionality of the common pattern does not determine the number of particle species.

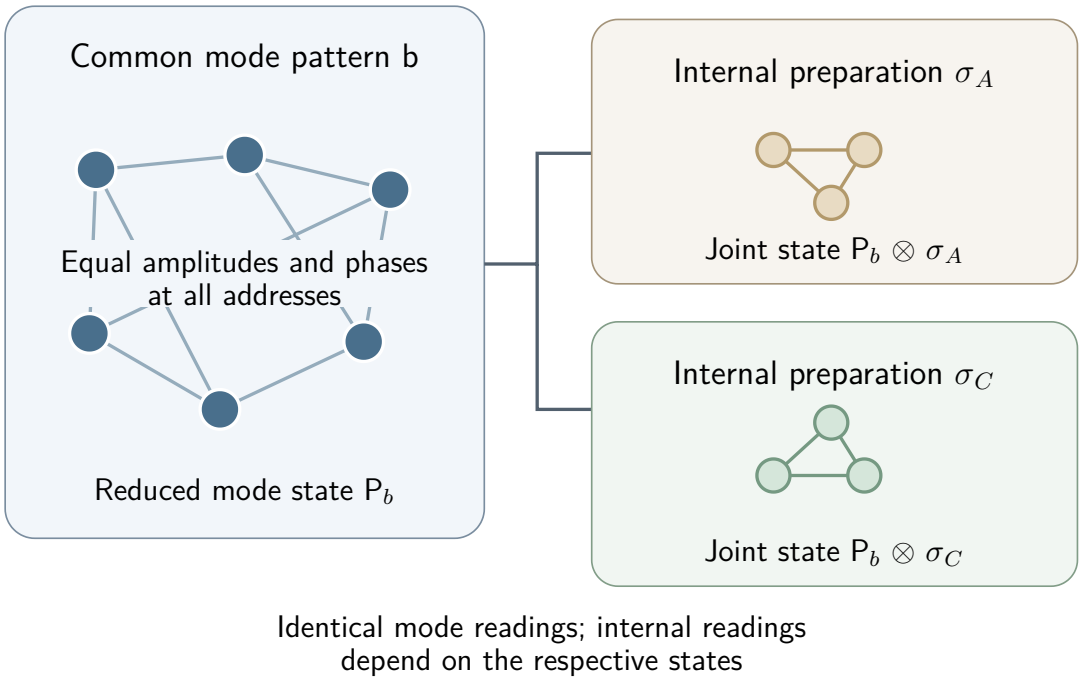


Figure 9-1 Common collective pattern and internal distinctions. Amplitudes at the addresses on the left jointly form b; the two internal preparations on the right can be paired with the same mode pattern. Lines indicate composition into a joint state under the specified tensor decomposition, not time evolution. Internal states and their readouts must be realized in actual carriers.

In the candidate nucleon identification, this division of roles allows a common collective scale to coexist with an internal mass difference: overall participation uses the same mode structure, while internal responses vary with species and exchange paths. A common mode does not require the complete generator to act only on the mode factor, and therefore does not require two internal states to have identical cycle phases. Specific charge and color readings can be calculated only after the corresponding internal representations have been fixed.

9.1.3 Cross-Shell Coherence

Following the common-final-state criterion of Section 2.5.4, formation must preserve the relative phases of the shells within one collective pattern. For a normalized joint component,

$$|\Psi\rangle = \sum_{j=0}^{K} |e_j\rangle \otimes |\xi_j\rangle, \qquad \sum_j \|\xi_j\|^2 = 1, \qquad \rho_{\text{mode}} = \sum_{j,\ell} \langle \xi_\ell | \xi_j \rangle |e_j\rangle\langle e_\ell|. \tag{9.6}$$

Actually producing P_b requires precisely

$$\boxed{\rho_{\mathrm{mode}} = P_b \quad \Longleftrightarrow \quad \xi_j = \sqrt{h_j}\,\xi \;\; (0 \le j \le K), \qquad \|\xi\| = 1.} \tag{9.7}$$

Then $\Psi = b \otimes \xi$, while the internal factor and remaining environment may still be correlated. If formation has several outcomes, the total formation share sums mutually exclusive outcomes satisfying this criterion. Orthogonal records of different shells yield only mixed restoration, as explained in Section 2.5.4.

This common final state has a finite energy-conserving realization. Take an isospectral reference copy of the shell-average space, set $H_R e_j^R = E_j e_j^R$, and prepare $b_R = \sum_j \sqrt{h_j} e_j^R$. Swap the two shell-average spaces while bypassing the direction space:

$$U_R(e_j \otimes e_k^R) = e_k \otimes e_j^R, \qquad [U_R, H_K + H_R] = 0, \qquad U_R(c \otimes b_R) = b \otimes c_R. \tag{9.8}$$

The joint energy before and after the swap is $E_j + E_k$. Both reference and shell-average states carry trivial group charge, so group charge is also balanced. Retaining color, flavor, spin, and spectator correlations in the internal factor, and tracing out the reference, gives

$$\boxed{(\mathcal{R} \otimes \mathrm{Id})(\rho_{\mathcal{C},\mathrm{int}}) = P_b \otimes \mathrm{Tr}_{\mathcal{C}}\, \rho_{\mathcal{C},\mathrm{int}}.} \tag{9.9}$$

This swap works for any input supported on the shell-average space, including its entanglement with the internal factor: the original mode information enters the reference output, the internal marginal is preserved, and the cross-shell coherence of b_R is transferred completely to the collective pattern. Thus with a coherent reference b_R, collective formation already has a finite energy-conserving realization. A mixed reference with the same occupations instead produces the corresponding mixed state.

After formation, the reference is generally c_R and must actually be prepared anew, or the swap must stop; it cannot immediately be reused unchanged. If both ends already contain b and b_R, the swap returns both ends together. A bare H_K free segment still changes relative shell phases, so the persistent action must next be connected to the same pattern. The internal representation and readout are carried by the same internal state below.

9.2 Baryon Quantum Numbers

The collective pattern supplies the common mode structure, while the internal color singlet determines which color components can form a whole invariant under changes of color reference. Nucleons must also have definite flavor and spin readings. Writing these parts as one candidate state allows their transformation properties to be checked directly, without inferring internal structure from a name.

9.2.1 The Three-Color Singlet

Use the color carrier $V_c = \mathbb{C}^3$ specified in Sections 5.2 and 6.3, with the fundamental SU(3) representation U as the color action. The species-space action $\mathrm{I}_2 \otimes \mathrm{U}$ of Section

5.2.3 and U and U⊗U⊗U here are representations of the same abstract group on different carriers. The complete matter representation must still embed color-singlet, three-dimensional, and conjugate carriers into the established construction compatibly with species labels, physical charges, and exchange vertices. In the specified three copies of the fundamental representation, the normalized singlet is

$$|\varepsilon_3\rangle = \frac{1}{\sqrt{6}} \sum_{a,b,c=1}^{3} \varepsilon^{abc} |a, b, c\rangle, \qquad U^{\otimes 3}|\varepsilon_3\rangle = \det(U)|\varepsilon_3\rangle = |\varepsilon_3\rangle. \tag{9.10}$$

The completely antisymmetric tensor uses each distinct label once among the three color indices. It spans the top exterior power of the fundamental representation, on which the action is the determinant representation and hence trivial under the special unitary group [27]. Section 6.3 proved that the SU(3)-invariant subspace of three fundamental representations is precisely this ray. One or two fundamental representations have no singlet, and allowed nonzero numbers of copies are positive multiples of 3. This result is restricted to the representation used and does not exclude a two-factor singlet formed from a fundamental and a conjugate representation.

Neutrality of the cyclic grading is a weaker condition. Averaging only over the cyclic subgroup of order three tests whether the gradings sum to zero. For example, gradings 1 and 2 sum to zero but still cannot form an SU(3) singlet in $V_c \otimes V_c$. A full color singlet must be tested against all color actions. Antisymmetry of the three copies follows from this invariant representation, without yet invoking Fermi statistics.

The three tensor positions here label the three factors of the internal representation. The color singlet determines only the color part; flavor, spin, and spatial parts must be organized separately. Having one of each color does not require three distinct flavor labels, so it is compatible with both uud and udd but does not by itself select these combinations.

9.2.2 Flavor and Spin Structure

Quark flavor names and fractional charges follow the usual quark-model conventions [28]. We now give an explicit carrier for the internal nucleon readings. Take the internal flavor space span$\{u, d\}$ spanned by u and d, with charges $2/3$ and $-1/3$ as adopted in Section 5.2. Independently take the two-dimensional SU(2) representation for spin, with basis $\uparrow$, $\downarrow$ and $\hbar$=1. The spin representation and fermionic exchange rule here are conditions of the physical identification; they do not follow from the binary CRT label alone.

On the three tensor positions, two normalized vectors with total spin projection $+1/2$ can be chosen as

$$\eta_\rho = \frac{|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{2}}, \qquad \eta_\lambda = \frac{2|\uparrow\uparrow\downarrow\rangle - |\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{6}}. \tag{9.11}$$

These vectors are orthogonal and both satisfy $S_z\eta = \eta/2$ and $S_+\eta = 0$. Using $S^2 = S_-S_+ + S_z(S_z + 1)$ gives $S^2\eta$=$3\eta/4$, so they indeed belong to spin-1/2 representations. Use the same permutation forms for the two flavor combinations:

$$\phi_{p,\rho} = \frac{|udu\rangle - |duu\rangle}{\sqrt{2}},$$
$$\phi_{p,\lambda} = \frac{2|uud\rangle - |udu\rangle - |duu\rangle}{\sqrt{6}}, \tag{9.12}$$
$$\phi_{n,a} = \phi_{p,a}\big|_{u\leftrightarrow d}$$
$$(a = \rho, \lambda).$$

Every flavor component of the proton candidate contains two u and one d; the neutron candidate has the reverse composition. Here ρ and λ label only two permutation vectors and are unrelated to the thermal density operator or the flip parameter of Chapter 8. Let

$$\Phi_{N,\uparrow} = \frac{\phi_{N,\rho} \otimes \eta_\rho + \phi_{N,\lambda} \otimes \eta_\lambda}{\sqrt{2}}, \qquad N = p, n. \tag{9.13}$$

When two flavor positions and their corresponding spin positions are exchanged together, both sets of ρ and λ vectors transform by the same real orthogonal matrix, leaving the inner-product combination above invariant. This uses the two-dimensional representation of the permutation group on three positions and its invariant tensor-product pairing [29]. Specifically, exchanging the first two positions gives diag(−1,1), and exchanging the last two gives the matrix below; together they generate every permutation of the three positions:

$$D_{23} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \qquad D_{23}^T D_{23} = I_2, \qquad P_\pi \Phi_{N,\uparrow} = \Phi_{N,\uparrow} \quad (\pi \in S_3). \tag{9.14}$$

Provided the common mode b transforms trivially under permutations of these three internal positions and the remaining spatial part is symmetric, normalized candidate states can be written as follows, omitting the additional symmetric spatial factor:

$$\boxed{|B_{N,\uparrow}\rangle = b \otimes |\varepsilon_3\rangle \otimes \Phi_{N,\uparrow}, \qquad N = p, n.} \tag{9.15}$$

Under simultaneous permutation of the three internal positions, the color part acquires the permutation sign while the flavor–spin part remains invariant. The complete internal state is therefore antisymmetric and satisfies the specified fermionic exchange rule. If the spatial part has another permutation type, the flavor–spin part must change accordingly. This construction proves the existence of compatible candidate states in the given representations. Their actual preparation and cycle maintenance must still be checked as in Sections 9.1 and 9.3.

9.2.3 Charge Structure

Write the charge operator on flavor space as q=diag(2/3,−1/3). The total charge of the three internal readings is the sum of its actions at the three positions. Since every candidate vector has fixed numbers of u and d,

$$Q = q \otimes I \otimes I + I \otimes q \otimes I + I \otimes I \otimes q, \qquad \boxed{Q\Phi_{p,\uparrow} = \Phi_{p,\uparrow}, \qquad Q\Phi_{n,\uparrow} = 0.} \tag{9.16}$$

Q acts as the identity on spin, color, and the common mode. An overall color singlet can therefore simultaneously have nonzero charge; the two conditions act on different representations. The relation of flavor labels u and d to the directions χ_1 and χ_5 of Chapter 8 still uses the physical identifications of Section 5.2. Character labels alone do not automatically produce this additive charge operator.

The reduced color state at any one position is $\mathrm{I}_3/3$. This follows by tracing the ϵ tensor term by term: each color appears in exactly two nonzero permutations, and off-diagonal terms vanish. For the globally pure color-singlet preparation here, the reduced mixed state reflects internal correlations. A local reading obtained by an internal projection does not mean that the corresponding component has separated from the whole and become an independently maintainable output.

9.3 Total Mass and Maintenance

Collective nucleon participation and internal actions jointly enter persistent evolution. Relative to the same background reference, collective participation supplies a reference frequency shared by both nucleons, while internal actions give their branches different phases. Return and exit shares determine survival of the original excitation. Mass and survival are read from the phase and return share of the same complete cycle, respectively.

9.3.1 Collective Mass Scale

Using the complete logarithmic spectrum, prescription based on the leading term of the variance, and common-background matching of Section 6.3, the lowest singlet of three fundamental color representations uses $\beta_B = 1 + 1/\sqrt{3}$. By Eq. (6.40), its participation and common energy scale are

$$W_B = \zeta(\beta_B),$$

$$\boxed{M_B = E_0\sqrt{\frac{W_B}{C^*}}\log W_B = 936.669312626\ldots\ \mathrm{MeV}}, \tag{9.17}$$

$$E_0 = 1\ \mathrm{GeV}.$$

This value inherits the spectral preparation and approximation conditions of Section 6.3. β_B uses the prescription that the leading term of the variance equals 3; an exact variance condition would give a different value. Finite address depth, the cutoff on the positive-integer labels, and the full ζ sum have separate definitions and do not automatically coincide because the same collective pattern is used.

Mass reads the phase slope of the returning branch relative to the common reference. As in Eq. (2.59)–Eq. (2.60), the physical duration τ_B of one round includes all processing and waiting. ϕ_N is the single-branch phase relative to the calibrated zero, unwrapped continuously along the evolution. If the total beat phase between two symmetric orientation branches is read, it must first be halved. Writing the common scale and internal frequency shifts within this candidate subspace gives

$$m_N = -\frac{\varphi_N}{\tau_B}, \qquad \varphi_N = -\tau_B(M_B + \delta_N), \qquad N = p, n. \tag{9.18}$$

δ_N contains the internal response relative to the common reference. Actual mass readout must retain coherence with an internal clock or another branch; a

pure-state projection alone does not display a common phase. The collective-mass identification here requires M_B to become the actual persistent frequency of pattern b. Color invariance and a constant pattern determine only state structure; frequency also depends on the action maintaining it.

Under this identification, uud and udd specify internal charge and flavor content, while the common collective energy scale is supplied by the participation weights of Section 6.3. Comparing component masses, interaction energy, and total mass requires a decomposition of the same generator and its expectation values. Binding properties likewise require an explicit description of components and interactions.

9.3.2 Maintenance Frequency and Collective Mass

The outward and return parts of the spectral path must both generate phase and preserve the common mode b and its internal color state. Maintenance must therefore cover the complete mode, port, and internal carrier, so that participation weights enter actual exchange matrix elements. The frequencies generated by these matrix elements are then compared with the background response in the same process.

Fix spectral cutoff $L \geq 2$ and $\beta > 1$, and let $w_n = n^{-\beta}$ and $Z_L = \sum_{n=1}^{L} w_n$. Retain orthogonal auxiliary ports $z, 1, \ldots, L$ outside the mode pattern b, connecting multiplicative paths T_n to their corresponding outputs. Let $J_b\psi = b \otimes \psi$, where ψ includes all ports and internal degrees of freedom; identity actions on internal factors are omitted. Following Eq. (2.38)–Eq. (2.39), apply coherent, simultaneous, equal-strength two-arm protection on this fixed complete embedding:

$$\begin{aligned} A_B &= (\log Z_L) \sum_{n=1}^{L} \sqrt{w_n}\big(T_n \otimes |n\rangle\langle z| + T_n^\dagger \otimes |z\rangle\langle n|\big), \\ S_b &= 2J_bJ_b^\dagger - I, \qquad \widehat{A}_B = \tfrac{1}{2}(A_B + S_bA_BS_b). \end{aligned} \tag{9.19}$$

Obtain the compressed matrix elements from $T_nb = b$, then use $S_bJ_b = J_b$ and $(I + S_b)/2 = J_bJ_b^\dagger$ to obtain the complete intertwining relation and bright block:

$$\begin{aligned} a_B &= (\log Z_L) \sum_n \sqrt{w_n}|n\rangle, \qquad X_B = |a_B\rangle\langle z| + |z\rangle\langle a_B|, \\ \boxed{\widehat{A}_BJ_b = J_bX_B, \qquad X_B|\pm_B\rangle = \pm r_B|\pm_B\rangle,} &\qquad r_B = \sqrt{Z_L}\log Z_L, \quad |\pm_B\rangle = \frac{|z\rangle \pm a_B/r_B}{\sqrt{2}}. \end{aligned} \tag{9.20}$$

Equation (9.20) preserves pattern b throughout the outward and return paths, while the ports and internal states evolve under X_B. Simultaneous protection covering every port promotes the compressed matrix elements to a complete intertwining relation. The specified port coherence, common coefficient $\log Z_L$, and path amplitudes $\sqrt{w_n}$ jointly give the dimensionless eigenvalue r_B on the bright subspace. Nonunitary pullbacks are realized here as matrix elements of a self-adjoint action; required energy compensation or control work is included as in Section 4.4.3.

Couple $g\widehat{A}_B$ to the existing self-adjoint unitary orientation operator Q and prepare the positive bright branch. Target displacement leaves the mode b, path ports, and color singlet as spectators, and the velocity orientation anticommutes with Q. The shared motion port therefore gives $gr_BQ + 2\omega ZD_x$ on the complete carrier. The dispersion during a constant active segment is $E_\pm(k) = \pm\sqrt{g^2r_B^2 + 4\omega^2\sin^2(k\ell_x)}$, and the rest mass reads the half-gap gr_B between the two orientation branches.

The negative bright and dark branches remain part of the complete space; selecting a positive bright preparation does not exclude the other states.

Thermal processing uses isospectral temporary storage as in Section 4.4.3. The entire mode, path ports, and internally correlated state must be stored, not just b. Let $\mathcal{C}_B$ be the space to be stored, a_R the copy's entrance state, and g_{aux} the processing-auxiliary preparation. On success, the inverse swap returns the original carrier, and every outcome satisfies

$$\boxed{K_\eta(\psi_{\mathcal{C}_B,\mathrm{int}} \otimes a_R \otimes g_{\mathrm{aux}}) = \sqrt{r_\eta}\,\psi_{\mathcal{C}_B,\mathrm{int}} \otimes a_R \otimes g_{\mathrm{aux}}.} \tag{9.21}$$

Successful copies and auxiliaries proved to be reversible can be reused; thermal outputs and history records remain retained. On failure, the stored state exits together with the output and is not reused as a new preparation. Active evolution, storage, thermal processing, the inverse swap, and free-phase compensation all count toward elapsed time. Processing segments accumulate no additional phase, or that phase is compensated by actual control.

The absolute frequency is then compared with the same global background. Let $r_{\mathrm{vac}} = \sqrt{W_{\mathrm{vac}}}\log W_{\mathrm{vac}}$, construct the vacuum-response port X_{vac} in the same way, and take its positive bright preparation. The dimensionless half-gaps on the bright subspaces of the background-gradient action G are $\sqrt{b_n}$ and $\sum_n b_n = C_{\mathrm{bg}}$. The collective endpoint and actual background vertex $gX_{\mathrm{vac}} \otimes G$ use independent resource units, a common drive, and the same active window. The background remains coherent across processing segments and continues after target exit until that history's final readout. For a delivered collective carrier that survives, averaging over the actual elapsed duration of the same complete history h shared with the background gives

$$\begin{aligned} \bar{g}_h &= \frac{1}{T_h}\int_h g(t)\,dt, \\ m_{B,h} &= \bar{g}_h r_B, \\ \nu_{\mathrm{ref},h} &= \bar{g}_h r_{\mathrm{vac}}, \\ \boxed{\frac{m_{B,h}}{\nu_{\mathrm{ref},h}} &= \frac{\sqrt{Z_L}\log Z_L}{\sqrt{W_{\mathrm{vac}}}\log W_{\mathrm{vac}}}.} \end{aligned} \tag{9.22}$$

The full-cycle background frequency is $\Omega_{n,h} = \nu_{\mathrm{ref},h}\sqrt{b_n}$. Retaining the squared-matching rule of Eq. (2.75), this section calibrates the vacuum endpoint with the actual background frequency $\nu_{\mathrm{ref},h} = \bar{g}_h r_{\mathrm{vac}}$. For a positive drive,

$$\begin{aligned} \frac{\sum_n \Omega_{n,h}^2}{E_0^2} = W_{\mathrm{vac}}\log^2 W_{\mathrm{vac}} \quad &\Longrightarrow \quad \bar{g}_h = \frac{E_0}{\sqrt{C_{\mathrm{bg}}}}, \\ \boxed{m_{B,h} &= E_0\sqrt{\frac{Z_L}{C_{\mathrm{bg}}}}\log Z_L.} \end{aligned} \tag{9.23}$$

Common matrix elements, simultaneous protection, and the positive bright vacuum preparation give both ends the same $\bar{g}_h$. Background squared matching calibrates this average drive, including pauses, rather than the time average of the squared instantaneous drive strength. The common calibration thus converts the

collective port's dimensionless response into an absolute rest-frame frequency. A finite realization uses Z_L and the chosen C_{bg}. Taking $\beta = \beta_B$ and using the complete spectral references $Z_L \to \zeta(\beta_B)$ and $C_{\mathrm{bg}} = C^*$ of Section 6.3 reduces Eq. (9.23) to the common nucleon energy scale in Eq. (9.17). Finite-truncation differences remain treated as in Section 6.3. Internal frequency shifts δ_N must be calculated separately from the nucleon branches' own actions.

This connection places the common mode's participation weights, actual maintenance phase, and common nucleon mass within one process.

9.3.3 Stability Conditions

Complete thermal return ensures conditional evolution of the selected carrier; absolute stability additionally requires examination of every exit outcome. Under the general criteria of Chapter 2 and Section 11.5, after fixing the environmental preparation, a candidate B should satisfy $R_r B = c_r B$, $D_s B = 0$, and $\sum_r |c_r|^2 = 1$. If only collinear return and $s_B = \sum_r |c_r|^2 < 1$ hold, the continuous-survival share under independent cycles with the same preparation is s_B^m. State-dependent or memory-dependent exits require composition of the actual survival subchannels. The common frequency in this section does not by itself determine nucleon lifetimes.

A vanishing diagonal response of the common pattern does not mean it cannot exchange with directional excitations. For example, take a nontrivial-direction vertex whose endpoint is the terminal shell, with normalized unit-shell vector ψ_χ. Then

$$V_\chi = |e_K\rangle\langle\psi_{\bar{\chi}}| - \chi(-1)|\psi_\chi\rangle\langle e_K|, \qquad \langle b|V_\chi|b\rangle = 0, \qquad V_\chi b = -\chi(-1)p^{-K/2}\psi_\chi \neq 0. \quad (9.24)$$

The first expression is the mode-side vertex; a conserving complete exchange must also include environmental charge. Character orthogonality makes the diagonal element vanish, but the zero-element share gives a nonzero component leaving the original pattern. Protecting the candidate state requires these components to cancel in the actual action, be excluded conditionally by the selected channel, or restore the original pattern through subsequent retrieval.

The collective energy scale of Section 6.3 increases with the allowed number of fundamental color factors in a singlet, showing only that three fundamental representations are lowest within that family. Complete final states may involve other internal representations, different particle numbers, or environmental outputs. Only excluding every actual exit channel allowed by joint conservation and energy conditions can make s_B exactly one. The "lowest three-color singlet" alone cannot prove absolute proton stability; nor can the sign of the mass difference alone determine the lifetime of an allowed neutron-to-proton conversion.

Both nucleons can therefore share the lowest collective scale while having different internal frequency shifts and return shares. The mass difference and maintenance duration test the phase and exit parts of the complete cycle, respectively, and must be calculated on the same internal states.

9.4 The Neutron–Proton Mass Difference

The proton and neutron candidates use the same b and color singlet, with their difference residing in the flavor–spin part. Subtracting the common baseline from

both masses isolates the internal phase and electromagnetic response. Under the stated evaluation identification, these terms have opposite signs and produce a mass difference far smaller than the common nucleon energy scale.

9.4.1 Nucleon Spectral Phase

Retain the generator 3 and χ_1, χ_5 conventions of Section 7.1, but evaluate at the nucleon spectral point β_B rather than the elementary-state point 1/2. Termwise conjugation and absolute convergence of $\beta_B > 1$ give $L(\beta_B, \chi_5) = \overline{L}(\beta_B, \chi_1)$. Denote the present principal argument by θ. Numerically,

$$\begin{aligned} L(\beta_B, \chi_1) &= 0.869968720699\ldots + 0.298172012211\ldots\, i, \\ \theta = \arg L(\beta_B, \chi_1) &= 0.330191432349\ldots, \\ \tan\theta &= 0.342738773379\ldots. \end{aligned} \tag{9.25}$$

The present spectral value lies in the right half-plane, and the conjugate direction has principal phase $-\theta$. Conjugacy here means complex conjugation of spectral data, not a particle–antiparticle identification. Specify multiplication of component spectral responses along the complete coherent connection. Every uud permutation contains two L_1 and one L_5, and every udd permutation the reverse. The permutation superposition of Section 9.2 therefore leaves this total phase unchanged:

$$\begin{aligned} a_p &= L_1^2 L_5 = |L_1|^2 L_1, \\ a_n &= L_1 L_5^2 = \overline{a_p}, \\ \Theta_p &= +\theta, \\ \Theta_n &= -\theta, \\ L_k &:= L(\beta_B, \chi_k). \end{aligned} \tag{9.26}$$

The two responses have equal moduli and real parts. Nucleon splitting therefore reads the conjugation-odd part relative to the same reference; below, their opposite phases are used for the mass identification.

9.4.2 Nucleon Mass Corrections

Adopt the phase–mass prescription $\delta_N^{(L)} = -(M_B^2/v)\tan\Theta_N$, with v taken from the electroweak energy scale. This function, sign, and scale constitute the physical identification used in this section; the conjugate spectra determine phase antisymmetry. Substitution gives

$$\delta_p^{(L)} = -\frac{M_B^2}{v}\tan\theta, \qquad \delta_n^{(L)} = +\frac{M_B^2}{v}\tan\theta, \qquad \boxed{\Delta m^{(L)} = \frac{2M_B^2}{v}\tan\theta.} \tag{9.27}$$

For the electromagnetic term, independently adopt a self-energy-difference prescription equally divided among six paths: the charged branch gains $\alpha_{\mathrm{EM}} M_B/6$ relative to the neutral branch. α_{EM} uses the complete response of Section 7.4, and the factor 6 is the independently specified number of equally weighted paths. Thus

$$\delta_p^{(\mathrm{EM})} - \delta_n^{(\mathrm{EM})} = \frac{\alpha_{\mathrm{EM}} M_B}{6}, \qquad \boxed{\Delta m^{(\mathrm{EM})} = -\frac{\alpha_{\mathrm{EM}} M_B}{6}.} \tag{9.28}$$

This term specifies the electromagnetic difference; denote the common shift by δ_0. The independent equal division among six paths and the matching denominator of $\alpha_{\mathrm{EM}} = w_2/N_{\mathrm{EM}}^{\mathrm{full}}$ jointly give the total denominator $6N_{\mathrm{EM}}^{\mathrm{full}}$.

Both terms use the same energy reference and compatible effective generators. When free mass readout preserves charge, off-diagonal elements between the proton and neutron branches vanish. Writing all undetermined common shifts as δ_0 gives

$$m_p = M_B + \delta_0 - \frac{M_B^2}{v}\tan\theta + \frac{\alpha_{\mathrm{EM}} M_B}{6}, \qquad m_n = M_B + \delta_0 + \frac{M_B^2}{v}\tan\theta. \tag{9.29}$$

The common term cancels exactly when the branches are subtracted. Additivity here comes from the adopted charge-preserving generator. Nucleon frequency shifts do not acquire an additional factor from the charged elementary-state frequency shift of Section 8.1.

Table 9-1 Shared structure and differences of the two candidate nucleon states. Mass corrections are relative to the common baseline $M_B + \delta_0$, with $A_B = (M_B^2/v)\tan\theta$ and $C_B = \alpha_{\mathrm{EM}} M_B/6$.

Reading	Proton candidate	Neutron candidate
Mode and color parts	b $\otimes$ ϵ_3	b $\otimes$ ϵ_3
Flavor composition	Two u, one d	One u, two d
Charge Q	+1	0
Selected spin	S=1/2, $S_z = +1/2$	S=1/2, $S_z = +1/2$
Selected spectral-product phase Θ_N	$+\theta$	$-\theta$
Mass correction in this section	$-A_B + C_B$	$+A_B$

For the distinction between per-round phase reading and the tangent prescription, and a conditional realization of division among six paths, see Supplementary Proof S14.

9.4.3 Numerical Nucleon Mass Difference

Using the unrounded M_B, v, and α_{EM} above gives the two contributions to the difference and their sum:

$$\begin{aligned} \Delta m^{(L)} &= +2.442828542637\ldots \text{ MeV}, \\ \Delta m^{(\mathrm{EM})} &= -1.139198043531\ldots \text{ MeV}, \\ \boxed{m_n - m_p &= 1.303630499106\ldots \text{ MeV}.} \end{aligned} \tag{9.30}$$

The CODATA 2022 table of constants [18] gives a mass difference of 1.29333251(38) MeV. This section's result is higher by 0.0102979891… MeV, or 0.79624% relative to the central value. This far exceeds experimental uncertainty. The digits indicate reproducibility of the stated prescription and cannot replace

theoretical error estimates for phase readout, self-energy allocation, and spectral preparation. The observed mass difference was not used to choose the parameters above.

The common shift δ_0 cancels from the difference, so this section directly tests the mass splitting of the two nucleon branches. Absolute proton and neutron masses additionally depend on the common collective scale M_B and the complete common frequency shift. The baseline of Chapter 6 and the difference in this section should be tested separately.

Using the electron mass from Section 8.2 for a kinematic check within the same model gives $\Delta m - m_e = 0.7926182479\ldots\,\mathrm{MeV}$, larger than the neutrino mass scale of Section 8.5. Thus $\mathrm{n}\to\mathrm{p}+\mathrm{e}^-+\bar{\nu}$ has positive available energy under this mass identification. Actual decay additionally requires an output channel with nonzero matrix element satisfying joint conservation. Its lifetime is determined by the exit share and cycle duration in Section 9.3; this section does not infer that share from the mass difference.

9.5 Resonance Between Nucleons

When two nucleons interact, exchange acts between their complete states. The common collective structure supplies the scope of participation, while internal representations determine how vertices connect initial and final states. The emission–propagation–absorption process of Chapter 3 continues to apply, with the nuclear-force response using the channels and weights fixed in Section 7.5.

9.5.1 Internucleon Exchange

Label the two nucleons A and B, with inputs consisting of the selected complete states and environmental preparation. If each has the actual two-level port, independent complete preparation, and resonance conditions required by Section 3.1.2, three-stage exchange and retrieval can restore the mediator while retaining each source and its auxiliaries. This conditional realization provides one complete coupling construction. Multimode nucleon exchange must still use its own actual endpoints; sharing the status of stable states does not justify importing a two-level matrix. Denote available exchange modes by μ, ν, emission operators by J_μ, absorption operators by A_ν, and the propagation matrix by $G_{\nu\mu}$. Under the chosen emission–propagation–absorption decomposition and product-state endpoints, the path amplitude for a specified complete output record g has the form below, with all operators and endpoints taking the corresponding components for that record:

$$\mathcal{A}_g = \sum_{\mu,\nu} \langle N'_B|A_\nu|N_B\rangle\, G_{\nu\mu}\, \langle N'_A|J_\mu|N_A\rangle. \tag{9.31}$$

The μ, ν paths not distinguished by the complete record are summed within one amplitude; mutually orthogonal final records are then summed at the probability level. If final states still overlap, the record-overlap matrix must be retained. This order places internal vertices, propagation, and actual statistics within one exchange process.

A common mode pattern does not imply automatic resonance of every nucleon transition. Phase matching involves the complete endpoint masses, internal energy-level differences, and exchange environment. Proton–neutron conversion must also

balance charge and other actual conserved quantities. Comparing only the common b would omit the internal frequency shifts of Section 9.4. Elastic and inelastic processes require separately selected endpoints and energy conditions.

9.5.2 Nuclear Response and Spatial Interaction

Following Section 7.5, nucleon exchange uses 42 distinguishable complete outputs and the node weights, giving coupling $\alpha_{\text{nuclear}} = (42/\mathcal{L}_7)w_1$. Actual probabilities are fully normalized as in Section 3.5.3; the πNN comparison retains the vertex and scattering definitions of Section 7.5.

Collective participation in the nuclear response and elementary endpoint matching in the color response produce different weights. Their relation in an effective theory is tested through the same scattering amplitude, elimination of degrees of freedom, and scale identification. The spatial interaction below remains determined by the endpoint vertices and propagation kernel.

Interaction range, attraction or repulsion, and spin dependence in space are contained in the propagation kernel $G_{\nu\mu}$ and endpoint vertices. A dimensionless coupling constant alone cannot determine a nuclear potential. A specific spatial interaction requires these operators to be connected to position readout and propagation conditions in observational spacetime. The common cycle specifies the calculation order, and Section 7.5 supplies the strength normalization under the present channel prescription.

9.6 Color Confinement and Independent Outputs

A nucleon's internal color singlet constrains the joint representation. To determine whether a colored component can become independent matter, one must additionally follow the actual separated output and its subsequent maintenance. This section first gives a process criterion for zero colored output, then distinguishes a global singlet from local colored outputs.

9.6.1 Conditions Excluding Colored Outputs

Fix the output object under discussion and the actual formation or separation instrument. Denote the input-support projection by Π_S, the output object's color-singlet projection by Π_{sing}, and its colored projection by $Q_{\text{col}} = I - \Pi_{\text{sing}}$. The colored projection is defined by the full SU(3) representation decomposition; vanishing of one cyclic label cannot replace this definition. Let the selected output process be the completely positive, trace-nonincreasing map $\mathcal{J}(\rho) = \sum_r K_r \rho K_r^\dagger$.

For any normalized input $\psi \in \operatorname{ran}\Pi_S$, the unnormalized output share entering colored representations is a sum of squared norms. This gives the necessary and sufficient criterion:

$$\operatorname{Tr}[Q_{\text{col}}\mathcal{J}(P_\psi)] = \sum_r \|Q_{\text{col}}K_r\psi\|^2,$$
$$\boxed{\operatorname{Tr}[Q_{\text{col}}\mathcal{J}(\rho)] = 0 \ \forall \rho = \Pi_S\rho\Pi_S \quad \Longleftrightarrow \quad Q_{\text{col}}K_r\Pi_S = 0 \ \forall r.} \tag{9.32}$$

The proof uses only nonnegativity of each term. If the colored share vanishes for every input, each K_r produces no colored output from any vector in the input sup-

port. Conversely, substitution gives zero, and mixed inputs follow by linearity. If $\mathcal{J}$ is a conditional process, its total success probability is Tr $\mathcal{J}(\rho)$, and the conditional colored share must also be divided by that nonzero number. The zero condition itself is unchanged.

Representation theory supplies a sufficient condition: the input support is a color-singlet space, and every actual K_r intertwines the input and output color representations. Then for every g∈SU(3),

$$U_{\text{out}}(g)K_r = K_r U_{\text{in}}(g), \qquad U_{\text{in}}(g)\Pi_S = \Pi_S \quad \Longrightarrow \quad Q_{\text{col}} K_r \Pi_S = 0. \tag{9.33}$$

Because $K_r\psi$ remains invariant under every color action, it must lie in the output singlet subspace. One checkable realization is that the complete joint action preserves the color representation, the environmental preparation lies in the trivial color representation, and every environmental output actually coupled to the process also lies in the trivial representation. Only then do the corresponding environmental matrix elements individually become intertwiners.

Intertwining by each K_r is stronger than covariance of the reduced channel as a whole. If the environment can carry compensating color, the K_r may mix as a nontrivial representation while the channel remains covariant, without satisfying the termwise condition above. The no-colored-output criterion must hold for the complete actual outcome set before colored outcomes have been filtered out.

9.6.2 Color Separation and Compensation

Preservation of a total color singlet does not suffice to exclude local colored representations. The normalized singlet of a fundamental and conjugate representation provides a direct test:

$$|\Omega_c\rangle = \frac{1}{\sqrt{3}} \sum_{a=1}^{3} |a\rangle \otimes |\bar{a}\rangle, \qquad (U \otimes \bar{U})|\Omega_c\rangle = |\Omega_c\rangle, \qquad \text{Tr}_2\, P_{\Omega_c} = \frac{I_3}{3}. \tag{9.34}$$

The local $\mathrm{I}_3/3$ is invariant as a density operator under color transformations but is supported in the three-dimensional fundamental representation, which has no singlet vector. Thus "group-invariant density operator" and "state supported in the singlet subspace" are different conditions. The same distinction applies to the one-position reduction of a nucleon's color tensor.

Taking this partial trace does not itself accomplish physical separation. An independent output also requires a definite carrier capable of continued cycle maintenance, together with examination of its interactions and records with the system carrying compensating color. If an actual process satisfies the termwise zero condition of Section 9.6.1, that process produces no colored output. Otherwise its output branches and subsequent survival must be calculated to determine whether an independent stable state results.

Universal confinement must cover every allowed separation process. Algebraic grading and relative suppression in the complete spectrum cannot replace examination of actual outputs and subsequent survival.

The collective baryon pattern, internal flavor–spin representation, mass readout, and exchange process admit a consistent state description. Common participation gives the nucleon energy scale; the adopted internal-phase and electromagnetic prescriptions give the mass difference; complete output records determine nuclear responses and maintenance shares. Absolute stability and universal confinement require coverage of all exit and separation processes, respectively. The next chapter examines exchange-channel excitations and collective oscillations to calculate bosonic states and energy readings.

§10 Bosons

Exchange between stable states requires actual propagating components, and perturbing a condensate can produce new oscillations. In records, these appear as delivery of energy and charge, or periodic changes relative to the background. Connecting such processes to photons, W and Z bosons, gluons, pions, and the Higgs requires the carrier, transformation properties, and cycle data read as mass to be determined for each excitation.

The mass inputs here come respectively from the condensate share acquired by a channel, the two-layer response of internal nucleon transitions, and the restoring action on radial perturbations. They share the rules for amplitude composition, record summation, and phase timing. Their respective carriers, representations, and actual actions determine the physical identities of the excitations. We first establish how these inputs become cycle readings, then calculate the different masses.

10.1 Channel Excitations and Condensate Oscillations

10.1.1 Channel Excitations

The resonance process of Chapter 3 consists of emission, propagation, and absorption. Part of an initial stable state's amplitude enters an exchange component, reaches the other endpoint through propagation, and forms an output jointly with the receiving stable state and environment. A channel specifies the charge and phase transformations permitted by this connection; the state actually propagating in that channel is the excitation. A vertex coefficient or group generator does not simultaneously specify its spatial waveform and frequency.

Section 3.1.2 already gave exchange with mediator restoration on the selected complete carrier. That mediator is an auxiliary that temporarily stores logical components within the process. Identifying one of its excitations as an independently propagating boson additionally requires a propagation space, energy, and readout; complete retrieval alone does not supply this identification.

After fixing input, output, and environmental preparation, the relevant exchange component can be written A G J: J produces the exchange component, G propagates it, and A receives it. Joint conservation in the complete process relates what the emitter loses, the exchange component carries, and the receiver gains. Orthogonal final records require probability summation, whereas paths reaching the same undistinguished record require amplitude summation first. The presence and frequency of an excitation must be read from these actual components.

Internucleon exchange involves the common collective mode and internal flavor changes; condensate oscillations involve departures and restoration of the background amplitude. They share earlier parameters with channel propagation, but their participating states, exchange actions, and readout conditions are determined by their respective processes.

10.1.2 Quantum Numbers and Mass

The internal channel carriers of this chapter retain the U(1), SU(2), and SU(3) identifications of Chapter 5. Electric and color charges act on these internal representations, while spin concerns rotations in observational space. These actions must be

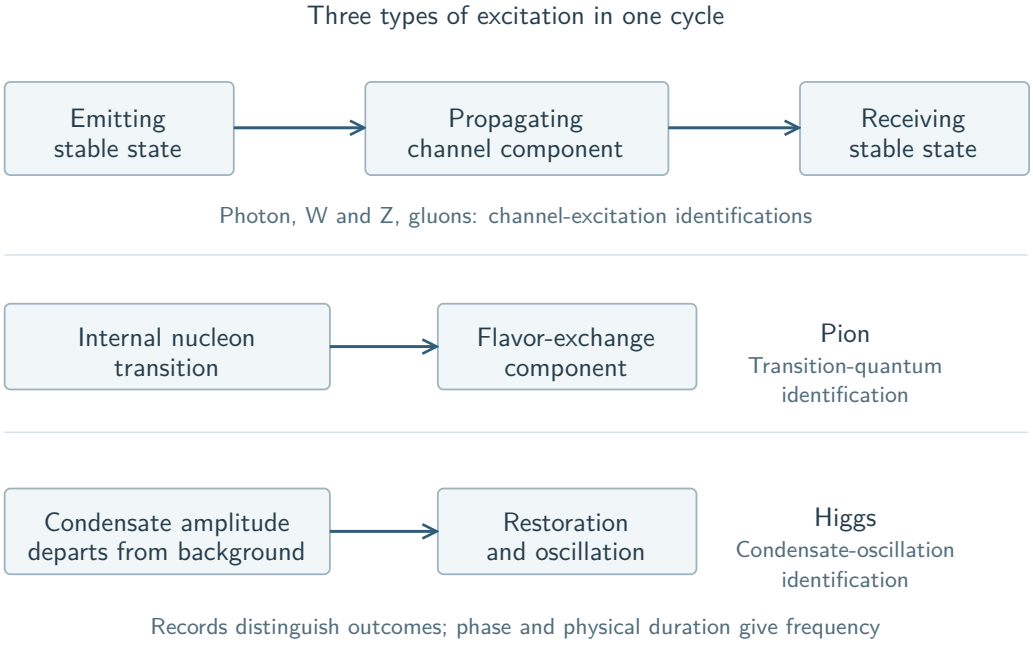


Figure 10-1 Process relations among three excitation types. Arrows indicate exchange or amplitude restoration; particle identifications use the states, representations, and readout conditions specified in the text.

specified separately. In a vector propagation representation, rotations act on three spatial components; for a scalar amplitude, rotations leave it unchanged. Character order or parity under negation does not replace these rotation representations.

Calling an excitation a boson also adopts symmetric exchange for multiexcitation states. For two identical excitations, the symmetric projection of the exchange operator S_{12} is $\Pi_+ = (I + S_{12})/2$, and the selected two-excitation state lies in its image. Integer spin, symmetric exchange, and concrete propagation equations each have their own content in this finite construction. This chapter does not infer a complete spin–statistics result directly from the existence of a single direction vector.

Rest mass is read from the coherent frequency as in Chapter 2. If a return amplitude is $\sqrt{s}e^{i\varphi}$, where s is the return share for that outcome, subtracting the reference phase and unwrapping it along the actual process gives frequency $-\varphi/\tau$. General reduced coherence also contains the reference branch and environmental overlap, so the squared modulus of its multiplier cannot directly be identified with a maintenance probability. Below, we first obtain oscillation frequencies from actual exchange, then express their mass squares.

Let the actual coupling from the entrance to the temporary-storage output be B, with endpoints and compensating auxiliaries already balanced, and use simultaneous mutually inverse exchange. The complete generator and entrance-amplitude equation are

$$\boxed{\mathbb{H}_B = \begin{pmatrix} 0 & B^\dagger \\ B & 0 \end{pmatrix}, \qquad \ddot{\psi}_{\mathrm{in}} = -B^\dagger B \psi_{\mathrm{in}}.} \tag{10.1}$$

If $B^\dagger B u_j = \sigma_j^2 u_j$ and $\sigma_j > 0$, then $B u_j/\sigma_j$ and the entrance span a two-dimensional bright subspace with generator $\sigma_j \sigma_x$ and eigenfrequencies $\pm\sigma_j$. Under the same rest-frame half-gap reading, the mass squared during activity is the actual $B^\dagger B$.

The entrance and bright output jointly form the oscillating carrier. The entrance's return amplitude is $\cos(\sigma_j t)$, the eigenfrequencies of the complete two-state system are $\pm\sigma_j$, and the angular frequency of the transfer probability $\sin^2(\sigma_j t)$ is $2\sigma_j$. This chapter uses the rest-frame half-gap as the mass phase reading and then converts it using total elapsed duration. These three readings must not be conflated. The protection projection must likewise contain the complete two-state system: protecting only the entrance would remove the required off-diagonal exchange under reflection averaging.

In the two-dimensional bright subspace, the positive-frequency eigenvector is $(u_j, B u_j/\sigma_j)/\sqrt{2}$. Labeling these vectors by u_j represents the generator on the positive-frequency branch as $\sqrt{B^\dagger B}$. The entrance kernel $\ker B$ and output kernel $\ker B^\dagger$ together form the complete zero-mode space. This spectral representation follows from the actual exchange above.

Thermal processing uses the complete temporary-storage return of Section 4.4.3, transferring the entrance, all bright outputs, dark directions, and reversible purification auxiliaries together into an isospectral copy. Each successful outcome returns the entire state with the same $\sqrt{r_\eta}$, preserving its oscillation and correlations with other carriers. Failure outputs come from the same complete instrument. Active interactions are switched off during thermal processing, with free-phase compensation and all elapsed duration included. The original thermal endpoints remain balanced by bare energy; this process does not require Gibbs equilibrium for the active eigenenergies $\pm\sigma_j$.

Following Eq. (2.59)–Eq. (2.60), if each round evolves under the active generator for t_H, all other processing gives only scalar returns, and the complete cycle period is τ, the readable rest-frame frequency is $M_j = (t_H/\tau)\sigma_j$. Define $\mathbb{M}^2$ using these frequencies converted by total elapsed duration. The positive-frequency branch then has the common form

$$\mathbb{M}^2|j\rangle = M_j^2|j\rangle, \qquad U_\tau = \exp(-i\tau\sqrt{\mathbb{M}^2}), \qquad U_\tau|j\rangle = e^{-i\tau M_j}|j\rangle. \tag{10.2}$$

Here $M_j \geq 0$; phase references and physical units follow Chapters 2 and 6. Under continuous activity, $\mathbb{M}^2 = B^\dagger B$. With processing intervals, the common activity share must be included in both matter frequencies and background calibration. Spatial propagation also changes energy: zero rest mass does not mean zero propagation energy or require unit return share.

10.1.3 Condensate Contributions to Channel Masses

Classical constructions in continuous gauge field theory describe how symmetry-breaking backgrounds give mass to gauge vector fields [30]. Chapter 6 obtains the electroweak energy scale v from the intrinsic metric constant C^* and effective participation. Channel-excitation masses use an identification with condensate energy shares: the common quadratic condensate weight is $\mathrm{v}^2 C^*$, a channel receives share ϖ_j, and its effective matching denominator N_j allocates that share to the selected mode. Accordingly, specify

$$\boxed{M_j^2 = v^2 C^* \frac{\varpi_j}{N_j}}, \qquad N_j > 0, \quad 0 \le \varpi_j \le 1, \quad v > 0,\ C^* > 0. \tag{10.3}$$

The coefficients in this expression are first specified by condensate and matching conditions; through the actual response endpoints below, they can become the same mass squared. As Chapter 7 explained, corrected N_j is the ratio of total preparation weight to matching weight, an effective matching denominator that need not be an integer. It cannot be interpreted as a noninteger number of orthogonal basis vectors. Its use requires participation, routing, and normalization to be specified first. $\varpi_j = 0$ makes this mass vanish, while $\varpi_j > 0$ gives positive mass. Any additional independent mass terms still depend on the choice of complete generator.

When participation response and endpoint matching are connected to the same entrance, the norms of the two response columns jointly determine the exchange frequency. The outward and return paths use the complete two-state system of Section 10.1.2, while processing and waiting enter the same cycle period. The absolute scale is supplied by the background squared-matching calibration of Section 2.5.3.

First give a direct realization for continuous activity with $f_{\mathrm{act}} = 1$. Let the positive participation be $W = \sum_\alpha w_\alpha > 1$. Preparation and endpoint phases give $|a\rangle = (\log W) \sum_\alpha \sqrt{w_\alpha} e^{i\phi_\alpha} |\alpha\rangle$, hence $\|a\|^2 = W \log^2 W$. Retain the purification of the actual matching preparation from Chapter 7, with acceptance column η_i satisfying $\|\eta_i\|^2 = \varpi_i / N_i$. If these complete output factors connect to the same entrance through a product vertex, use the actual action

$$B_i = E_0 |a\rangle \otimes |\eta_i\rangle\langle i|, \qquad \boxed{B_i^\dagger B_i = E_0^2 W \log^2 W \frac{\varpi_i}{N_i} P_i = v_E^2 C_{\mathrm{bg}} \frac{\varpi_i}{N_i} P_i.} \tag{10.4}$$

Here P_i is the entrance projection. The product vertex of the response and matching columns, together with its inverse return, produces the actual oscillation. The last equality uses the background squared-matching relation in Eq. (2.75); on the complete background, this is the energy-scale relation in Eq. (6.13). Taking $v_E = v$ and $C_{\mathrm{bg}} = C^*$ gives Eq. (10.3) as the mass squared of this specified vertex.

If the direct realization includes processing intervals, rename the active vertex coefficient E_0 in Eq. (10.4) as g_B. The mass squared over the total elapsed duration is $f_{\mathrm{act}}^2 B_i^\dagger B_i$. To use the same physical calibration E_0 while retaining Eq. (10.3), this endpoint's common-background matching must satisfy $f_{\mathrm{act}} g_B = E_0$. This relation constrains the chosen drive and actual activity share. If the active coefficient remains fixed at E_0, the mass squared retains the additional factor f_{act}^2.

The acceptance column uses a matching projection. If the actual endpoint also contains the spectral vertex R_i of Chapter 7, the mass squared acquires an additional $|R_i|^2$ that cannot be removed. Purification auxiliaries must participate in the coherent return; irreversible records already formed cannot be omitted. Different activity times for matter and background also require their ratio to be retained.

The same mass squared also admits a common-drive realization. Let the background-gradient blocks be $\sqrt{b_r}\sigma_x$ and $\sum_r b_r = C_{\mathrm{bg}}$, and additionally couple them to a common entrance and orthogonal outputs to form $c = \sum_r \sqrt{b_r} |r\rangle$. Connecting to the matching column above gives $\|c \otimes \eta_i\|^2 = C_{\mathrm{bg}} \varpi_i / N_i$. Matter and background use the two-arm protection with a common source of Section 2.3.5, a common drive, and the complete activity share. The calibrated drive rate $\nu = v$

then gives $M_i^2 = \nu^2 C_{\mathrm{bg}} \varpi_i / N_i$. This is a different coupling structure with an actual common entrance and inverse return, requiring endpoint energy and charge to be balanced. The two realizations each use their own background calibration. Section 10.6.3 uses this common drive for the Higgs bright mode. The two entrances of the neutral sector may participate in the same condensate response. Whether they retain coherence depends on the connections and records of the complete output. Mass contributions must be obtained from this joint action; each channel's condensate share and effective matching denominator are separately specified in its evaluation.

10.2 The Photon

10.2.1 Zero Mode of the Neutral Sector

The neutral weak and Abelian directions can jointly participate in condensate exchange. Here both entrances connect to the same complete response output, with coherence between the two paths retained during return.

Let $|\mathrm{w}\rangle$ and $|\mathrm{y}\rangle$ be orthonormal neutral-weak and Abelian basis directions with canonically normalized kinetic terms. Choose real mixing parameters s_v and c_v satisfying $s_v^2 + c_v^2 = 1$; Section 10.3 fixes them from the vertex-response ratio. Take the actual response column to be $B_Z = |a_Z\rangle(c_v\langle w| - s_v\langle y|)$, with $\|a_Z\|^2$ determined by the matching of Section 10.1. To describe this joint action, define two orthogonal combinations:

$$|\gamma\rangle = s_v|w\rangle + c_v|y\rangle, \qquad |Z\rangle = c_v|w\rangle - s_v|y\rangle, \qquad \langle\gamma|Z\rangle = 0. \tag{10.5}$$

Under the mass reading of Section 10.1, the response column above gives the following neutral mass-squared matrix. This section adopts this condensate contribution as the complete rest-mass term in the sector:

$$\mathbb{M}^2_{\mathrm{neutral}} = M_Z^2 |Z\rangle\langle Z| = M_Z^2 \begin{pmatrix} c_v^2 & -c_v s_v \\ -c_v s_v & s_v^2 \end{pmatrix}, \qquad \boxed{\mathbb{M}^2_{\mathrm{neutral}}|\gamma\rangle = 0, \quad M_\gamma = 0.} \tag{10.6}$$

Matrix multiplication verifies spectral values 0 and M_Z^2. The matrix is positive semidefinite because its quadratic form on any ξ is $M_Z^2|\langle Z|\xi\rangle|^2$. Zero mass is given by the kernel of this explicit matrix. For general output columns a_1, a_2, the Gram determinant is $\|a_1\|^2\|a_2\|^2 - |\langle a_1|a_2\rangle|^2$; with nonzero columns, it vanishes precisely when they are collinear. Retaining orthogonal path records generally removes this interference. Thus neither a two-dimensional neutral space nor the existence of a nontrivial real character guarantees a zero mode.

Under the selected unbroken U(1), the generator's adjoint action is trivial, so this Abelian channel excitation carries no charge of that channel. It can still exchange with charged endpoints. The Abelian bracket has no direct noncommutative self-coupling term, but does not exclude effective photon scattering through charged intermediate processes.

10.2.2 Propagation and Two Polarizations

Polarization describes how the propagation amplitude varies relative to spatial directions. For an explicit readout, take a nonzero real propagation vector $\mathrm{k} \in \mathbb{R}^3$, a

complex three-dimensional vector amplitude, and impose $\mathrm{k}\cdot\epsilon$=0. The transverse space is the kernel of a nonzero linear functional and hence has complex dimension two. Choose a real orthonormal transverse basis e_1, e_2 and fix the active-rotation convention around k by $\mathrm{R}(\vartheta)\mathrm{e}_1$=cos$\vartheta$ e_1+sinϑ e_2. Then

$$\mathcal{T}_k = \{\varepsilon \in \mathbb{C}^3 : k\cdot\varepsilon = 0\}, \quad \dim_{\mathbb{C}} \mathcal{T}_k = 2, \qquad \varepsilon_\pm = \frac{e_1 \pm i e_2}{\sqrt{2}}, \quad R(\vartheta)\varepsilon_\pm = e^{\mp i\vartheta}\varepsilon_\pm. \tag{10.7}$$

The rotation action distinguishes the two senses of rotation; linear polarizations are their coherent combinations. The transverse constraint, vector representation, and identification between propagation vector and observed displacement are conditions of this realization. Taking χ_1 and χ_5 as internal labels for the two senses makes conjugate pairing compatible with them. Two conjugate characters alone do not conversely imply transversality.

In records, photon detection appears as a change in the receiving endpoint's energy and state. Zero rest mass allows the corresponding propagation model to have no mass threshold, but the propagation kernel, speed, and spatial range must still follow from the actual phase–displacement relation. The two polarizations obtained here belong to the stated vector-propagation realization.

10.3 W and Z Bosons

10.3.1 Charged Weak Bosons

The weak channel uses an internal SU(2) action. Let T_1, T_2, T_3 be its generators, with $T_\pm = T_1 \pm iT_2$ satisfying $[T_3, T_\pm] = \pm T_\pm$. Under the chosen charge normalization and Abelian-component identification, W^+ and W^- follow these two charged connections, while the third weak generator enters neutral mixing. The $\pm$ labels here come from the charge action and belong to degrees of freedom distinct from spatial-polarization $\pm$ labels.

For channel excitations using a vector rotation representation, a massive mode has a three-dimensional rest-frame spin space. In a four-vector realization with the corresponding propagation constraint, removing the time component in the rest frame leaves three spatial components, compatible with the spin-1 interpretation. The three internal SU(2) generators do not themselves prove spatial spin 1; the physical identification specifies these separately.

Take the condensate contribution $\varpi_W = 1$. This chapter abbreviates the effective denominator of Chapters 6 and 7 as $N_W := N_W^{\mathrm{full}} = N_W^{\mathrm{eff}}$, distinguishing it from the earlier bare count 18. This effective matching denominator is $N_W = 18 - 3w_1 + w_2 + 2w_3$, where $w_k = |R_k(1/2)|^2$. Substitution into the common mass formula gives

$$N_W = 18 - 3w_1 + w_2 + 2w_3 = 18.246178099119\ldots,$$
$$\boxed{M_W = v\sqrt{\frac{C^*}{N_W}} = 80.5733091570\ldots\ \mathrm{GeV}.} \tag{10.8}$$

The values v=246.191373933… GeV and C^*=1.954378366538… are retained from Chapter 6. The two charged members have equal masses under this charge-conjugation-symmetric mass identification. Their mass is jointly determined by the condensate weight and specified effective matching denominator. Decay widths

must instead be calculated from exit outcomes and physical durations and cannot be replaced by that same numerical value.

10.3.2 Z-Boson Mass

The neutral-basis angle uses the Born-level vertex-response ratio, rather than the corrected mass ratio. Write $\alpha_{\mathrm{EM}}^{(0)} = w_2/6$ and $\alpha_W^{(0)} = w_3/18$, and define, under the vertex-charge identification,

$$s_v^2 := \sin^2\theta_v = \frac{\alpha_{\mathrm{EM}}^{(0)}}{\alpha_W^{(0)}} = \frac{3w_2}{w_3} = 0.231464478899\ldots, \qquad c_v^2 = 1 - s_v^2. \tag{10.9}$$

Under this angle identification, the γ and Z combinations of Section 10.2 are fixed. Diagonal corrections to channel weights need not preserve the eigenbasis of an arbitrary physical mass-squared matrix. This section retains the Born mixing basis and explicitly puts the corrections in the effective matching denominator. The resulting values belong to this specified neutral process.

The Z effective matching denominator has three terms. The first is the specified projected contribution $18c_v^2$ of the Born-level denominator 18 onto the massive combination. The second retains the redistribution correction $-3\mathrm{w}_1{+}\mathrm{w}_2{+}2\mathrm{w}_3$ of the weak effective denominator. The third comes from two additional mixed routes for the neutral member, $\chi_2\chi_3{=}\chi_5$ and $\chi_4\chi_3{=}\chi_1$. If these endpoints leave distinguishable records and each contributes w_2 to the same effective matching denominator, their total addition is $2\mathrm{w}_2$. Thus

$$\boxed{N_Z = 18c_v^2 + (-3w_1 + w_2 + 2w_3) + 2w_2 = 14.174941350700\ldots.} \tag{10.10}$$

Character multiplication proves the destinations of the two connections, but does not alone ensure that they are actual processes exclusive to the neutral member. Neutral projection, record orthogonality, and each contribution to the same effective matching denominator N_Z remain routing conditions of this expression. To preserve the zero mode of Section 10.2 simultaneously, records may distinguish only routes; both entrances within each route must still merge coherently into its output through the same combination $c_v\langle w| - s_v\langle y|$. If final states combine coherently, or the charged member also opens the same routes, the weights must be recalculated.

For the massive combination of Section 10.2, take $\varpi_Z = 1$. Projection effects are already included in N_Z, so c_v^2 is not multiplied into the numerator again. Substitution gives

$$\boxed{M_Z = v\sqrt{\frac{C^*}{N_Z}} = 91.4148013904\ldots\ \mathrm{GeV}.} \tag{10.11}$$

This value and the rank-one matrix of Section 10.2 jointly determine the neutral rest-mass spectrum. The result uses the collinear response columns of Section 10.2 and requires this section's participation, projection, and routing to be realized within the same joint process.

10.3.3 Vertex Angle and Mass-Ratio Angle

Both massive excitations use the same $\mathrm{v}^2 C^*$, so their ratio cancels the absolute energy scale. Defining the mass-ratio angle $s_M^2 = 1 - M_W^2/M_Z^2$ gives directly

$$\boxed{\frac{M_Z}{M_W} = \sqrt{\frac{N_W}{N_Z}} = 1.134554387139\ldots}, \qquad s_M^2 = 1 - \frac{N_Z}{N_W} = 0.223128193000\ldots. \tag{10.12}$$

s_v^2 reads the Born vertex-response ratio, while s_M^2 reads the two corrected masses. They use different process quantities and differ by approximately 0.00834 under the present specifications. Before comparing them with experimental definitions such as effective mixing angles or on-shell mass ratios, their vertices, scales, and schemes must be matched. Similar names do not make their numerical values interchangeable.

As a separate mass-ratio test, using $M_Z = 91.1880\mathrm{GeV}$ from the PDG 2025 gauge and Higgs boson summary [17] with only the ratio above gives $M_W = 80.3734056593\ldots\mathrm{GeV}$. This calculation already uses an experimental mass as its scale and tests something different from independently calculating both absolute masses from v and C^*. Its smaller residual cannot replace the approximately 0.25% deviations of the latter calculation.

The cited W and Z measurements use Breit–Wigner mass definitions with mass-dependent widths. Comparing cycle rest-frame frequencies with such resonance masses still requires matching mass schemes. The detailed PDG W listing [17] states that its summary excludes the CDF 2022 measurement, while the Z listing [17] also distinguishes this mass from the complex-pole definition. The same summary gives $M_W = 80.3692 \pm 0.0133\mathrm{GeV}$ and $M_Z = 91.1880 \pm 0.0020\mathrm{GeV}$. The present absolute evaluations are higher by 0.25396% and 0.24872%, respectively, exceeding the listed experimental uncertainties. Deviations in the same direction suggest checking the energy scale and mass-share identification together. Numerical convergence supplies no theoretical error bounds on these identifications.

10.4 Gluons

10.4.1 The Eight-Dimensional Color Channel

Color exchange acts on the three-dimensional internal carrier $V_c = \mathbb{C}^3$. A local color response is represented by an operator in $\mathrm{End}(V_c)$. Separating trace and traceless parts gives

$$\mathrm{End}(\mathbb{C}^3) = \mathbb{C} I_3 \oplus \mathfrak{sl}_3(\mathbb{C}), \qquad \mathcal{C}_8 := \{X = X^\dagger : \mathrm{Tr}\, X = 0\}, \qquad \boxed{\dim_{\mathbb{R}} \mathcal{C}_8 = 8.} \tag{10.13}$$

A self-adjoint three-by-three matrix has nine real parameters; imposing one trace condition leaves eight. A concrete basis consists of three pairs of off-diagonal matrices $E_{ab} + E_{ba}$ and $-i(E_{ab} - E_{ba})$, together with two independent traceless diagonal matrices. Multiplication by i gives the anti-self-adjoint convention for su(3); both conventions describe the same eight-dimensional real representation.

Color transformations act by $\mathrm{X} \mapsto \mathrm{U\ X\ U}^\dagger$. Identifying the internal modes of channel excitations with this adjoint representation yields eight internal color directions.

Propagation polarizations are counted separately through the spatial propagation representation. Whether these modes can be prepared in isolation must be tested through actual separation processes and subsequent maintenance conditions.

The trace part commutes with every color action and is a color-blind response. Its decomposition from traceless exchange follows from the representation structure. General traceless directions have nonzero commutators, so composing two color responses can again produce a color response, unlike the vanishing Abelian bracket. Concrete self-interaction strengths and occurrence rates still require actual vertices and records.

10.4.2 Mass Contributions and Independent Outputs

Under the factor identification of Chapter 5, the quadratic character χ_3 lies in the binary factor, while χ_2 and χ_4 are trivial on it. Specify that symmetry breaking acts only on that factor and that the color channel acquires no condensate mass share. Then $\varpi_g = 0$ makes the mass contribution in Section 10.1 vanish. Preservation of the color representation alone allows positive mass: on the irreducible adjoint representation, an invariant positive mass-squared operator may be $\mu^2 \mathrm{I}_8$ for any $\mu^2 \geq 0$. Thus $M_g = 0$ depends on the chosen mass mechanism and the absence of additional mass terms.

Gluon modes have a definite role in internal nucleon exchange. The color response of Chapter 7 and internucleon exchange of Chapter 9 already use the corresponding vertices; the adjoint representation here specifies how the excitations carry and transfer color information. Being colored and being readable are different questions, and self-adjoint color operators can contain superpositions of opposite gradings.

For a separated independent gluon output, Section 9.6 tests the complete output and subsequent maintenance on the adjoint carrier. The eight-dimensional color structure, zero mass under the adopted condensate mechanism, and conditions of independent existence each act on a clearly specified object.

10.5 Pions

10.5.1 Pion Quantum Numbers

The nucleon exchange of Chapter 9 can change internal flavor while preserving the common collective pattern. Here one color-singlet, flavor-nontrivial exchange component is identified with the pion. Its carrier includes a fundamental color representation and a conjugate representation, allowing a two-factor color singlet, unlike the three-fundamental-representation structure of a baryon.

Let $|\bar{a}\rangle$ carry the conjugate of the fundamental representation. Use the normalized color vector Ω_c and the following flavor and spin components:

$$|\Omega_c\rangle = \frac{1}{\sqrt{3}} \sum_{a=1}^{3} |a\rangle|\bar{a}\rangle, \qquad (U \otimes \bar{U})|\Omega_c\rangle = |\Omega_c\rangle, \qquad \eta_0 = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{10.14}$$

$$
\begin{aligned}
\phi_{\pi^+} &= |u\bar{d}\rangle, \\
\phi_{\pi^-} &= |d\bar{u}\rangle, \\
\phi_{\pi^0} &= \frac{|u\bar{u}\rangle - |d\bar{d}\rangle}{\sqrt{2}}, \\
Q\phi_{\pi^q} &= q\phi_{\pi^q}, \\
q &= +1, -1, 0.
\end{aligned}
\tag{10.15}
$$

The antiparticle representation has the opposite charge, so the charges in these expressions add directly. Introducing an antiparticle factor is part of the physical representation identification; arithmetic $\bar{\chi}$ alone is not treated as an already constructed antiparticle. The spin state η_0 is invariant under total SU(2), giving S=0. For relative orbital angular momentum L=0, the total angular momentum is J=0.

Spatial inversion must be specified separately. Adopting intrinsic parities whose product for component and anticomponent is -1 gives negative total parity at L=0. The pion is then read as $J^P = 0^-$, a pseudoscalar. Evenness of a character under negation is not spatial parity and cannot identify the pion with the positive-parity scalar Higgs. The pion quantum numbers listed in the PDG 2025 meson summary [17] correspond to this representation realization.

These vectors provide testable internal carriers. Actual emission and absorption must still connect to nucleon endpoints, joint conservation, and complete propagation. Writing a color singlet does not by itself calculate the transition amplitude or lifetime.

10.5.2 Pion Mass

The pion-mass identification combines the common nucleon energy scale with an internal flavor response. Retain $M_B = 936.669312626\ldots$ MeV from Section 9.3 and take the internal inputs m_u and m_d from the same mass readings of Chapter 8. This expression uses the model's collective baseline M_B, without introducing the undetermined common shift δ_0 of Section 9.4 or the experimental proton mass.

To distinguish multiplication from addition, first nondimensionalize the inputs with the same energy unit E_0. Specify the collective-side reading $b_B = M_B/E_0$ and the two orthogonal flavor-outcome readings $b_u = m_u/E_0$ and $b_d = m_d/E_0$, using a product preparation of the collective and flavor sides. The orthogonal sum gives $b_u + b_d$; only the product preparation makes the evaluations on both sides multiply. Then select three equally strong color-rearrangement routes distinguishable in the complete output, giving total response $3b_B(b_u + b_d)$.

These numbers are specified positive response values and become event probabilities only after actual normalization is completed. Identifying the amplitude-level reading of the total response with the pion mass gives the conditional mass relation

$$
\left(\frac{m_\pi}{E_0}\right)^2 = 3\frac{M_B}{E_0}\left(\frac{m_u + m_d}{E_0}\right), \qquad \boxed{m_\pi^2 = 3M_B(m_u + m_d).}
\tag{10.16}
$$

The three routes here are separately specified reversible orthogonal exchange outputs; the three coherent components of the color singlet do not directly serve as those routes. The complete vertex column must independently give $S_\pi = 3b_B(b_u + b_d)$, and

the effective drive over the complete cycle period must be $\nu = E_0$, before $m_\pi = \nu\sqrt{S_\pi}$ yields the expression above. For the conditional connection of response columns, coherent return, and actual emission–propagation–absorption, see Supplementary Proof S4.4.

Using the unrounded inputs above, with $m_u + m_d = 6.805667343664\ldots\,\mathrm{MeV}$, gives

$$\boxed{m_\pi = 138.289476310587\ldots\ \mathrm{MeV}}, \qquad \frac{m_\pi^2}{M_B(m_u + m_d)} = 3. \tag{10.17}$$

This gives a common pion mass before electromagnetic splitting. PDG 2025 gives $m_{\pi^\pm} = 139.57039 \pm 0.00018\mathrm{MeV}$ and $m_{\pi^0} = 134.9768 \pm 0.0005\mathrm{MeV}$; the present common reading differs from them by -0.91775% and $+2.45426\%$, respectively. Splitting between charged and neutral members must be calculated separately.

10.6 Condensate Oscillations and the Higgs

10.6.1 Radial Oscillations

Condensate oscillations change the amplitude of the collective background. Let v denote the selected background scale and h a nearby real radial displacement; restoration and phase evolution of h constitute a candidate oscillation. Real h describes an amplitude coordinate and does not require every participating complex restoration mode to be a real character. The actual oscillation frequency must be read from the response or generator near that background.

In the original Higgs model, amplitude fluctuations around a symmetry-breaking background include a massive scalar excitation [31]. This chapter selects an electrically neutral amplitude h that is invariant under rotations of observational space and even under spatial inversion, and identifies it with a Higgs candidate of $J^P = 0^+$. The trivial internal representation preserves the corresponding internal labels, while the spatial scalar property is specified by the rotation realization of h. The condensate pattern b, the trivial pole in the logarithmic spectrum, and preparation of a concrete maintainable background retain their separate definitions from Chapters 2 and 9.

This section calculates the actual return frequency of the selected radial process; Section 11.4 discusses this branch's coverage.

10.6.2 Condensate Restoration Probability

This section identifies the two all-equal orientation outcomes with restoration of the same radial mode. Each point pair's orientation is stored separately in an auxiliary, and accepted outcomes connect to that mode through actual retrieval. This is a connection condition of the chosen restoration process. For any odd prime p, the species layer has $m = (p-1)/2$ nonzero pairs under negation, each containing one even and one odd direction. These point pairs form a direct sum in address space; their orthogonality does not itself supply the independent preparation required below.

Use the following finite-register realization: assign a binary factor $\mathcal{A}_j \cong \mathbb{C}^2$ to each of the m checks and prepare independent uniform states $\mathrm{I}_2/2$. Here $\mathcal{A}$ is an auxiliary record carrier within the joint process, not a reinterpretation of the original

address space as a tensor product. In each factor, choose orthogonal vectors $|+\rangle$ and $|-\rangle$ corresponding to even and odd check outcomes. The event projection for the two all-equal orientation outcomes is

$$\mathcal{A} = \bigotimes_{j=1}^{m} \mathbb{C}^2, \qquad \rho_{\mathcal{A}} = \bigotimes_{j=1}^{m} \frac{I_2}{2}, \qquad P_{\pm} = \bigotimes_{j=1}^{m} |\pm\rangle\langle\pm|, \qquad P_+P_- = 0. \tag{10.18}$$

A normalized vector in each factor has probability $1/2$ under $\mathrm{I}_2/2$. Product preparation and factorization of the trace therefore give probability 2^{-m} for each term. Defining the single-orientation share λ_H gives

$$\boxed{\lambda_H = \mathrm{Tr}(\rho_{\mathcal{A}} P_+) = \mathrm{Tr}(\rho_{\mathcal{A}} P_-) = 2^{-m}}, \qquad \mathrm{Tr}[\rho_{\mathcal{A}}(P_+ + P_-)] = 2^{1-m} \leq 1. \tag{10.19}$$

The two terms add because they are mutually exclusive input events. Each can be retrieved into the same normalized radial mode $|\mathrm{h}_0\rangle$ while retaining orthogonal environmental records r_+ and r_-:

$$K_{\pm} = |h_0, r_{\pm}\rangle\langle\pm, \ldots, \pm|, \qquad K_{\pm}^{\dagger} K_{\pm} = P_{\pm}, \qquad K_0^{\dagger} K_0 = I - P_+ - P_-. \tag{10.20}$$

The complementary outcome K_0 can be mapped to an orthogonal failure-output space. These effects then satisfy completeness, and the corresponding Kraus actions form a complete instrument. This proves a finite-dimensional realization of the required two-path share. Interpreting the two all-equal orientation outcomes as actual orientations of the selected condensate-restoration process is an additional process identification; the associated energy, charges, and compensating quantities must still be checked in the joint realization. If only event P_+ exists, attaching two different records merely divides its original probability: the sum remains λ_H and cannot become $2\lambda_H$.

This independent uniform preparation and selection of all-equal orientation outcomes determine the accepted share. Correlated preparations require direct calculation of $\mathrm{Tr}(\rho_{\mathcal{A}} P_{\pm})$ and no longer guarantee 2^{-m}. For p=7, m=3, the single-path share is $\lambda_H = 1/8$ and the two-path total is q=1/4.

10.6.3 Higgs Mass

Converting the two-path share into a frequency additionally requires a return endpoint. Take $g_H > 0$, first calculate the active half-gap, then include thermal processing and waiting. Retain the mixed preparation $\rho_{\mathcal{A}} = I/2^m$, and let $P = P_+ + P_-$ and $q = 2^{1-m}$. If only $g_H P \otimes \sigma_x$ is implemented, the output signal is $q \sin^2(g_H t)$: the share changes the amplitude, while the accepted-branch frequency remains g_H.

This section uses a reversible purified preparation to realize a share-dependent frequency. Purify the same mixed preparation as $\chi = 2^{-m/2} \sum_z |z, R_z\rangle$, retain operational access to all purification auxiliaries, and select the single entrance i. The actual bidirectional vertex and its Gram value are

$$B_H = g_H (P \otimes I_R)|\chi\rangle\langle i|, \qquad B_H^{\dagger} B_H = g_H^2 q P_i, \qquad \boxed{\mu_H = g_H \sqrt{q} = g_H \, 2^{(1-m)/2}.} \tag{10.21}$$

Here μ_H denotes the rest-frame half-gap during activity. The normalized bright vector is $(P \otimes I_R)\chi/\sqrt{q}$. Together with the entrance, it forms a two-dimensional invariant space with coupling strength $g_H\sqrt{q}$, so the half-gap above arises from actual exchange. The entrance-to-bright-output transfer probability is $\sin^2(g_H\sqrt{q}\,t)$. For $m > 1$, it agrees with the mixed-preparation scheme at second order in short time but differs at fourth order; for $m = 1$, both have $q = 1$. Acceptance labels must remain reversible auxiliaries throughout the round trip; permanently distinguishing the two outputs changes this closed oscillation. g_H is the active coupling strength. The effective frequency over the complete cycle period follows from the common time accounting below.

A joint action can realize a common strength for matter and background. Let the isometry J act only on the formed source and its fixed auxiliaries. Let the actual source operator $A = A^\dagger$ satisfy $J^\dagger AJ = aI$ and $a > 0$, and take $S = 2JJ^\dagger - I$. The dimensionless self-adjoint operator K_H uses $(P \otimes I_R)\chi$ as its outward column and contains the complete inverse exchange. The dimensionless generator G on an independent background factor retains the actual gradient block $\sqrt{b_r}\sigma_x$ of Section 2.5.3 and satisfies $\sum_r b_r = C_{\mathrm{bg}}$. Choose the common drive $g(t) \geq 0$, set both relative coupling coefficients to one, and activate both protection arms simultaneously according to Eq. (2.38)–Eq. (2.39):

$$\begin{aligned} L &= K_H \otimes I + I \otimes G, \qquad H(t) = \frac{g(t)}{2}(A \otimes L + SAS \otimes L), \\ H(t)(J \otimes I) &= (J \otimes I)ag(t)L, \qquad \boxed{\nu = \frac{a\int g(t)\,dt}{T}, \quad m_H = \nu\sqrt{q}.} \end{aligned} \tag{10.22}$$

Here $J \otimes I$ embeds the joint source, matter, and background state. The intertwining relation follows from $SJ = J$ and $(I+S)AJ = 2Ja$. Matter and background occupy different tensor factors of L and are therefore driven by the same coefficient $ag(t)$. T includes every active, return, and waiting interval of the selected completed history, while the integral accumulates only actual active action. Thus the instantaneous Higgs coefficient is $ag(t)$, and its effective drive strength over total elapsed duration equals the background's ν.

Below we use the common drive rate ν for a fixed complete cycle period. Completed histories of different durations have their phase rates calculated separately as in Section 2.5.3.

This coherent bright mode uses a common source, relative unit coupling, and reversible purification auxiliaries. The joint action and complete return above give $m_H/\nu = \sqrt{q}$ on the same history. The background uses W_{vac}, E_0, and the complete squared-response quantity $C_{\mathrm{bg}} = C^*$ of Chapter 6. Calibrating $\nu = v$ by the squared-matching rule of Eq. (2.75) and the energy-scale relations Eq. (6.13)–Eq. (6.14) gives

$$\begin{aligned} \left(\frac{m_H}{v}\right)^2 &= 2\lambda_H = 2^{1-m}, \\ m_H &= v\,2^{(1-m)/2}, \\ \boxed{p = 7: \quad m_H = \frac{v}{2} = 123.095686966710\ldots\ \mathrm{GeV}.} \end{aligned} \tag{10.23}$$

The PDG 2025 summary gives a Higgs mass of 125.20±0.11 GeV. This section's result is lower by 1.68076% and does not reach that experimental precision. The listed mass uses the background preparation, orientation share, and frequency identification above; its theoretical error has not been quantified. Different actual backgrounds, joint preparations, or orientation outputs require the frequency to be recalculated from the process. The remaining discrepancy cannot simply be absorbed into an undefined correction term.

The relation of the radial condensate amplitude to other masses can be calculated explicitly with dimensionless spectral data fixed. If the mass functions are $m_f(v) = a_f v$ and $M_V^2(v) = b_V v^2$, inserting the same normalized radial change $v \to v + h$ gives

$$\left.\frac{\partial m_f}{\partial h}\right|_{h=0} = \frac{m_f}{v}, \qquad \left.\frac{\partial M_V^2}{\partial h}\right|_{h=0} = \frac{2M_V^2}{v}, \qquad M_V^2(v+h) = M_V^2\left(1 + \frac{2h}{v} + \frac{h^2}{v^2}\right). \quad (10.24)$$

For fixed a_f, b_V, radial responses have definite proportions to the existing masses, with the coefficients above for fermion mass terms and vector mass-squared terms, respectively. If these coefficients vary with the background, their derivatives contribute additional terms. Connecting the resulting responses to Higgs transition couplings additionally requires a common dynamical normalization of h and a relation between the mass operator and actual transition vertices.

Combined with the W mass, the current identification gives $M_W/m_H = 2\sqrt{C^*/N_W} = 0.654558345158\ldots$: the common v cancels, while C^* remains. This tests compatibility between channel mass shares and the radial share, unlike the W/Z test that cancels C^* completely. λ_H is defined here first as the single-orientation event share. Further identification with a quartic coefficient in an effective potential requires a separate correspondence between the potential and process.

10.6.4 Comparison of Boson Evaluations

Table 10-1 places this chapter's structural realizations alongside their mass readings. W, Z, Higgs, and pion masses use the same set of unrounded inputs; parenthetical deviations compare only the stated PDG 2025 central values. Zero-mass entries denote zero eigenvalues of the present mass mechanism. Quantum numbers are read from the rotation, parity, and internal representations specified in this chapter.

Table 10-1 Boson identifications and conditional mass readings. Mass units are indicated by row; experimental comparisons use the PDG 2025 summaries cited in this chapter.

Object	Structure and quantum numbers	Mass reading	Main conditions
γ	Neutral zero mode; two transverse polarizations	0	Rank-one neutral mass matrix; vector and transverse-propagation identifications
W^+, W^-	Weak channels; charged vectors	80.5733 GeV (+0.25396%)	Condensate share 1; effective matching denominator N_W
Z	Massive neutral combination; vector	91.4148 GeV (+0.24872%)	Born mixing basis; neutral projection and two-path records

Object	Structure and quantum numbers	Mass reading	Main conditions
Gluons	Eight-dimensional adjoint color representation; vector identification	Present condensate mass contribution 0	Color share 0 and no other mass terms; independent colored outputs subject to the process criterion of Section 9.6
π	Color-singlet transition; 0^-	138.2895 MeV (-0.91775% against $\pi\pm$; $+2.45426\%$ against π^0)	Two-layer product preparation and three-path sum; electromagnetic splitting not calculated
Higgs	Neutral radial oscillation; 0^+	123.0957 GeV (-1.68076%)	Retained purification of independent registers, coherent bright-mode return, and strength from a common source

The three mass types in this chapter read condensate participation, collective and flavor responses, and radial restoration frequencies, respectively. Actual channel and radial frequencies have been obtained from the specified bidirectional exchanges and returns, with absolute calibration through the common background. The pion mass uses the mass identification of two-layer responses. They share one background scale, and their respective preparations, vertices, and physical inputs are given in the corresponding constructions.

Propagation, decay, and spin–statistics properties must still be tested for each excitation type. Together with elementary states and baryons, the coverage of the resulting matter and excitation identifications is examined next through allowed representations, actual returns, and the complete fluctuation spectrum.

§11 Spectral Completeness

The preceding chapters gave concrete representations and evaluations for elementary-state species patterns, collective baryon structures, and exchange excitations. Bringing them together raises a further question: have all allowed structures been listed, and does every listed structure correspond to a mode that can actually form and persist? Only after both ranges are established does completeness of the matter and force inventory have a definite meaning.

This chapter tests the inventory at three levels of the same objects: whether species bases and representations cover the stated space, whether actual cycles preserve candidate carriers, and whether unlisted modes can form, persist, and be read. Construction parameters remain an odd prime p and finite depth K, with the physical identifications for $p = 7$ following Chapter 5. Completeness of structural classification and exhaustion of the physical spectrum are assessed separately at these levels.

11.1 Elementary States and Generation Structure

11.1.1 Species Layer and Complete Direction Space

Character orthogonality within shells in Section 2.2.1 gives $p-1$ orthonormal directions in the modulo-p species layer, whose trivial direction is the unit-shell average state e_0. Thus

$$\mathcal{H}_{\mathrm{sp}} = \mathbb{C}e_0 \oplus \mathcal{M}_{\mathrm{sp}}, \qquad \dim \mathcal{H}_{\mathrm{sp}} = p-1, \qquad \dim \mathcal{M}_{\mathrm{sp}} = p-2. \tag{11.1}$$

The complete state space contains every valuation shell. The decomposition in Eq. (2.13) gives

$$\dim \mathcal{C} = K+1, \qquad \boxed{\dim \mathcal{M} = \sum_{r=1}^{K} \big(\varphi(p^r) - 1\big) = p^K - K - 1}, \qquad \dim \mathcal{H} = p^K. \tag{11.2}$$

For $p = 7$, the five nontrivial labels exhaust the species layer $\mathcal{M}_{\mathrm{sp}}$, while the complete direction space has dimension $7^K - K - 1$. Actual modes in the remaining shells and higher-conductor directions are tested by the return criterion of Section 11.5.

11.1.2 Conjugacy Orbits and Generation Groups

Following Sections 2.2.3 and 5.1, nontrivial species split into one real direction and several complex-conjugate pairs. Writing m=(p−1)/2 gives

$$p - 2 = 1 + 2\frac{p-3}{2}, \qquad N_{\mathrm{conj}} = 1 + \frac{p-3}{2} = \frac{p-1}{2} = m. \tag{11.3}$$

The negation involution of Section 2.2.3 gives m point-pair groups among modulo-p addresses, while all nonzero addresses at depth K form $(p^K - 1)/2$ point pairs. Species conjugacy and address point-pair grouping act on different objects; for $p = 7$ they give, respectively,

$$\{\chi_3\}, \quad \{\chi_2, \chi_4\}, \quad \{\chi_1, \chi_5\}; \qquad \{1, 6\}, \quad \{3, 4\}, \quad \{2, 5\}. \tag{11.4}$$

Conjugacy orbits distinguish species phases, whereas address point-pair groups supply generation readings of a given formation pattern. Under the physical identifications of Chapter 5, Chapter 8 lists three mass readings each for neutrinos, charged leptons, up-type quarks, and down-type quarks, twelve in total. Antiparticles, color, and spin enter through the complete representations. Table 11-1 gives the actual mode coverage of this inventory.

11.1.3 Pattern-Preserving Modes

First classify the pattern-preserving modes of the shell-preserving spectral response T. Suppose it is diagonal in the character basis of the space under consideration, $Tf_a = R_{\chi_a} f_a$. Writing an arbitrary vector as $\psi = \sum_a c_a f_a$ gives

$$T\psi = \lambda\psi \quad \Longleftrightarrow \quad (R_{\chi_a} - \lambda)c_a = 0 \quad (\forall a); \qquad \lambda \neq 0. \tag{11.5}$$

Comparing coefficients shows that nonzero pattern-preserving states lie precisely in eigenspaces of nonzero eigenvalues. Components with different complex eigenvalues cannot mix within one mode; components with the same eigenvalue may be coherently superposed arbitrarily. Equal moduli with different phases do not suffice to preserve the whole pattern. Although the zero eigenspace belongs to the expansion basis, it does not satisfy the nonzero-return condition used here.

An eigenspace of dimension two or greater contains infinitely many distinct pure-state rays, but each ray does not introduce another particle species. Physical species also depend on symmetry representations, readable distinctions, and complete cycles. For a simple spectrum, each nonzero eigenspace is one character direction; with degeneracy, the proper object of classification is the whole eigenspace.

These eigenspaces exhaust nonzero pattern-preserving modes of the given spectral response T. The complete return also includes internal-mode mixing and transfer to the background. Its classification uses the common-eigenvector conditions of Section 11.5, applied to the complete joint carrier or, once intertwining holds, to the corresponding logical space.

11.2 The Emergent-State Sequence

Chapter 9 proved that the common fixed pattern is b and that a pure reduced mode state corresponds to the joint state $P_b \otimes \sigma$. Different internal representations and return actions can distinguish collective members on the same pattern. The unit-shell direction e_0 and full-space pattern b enter the inventory separately. Uniqueness of a fixed direction or spectral pole does not determine the number of internal matter families.

Let $V_c = \mathbb{C}^{d_c}$, with $d_c \geq 2$, carry the fundamental representation of $\mathrm{SU}(d_c)$. For n such copies, the central phase and completely antisymmetric tensor in Section 6.3 give the necessary and sufficient condition for a color singlet:

$$\boxed{\mathrm{Inv}_{SU(d_c)}(V_c^{\otimes n}) \neq \{0\} \quad \Longleftrightarrow \quad d_c \mid n}, \qquad n \geq 1. \tag{11.6}$$

For $d_c = 3$, the allowed levels of purely fundamental representations are $3, 6, 9, \ldots$. The lowest three-copy color-singlet space is one-dimensional; higher levels may contain several internal directions. With conjugate representations included, $V_c \otimes \bar{V}_c$ already contains the color singlet used in Section 10.5. Here n counts tensor positions; each level's carriers, internal representations, and actual returns are constructed separately.

The energy-conserving reference swap of Section 9.1 and persistent action of Section 9.3 already provide formation and maintenance of the selected nucleon carrier. Higher allowed levels provide new candidate carriers. Their internal representations, participation weights, and maintenance frequencies are calculated from the actual process at that level; one energy scale at the lowest level does not replace all higher-level spectra.

11.3 Forces and Channel Excitations

11.3.1 Complete Classification of Factor-Preserving Actions

Elementary resonance channels are organized by the single-factor effective actions of Chapter 3. Write $p-1 = \prod_i n_i$, where n_i are pairwise coprime prime powers and $r = \omega(p-1)$. With the trivial direction retained, the species layer becomes $\bigotimes_i \mathbb{C}^{n_i}$. Multiplying each factor's traceless self-adjoint generators by i gives $\mathfrak{su}(n_i)$, while the scalar actions of all factors coincide as one $\mathfrak{u}(1)$. Thus

$$\mathfrak{g}_{\text{fact}} \simeq \mathfrak{u}(1) \oplus \bigoplus_{i=1}^{r} \mathfrak{su}(n_i), \qquad N_{\text{factor}} = r + 1, \qquad \dim_{\mathbb{R}} \mathfrak{g}_{\text{fact}} = 1 + \sum_{i=1}^{r} (n_i^2 - 1). \quad (11.7)$$

Local actions on different factors commute, and commutators within each factor remain in that factor, so this action class is closed under the Lie bracket. For p=7, there is one Abelian factor, one SU(2) factor, and one SU(3) factor: three structural factors and twelve real generator directions. The central quotient of the global group is fixed as in Section 3.2 and adds no local generator directions.

This classification covers factor-preserving generating actions. A general six-by-six self-adjoint operator has 36 real dimensions; besides the 12 above, there are 24 dimensions spanned by tensor products of traceless operators on both factors. Actual actions determine whether such direct joint generator terms are allowed. The classification of Chapter 3 uses the stated factor-preserving action class.

Sequential products of local operators can also contain such two-factor tensor products. This establishes a composite process without automatically making it a new independent interaction. Factor counts, single-step generator terms, and complete composite paths are treated separately as in Chapter 3. Using the central U(1) as an elementary response also requires implementation of its action relative to an internal reference and its charge representation as in Section 5.2.

11.3.2 Five Interaction Types and Their Excitations

After adopting the matter representations and interaction identifications above, the three elementary structural factors provide electromagnetic, weak, and color responses. Resonance between collective states is organized separately as in Sections 3.5 and 9.5. Metric action arises as in Chapter 4 when a matter source changes the

background and background propagation changes maintenance. Under the process classification adopted here, the total is

$$\boxed{\text{interaction types: } 3+1+1=5} \qquad \text{electromagnetic, weak, color, nuclear, metric} \tag{11.8}$$

This counts five response types, not a separate force for every generator, record route, or propagation mode. Section 9.5 explained the relation between internucleon resonance and internal color exchange. The five-type reading does not require five mutually unrelated elementary microscopic actions.

The same common pattern permits different internal phases and masses among nucleons. Resonance is still determined by endpoint frequencies, energy balancing, and propagation; equal b does not guarantee automatic matching. Metric action is transmitted by the common background. Source writing, the propagation kernel, and maintenance response are calculated within the same process of Chapter 4, with observational geometry supplying the corresponding reading structure.

Chapter 10 supplied excitation realizations for these interactions: the photon is a zero direction of the chosen neutral mass-squared matrix; W and Z arise from weak and neutral massive modes; gluons use the eight-dimensional adjoint color representation; and pions represent specified nucleon transitions. In the current construction, metric action is described by carrier momentum response to background changes; a gravitational propagation quantum has not been separately constructed. Excitations are counted by actual representations and propagation equations, without forcing a one-to-one correspondence with the five interaction types.

The eight internal color directions and spatial polarizations are counted separately, and the two charged W members do not increase the number of weak-action factors. The pion is an identification of color-singlet internal exchange; not every allowed composite color singlet can be collapsed into this one mode.

11.4 Condensate Oscillations

Under reversible purification, coherent bidirectional exchange, and common complete time accounting, Section 10.6.3 constructed a two-dimensional invariant space consisting of the entrance and bright output, with frequency $m_H = v\sqrt{q}$. The selected p=7 radial branch gives $m_H/v = 1/2$. Fixing dimensionless spectral data and using the same normalized radial perturbation also gives the corresponding mass responses. These results cover that restoration process. Whether the complete physical fluctuation spectrum has additional normal modes of the same type must be tested through the actual restoring action and its readout.

11.5 Completeness and Testable Conclusions

11.5.1 Scope of the Classification

The classifications above can be brought together by their objects. “Coverage” in Table 11-1 means the range established by bases, representations, or specified processes; physical names retain the complete identifications given earlier. Internal dimensions, particle-species counts, and interaction types are counted separately.

Table 11-1 Scope of structural and physical-spectrum correspondences. Counts in different columns refer to different objects and cannot be added across levels.

Object	Established structure	Physical identification in this paper	Completeness conditions
Elementary states	p−2 nontrivial species-layer directions; 5 for p=7	Four classes, each with three mass readings	Complete representations, endpoints, and actual branches; conditions covering modes in other shells and higher-conductor directions
Generation readout	Nonzero modulo-p addresses form m point-pair groups	Three group readings across generations for p=7	Realization and exhaustion of formation and mass actions on these groups
Collective states	Common pattern b; a purely fundamental color singlet requires d_c to divide n	Nucleons and corresponding internal-state families	Selected formation and maintenance conditions and coverage of complete internal representations
Resonance and metric action	r+1 structural factors for factor-preserving actions, plus collective resonance and metric response	Five interaction types in this paper	Allowed action class, actual exchange, and geometric response
Exchange excitations	Selected neutral, weak, color, and color-singlet propagation realizations	γ, W, Z, gluons, and π	Propagation representations, mass mechanisms, and independent outputs
Condensate oscillations	Two-dimensional invariant space of the constructed bright mode and its frequency	Selected radial Higgs process	Conditions of the constructed frequency mechanism and coverage of the complete fluctuation spectrum

Each completeness claim in the table is restricted to the stated space, representation, or process. Additional generations, independent interactions, and Higgs-like modes test coverage of mass branches of the same kind, allowed actions, and the physical restoration spectrum, respectively. Actual return modes are determined by the common-eigenvector criterion in the next section.

11.5.2 Coverage Criterion for Return Modes

The coverage criterion must apply to the object that actually persists. For the construction in Chapter 2, one may calculate on the complete joint carrier or move to its logical space when the intertwining relation holds. Retained sources, auxiliaries, and permanent records must be consistent with that representation. Fix such a finite carrier, the environmental preparation per round, and the return outcomes under consideration. Denote the return Kraus operators by R_a. Their sum defines the completely positive trace-nonincreasing map $\mathcal{Q}(\rho) = \sum_a R_a \rho R_a^\dagger$. For a normalized

pure state ψ, the returned state preserves the same ray with nonzero return share if and only if

$$\boxed{\mathcal{Q}(P_\psi) = s_\psi P_\psi,\ s_\psi > 0 \quad \Longleftrightarrow \quad R_a\psi = c_a\psi\ \ (\forall a), \quad s_\psi = \sum_a |c_a|^2 > 0.} \tag{11.9}$$

The proof uses positivity of each outcome. Projecting the return output onto the orthogonal complement of ψ gives trace $\sum_a \|(I-P_\psi)R_a\psi\|^2$. This vanishes precisely when every term vanishes, meaning every $R_a\psi$ is collinear with ψ. Taking the trace then gives s_ψ. The converse follows by substitution. This criterion does not require different R_a to commute.

All pattern-preserving pure return modes therefore arise from common eigenspaces $E_c = \bigcap_a \ker(R_a - c_a I)$, with nonzero vector c. Covering every such pure return mode requires the candidate carriers to contain all nonzero common eigenspaces; classification of physical objects additionally depends on actual observables. Mass also requires reading phase relative to a reference. Preservation of a pure state by a reduced map does not alone supply this comparable phase.

The formation cycle of Section 2.3.7 already has a concrete spectral result (Supplementary Proof S3): the cycle eigenvalues on coherence components within the code subspace are $q_T e^{-i(E_a-E_b)t_H}$, and the remaining transient spectral radius is at most $q_T\sqrt{\kappa} < q_T$. After normalization by survival, the selected transient domain therefore approaches the persistent orbit. This separates modes within that formation process. It allows input from outside the code and does not exclude other carriers not included in the construction.

§12 Conclusions

This paper connects matter, forces, and observational spacetime through the persistence of the same excitation: background exchange forms and maintains patterns, relative persistent phases provide mass readings, carrier exchange and background response produce interactions, and internal records and calibration make these changes into events. All evaluations share a state space and scales while retaining explicit preparations, actions, and physical identifications.

12.1 Common Dynamics and Evaluations

Stable states form through background exchange and preserve their patterns during subsequent propagation and return. Formation convergence, continuous survival, and phase evolution describe pattern establishment, lifetime, and frequency, respectively. A finite lifetime does not prevent the surviving component from preserving its pattern. In natural units $\hbar = c = 1$, continuously unwrapping phase relative to an internal reference and using the complete cycle period τ gives the mass difference

$$\boxed{m_\nu - m_\mu = -\frac{\Delta\widetilde{\varphi}_{\nu\mu}}{\tau}}. \tag{12.1}$$

Path responses enter the family mass matrix M_f through actual exchange. Its eigenvalues $\mu_{f,j}$ give $m_{f,j}^{\mathrm{obs}} = f_{\mathrm{act}}|\mu_{f,j}|$, where $f_{\mathrm{act}} = t_H/\tau$ retains all processing times. Once the shared motion port is connected, the same mass parameter controls both the rest-frame half-gap and inertia at the origin. A formation share becomes a frequency coefficient only through this actual action.

Paths, nodes, and the formation pattern jointly determine scales between charged elementary-state families and hierarchies within them. For the factorizable branches constructed in this paper,

$$\boxed{M_{f,j} = vA_f\,\mathrm{Gen}_{f,j}(1-\alpha_{\mathrm{eff}}).} \tag{12.2}$$

A_f collects path spectral responses and node overlaps, $\mathrm{Gen}_{f,j}$ distinguishes branches within a family, and v supplies the common background scale. Charged-lepton generation ratios arise from whole-group readings $\varrho_j = p_j/p_0$ of the same formation pattern. Quark hierarchies additionally contain segmented propagation, flip links, and accumulated returns. Coherent paths combine amplitudes first, whereas distinguishable outputs contribute summed shares; this distinction determines powers of spectral factors and relations across generations.

Neutrinos use indirect propagation while retaining the complete nondiagonal mass structure:

$$\boxed{m_{\nu_i} = \frac{v|L_2|^8}{4N^2}\,\varepsilon_i, \qquad \varepsilon_i \in \mathrm{spec}(C_\nu Q_\nu C_\nu^T).} \tag{12.3}$$

Here the $p = 7$ identification of Chapter 8 is retained, with $L_2 = L(1/2, \chi_2)$ and $N = 2058$. Two background matches suppress the common scale, while chain-kernel eigenvalues distinguish the three mass branches. Both transport stages, the coherent relay, nodes, and complete inverse connection form the same persistent process, with matter and background sharing the same complete-cycle timing. Proofs for the mass

matrix and all return records appear in Supplementary Proof S13. This neutral routing carries no charged frequency-shift factor, and its mass spectrum does not also determine mixing angles.

Squared-response matching of the common background relates different participating quantities to one energy-scale function:

$$\boxed{E(W) = E_0\sqrt{\frac{W}{C^*}}\log W}, \qquad C^* = 2\sum_{n=1}^{\infty}\log^2\!\left(1+\frac{1}{n}\right), \quad W > 1. \tag{12.4}$$

Elementary states use $v = E(W_{\mathrm{ref}})$, while collective states use their total participation. The nucleon baseline is $M_B = E(W_B)$, where $W_B = \zeta(1 + 1/\sqrt{3})$ retains the prescription based on the leading term of the variance. Internal phases and electromagnetic response give the neutron–proton mass difference, while the common shift remains undetermined. Collective mass thus arises from the total persistent response and cannot be directly replaced by a sum of component masses.

Condensate responses of channel bosons form a mass-squared structure:

$$\boxed{M_j^2 = v^2 C^* \frac{\varpi_j}{N_j}.} \tag{12.5}$$

ϖ_j is the channel share, and N_j the effective matching denominator. The neutral matrix's zero mode corresponds to the photon; charged and neutral massive branches give W, Z, and their mass ratio. Radial condensate oscillation gives $m_H = v\,2^{(1-m)/2}$, where $m = (p-1)/2$. The common pion mass satisfies $m_\pi^2 = 3M_B(m_u + m_d)$, linking the collective scale to internal flavor response. These masses are read through their respective persistent exchanges or response identifications and share the physical calibration E_0.

Resonance between stable states is realized by exchange of complete carriers: endpoint sources and auxiliaries are preserved, the mediator is retrieved, and target states are transferred and may become entangled. Equal-weight preparation and single-direction matching give Born-level spectral couplings. In the resonance exchange of Chapter 3, the actual response then enters the exchange angle:

$$\alpha_\chi^{(0)} = \frac{|R_\chi|^2}{N_\chi}, \qquad \boxed{\gamma = g_B\beta_B T_B, \quad P_{10\to01} = \sin^2\gamma.} \tag{12.6}$$

Here β_B is obtained from actual ports and auxiliary preparation. For the stated electromagnetic single-address effect, it equals $\alpha_{\mathrm{EM}}^{(0)}$. This action relates spectral coupling, exchange angle, and event probability while retaining them as distinct readings. Under the listed preparations and coupling structures, weak endpoint matching and background remainder follow from the same occupation output. The all-order electromagnetic response has a joint static realization, and its coupling normalization is obtained under an explicit terminal identification. Connecting that terminal identification to the original virtual-port action remains conditional. Exchange between collective states further gives emergent resonance forces.

Metric action arises from matter changing the common background. A fixed source writes a local occupation bias; background exchange carries it to other positions and changes the test carrier's maintenance action through the available share.

These stages compose within the same resource–carrier channel. The momentum change over one complete round gives the metric force:

$$\boxed{\overline{F}_{t,x} = \frac{\Delta\langle \Pi_x \rangle}{\tau}}. \tag{12.7}$$

Within controlled averaging, weak-field, low-momentum, and slowly varying regimes, the same maintenance action gives $V_t = m_t\Phi$, $\Phi = -2\xi_*$, and $F_{t,x} = -m_t\partial_x\Phi$. The inertial parameter is m_t/c_0^2, so the test mass cancels from the leading acceleration.

In the far region of the common three-dimensional nearest-neighbor positions, the Green response outside the source gives a $1/r$ potential and inverse-square force. Its coefficient is determined by source writing, return, and the complete-cycle timing. The same matching gives source phase and background occupation a common origin and makes local matter and background share a maintenance factor. Chapter 4 calculates the actual leading-order strength on this basis and adopts one physical calibration by the inverse square of the spectral scale. Finite-source-angle and field-establishment corrections continue to follow from the complete process.

The paper thereby connects formation patterns, mass frequencies, carrier exchange, and background response within one state space. Spectral inputs determine participation structures, and common scales connect masses and couplings. Physical identifications are stated when first used and collected in Appendix A.

12.2 Numerical Summary

Table 12-1 retains the process identifications of each chapter, $p = 7$ selected under the condition on the complete set of single-factor CRT labels, and physical calibration $E_0 = 1$ GeV. The two metric entries additionally use K=41 and the strength calibration of Section 4.5.5. Main dependencies are listed in Appendix A. Deviations are defined as 100×(framework value/reference central value−1) and calculated from unrounded results. Digits support reproducibility; reference definitions, scales, and uncertainties appear in the cited sections.

Constant comparisons retain CODATA 2022 [18]. The Fermi scale is calculated as in Section 6.1 using the PDG 2025 electroweak review [15]. Particle masses and the strong coupling use PDG 2025 [17], and the nuclear response uses the Roy–Steiner πNN coupling extraction [22]. Neutrino mass-squared differences retain the NuFIT 6.0 normal-ordering IC24 combination including SK atmospheric data from Section 8.6. The table lists reference central values; uncertainties and definitions appear in the indicated sections, with charged elementary-state mass uncertainties and schemes collected in Section 8.6.1.

Table 12-1 Evaluations under the specified processes and physical identifications.

Quantity (unit)	Calculated value	Reference definition and central value	Deviation (%)	Section
Energy scales				
v (GeV)	246.191374	246.219672 (Fermi)	-0.011493	6.1
M_{met}(GeV)	$1.22144637 \times 10^{19}$	$1.22089013 \times 10^{19}$	+0.045560	6.2

Quantity (unit)	Calculated value	Reference definition and central value	Deviation (%)	Section
M_B(MeV)	936.669313	Collective baseline; δ_0 undetermined	—	6.3, 9.3
Couplings and responses				
$\alpha_{\mathrm{EM}}^{-1}$	137.036359034	137.035999177 (low energy)	+0.0002626	7.4
α_W^{-1}	29.5988940685	Scheme and scale to be fixed	—	7.4
α_s	0.117788996	0.1180 (M$\bar{S}$, M_Z)	-0.178817	7.4
$\alpha_{\mathrm{nuclear}}$	14.0972078715	13.7 (πNN)	+2.899328	7.5
$G_E(\mathrm{GeV}^{-2})$	$6.70272181 \times 10^{-39}$	$6.70883075 \times 10^{-39}$	-0.091058	7.6
Charged elementary-state masses (MeV)				
e	0.5110122512	0.51099895000	+0.002603	8.2
μ	105.6553711	105.6583755	-0.002844	8.2
τ	1776.926974	1776.93	-0.0001703	8.2
d	4.651213747	4.70	-1.038005	8.3
s	92.10919837	93.5	-1.487488	8.3
b	4185.420021	4183	+0.057854	8.3
u	2.154453597	2.16	-0.256778	8.4
c	1262.842920	1273.0	-0.797885	8.4
t	173055.3957	172560	+0.287086	8.4
Neutrino masses and mass-squared differences				
m_{ν_1}(meV)	5.070653857	No absolute-mass central value adopted	—	8.5
m_{ν_2}(meV)	9.961106144	No absolute-mass central value adopted	—	8.5
m_{ν_3}(meV)	50.63175635	No absolute-mass central value adopted	—	8.5
$\sum m_\nu$(meV)	65.66351635	No total-mass reference adopted	—	8.5
$\Delta m_{21}^2(\mathrm{eV}^2)$	$7.351210508 \times 10^{-5}$	7.49×10^{-5}	-1.852997	8.6
$\Delta m_{31}^2(\mathrm{eV}^2)$	$2.537863220 \times 10^{-3}$	2.513×10^{-3}	+0.989384	8.6
Nucleon mass difference and boson masses				
$m_n - m_p$(MeV)	1.303630499	1.29333251	+0.796237	9.4
M_W(GeV)	80.57330916	80.3692	+0.253964	10.3

Quantity (unit)	Calculated value	Reference definition and central value	Deviation (%)	Section
M_Z(GeV)	91.41480139	91.1880	+0.248718	10.3
Common m_π(MeV)	138.2894763	139.57039($\pi^\pm$)	-0.917755	10.5
Same common m_π(MeV)	138.2894763	134.9768 (π^0)	+2.454256	10.5
m_H(GeV)	123.0956870	125.20	-1.680761	10.6

Under the table's comparison definitions and scales, deviations from central values are below 1.49% for all nine charged elementary-state masses and below 0.003% for the three charged leptons. W and Z deviations are approximately 0.25%, and the Higgs deviation approximately 1.68%. The common Fermi scale differs by approximately 0.0115%, the metric energy scale by 0.0456%, and the nuclear response from the selected πNN central value by 2.90%. Together these quantities constrain spectral shapes, channel weights, and the common energy scale; deviations must still be assessed with their respective uncertainties and definitions.

Matching model frequencies to experimental mass definitions, renormalization schemes, and scales remains to be tested as in Sections 8.6.1 and 10.3.3.

M_{met} and G_E are two expressions of the same comparison, and the two common-pion-mass rows are likewise not independent. The nucleon baseline still has an undetermined common shift δ_0 and is therefore not treated as the complete proton mass. Current deviations of the inverse electromagnetic coupling, electron, muon, W, Z, and Higgs masses, Fermi scale, and Newton constant exceed their respective cited experimental uncertainties. Theoretical errors have not been uniformly quantified. These central-value differences can neither be combined into a joint goodness-of-fit measure nor regarded as already covered by theoretical errors. Under the corresponding physical identifications, the present values do not yet reach the precision of those measurements.

12.3 Testable Relations

Mass ratios primarily constrain patterns, paths, and nodes, while absolute values additionally test background calibration. Separation, decay, and additional modes test the outputs and coverage of complete processes. The following relations retain their chapters' process conditions, and the same data are not counted repeatedly as independent evidence.

12.3.1 Mass Spectra and Mixing

With the common phase and positive-amplitude branch of Section 8.2, charged leptons satisfy

$$\boxed{\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}}. \tag{12.8}$$

The cross-family ratios $m_b/m_\tau = 2.35542601468$, $m_c/m_\tau = 0.710689261785$, and $m_t m_u/m_c^2 = 0.233788757123$ also cancel the common energy scale and test

whether different paths can hold simultaneously. Quark comparisons require the common mass schemes of Section 8.6. Down-type links give $\lambda = 0.224714785\ldots$ and $m_d/m_s = \lambda^2$. Using them to test mixing additionally requires connection to all entrances, phases, and outputs of the actual mixing matrix.

12.3.2 Neutrino Spectrum

The selected matrix and normal-ordering identification of Section 8.5 give the three positive spectral values in Table 12-1. The mass-squared-difference ratio 34.52306552 contains no common scale and tests relative spacings. Absolute- and total-mass experiments additionally depend on the corresponding mixing weights or cosmological identifications. Normal ordering is the physical identification currently adopted; numbering eigenvalues does not exclude another oscillation identification. The two mass-squared differences and their ratio arise from the same spectrum and are not three independent measurements.

12.3.3 The W–Z Mass Relation

The common condensate share cancels the energy scale and squared background quantity from the ratio:

$$\begin{aligned} \frac{M_Z}{M_W} &= \sqrt{\frac{N_W}{N_Z}} = 1.134554387139\ldots, \\ \sin^2\theta_{\text{mass}} &= 1 - \frac{N_Z}{N_W} = 0.223128193000\ldots. \end{aligned} \tag{12.9}$$

The vertex-side $\sin^2\theta_{\text{vertex}} = 0.231464478899$ uses a different reading definition and must be compared in a compatible experimental scheme. Using the experimental $M_Z = 91.1880$ GeV as input gives $M_W = 80.37340566$ GeV from the mass ratio. This tests the ratio and does not replace the absolute evaluations in Table 12-1.

12.3.4 Collective Mass Relations

The common pattern and internal representations of Chapter 9 allow different charges and mass differences at the same collective scale. The nucleon difference cancels the undetermined common shift and tests internal phases and electromagnetic response. The relation $m_\pi^2 = 3M_B(m_u + m_d)$ of Chapter 10 connects collective and elementary-state readings. These results do not yet determine the complete common nucleon mass or electromagnetic pion splitting.

12.3.5 Condensate Oscillations

$m_H/v = 1/2$ is a conditional result under the coherent bright-mode return, strength from a common source, and common calibration selected in Section 10.6.3. Fixing dimensionless spectral data and using the same canonically normalized radial perturbation $v \mapsto v + h$ also gives mass responses $\partial_h m_f = m_f/v$ and $\partial_h M_V^2 = 2M_V^2/v$, both evaluated at $h = 0$. Comparisons with Higgs production and decay require these responses to enter actual transition vertices while preserving the same field normalization.

12.3.6 Stability and Mode Coverage

Stability is tested through actual exit and separation processes. Section 9.6.1 tracks complete separation outputs of colored objects, subsequent maintenance, and environmental compensation, while proton stability depends on every exit channel acting on that state. Additional generations, interactions, or Higgs-like oscillations are assessed against the coverage of Section 11.5. Experimental discovery of such additional objects would test these physical identifications.

12.4 Closing Remarks

This paper connects the persistence of matter with its interactions. Stable states form through exchange with the background and sustain themselves through subsequent propagation and return. Their masses and modes of participation in interactions are studied on these persistently evolving objects. Matter's structure, mass, and forces thereby acquire a common dynamical description, with mass-spectrum and coupling calculations giving this connection quantitative content. Internal references and records further connect these processes to observational spacetime, allowing matter, the background, and their observable changes to be described within one state space.

This common description rests on three basic relations: excitations are countable, energies of independent systems add, and independent states compose multiplicatively. Combining countable objects corresponds to addition, independent pairing to multiplication, and distributivity expresses compatibility of these two forms of composition. Unique factorization of natural numbers and the logarithmic energy spectrum further connect compositional structure with energy:

$$n = \prod_p p^{a_p}, \qquad H = \log n = \sum_p a_p \log p. \tag{12.10}$$

The Bost–Connes system combines this multiplicative structure with time evolution, while finite p-adic reduction provides a concrete state space for excitations, propagation, and exchange. Countability, additivity, and multiplicative composition thus run through the combination of excitations and their energy description, supplying a common arithmetic foundation for this paper's connection of matter, forces, and spacetime.

Appendix A. Index of Constructions and Dependencies

This table retains Q01–Q22 as dependency identifiers throughout the paper and summarizes the constructions and physical identifications used. The identifiers do not count independent parameters or unproved propositions; conditions are stated where each item is used. A complete "adopted construction–result–identification–scope" index appears in the accompanying Supplementary Proofs, S0, with detailed proofs in S1–S14. Both documents use the same companion version.

Masses scaled by v retain Q06 and Q07; use of corrected couplings inherits the corresponding conditions of Q10. Actual force strengths follow from the source–background–test process in Section 4.5. Complete time accounting and spectral matching for the active-refresh process appear in Supplementary Proof S9; common-source configurations, shared carriers, and spatial outward and return paths in S6.4, S7, S10; and matching normalization of the output state in S8. Assumptions of different thermal and controlled cycles are used separately, as are the definitions of spectral responses, event shares, and positive thermal weights.

A.1 Objects and Spectral Inputs

Item	Construction and identification used	Main text and supplementary proofs
Q01	Single-factor label selection and identification of three groups with three generations	5.2.1
Q02	Correspondence between species-space representations and actual matter representations	5.2.2–5.2.3, 9.2
Q03	Distinction between the analytic spectral reference, positive-spectrum preparation, and shell thermal preparation	7.1; S5
Q04	Independent equal-weight labels, matching effects, and actual thermal shares	3.3, 7.2; S5.1

A.2 Formation and Common Readings

Item	Construction and identification used	Main text and supplementary proofs
Q12	Coherent delivery, actual return, and unit spectral-kernel identification	2.3.2–2.3.4, 2.4.5, 8.1; S2
Q05	Complete preparation and thermal interface; within reflection-based two-arm actions with nonzero cross-block coupling, preservation at all times implies equal arms	2.3.5–2.4.7; S1.5, S3; Supplementary Proof S7

Item	Construction and identification used	Main text and supplementary proofs
Q06	Common reference and one squared matching; complete transport implies a common multiplier, a common period implies an inverse-frequency ratio, and equal fully gated windows constrain accumulation	2.5, 4.5.5, Chapter 6, 9.3.2, 10.6.3; S6.3, S9.6; Supplementary Proofs S7, S10

A.3 Participation and Resonance Responses

Item	Construction and identification used	Main text and supplementary proofs
Q07	Four-layer participation, role embedding, spectral control, and matching preservation	6.1, 7.3; S6.3; Supplementary Proof S6.1
Q08	Formation of configurations from a common source and complete matching; identifications of participating configurations, layer weights, and spectral calibration retained, with the actual force obtained from the same procedure	6.2, 7.6, 4.5; S6, S9; Supplementary Proofs S6.4, S7, S10
Q09	Identification of the lowest color singlet with the spectral preparation based on the leading term of the variance	6.3
Q10	Identification of the actual terminal stage with the original virtual process; exit balance under a fixed auxiliary preparation implies finite normalization	7.3–7.4; Supplementary Proof S8
Q11	Orthogonal complete outputs of nuclear response, per-route weights, and matching shares	7.5, 9.5.2

A.4 Elementary-State Mass Spectrum

Item	Construction and identification used	Main text and supplementary proofs
Q13	Conjugate lepton spectral paths, coherent nodes, and a reference from the same source	2.4, 2.5.3, 8.2
Q14	Path readouts of down-type branches and the finite-amplitude prescription	8.3
Q15	Finite up-type loading, persistent frequencies, and total elapsed duration	8.4; Supplementary Proof S12
Q16	Two neutral filtering stages, nodes, and ports; the complete round trip and common calibration give the shared scale	2.5.3, 8.5; S13
Q17	Odd-sector transport and three-chain connections; the persistent generator gives a linear mass spectrum	8.5–8.6; S13

A.5 Collective States and Physical Comparisons

Item	Construction and identification used	Main text and supplementary proofs
Q18	Nucleon phase prescription, electromagnetic allocation, and common energy reference	9.4; Supplementary Proof S14
Q19	Separately calibrated weak vertices, complete returns, and collinear neutral outputs	10.1.3, 10.2.1, 10.3, 10.6.3; Supplementary Proof S4.4
Q20	Actual product-response columns, reversible routing through a common entrance, and mass identification of the complete effective drive	10.5.2; Supplementary Proof S4.4
Q21	Independent checks, purified return, and calibration of the common effective drive rate	10.6
Q22	Scales, mass definitions, data correlations, and actual strength matching for the complete schedule	4.5, 7.4–7.6, 8.6, 10.3.3, 12.2; S9

The scope of these items is fixed by their corresponding propositions; this index does not replace the spectral-coverage and exclusion tests of Section 11.5.

References

Online sources accessed and checked on 2026-09-27 and 2026-10-04.

[1] J.-B. Bost and A. Connes. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. Selecta Mathematica (New Series) 1, 411–457 (1995). DOI:10.1007/BF01589495.

[2] Shao-Wu Hu. Dynamics of the Bost–Connes State Space. Version v1. Zenodo preprint, 2026-09-22. Version DOI: 10.5281/zenodo.22896932.

[3] F. Q. Gouvêa. p-adic Numbers: An Introduction. 3rd ed. Universitext. Springer, Cham (2020). DOI:10.1007/978-3-030-47295-5.

[4] J.-P. Serre. Local Fields. Graduate Texts in Mathematics 67. Springer-Verlag, New York (1979). DOI:10.1007/978-1-4757-5673-9.

[5] A. Terras. Fourier Analysis on Finite Groups and Applications. London Mathematical Society Student Texts 43. Cambridge University Press (1999). DOI:10.1017/CBO9780511626265.

[6] T. M. Apostol. Introduction to Analytic Number Theory. Undergraduate Texts in Mathematics. Springer-Verlag, New York (1976). DOI:10.1007/978-1-4757-5579-4.

[7] L. Bruneau, A. Joye and M. Merkli. Asymptotics of repeated interaction quantum systems. Journal of Functional Analysis 239(1), 310–344 (2006). DOI:10.1016/j.jfa.2006.02.006.

[8] W. F. Stinespring. Positive functions on C*-algebras. Proceedings of the American Mathematical Society 6(2), 211–216 (1955). DOI:10.1090/S0002-9939-1955-0069403-4.

[9] B.-G. Englert. Fringe visibility and which-way information: An inequality. Physical Review Letters 77(11), 2154–2157 (1996). DOI:10.1103/PhysRevLett.77.2154.

[10] E. B. Davies and J. T. Lewis. An operational approach to quantum probability. Communications in Mathematical Physics 17, 239–260 (1970). DOI:10.1007/BF01647093.

[11] K. Kraus. General state changes in quantum theory. Annals of Physics 64(2), 311–335 (1971). DOI:10.1016/0003-4916(71)90108-4.

[12] M.-D. Choi. Completely positive linear maps on complex matrices. Linear Algebra and its Applications 10(3), 285–290 (1975). DOI:10.1016/0024-3795(75)90075-0.

[13] R. Haag, N. M. Hugenholtz and M. Winnink. On the equilibrium states in quantum statistical mechanics. Communications in Mathematical Physics 5, 215–236 (1967). DOI:10.1007/BF01646342.

[14] J. L. Alperin. Local Representation Theory: Modular Representations as an Introduction to the Local Representation Theory of Finite Groups. Cambridge Studies in Advanced Mathematics 11. Cambridge University Press (1986). DOI:10.1017/CBO9780511623592.

[15] J. de Blas, S. Dittmaier and R. Kogler. Electroweak Model and Constraints on New Physics. Review of Particle Physics, 2025 update, revised November 2025. The Fermi constant and associated electroweak definitions are from Section 10.3.2, Eq. (10.24). Topical review.

[16] E. Majorana. Teoria simmetrica dell'elettrone e del positrone. Il Nuovo Cimento 14, 171–184 (1937). DOI:10.1007/BF02961314.

[17] S. Navas et al. (Particle Data Group). Review of Particle Physics. Physical Review D 110, 030001 (2024), including the 2025 update. DOI:10.1103/PhysRevD.110.030001; Lepton summary; Quark summary; Gauge and Higgs boson summary; Meson summary; Detailed W listing; Detailed Z listing; Summary-table index.

[18] P. J. Mohr, D. B. Newell, B. N. Taylor and E. Tiesinga. CODATA recommended values of the fundamental physical constants: 2022. Reviews of Modern Physics 97, 025002

(2025). Numerical values use the complete NIST table of the 2022 adjustment. DOI:10.1103/RevModPhys.97.025002; Complete table of constants.

[19] A. Connes and M. Marcolli. Noncommutative Geometry, Quantum Fields and Motives. American Mathematical Society Colloquium Publications 55. American Mathematical Society, Providence, RI; Hindustan Book Agency, New Delhi (2008). DOI:10.1090/coll/055.

[20] F. W. J. Olver et al., eds. NIST Digital Library of Mathematical Functions. Version 1.2.8, 2026-09-15. The Hurwitz ζ expansion and Dirichlet L functional equation are from Section 25.15. DLMF; Eq. 25.15.3; Eq. 25.15.5.

[21] J. Huston, K. Rabbertz and G. Zanderighi. Quantum Chromodynamics. Review of Particle Physics, 2025 update, revised August 2025. The strong-coupling average and scheme are from Section 9.4.8, Eq. (9.25). Topical review.

[22] M. Hoferichter, J. Ruiz de Elvira, B. Kubis and U.-G. Meißner. Roy–Steiner-equation analysis of pion–nucleon scattering. Physics Reports 625, 1–88 (2016). Section 3.2 gives the GMO coupling input used here. DOI:10.1016/j.physrep.2016.02.002; arXiv:1510.06039.

[23] K. Ireland and M. Rosen. A Classical Introduction to Modern Number Theory. 2nd ed. Graduate Texts in Mathematics 84. Springer-Verlag, New York (1990). DOI:10.1007/978-1-4757-2103-4.

[24] F. Canelli, A. Knue and F. Maltoni. Top Quark. Review of Particle Physics, 2025 update, revised September 2025. Section 61.1.1 discusses the relation between directly reconstructed and field-theoretic mass definitions. Topical review.

[25] Y. Koide. Fermion-boson two-body model of quarks and leptons and Cabibbo mixing. Lettere al Nuovo Cimento 34, 201–205 (1982). DOI:10.1007/BF02817096.

[26] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro and T. Schwetz. NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations. Journal of High Energy Physics 12 (2024), 216. Numerical values use the NuFIT 6.0 normal-ordering combination with IC24 and SK atmospheric data. DOI:10.1007/JHEP12(2024)216; arXiv:2410.05380; NuFIT 6.0 data; Parameter table.

[27] W. Fulton and J. Harris. Representation Theory: A First Course. Graduate Texts in Mathematics 129. Springer, New York (1991). DOI:10.1007/978-1-4612-0979-9.

[28] M. Gell-Mann. A schematic model of baryons and mesons. Physics Letters 8(3), 214–215 (1964). DOI:10.1016/S0031-9163(64)92001-3.

[29] J.-P. Serre. Linear Representations of Finite Groups. Graduate Texts in Mathematics 42. Springer-Verlag, New York (1977). DOI:10.1007/978-1-4684-9458-7.

[30] F. Englert and R. Brout. Broken Symmetry and the Mass of Gauge Vector Mesons. Physical Review Letters 13, 321–323 (1964). DOI:10.1103/PhysRevLett.13.321.

[31] P. W. Higgs. Broken Symmetries and the Masses of Gauge Bosons. Physical Review Letters 13, 508–509 (1964). DOI:10.1103/PhysRevLett.13.508.